

**Mathematical Programming and
Operations Research**
Modeling, Algorithms, and Complexity
Examples in Excel and Python
(Work in progress)

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Preface

This entire book is a working manuscript. The first draft of the book is yet to be completed.

This book is being written and compiled using a number of open source materials. We will strive to properly cite all resources used and give references on where to find these resources. Although the material used in this book comes from a variety of licences, everything used here will be CC-BY-SA 4.0 compatible, and hence, the entire book will fall under a CC-BY-SA 4.0 license.

MAJOR ACKNOWLEDGEMENTS

I would like to acknowledge that substantial parts of this book were borrowed under a CC-BY-SA license. These substantial pieces include:

- "A First Course in Linear Algebra" by Lyryx Learning (based on original text by Ken Kuttler). A majority of their formatting was used along with selected sections that make up the appendix sections on linear algebra. We are extremely grateful to Lyryx for sharing their files with us. They do an amazing job compiling their books and the templates and formatting that we have borrowed here clearly took a lot of work to set up. Thank you for sharing all of this material to make structuring and forming this book much easier! See subsequent page for list of contributors.
- "Foundations of Applied Mathematics" with many contributors. See <https://github.com/Foundations-of-Applied-Mathematics>. Several sections from these notes were used along with some formatting. Some of this content has been edited or rearranged to suit the needs of this book. This content comes with some great references to code and nice formatting to present code within the book. See subsequent page with list of contributors.
- "Linear Inequalities and Linear Programming" by Kevin Cheung. See <https://github.com/dataopt/lineqlpbook>. These notes are posted on GitHub in a ".Rmd" format for nice reading online. This content was converted to $\text{\LaTeX}$ using Pandoc. These notes make up a substantial section of the Linear Programming part of this book.
- Linear Programming notes by Douglas Bish. These notes also make up a substantial section of the Linear Programming part of this book.

I would also like to acknowledge Laurent Porrier and Diego Moran for contributing various notes on linear and integer programming.

I would also like to thank Jamie Fravel for helping to edit this book and for contributing chapters, examples, and code.

Contents

Introduction

Letter to instructors

This is an introductory book for students to learn optimization theory, tools, and applications. The two main goals are (1) students are able to apply the of optimization in their future work or research (2) students understand the concepts underlying optimization solver and use this knowledge to use solvers effectively.

This book was based on a sequence of course at Virginia Tech in the Industrial and Systems Engineering Department. The courses are *Deterministic Operations Research I* and *Deterministic Operations Reserach II*. The first course focuses on linear programming, while the second course covers integer programming an nonlinear programming. As such, the content in this book is meant to cover 2 or more full courses in optimization.

The book is designed to be read in a linear fashion. That said, many of the chapters are mostly independent and could be rearranged, removed, or shortened as desited. This is an open source textbook, so use it as you like. You are encouraged to share adaptations and improvements so that this work can evolve over time.

Letter to students

This book is designed to be a resource for students interested in learning what optimizaiton is, how it works, and how you can apply it in your future career. The main focus is being able to apply the techniques of optimization to problems using computer technology while understanding (at least at a high level) what the computer is doing and what you can claim about the output from the computer.

Quite importantly, keep in mind that when someone claims to have *optimized* a problem, we want to know what kind of gaurantees they have about how good their solution is. Far too often, the solution that is provided is suboptimal by 10%, 20%, or even more. This can mean spending excess amounts of money, time, or energy that could have been saved. And when problems are at a large scale, this can easily result in millions of dollars in savings.

For this reason, we will learn the perspective of *mathematical programming* (a.k.a. mathematical optimization). The key to this study is that we provide gaurantees on how good a solution is to a given problem. We will also study how difficult a problem is to solve. This will help us know (a) how long it might take to solve it and (b) how good a of a solution we might expect to be able to find in reasonable amount of time.

We will later study *heuristic* methods - these methods typically do not come with gaurantees, but tend to help find quality solutions.

Note: Although there is some computer programming required in this work, this is not a course on programming. Thanks to the fantastic modelling packages available these days, we are able to solve complicated problems with little programming effort. The key skill we will need to learn is *mathematical modeling*: converting words and ideas into numbers and variables in order to communicate problems to a computer so that it can solve a problem for you.

As a main element of this book, we would like to make the process of using code and software as easy as possible. Attached to most examples in the book, there will be links to code that implements and solves the problem using several different tools from Excel and Python. These examples can should make it easy to solve a similar problem with different data, or more generally, can serve as a basis for solving related problems with similar structure.

How to use this book

Skim ahead. We recommend that before you come across a topic in lecture, that you skim the relevant sections ahead of time to get a broad overview of what is to come. This may take only a fraction of the time that it may take for you to read it.

Read the expected outcomes. At the beginning of each section, there will be a list of expected outcomes from that section. Review these outcomes before reading the section to help guide you through what is most relevant for you to take away from the material. This will also provide a brief look into what is to follow in that section.

Read the text. Read carefully the text to understand the problems and techniques. We will try to provide a plethora of examples, therefore, depending on your understanding of a topic, you many need to go carefully over all of the examples.

Explore the resources. Lastly, we recognize that there are many alternative methods of learning given the massive amounts of information and resources online. Thus, at the end of each section, there will be a number of superb resources that available on the internet in different formats. There are other free textbooks, informational websites, and also number of fantastic videos posted to youtube. We encourage to explore the resources to get another perspective on the material or to hear/read it taught from a differnt point of view or in presentation style.

Outline of this book

This book is divided in to 4 Parts:

Part I Linear Programming,

Part II Integer Programming,

Part III Discrete Algorithms,

Part IV Nonlinear Programming.

There are also a number of chapters of background material in the Appendix.

The content of this book is designed to encompass 2-3 full semester courses in an industrial engineering department.

Work in progress

This book is still a work in progress, so please feel free to send feedback, questions, comments, and edits to Robert Hildebrand at open.optimization@gmail.com.

1. Resources and Notation

Here are a list of resources that may be useful as alternative references or additional references.

FREE NOTES AND TEXTBOOKS

- Mathematical Programming with Julia by Richard Lusby & Thomas Stidsen
- Linear Programming by K.J. Mtetwa, David
- A first course in optimization by Jon Lee
- Introduction to Optimizaiton Notes by Komei Fukuda
- Convex Optimization by Bord and Vandenberghe
- LP notes of Michel Goemans from MIT
- Understanding and Using Linear Programming - Matousek and Gärtner [Downloadable from Springer with University account]
- Operations Research Problems Statements and Solutions - Raúl PolerJosefa Mula Manuel Díaz-Madroñero [Downloadable from Springer with University account]

NOTES, BOOKS, AND VIDEOS BY VARIOUS SOLVER GROUPS

- AIMMS Optimization Modeling
- Optimization Modeling with LINGO by Linus Schrage
- The AMPL Book
- Microsoft Excel 2019 Data Analysis and Business Modeling, Sixth Edition, by Wayne Winston - Available to read for free as an e-book through Virginia Tech library at Orielly.com.
- Lesson files for the Winston Book
- Video instructions for solver and an example workbook
- youtube-OR-course

GUROBI LINKS

- Go to <https://github.com/Gurobi> and download the example files.
- Essential ingredients
- Gurobi Linear Programming tutorial
- Gurobi tutorial MILP
- GUROBI - Python 1 - Modeling with GUROBI in Python
- GUROBI - Python II: Advanced Algebraic Modeling with Python and Gurobi
- GUROBI - Python III: Optimization and Heuristics
- Webinar Materials
- GUROBI Tutorials

HOW TO PROVE THINGS

- Hammack - Book of Proof

STATISTICS

- Open Stax - Introductory Statistics

LINEAR ALGEBRA

- Beezer - A first course in linear algebra
- Selinger - Linear Algebra
- Cherney, Denton, Thomas, Waldron - Linear Algebra

REAL ANALYSIS

- Mathematical Analysis I by Elias Zakon

DISCRETE MATHEMATICS, GRAPHS, ALGORITHMS, AND COMBINATORICS

- Levin - Discrete Mathematics - An Open Introduction, 3rd edition
- Github - Discrete Mathematics: an Open Introduction CC BY SA
- Keller, Trotter - Applied Combinatorics (CC-BY-SA 4.0)
- Keller - Github - Applied Combinatorics

MIXED INTEGER NONLINEAR PROGRAMMING

- Mixed-Integer Nonlinear Optimization Pietro Belotti, Christian Kirches, Sven Leyffer, Jeff Linderoth, Jim Luedtke, and Ashutosh Mahajan

PROGRAMMING WITH PYTHON

- A Byte of Python
- Github - Byte of Python (CC-BY-SA)

Also, go to <https://github.com/open-optimization/open-optimization-or-examples> to look at more examples.

Notation

- $\mathbf{1}$ - a vector of all ones (the size of the vector depends on context)
- $\forall$ - for all
- $\exists$ - there exists
- $\in$ - in
- $\therefore$ - therefore
- $\Rightarrow$ - implies
- s.t. - such that (or sometimes "subject to".... from context?)
- $\{0, 1\}$ - the set of numbers 0 and 1
- $\mathbb{Z}$ - the set of integers (e.g. $1, 2, 3, -1, -2, -3, \dots$)
- $\mathbb{Q}$ - the set of rational numbers (numbers that can be written as p/q for $p, q \in \mathbb{Z}$ (e.g. $1, 1/6, 27/2$)
- $\mathbb{R}$ - the set of all real numbers (e.g. $1, 1.5, \pi, e, -11/5$)
- $\setminus$ - setminus, (e.g. $\{0, 1, 2, 3\} \setminus \{0, 3\} = \{1, 2\}$)
- $\cup$ - union (e.g. $\{1, 2\} \cup \{3, 5\} = \{1, 2, 3, 5\}$)

- $\cap$ - intersection (e.g. $\{1, 2, 3, 4\} \cap \{3, 4, 5, 6\} = \{3, 4\}$)
- $\{0, 1\}^4$ - the set of 4 dimensional vectors taking values 0 or 1, (e.g. $[0, 0, 1, 0]$ or $[1, 1, 1, 1]$)
- $\mathbb{Z}^4$ - the set of 4 dimensional vectors taking integer values (e.g., $[1, -5, 17, 3]$ or $[6, 2, -3, -11]$)
- $\mathbb{Q}^4$ - the set of 4 dimensional vectors taking rational values (e.g. $[1.5, 3.4, -2.4, 2]$)
- $\mathbb{R}^4$ - the set of 4 dimensional vectors taking real values (e.g. $[3, \pi, -e, \sqrt{2}]$)
- $\sum_{i=1}^4 i = 1 + 2 + 3 + 4$
- $\sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2$
- $\sum_{i=1}^4 x_i = x_1 + x_2 + x_3 + x_4$
- $\square$ - this is a typical Q.E.D. symbol that you put at the end of a proof meaning "I proved it."
- For $x, y \in \mathbb{R}^3$, the following are equivalent (note, in other contexts, these notations can mean different things)

– $x^\top y$ *matrix multiplication*

– $x \cdot y$ *dot product*

– $\langle x, y \rangle$ *inner product*

and evaluate to $\sum_{i=1}^3 x_i y_i = x_1 y_1 + x_2 y_2 + x_3 y_3$.

A sample sentence:

$$\forall x \in \mathbb{Q}^n \exists y \in \mathbb{Z}^n \setminus \{0\}^n \text{ s.t. } x^\top y \in \{0, 1\}$$

"For all non-zero rational vectors x in n -dimensions, there exists a non-zero n -dimensional integer vector y such that the dot product of x with y evaluates to either 0 or 1."

2. Mathematical Programming

Outcomes

- *Consider an introductory problem*
- *Identify reasons for studying operations research*
- *Define "Mathematical Programming"*
- *Learn about different applications of the tools in this book*
- *Explore the different types of optimization models and what types we will see in this book*

2.1 Warm-Up Scenario

Imagine you're starting a new clothing company and decide to sell shirts. You've opened a store in Roanoke and, based on market research, you estimate the demand for the upcoming weekend:

- **Friday:** 5 shirts
- **Saturday:** 10 shirts
- **Sunday:** 7 shirts

Your supplier can deliver shirts in one day, meaning an order placed on Thursday will arrive on Friday, and so on. As a business owner, you need to decide how many shirts to order and when to place each order to ensure you meet the daily demand without overstocking.

*Defining the Problem To tackle this challenge, you'll need to think strategically about two main factors:

- **Ordering Quantities:** How many shirts to order on each day.
- **Inventory Levels:** How many shirts to keep in stock at the end of each day.

You can define these decisions using variables:

- Let the number of shirts you order on a specific day be a decision variable. For example, you could track the number of shirts ordered on Thursday, Friday, and Saturday.
- Define variables to represent the number of shirts you have in stock at the end of each day.

Your goal is to meet the projected demand while managing costs. Costs could include:

- The expense of ordering shirts.

- The cost of holding inventory.
- Penalties for running out of stock.

Refining the Scenario

As your business grows, additional complexities may arise. For instance:

- The cost of ordering shirts might vary by day due to supplier pricing.
- Holding inventory could incur storage costs.
- A budget might limit how much you can spend on orders.
- You might expand your product line to include pants and socks, each with its own demand and costs.

By clearly defining variables and constraints for these decisions, you can systematically approach the problem and develop an optimal strategy. This step-by-step process of defining the problem, incorporating real-world details, and refining the solution is a fundamental aspect of optimization.

Model Development Cycle

This multi-stage example illustrates the iterative cycle of modeling:

1. **Initial Problem:** Determine ordering quantities to meet demand.
2. **Modeling:** Formulate an optimization model based on available information.
3. **Solution:** Solve the model to find the optimal ordering plan.
4. **New Information:** Incorporate additional real-world complexities (varying costs, inventory costs, budget constraints, additional products, fixed ordering costs).
5. **Refinement:** Update the model to reflect new information and solve again.

Further Extensions

Looking ahead, there are several ways to expand this scenario:

- **Expanding to New Stores:** Opening additional stores introduces location-based demand, transportation costs, and coordination of inventory across multiple sites.
- **Hiring New Staff:** Incorporate labor costs, scheduling constraints, and productivity rates into the model.
- **Producing Clothing In-House:** Transitioning to manufacturing involves modeling production processes, including purchasing raw materials, machine capacities, and processing times.
- **Uncertain Demand:** In reality, demand may be uncertain. Advanced models can incorporate stochastic elements or use probabilistic methods to handle randomness.

In this course, we will focus on deterministic models where all data is known and fixed. Building a strong foundation with these models will prepare us to tackle more complex problems involving uncertainty and randomness in future studies.

2.2 Why Study Operations Research?

Operations Research (OR) is a discipline that applies advanced analytical methods to help make better decisions. In a world where resources are limited and organizations face complex challenges, OR provides the tools and methodologies to model, analyze, and solve problems involving the optimal allocation of scarce resources. Studying OR equips individuals with a systematic and quantitative approach to decision-making, which is invaluable in industries such as logistics, finance, healthcare, and manufacturing.

By understanding OR, you gain the ability to construct mathematical models of real-world systems, allowing for the simulation and optimization of different scenarios. This leads to improved efficiency, reduced costs, and enhanced performance. Moreover, the skills developed through studying OR—such as critical thinking, problem-solving, and data analysis—are highly transferable and sought after in today's data-driven economy.

Operations Research also bridges the gap between theory and practice. It transforms abstract mathematical concepts into actionable strategies that can have a tangible impact on organizational success. Whether it's optimizing supply chains, scheduling flights, or managing portfolios, the applications of OR are vast and impactful.

2.3 What is Mathematical Programming?

Mathematical Programming is a core area within Operations Research that focuses on the formulation and solution of optimization problems. It involves creating mathematical models that represent the decision-making process of a system, where the goal is to find the best possible outcome under given constraints. The term "programming" in this context refers to planning or scheduling, not computer programming.

At the heart of Mathematical Programming is the objective function—a mathematical expression that defines the criterion to be optimized, such as minimizing costs or maximizing profits. The decision variables represent the choices available, and the constraints reflect the limitations or requirements of the problem, such as resource capacities or regulatory conditions.

There are various types of mathematical programming, each suited to different kinds of problems:

- **Linear Programming (LP):** Deals with problems where both the objective function and constraints are linear.
- **Integer Programming (IP):** Involves decision variables that must take on integer values.

- **Nonlinear Programming (NLP):** Handles problems with nonlinear objective functions or constraints.
- **Dynamic Programming (DP):** Breaks down problems into simpler subproblems and is particularly useful for multistage decision processes.

Mathematical Programming provides a powerful framework for optimizing complex systems and making informed decisions. It combines elements of mathematics, economics, and computer science to develop solutions that are both efficient and effective.

2.4 Applications

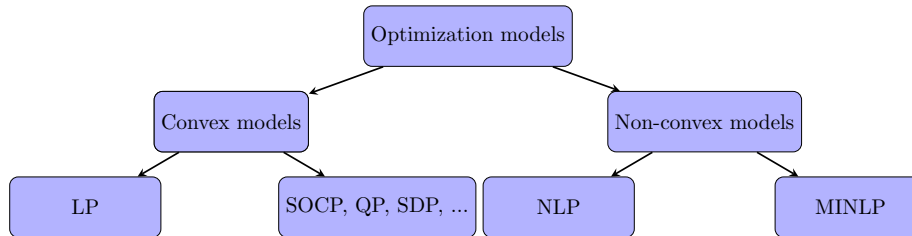
Operations Research and Mathematical Programming have a wide range of applications across various industries and sectors:

- **Transportation and Logistics:** Optimizing routing and scheduling for delivery vehicles, reducing transportation costs, and improving supply chain efficiency.
- **Manufacturing:** Streamlining production processes, managing inventory levels, and scheduling machinery to maximize productivity and minimize waste.
- **Finance:** Portfolio optimization, risk assessment, and capital budgeting to enhance financial performance and manage uncertainties.
- **Healthcare:** Allocating resources such as staff and equipment, scheduling surgeries, and managing patient flow to improve service quality and reduce wait times.
- **Energy Sector:** Optimizing power generation and distribution, planning for renewable energy integration, and managing energy consumption.
- **Telecommunications:** Network design and optimization, bandwidth allocation, and improving the reliability and efficiency of communication systems.
- **Agriculture:** Planning crop rotations, optimizing the use of fertilizers and water, and managing supply chains for agricultural products.
- **Public Sector:** Urban planning, emergency response logistics, and policy analysis to enhance public services and resource utilization.

These applications demonstrate the versatility and impact of OR and Mathematical Programming. By leveraging these tools, organizations can make data-driven decisions that lead to significant improvements in performance, cost savings, and competitive advantage. As global challenges become more complex, the importance of skilled professionals in Operations Research continues to grow.

2.5 Types of Optimization problems

We will state main general problem classes to be associated with in these notes. These are Linear Programming (LP), Mixed-Integer Linear Programming (MILP), Non-Linear Programming (NLP), and Mixed-Integer Non-Linear Programming (MINLP).



Along with each problem class, we will associate a complexity class for the general version of the problem. Complexity classes are discussed in a later portion of the book. Although we will often state that input data for a problem comes from $\mathbb{R}$ (as a real number), when we discuss complexity of such a problem, we actually mean that the data is rational, i.e., from $\mathbb{Q}$ (rational numbers that are encoded in the computer typically), and is given in binary encoding.

CAUTION!

In the following subsections, we will use advanced notation that may be unfamiliar at this point. These mathematical notations will be developed throughout the book.

Furthermore, don't be too concerned about the specific definitions of the complexity classes at this point. For now, you can read that *Polynomial time (P)* means it should be relatively easy to solve, while the other classes might be much more difficult to solve.

2.6 Linear Programming (LP)

Some linear programming background, theory, and examples will be provided in ??.

Linear Programming (LP):

Polynomial time (P)

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *linear programming* problem is

$$\begin{aligned}
 \max \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b \\
 & x \geq 0
 \end{aligned} \tag{2.1}$$

Linear programming can come in several forms, whether we are maximizing or minimizing, or if the constraints are $\leq$, $=$ or $\geq$. One form commonly used is *Standard Form* given as

Linear Programming (LP) Standard Form:

Polynomial time (P)

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *linear programming* problem in *standard form* is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned} \tag{2.2}$$

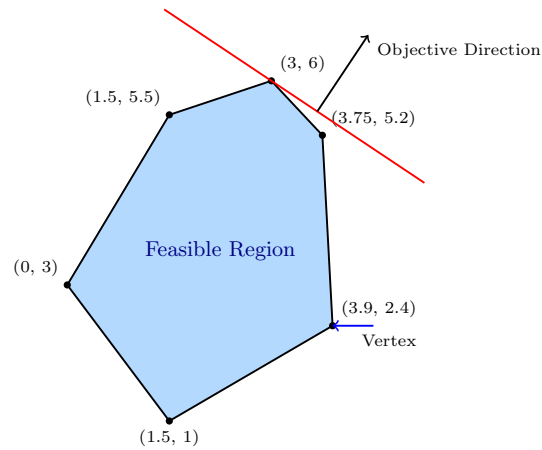


Figure 2.1: Linear programming constraints and objective.

2.7 Mixed-Integer Linear Programming (MILP)

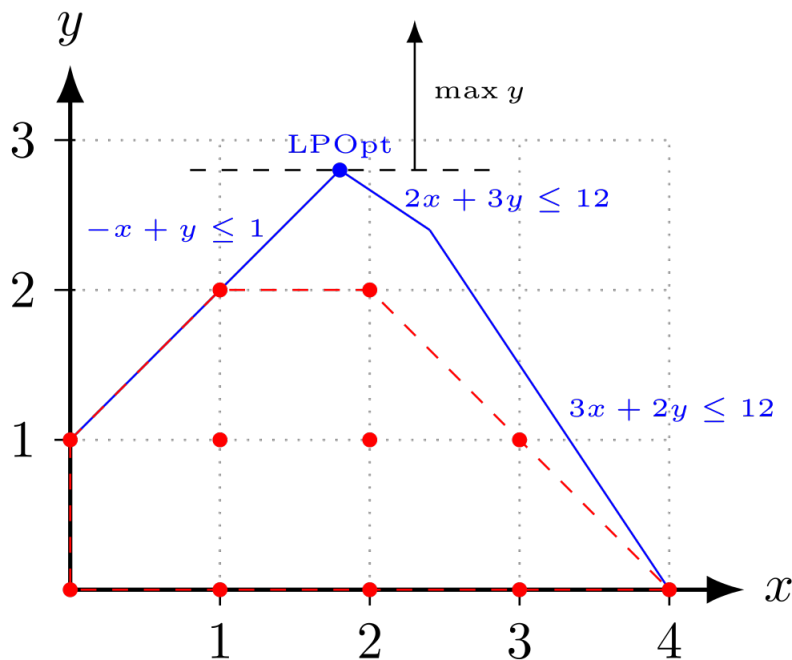
Mixed-integer linear programming will be the focus of Sections ??, ??, ??, and ??. Recall that the notation $\mathbb{Z}$ means the set of integers and the set $\mathbb{R}$ means the set of real numbers. The first problem of interest here is a *binary integer program* (BIP) where all n variables are binary (either 0 or 1).

Binary Integer programming (BIP):*NP-Complete*

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *binary integer programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \{0, 1\}^n \end{aligned} \tag{2.1}$$

A slightly more general class is the class of *Integer Linear Programs* (ILP). Often this is referred to as *Integer Program* (IP), although this term could leave open the possibility of non-linear parts.



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Figure 2.2: Comparing the LP relaxation to the IP solutions.

Figure 2.2

Integer Linear Programming (ILP):*NP-Complete*

¹IP polytope with LP relaxation, from https://en.wikipedia.org/wiki/Integer_programming#/media/File:IP_polytope_with_LP_relaxation.svg. Fanosta CC BY-SA 4.0., 2019.

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *integer linear programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{Z}^n \end{aligned} \tag{2.2}$$

An even more general class is *Mixed-Integer Linear Programming (MILP)*. This is where we have n integer variables $x_1, \dots, x_n \in \mathbb{Z}$ and d continuous variables $x_{n+1}, \dots, x_{n+d} \in \mathbb{R}$. Succinctly, we can write this as $x \in \mathbb{Z}^n \times \mathbb{R}^d$, where $\times$ stands for the *cross-product* between two spaces.

Below, the matrix A now has $n + d$ columns, that is, $A \in \mathbb{R}^{m \times (n+d)}$. Also note that we have not explicitly enforced non-negativity on the variables. If there are non-negativity restrictions, this can be assumed to be a part of the inequality description $Ax \leq b$.

Mixed-Integer Linear Programming (MILP):

NP-Complete

Given a matrix $A \in \mathbb{R}^{m \times (n+d)}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^{n+d}$, the *mixed-integer linear programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{Z}^n \times \mathbb{R}^d \end{aligned} \tag{2.3}$$

2.8 Non-Linear Programming (NLP)

NLP:

NP-Hard

Given a function $f(x): \mathbb{R}^d \rightarrow \mathbb{R}$ and other functions $f_i(x): \mathbb{R}^d \rightarrow \mathbb{R}$ for $i = 1, \dots, m$, the *nonlinear programming* problem is

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & f_i(x) \leq 0 \quad \text{for } i = 1, \dots, m \\ & x \in \mathbb{R}^d \end{aligned} \tag{2.1}$$

Nonlinear programming can be separated into convex programming and non-convex programming. These two are very different beasts and it is important to distinguish between the two.

2.8.1. Convex Programming

Here the functions are all **convex**!

Convex Programming:

Polynomial time (P) (typically)

Given a convex function $f(x): \mathbb{R}^d \rightarrow \mathbb{R}$ and convex functions $f_i(x): \mathbb{R}^d \rightarrow \mathbb{R}$ for $i = 1, \dots, m$, the *convex programming* problem is

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & f_i(x) \leq 0 \quad \text{for } i = 1, \dots, m \\ & x \in \mathbb{R}^d \end{aligned} \tag{2.2}$$

Observe that convex programming is a generalization of linear programming. This can be seen by letting $f(x) = c^\top x$ and $f_i(x) = A_i x - b_i$.

2.8.2. Non-Convex Non-linear Programming

When the function f or functions f_i are non-convex, this becomes a non-convex nonlinear programming problem. There are a few complexity issues with this.

IP AS NLP As seen above, quadratic constraints can be used to create a feasible region with discrete solutions. For example

$$x(1 - x) = 0$$

has exactly two solutions: $x = 0, x = 1$. Thus, quadratic constraints can be used to model binary constraints.

Binary Integer programming (BIP) as a NLP:

NP-Hard

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *binary integer programming* problem

is

$$\begin{aligned}
 \max \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b \\
 & \cancel{x \in \{0,1\}^n} \\
 & x_i(1-x_i) = 0 \quad \text{for } i = 1, \dots, n
 \end{aligned} \tag{2.3}$$

Alternatively, consider the transformation where $x_i \in \{-1, 1\}$.

$$\begin{aligned}
 \min \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b \\
 & x \in \{-1, 1\}^n
 \end{aligned}$$

This can be reformulated with a single nonconvex constraint as

$$\begin{aligned}
 \min \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b \\
 & -1 \leq x_j \leq 1, \quad 1 \leq j \leq n, \\
 & \|x\|^2 \geq n.
 \end{aligned}$$

2.8.3. Machine Learning

Machine learning problems are often cast as continuous optimization problems, which involve adjusting parameters to minimize or maximize a particular objective. Frequently they are convex optimization problems, but many turn out to be nonconvex. Here are two examples of how these problems arise at a glance. We will see examples in greater detail later in the book.

Loss Function Minimization

In supervised learning, this objective is typically a loss function L that quantifies the discrepancy between the predictions of a model and the true data labels. The aim is to adjust the parameters θ of the model to minimize this loss. Mathematically, this can be represented as:

$$\min_{\theta} L(\theta) = \min_{\theta} \frac{1}{N} \sum_{i=1}^N l(y_i, f(x_i; \theta)) \tag{2.4}$$

where N is the number of data points, l is a per-data-point loss (e.g., squared error for regression or cross-entropy for classification), y_i is the true label for the i -th data point, and $f(x_i; \theta)$ is the model's prediction for the i -th data point with parameters θ .

Clustering Formulation

Clustering, on the other hand, seeks to group or partition data points such that data points in the same group are more similar to each other than those in other groups. One popular method is the k-means clustering algorithm. The objective of k-means is to partition the data into k clusters by minimizing the within-cluster sum of squares (WCSS). The mathematical formulation can be given as:

$$\min_{\mathbf{c}_1, \dots, \mathbf{c}_k} \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mathbf{c}_j\|^2 \quad (2.5)$$

where C_j represents the j -th cluster and $\mathbf{c}_j$ is the centroid of that cluster.

This encapsulation presents a glimpse into how ML problems are framed mathematically. In practice, numerous algorithms, constraints, and regularizations add complexity to these basic formulations.

2.9 Mixed Integer Non-Linear Programming (MINLP)

MINLP:

NP-Hard

Mixed Integer Nonlinear Programming (MINLP) combines elements of integer programming and non-linear programming. In MINLP, some or all of the decision variables are constrained to be integers, and the objective function or constraints are nonlinear. A general MINLP problem can be formulated as:

$$\begin{aligned} \min \quad & f(x, y) \\ \text{s.t.} \quad & g_i(x, y) \leq 0 \quad \text{for } i = 1, \dots, m \\ & h_j(x, y) = 0 \quad \text{for } j = 1, \dots, p \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^k \end{aligned} \quad (2.1)$$

where f is the objective function, g_i and h_j are constraint functions, x is a vector of continuous variables, and y is a vector of integer variables.

MINLP is particularly challenging due to the nonlinearity in the objective and/or constraints and the discrete nature of some decision variables. It finds applications in various fields such as industry for process optimization, computational geometry, and machine learning for hyperparameter tuning.

2.9.1. Convex MINLP

In the convex case, both the objective function and the constraints are convex functions. This subclass is easier to solve compared to its non-convex counterpart.

Convex MINLP:

NP-Hard, but *Polynomial time (P)* in fixed dimension (typically)

For a convex MINLP, the problem is defined as:

$$\begin{aligned} \min \quad & f(x,y) \quad (\text{convex}) \\ \text{s.t.} \quad & g_i(x,y) \leq 0 \quad (\text{convex constraints}) \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^k \end{aligned} \tag{2.2}$$

Convex MINLPs, while still challenging, are typically more tractable due to the properties of convexity, which allow for more efficient solution methods.

2.9.2. Non-Convex MINLP

Non-convex MINLPs are significantly harder due to the presence of non-convex functions, which can lead to multiple local minima.

Non-Convex MINLP:

NP-Hard (in fact, undecidable)

The non-convex MINLP problem is formulated as:

$$\begin{aligned} \min \quad & f(x,y) \quad (\text{non-convex}) \\ \text{s.t.} \quad & g_i(x,y) \leq 0 \quad (\text{possibly non-convex constraints}) \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^k \end{aligned} \tag{2.3}$$

Non-convex MINLPs pose significant computational challenges due to the possibility of multiple local optima and the inherent complexity of integer constraints. These problems are common in real-world applications where decisions are discrete, and the system behavior is non-linear and complex.

Complexity and Applications

MINLP problems are known for their computational complexity, primarily due to the combination of non-linearity and integrality. The non-convex variants, in particular, are *NP-Hard*, making them some of the most challenging problems in optimization.

In practice, MINLP models find extensive applications across various domains. In industry, they are used for complex decision-making processes like supply chain optimization and production planning. In computational geometry, MINLP techniques help in solving problems like optimal packing or layout design. Furthermore, in the realm of machine learning, MINLPs are crucial for tasks like feature selection and hyperparameter optimization where discrete choices and nonlinear relationships are involved.

The versatility of MINLP models, combined with their inherent complexity, makes them a fascinating and essential area of study in the field of optimization.

Category	Software (License)
Linear programming (free)	CDD, CLP, GLPK, LPSOLVE, OSQP, SCIP
Mixed Integer Linear programming (free)	CBC, GLPK, LPSOLVE, SCIP
Linear programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), KNITRO, LINPROG, MOSEK (free for academia), XPRESS (free for academia)
Mixed Integer Linear programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), INTLINPROG, KNITRO, MOSEK (free for academia), XPRESS (free for academia)
Quadratic programming (free)	OSQP, CLP, DAQP (mixed-binary), OOQP, QPC, QPOASES, QUADPROGGBB (nonconvex QP)
Quadratic programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), KNITRO, MOSEK (free for academia), QUADPROG, XPRESS (free for academia)
Mixed Integer Quadratic programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), KNITRO, MOSEK (free for academia), XPRESS (free for academia)
Second-order cone programming (free)	ECOS, SCS, SDPT3, SEDUMI
Second-order cone programming (commercial)	COPT (free for academia), CPLEX (free for academia), CONEPROG, GUROBI (free for academia), MOSEK (free for academia), XPRESS (free for academia)
Mixed Integer Second-order cone programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), MOSEK (free for academia), XPRESS (free for academia)
Semidefinite programming (free)	CSDP, DSDP, LOGDETTPPA, PENLAB, SCS, SDPA, SDPLR, SDPT3, SDPNAL, SEDUMI
Semidefinite programming (commercial)	COPT (free for academia), LMILab (not recommended), MOSEK (free for academia), PENBMI, PENSDP (free for academia)
General nonlinear programming and other solvers	BARON, FILTERSD, FMINCON, GPPSOY, IPOPT, KNITRO, LMIRank, MPT, NOMAD, PENLAB, SNOPT, SPARSEPOP

Table 2.1: Optimization Software and their Licenses

2.10 Optimization Under Uncertainty

Optimization problems often involve uncertainty in their parameters, whether due to measurement errors, variability in input data, or unpredictable future events. Addressing uncertainty is crucial for developing solutions that are effective in real-world settings. Two primary approaches to optimization under uncertainty are stochastic optimization and robust optimization.

2.10.1. Stochastic Optimization

Stochastic optimization deals with problems where some parameters are uncertain but can be described probabilistically. In this framework, the optimization process incorporates the expected values, variances, or other statistical characteristics of uncertain parameters. Solutions are often evaluated based on their performance over a distribution of possible scenarios.

For example, in supply chain management, uncertain demand can be modeled using probability distributions. A company may aim to minimize costs while ensuring a high service level under varying demand conditions. Similarly, in finance, portfolio optimization often accounts for uncertain future returns by maximizing expected returns while managing risk.

2.10.2. Robust Optimization

Robust optimization, on the other hand, focuses on creating solutions that perform well across the worst-case realizations of uncertainty. Rather than assuming a specific probability distribution, robust optimization works with uncertainty sets, which define the range of possible values that uncertain parameters can take. The goal is to find solutions that are feasible and effective for all possible scenarios within the uncertainty set.

Applications of robust optimization include power grid operations, where uncertainties in demand and renewable energy generation need to be managed, and scheduling problems, such as airline crew scheduling, where disruptions must be minimized under varying operational conditions.

2.10.3. Applications

Optimization under uncertainty has a wide array of applications across industries:

- **Healthcare:** Designing vaccination strategies to minimize the spread of diseases while accounting for uncertain infection rates and patient behaviors.
- **Transportation:** Managing traffic flow or public transit schedules under uncertain demand and travel times.
- **Energy:** Planning renewable energy investments and operations under uncertain weather conditions and energy prices.

- **Manufacturing:** Ensuring production schedules are resilient to supply chain disruptions and variability in raw material quality.

Addressing the complexity of uncertain data typically makes these problems much harder than their deterministic counterparts. While these methods are powerful and applicable to a wide range of real-world scenarios, we will not focus on optimization under uncertainty in this textbook.

Part I

Linear Programming

3. Modeling: Linear Programming

Outcomes

1. Define what a linear program is
2. Understand how to model a linear program
3. View many examples and get a sense of what types of problems can be modeled as linear programs.

Resources

- Youtube! - Modeling with Linear Programming - Watch 0:00 - 4:19
- Youtube! - Watch 0:00 - 1:43

Linear Programming, also known as Linear Optimization, is the starting point for most forms of optimization. It is the problem of optimization a linear function over linear constraints.

In this chapter, we will define what this means, how to setup a linear program, and discuss many examples. Examples will be connected with code in Excel and Python (using with PuLP or Gurobipy modeling tools) so that you can easily start solving optimization problems. Tutorials on these tools will come in later chapters.

As a warm up, consider the following word problem.

Question: Find the age of Justin and Brad, given that the sum of their ages is equal to 76 and after 7 years, the age of Brad will be twice the age of Justin.

Solution: Let x and y denote the age of Justin and Brad, respectively. Then

$$x + y = 76 \tag{1}$$

$$2(x + 7) = y + 7. \tag{2}$$

On solving the foregoing equations we get

$$x = 23 \quad \text{and} \quad y = 53.$$

We will need to think about assigning variables to decisions and instead of equations guiding these variables, we may have inequalities that bound these variables in different ways. Thus, there can be many possible solutions.

3.1 Introductory LP Example

We begin with a simple example.

Example 3.1: Toy Maker

Excel PuLP Gurobipy

Consider the problem of a toy company that produces toy planes and toy boats. The toy company can sell its planes for \$10 and its boats for \$8 dollars. It costs \$3 in raw materials to make a plane and \$2 in raw materials to make a boat. A plane requires 3 hours to make and 1 hour to finish while a boat requires 1 hour to make and 2 hours to finish. The toy company knows it will not sell anymore than 35 planes per week. Further, given the number of workers, the company cannot spend anymore than 160 hours per week finishing toys and 120 hours per week making toys. The company wishes to maximize the profit it makes by choosing how much of each toy to produce.

We can represent the profit maximization problem of the company as a linear programming problem. Let x_1 be the number of planes the company will produce and let x_2 be the number of boats the company will produce. The profit for each plane is $\$10 - \$3 = \$7$ per plane and the profit for each boat is $\$8 - \$2 = \$6$ per boat. Thus the total profit the company will make is:

$$z(x_1, x_2) = 7x_1 + 6x_2 \quad (3.1)$$

The company can spend no more than 120 hours per week making toys and since a plane takes 3 hours to make and a boat takes 1 hour to make we have:

$$3x_1 + x_2 \leq 120 \quad (3.2)$$

Likewise, the company can spend no more than 160 hours per week finishing toys and since it takes 1 hour to finish a plane and 2 hour to finish a boat we have:

$$x_1 + 2x_2 \leq 160 \quad (3.3)$$

Finally, we know that $x_1 \leq 35$, since the company will make no more than 35 planes per week. Thus the complete linear programming problem is given as:

$$\begin{aligned} \max \quad & z(x_1, x_2) = 7x_1 + 6x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \leq 120 \\ & x_1 + 2x_2 \leq 160 \\ & x_1 \leq 35 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned} \quad (3.4)$$

Exercise 3.1: Chemical Manufacturing

A chemical manufacturer produces three chemicals: A, B and C. These chemical are produced by two processes: 1 and 2.

- Running process 1 for 1 hour costs \$4 and yields 3 units of chemical A, 1 unit of chemical B and 1 unit of chemical C.
- Running process 2 for 1 hour costs \$1 and produces 1 units of chemical A, and 1 unit of chemical B (but none of Chemical C).

To meet customer demand,

- at least 10 units of chemical A,
- at least 5 units of chemical B and
- at least 3 units of chemical C

must be produced daily.

Assume that the chemical manufacturer wants to minimize the cost of production.

Develop a linear programming problem describing the constraints and objectives of the chemical manufacturer.

[Hint: Let x_1 be the amount of time Process 1 is executed and let x_2 be amount of time Process 2 is executed. Use the coefficients above to express the cost of running Process 1 for x_1 time and Process 2 for x_2 time. Do the same to compute the amount of chemicals A, B, and C that are produced.]

3.2 Modeling and Assumptions in Linear Programming

Outcomes

1. Address crucial assumptions when chosing to model a problem with linear programming.

A Generic Linear Program (LP)

Linear programming (LP) is a mathematical optimization framework widely used in decision-making and resource allocation. In its most basic form, an LP seeks to optimize (maximize or minimize) a linear objective function subject to a set of linear constraints. These constraints typically represent limitations or requirements in real-world scenarios, such as budgets, capacities, or resource availability.

Decision Variables:

The decision variables represent the quantities we want to determine. For example:

- x_i : Continuous variables ($x_i \in \mathbb{R}$, i.e., a real number), $\forall i = 1, \dots, n$.

These variables could represent amounts of products to produce, resources to allocate, or investments to make. We can use any letters to represent these variables. Typically we use x_i , but frequently will use y_i or other letters depending on the context and what makes reading the problem most convenient.

Parameters (Known Input Parameters):

Parameters are the given data in the problem, which are assumed to be known in advance:

- c_i : Cost coefficients for each decision variable, $\forall i = 1, \dots, n$. These values represent the contribution of each variable to the objective function.
- a_{ij} : Constraint coefficients, $\forall i = 1, \dots, n$ and $j = 1, \dots, m$. These define the relationships between decision variables and the constraints.
- b_j : Right-hand side values of the constraints, $\forall j = 1, \dots, m$. These represent limits or requirements for each constraint.

The general form of a linear program can be written as follows:

$$\begin{aligned}
 \max \quad & z = c_1x_1 + \cdots + c_nx_n \\
 \text{s.t.} \quad & a_{11}x_1 + \cdots + a_{1n}x_n \leq b_1 \\
 & \vdots \\
 & a_{m1}x_1 + \cdots + a_{mn}x_n \leq b_m \\
 & x_i \geq 0 \quad \forall i = 1, \dots, n.
 \end{aligned} \tag{3.1}$$

Features of a Linear Program

- **Objective Function:** The linear function $z = c_1x_1 + \cdots + c_nx_n$ represents the goal to optimize. This could involve maximizing profit, minimizing cost, or achieving some other measurable outcome.
- **Constraints:** The system of inequalities (or equalities) restricts the values of the decision variables. Each constraint reflects a real-world limitation, such as resource availability or capacity limits.
- **Feasibility:** The set of all possible solutions that satisfy the constraints is called the feasible region. The optimal solution must lie within this region.
- **Linear Relationships:** Both the objective function and constraints are linear, meaning the relationships between variables are additive and proportional.
- **Sets:** For specific optimization problem, it can be useful to define sets of variables or parameters to help write down the problem in a generic formulation.

Example of a Linear Program

To illustrate, consider a specific example with three decision variables and four constraints. This includes a mix of constraint types ($\leq$, $\geq$, and $=$), showcasing the versatility of linear programming:

Decision Variables:

x_i : Continuous variables ($x_i \in \mathbb{R}$, i.e., a real number), $\forall i = 1, \dots, 3$.

Parameters (Known Input Parameters):

- c_i : Cost coefficients, $\forall i = 1, \dots, 3$.
- a_{ij} : Constraint coefficients, $\forall i = 1, \dots, 3$ and $j = 1, \dots, 4$.
- b_j : Right-hand side values for each constraint, $\forall j = 1, \dots, 4$.

The linear program can be written as:

$$\text{Minimize } z = c_1x_1 + c_2x_2 + c_3x_3 \quad (3.2)$$

$$\text{Subject to } a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \geq b_1 \quad (3.3)$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2 \quad (3.4)$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \quad (3.5)$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 \geq b_4 \quad (3.6)$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \text{ unrestricted.} \quad (3.7)$$

A key feature of linear programming is the use of linear functions, defined as follows:

Definition 3.2: Linear Function

A function $z : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear if there are constants $c_1, \dots, c_n \in \mathbb{R}$ such that:

$$z(x_1, \dots, x_n) = c_1x_1 + \dots + c_nx_n. \quad (3.8)$$

Linear functions are characterized by their simplicity and additivity, making them well-suited for optimization problems where proportionality is assumed. This simplicity allows linear programs to be solved efficiently, even for large-scale problems.

3.2.1. Assumptions

Inspecting Example 3.1 (or the more general problem in Equation (3.1)) we can see there are several assumptions that must be satisfied when using a linear programming model. We enumerate these below:

Proportionality Assumption A problem can be phrased as a linear program only if the contribution to the objective function *and* the left-hand-side of each constraint by each decision variable ($x_1, \dots, x_n$) is proportional to the value of the decision variable.

Additivity Assumption A problem can be phrased as a linear programming problem only if the contribution to the objective function *and* the left-hand-side of each constraint by any decision variable x_i ($i = 1, \dots, n$) is completely independent of any other decision variable x_j ($j \neq i$) and additive.

Divisibility Assumption A problem can be phrased as a linear programming problem only if the quantities represented by each decision variable are infinitely divisible (i.e., fractional answers make sense).

Certainty Assumption A problem can be phrased as a linear programming problem only if the coefficients in the objective function and constraints are known with certainty.

The first two assumptions simply assert (in English) that both the objective function and functions on the left-hand-side of the (in)equalities in the constraints are linear functions of the variables $x_1, \dots, x_n$.

The third assumption asserts that a valid optimal answer could contain fractional values for decision variables. It's important to understand how this assumption comes into play—even in the toy making example. Many quantities can be divided into non-integer values (ounces, pounds etc.) but many other quantities cannot be divided. For instance, can we really expect that it's reasonable to make $\frac{1}{2}$ a plane in the toy making example? When values must be constrained to true integer values, the linear programming problem is called an *integer programming problem*. There is a *vast* literature dealing with these problems. For many problems, particularly when the values of the decision variables may become large, a fractional optimal answer could be obtained and then rounded to the nearest integer to obtain a reasonable answer. For example, if our toy problem were re-written so that the optimal answer was to make 1045.3 planes, then we could round down to 1045.

The final assumption asserts that the coefficients (e.g., profit per plane or boat) is known with absolute certainty. In traditional linear programming, there is no lack of knowledge about the make up of the objective function, the coefficients in the left-hand-side of the constraints or the bounds on the right-hand-sides of the constraints. There is a literature on *stochastic programming* that relaxes some of these assumptions, but this too is outside the scope of the course.

Exercise 3.3

In a short sentence or two, discuss whether the problem given in Example 3.1 meets all of the assumptions of a scenario that can be modeled by a linear programming problem. Do the same for Exercise 3.1.

[Hint: Can you make $\frac{2}{3}$ of a toy? Can you run a process for $\frac{1}{3}$ of an hour?]

3.3 Examples

Outcomes

1. Identify decision variables and parameters
2. Learn how to format a linear optimization problem.
3. Identify and understand common classes of linear optimization problems.

We will begin with a few examples, and then discuss specific problem types that occur often.

Example 3.2: Production Planning

Excel PuLP Gurobipy

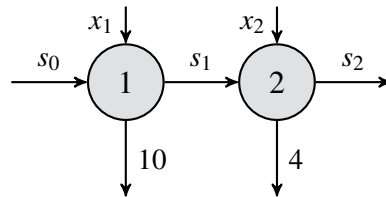
A manufacturing firm seeks to optimize its production schedule over a two-day planning horizon to meet daily demand while minimizing costs. The firm starts with an initial inventory of s_0 units. The costs associated with production and inventory are as follows:

- **Production costs:** \$10 per unit on Day 1, and \$16 per unit on Day 2.
- **Inventory cost:** \$5 per unit for carrying excess inventory from one day to the next.

Production

Inventory

Demand



The goal is to determine the number of units to produce each day and the inventory levels to minimize total costs while satisfying the demand for each day.

Solution. Sets:

- Days of production = $\{1, 2\}$.

Parameters:

- c_{p1} : Per unit production cost on Day 1 (\$10).
- c_{p2} : Per unit production cost on Day 2 (\$16).
- c_{inv} : Per unit inventory cost (\$5).
- d_1 : Demand on Day 1 (10 units).
- d_2 : Demand on Day 2 (4 units).
- s_0 : Inventory at the start of Day 1.

Decision variables:

- x_1 : Number of units produced on Day 1.
- x_2 : Number of units produced on Day 2.
- s_1 : Inventory at the end of Day 1.
- s_2 : Inventory at the end of Day 2.

Objective and Constraints:

$$\begin{array}{ll}
 \min & z = 10x_1 + 16x_2 + 5s_1 + 5s_2 & \text{(Total cost)} \\
 \text{s.t.} & s_0 + x_1 = 10 + s_1 & \text{(Day 1 balance)} \\
 & s_1 + x_2 = 4 + s_2 & \text{(Day 2 balance)} \\
 & x_1, x_2, s_0, s_1, s_2 \geq 0 & \text{(Non-negativity constraints).}
 \end{array}$$

**Definition 3.4: Optimization model components**

When defining an optimization model, it is essential to identify and clearly articulate the following five components:

- **Sets:** What lists of data do you need to write down? These could include indices, categories, or groups relevant to the problem.
- **Parameters (Data):** What is the actual data for the problem? These are the fixed numerical values such as costs, capacities, or resource availabilities.
- **Decision Variables:** What decisions need to be made to provide a solution? These variables represent the quantities that can be controlled or adjusted, such as production levels or resource allocations.
- **Objective Function:** The expression that we want to maximize or minimize. This is the goal of the optimization, such as minimizing cost or maximizing profit.
- **Constraints:** Certain rules that those decision variables should satisfy. These include limitations or requirements, such as resource capacities or demand fulfillment.

Each of these components plays a crucial role in formulating a well-structured optimization model that accurately represents the problem and provides actionable solutions. The role of Sets will become more important as we get into more complicated optimization models.

Example 3.3: Bakery Production

Excel PuLP Gurobipy

A small bakery produces three products each day: cakes, cookies, and muffins. Each item requires a specific amount of ingredients and baking time, and each contributes to the bakery's profit. The bakery operates with daily limits on flour, sugar, and baking time due to ingredient supply and staffing constraints.

Each cake earns a profit of \$6.00, requires 500 grams of flour, 200 grams of sugar, and 60 minutes of baking time. Each cookie earns a profit of \$1.50, requires 100 grams of flour, 50 grams of sugar, and 10 minutes of baking time. Each muffin earns a profit of \$2.50, requires 200 grams of flour, 80 grams of sugar, and 20 minutes of baking time.

The bakery has at most 4,800 grams of flour, 2,060 grams of sugar, and 480 minutes of baking time available each day.

Determine how many cakes, cookies, and muffins the bakery should produce each day in order to maximize its total profit, while staying within the limits on flour, sugar, and baking time.

Linear Programming Model

$$\begin{aligned}
 &\text{Maximize} && 6x_1 + 1.5x_2 + 2.5x_3 \\
 &\text{subject to} && 500x_1 + 100x_2 + 200x_3 \leq 4800 \quad (\text{Flour}) \\
 &&& 200x_1 + 50x_2 + 80x_3 \leq 2060 \quad (\text{Sugar}) \\
 &&& 60x_1 + 10x_2 + 20x_3 \leq 480 \quad (\text{Baking time}) \\
 &&& x_1, x_2, x_3 \geq 0 \\
 &&& x_1, x_2, x_3 \in \mathbb{Z} \quad (\text{if integer production is required})
 \end{aligned}$$

Example 3.4: Production with welding robot

Excel PuLP Gurobipy

You have 21 units of transparent aluminum alloy (TAA), LazWeld1, a joining robot leased for 23 hours, and CrumCut1, a cutting robot leased for 17 hours of aluminum cutting. You also have production code for a bookcase, desk, and cabinet, along with commitments to buy any of these you can produce for \$18, \$16, and \$10 apiece, respectively. A bookcase requires 2 units of TAA, 3 hours of joining, and 1 hour of cutting, a desk requires 2 units of TAA, 2 hours of joining, and 2 hour of cutting, and a cabinet requires 1 unit of TAA, 2 hours of joining, and 1 hour of cutting.

Formulate an LP to maximize your revenue given your current resources.

Solution.

Sets:

- The types of objects = { bookcase, desk, cabinet }.

Parameters:

- Purchase cost of each object
- Units of TAA needed for each object
- Hours of joining needed for each object
- Hours of cutting needed for each object
- Hours of TAA, Joining, and Cutting available on robots

Decision variables:

x_i : number of units of product i to produce,
for all i =bookcase, desk, cabinet.

Objective and Constraints:

$$\begin{aligned}
 \max \quad & z = 18x_1 + 16x_2 + 10x_3 && \text{(profit)} \\
 \text{s.t.} \quad & 2x_1 + 2x_2 + 1x_3 \leq 21 && \text{(TAA)} \\
 & 3x_1 + 2x_2 + 2x_3 \leq 23 && \text{(LazWeld1)} \\
 & 1x_1 + 2x_2 + 1x_3 \leq 17 && \text{(CrumCut1)} \\
 & x_1, x_2, x_3 \geq 0.
 \end{aligned}$$



Example 3.5: The Diet Problem

Excel PuLP Gurobipy

In the future (as envisioned in a bad 70's science fiction film) all food is in tablet form, and there are four types, green, blue, yellow, and red. A balanced, futuristic diet requires, at least 20 units of Iron, 25 units of Vitamin B, 30 units of Vitamin C, and 15 units of Vitamin D. Formulate an LP that ensures a balanced diet at the minimum possible cost.

Tablet	Iron	B	C	D	Cost (\$)
Chem 1	6	6	7	4	1.25
Chem 2	4	5	4	9	1.05
Chem 3	5	2	5	6	0.85
Chem 4	3	6	3	2	0.65

Solution. Sets:

- Set of tablets {1,2,3,4}

Parameters:

- Iron in each tablet
- Vitamin B in each tablet
- Vitamin C in each tablet
- Vitamin D in each tablet
- Cost of each tablet

Decision variables:

x_i : number of tablet of type i to include in the diet, $\forall i \in \{1, 2, 3, 4\}$.

Objective and Constraints:

$$\begin{array}{ll}
 \text{Min} & z = 1.25x_1 + 1.05x_2 + 0.85x_3 + 0.65x_4 \\
 \text{s.t.} & 6x_1 + 4x_2 + 5x_3 + 3x_4 \geq 20 \\
 & 6x_1 + 5x_2 + 2x_3 + 6x_4 \geq 25 \\
 & 7x_1 + 4x_2 + 5x_3 + 3x_4 \geq 30 \\
 & 4x_1 + 9x_2 + 6x_3 + 2x_4 \geq 15 \\
 & x_1, x_2, x_3, x_4 \geq 0.
 \end{array}$$

**Example 3.6: The Next Diet Problem**

Excel PuLP Gurobipy

Progress is important, and our last problem had too many tablets, so we are going to produce a single, purple, 10 gram tablet for our futuristic diet requires, which are at least 20 units of Iron, 25 units of Vitamin B, 30 units of Vitamin C, and 15 units of Vitamin D, and 2000 calories. The tablet is made from blending 4 nutritious chemicals; the following table shows the units of our nutrients per, and cost of, grams of each chemical.

Formulate an LP that ensures a balanced diet at the minimum possible cost.

Tablet	Iron	B	C	D	Calories	Cost (\$)
Chem 1	6	6	7	4	1000	1.25
Chem 2	4	5	4	9	250	1.05
Chem 3	5	2	5	6	850	0.85
Chem 4	3	6	3	2	750	0.65

Solution.Sets:

- Set of chemicals $\{1, 2, 3, 4\}$

Parameters:

- Iron in each chemical
- Vitamin B in each chemical
- Vitamin C in each chemical
- Vitamin D in each chemical
- Cost of each chemical

Decision variables:

x_i : grams of chemical i to include in the purple tablet, $\forall i = 1, 2, 3, 4$.

Objective and Constraints:

$$\begin{aligned}
 \min \quad & z = 1.25x_1 + 1.05x_2 + 0.85x_3 + 0.65x_4 \\
 \text{s.t.} \quad & 6x_1 + 4x_2 + 5x_3 + 3x_4 \geq 20 \\
 & 6x_1 + 5x_2 + 2x_3 + 6x_4 \geq 25 \\
 & 7x_1 + 4x_2 + 5x_3 + 3x_4 \geq 30 \\
 & 4x_1 + 9x_2 + 6x_3 + 2x_4 \geq 15 \\
 & 1000x_1 + 250x_2 + 850x_3 + 750x_4 \geq 2000 \\
 & x_1 + x_2 + x_3 + x_4 = 10 \\
 & x_1, x_2, x_3, x_4 \geq 0.
 \end{aligned}$$

**Example 3.7: Work Scheduling Problem**

Excel PuLP Gurobipy

You are the manager of LP Burger. The following table shows the minimum number of employees required to staff the restaurant on each day of the week. Each employees must work for five consecutive days. Formulate an LP to find the minimum number of employees required to staff the restaurant.

Day of Week	Workers Required
1 = Monday	6
2 = Tuesday	4
3 = Wednesday	5
4 = Thursday	4
5 = Friday	3
6 = Saturday	7
7 = Sunday	7

Solution. This problem has multiple optimal solutions for which days workers begin working on, all of which result in 8 total workers hired.

Decision variables:

x_i : the number of workers that start 5 consecutive days of work on day i , $i = 1, \dots, 7$

Objective and Constraints:

$$\begin{array}{ll}
 \text{Min} & z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \\
 \text{s.t.} & x_1 + x_4 + x_5 + x_6 + x_7 \geq 6 \\
 & x_2 + x_5 + x_6 + x_7 + x_1 \geq 4 \\
 & x_3 + x_6 + x_7 + x_1 + x_2 \geq 5 \\
 & x_4 + x_7 + x_1 + x_2 + x_3 \geq 4 \\
 & x_5 + x_1 + x_2 + x_3 + x_4 \geq 3 \\
 & x_6 + x_2 + x_3 + x_4 + x_5 \geq 7 \\
 & x_7 + x_3 + x_4 + x_5 + x_6 \geq 7 \\
 & x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0.
 \end{array}$$

One solution is as follows:

LP Solution	IP Solution
$z_{LP} = 7.333$	$z_I = 8.0$
$x_1 = 0$	$x_1 = 0$
$x_2 = 0.333$	$x_2 = 0$
$x_3 = 1$	$x_3 = 0$
$x_4 = 2.333$	$x_4 = 3$
$x_5 = 0$	$x_5 = 0$
$x_6 = 3.333$	$x_6 = 4$
$x_7 = 0.333$	$x_7 = 1$



Example 3.8: LP Burger - extended

Excel PuLP Gurobipy

LP Burger has changed its policy, and allows, at most, two part time workers, who work for two consecutive days in a week. Formulate this problem.

Solution. Decision variables:

x_i : the number of workers that start 5 consecutive days of work on day i , $i = 1, \dots, 7$

y_i : the number of workers that start 2 consecutive days of work on day i , $i = 1, \dots, 7$.

Objective and Constraints:

Min

$$z = 5(x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7) \\ + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7)$$

s.t.

$$x_1 + x_4 + x_5 + x_6 + x_7 + y_1 + y_7 \geq 6$$

$$x_2 + x_5 + x_6 + x_7 + x_1 + y_2 + y_1 \geq 4$$

$$x_3 + x_6 + x_7 + x_1 + x_2 + y_3 + y_2 \geq 5$$

$$x_4 + x_7 + x_1 + x_2 + x_3 + y_4 + y_3 \geq 4$$

$$x_5 + x_1 + x_2 + x_3 + x_4 + y_5 + y_4 \geq 3$$

$$x_6 + x_2 + x_3 + x_4 + x_5 + y_6 + y_5 \geq 7$$

$$x_7 + x_3 + x_4 + x_5 + x_6 + y_7 + y_6 \geq 7$$

$$y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 \leq 2$$

$$x_i \geq 0, y_i \geq 0, \forall i = 1, \dots, 7.$$



3.3.1. Knapsack Problem

We consider now a simple example where the variables can only take the values of 0 or 1. We call these *binary variables*.

Example 3.9: Camping Trip

Excel PuLP Gurobipy

Imagine you are preparing for a week-long camping trip to the mountains. You have a backpack with a weight capacity of 20 kilograms. Your goal is to pack the most valuable items to ensure a comfortable and safe trip without exceeding the backpack's weight limit. Each item has a weight and a value associated with it, representing its importance and utility for the trip. The table below lists the potential items you can take, their weights in kilograms, and their values on a scale from 1 to 10 (with 10 being the most valuable).

Formulate an Integer Program to find the optimal items you should bring on your trip with you.

Item	Index	Weight (kg)	Value
Tent	A	4	10
Stove	B	2	9
Sleeping Bag	C	3	8
First Aid Kit	D	1	7
Food Supplies	E	5	9
Water Filter	F	1	6
Clothes	G	4	5
Map and Compass	H	0.5	7

Solution. Decision variables:

x_i : whether to include item i in the backpack, where $i \in \{A, B, C, D, E, F, G, H\}$, $x_i = 1$ if item i is included, and $x_i = 0$ otherwise.

Objective and Constraints:

$$\begin{aligned}
 \text{Max } z &= 10x_A + 9x_B + 8x_C + 7x_D + 9x_E + 6x_F + 5x_G + 7x_H \\
 \text{s.t. } &4x_A + 2x_B + 3x_C + 1x_D + 5x_E + 1x_F + 4x_G + 0.5x_H \leq 20 \\
 &x_A, x_B, x_C, x_D, x_E, x_F, x_G, x_H \in \{0, 1\}.
 \end{aligned}$$



3.3.2. Capital Investment

We next see a similar application of binary variables.

Example 3.10: Capital Allocation Problem

Excel PuLP Gurobipy

You are a financial planner for an investment firm. The firm has \$100,000 to invest in a portfolio of projects. Each project has a projected return and requires an initial investment. The goal is to maximize the total return of the portfolio while not exceeding the available capital. The following table lists potential projects, their required investments, and their projected returns.

Formulate an LP to maximize the total return of the portfolio while not exceeding the available investment capital.

Project	Investment Required (\$)	Projected Return (\$)
A	20,000	30,000
B	30,000	40,000
C	50,000	75,000
D	10,000	15,000
E	40,000	60,000

Solution.

Sets:

- Projects: $P = \{A, B, C, D, E\}$

Parameters:

- inv_i : investment required for project i
- ret_i : projected return from project i
- $B = 100,000$: total available capital

Decision Variables:

$x_i \in \{0, 1\}$: 1 if project i is selected, 0 otherwise

Objective and Constraints:

$$\begin{aligned}
 \max \quad & z = 30,000x_A + 40,000x_B + 75,000x_C + 15,000x_D + 60,000x_E \\
 \text{s.t.} \quad & 20,000x_A + 30,000x_B + 50,000x_C + 10,000x_D + 40,000x_E \leq 100,000 \\
 & x_A, x_B, x_C, x_D, x_E \in \{0, 1\}
 \end{aligned}$$

Compact Form:

$$\begin{aligned}
& \max \quad \sum_{i \in P} \text{ret}_i \cdot x_i \\
& \text{s.t.} \quad \sum_{i \in P} \text{inv}_i \cdot x_i \leq B \\
& \quad x_i \in \{0, 1\} \quad \forall i \in P
\end{aligned}$$



3.3.3. Transportation problem

Example 3.11: Solar Panel Shipping

Excel PuLP Gurobipy

SolTranz Inc. manufactures high-efficiency solar panels at three factories located in Reno (R), Boulder (B), and Tucson (T). The maximum production at each facility is 4,500, 5,500, and 7,000 panels, respectively. These panels must be shipped to five warehouses (labeled 1 through 5), each with a fixed demand.

The cost of producing a solar panel depends on the factory, and the shipping costs to each warehouse are also known. However, no more than 2,000 panels can be shipped from a factory to a single warehouse. The table below summarizes the data.

Formulate a linear program that minimizes the total cost of production and shipping to satisfy all warehouse demands.

Factory	Prod. Cost (\$)	WH1	WH2	WH3	WH4	WH5
Reno (R)	150	60	70	75	65	50
Boulder (B)	140	80	55	65	70	60
Tucson (T)	130	85	60	55	50	65

Shipping cost from each factory to each warehouse (in \$ per panel).

Warehouse Demands:

WH1: 3000 WH2: 3500 WH3: 2500 WH4: 4000 WH5: 3000

Solution.

Sets:

- Factories: $F = \{R, B, T\}$
- Warehouses: $W = \{1, 2, 3, 4, 5\}$

Parameters:

- C_f : production cost per panel at factory f
- S_f : supply capacity of factory f
- D_w : demand at warehouse w
- $T_{f,w}$: shipping cost per panel from factory f to warehouse w
- $M = 2000$: max panels shipped from any factory to any warehouse

Decision Variables:

$x_{f,w}$: number of solar panels shipped from factory f to warehouse w

Objective and Constraints:

$$\begin{aligned}
 \min \quad & \sum_{f \in F} \sum_{w \in W} (C_f + T_{f,w}) \cdot x_{f,w} \\
 \text{s.t.} \quad & \sum_{w \in W} x_{R,w} \leq 4500 && \text{(Reno capacity)} \\
 & \sum_{w \in W} x_{B,w} \leq 5500 && \text{(Boulder capacity)} \\
 & \sum_{w \in W} x_{T,w} \leq 7000 && \text{(Tucson capacity)} \\
 & \sum_{f \in F} x_{f,1} = 3000 && \text{(WH1 demand)} \\
 & \sum_{f \in F} x_{f,2} = 3500 && \text{(WH2 demand)} \\
 & \sum_{f \in F} x_{f,3} = 2500 && \text{(WH3 demand)} \\
 & \sum_{f \in F} x_{f,4} = 4000 && \text{(WH4 demand)} \\
 & \sum_{f \in F} x_{f,5} = 3000 && \text{(WH5 demand)} \\
 & 0 \leq x_{f,w} \leq 2000 && \forall f \in F, \forall w \in W
 \end{aligned}$$



3.3.4. Assignment Problem

Consider the assignment of 3 employees to 3 tasks, where each employee has a cost to do each task. How to figure out the best assignment to the tasks?

Example 3.12: Hiring for tasks

Excel PuLP Gurobipy

In this assignment problem, we need to hire three people (Person 1, Person 2, Person 3) to three tasks (Task 1, Task 2, Task 3). In the table below, we list the cost of hiring each person for each task, in dollars. Since each person has a different cost for each task, we must make an assignment to minimize

	Cost	Task 1	Task 2	Task 3
our total cost.	Person 1	40	47	80
	Person 2	72	36	58
	Person 3	24	61	71

Given the specific costs of assigning three people to three tasks, we can write out the mathematical model explicitly using the given numbers.

Solution.

Sets: Set of tasks and set of persons

Parameters: Cost c_{ij} for person i to complete task j .

Variables: Let $x_{ij} = 1$ if person i is assigned to task j and 0 otherwise.

Objective Function:

Minimize the total cost of assignments:

$$z = 40x_{11} + 47x_{12} + 80x_{13} + 72x_{21} + 36x_{22} + 58x_{23} + 24x_{31} + 61x_{32} + 71x_{33} \quad (3.1)$$

Subject to Constraints:

Each person is assigned to exactly one task:

$$x_{11} + x_{12} + x_{13} = 1 \quad (3.2)$$

$$x_{21} + x_{22} + x_{23} = 1 \quad (3.3)$$

$$x_{31} + x_{32} + x_{33} = 1 \quad (3.4)$$

Each task is assigned to exactly one person:

$$x_{11} + x_{21} + x_{31} = 1 \quad (3.5)$$

$$x_{12} + x_{22} + x_{32} = 1 \quad (3.6)$$

$$x_{13} + x_{23} + x_{33} = 1 \quad (3.7)$$

Binary constraints on the variables:

$$x_{ij} \in \{0, 1\} \quad \forall i \in \{1, 2, 3\}, \forall j \in \{1, 2, 3\} \quad (3.8)$$



3.4 Exercises

Exercise 3.13 Which ones are linear functions?

1. $x^2 + 3y + 2z$
2. $\sqrt{x_1 x_2 x_3}$
3. $\sum_{i=1}^n a_i 2^{x_i}$, where $a_1, a_2, \dots, a_n$ are constant
4. $\sum_{i=1}^n a_i x_i$, where $a_1, a_2, \dots, a_n$ are constant

Exercise 3.14 Which one is a linear function?

1. $2x_1 + x_2 x_3 - 7x_3$
2. $\sqrt{2}x_1 + x_2 + 3^{1.6}x_3$

Exercise 3.15 Which ones are linear inequalities? Can any be made into linear inequalities?

1. $\frac{x_1}{x_1 + x_2 + x_3} \geq 0.5$
2. $\sin x_1 + x_2 + x_3 = 0.5$
3. $\sum_{i=1}^n a_i 2^{x_i} \geq b$, where $a_1, a_2, \dots, a_n, b$ are constant
4. $\sum_{i=1}^n a_i x_i = b$, where $a_1, a_2, \dots, a_n, b$ are constant
5. $\sum_{i=1}^n a_i x_i \geq b$, where $a_1, a_2, \dots, a_n, b$ are constant

Exercise 3.16 An appliance store sells three types of ovens: small, medium, and large. The small, medium, and large ovens require, respectively, 3, 5, and 6 cubic feet of storage space; at most 100 cubic feet of storage is available. Each oven takes 1 hour of sales time; there is a maximum of 200 hours of sales labor time available. The small, medium, and large ovens require, respectively, 1, 1, and 2 hours of installation time; at most 280 hours of installer labor are available monthly.

If the profit (in dollars) from small, medium, and large ovens is 50, 100, and 150 respectively, how many of each type of oven should be sold to maximize profit, and what is the maximum profit?

Exercise 3.17 A factory manufactures three products, A, B, and C. Each product requires the use of two machines (Machine I and Machine II). The total hours available per month on Machine I and Machine II

are 180 and 300, respectively. The table below shows the time requirements (in hours per unit) and profit per unit for each product.

	A	B	C
Machine I	1	2	2
Machine II	2	2	4
Profit (dollars)	20	30	40

How many units of each product should be manufactured to maximize profit, and what is the maximum profit?

Exercise 3.18 A company produces three products, A, B, and C, at two factories, Factory I and Factory II. Daily production of each factory for each product is listed below:

	Factory I	Factory II
Product A	10	20
Product B	20	20
Product C	20	10

The company must produce at least 1000 units of product A, 1600 units of product B, and 700 units of product C. If the cost of operating Factory I is \$4000 per day and the cost of operating Factory II is \$5000 per day, how many days should each factory operate to complete the order at a minimum cost, and what is the minimum cost?

Exercise 3.19 Professor Wright gives three types of quizzes (objective, recall, and recall-plus) and decides to give at least 20 quizzes next quarter. Each type of quiz requires the students to spend extra preparation time:

Objective: 10 minutes per quiz,

Recall: 30 minutes per quiz,

Recall-plus: 60 minutes per quiz.

He wants students to spend at least 12 hours (720 minutes) in total preparation for these quizzes (beyond normal study time).

Average quiz scores are:

Objective: 5, Recall: 6, Recall-plus: 7.

He wants the total of all quiz scores to be at least 130 points.

It takes the professor:

1 minute to grade each objective quiz,

2 minutes to grade each recall quiz,

3 minutes to grade each recall-plus quiz.

How many of each type should he give in order to minimize his own grading time?

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4. Software - Excel

Outcomes

- *Learn relevant excel functions*
- *Setup linear programs in excel*
- *Solve linear programs in excel*

4.1 Using Excel Solver

Excel Solver is a powerful tool for solving optimization problems with linear and nonlinear constraints. Before diving into using Solver, it is essential to understand some basic Excel functions and techniques that are commonly used when setting up optimization problems. These functions help organize data, compute intermediate values, and define objective functions and constraints.

4.1.1. Useful Excel Functions and Techniques

USING FORMULAS IN CELLS In Excel, formulas begin with an equals sign (=). For example, typing =A1+B1 in a cell will calculate the sum of the values in cells A1 and B1.

SUM() The SUM() function adds a range of numbers. For example, =SUM(A1:A10) calculates the sum of all values in the range A1 to A10.

SUMPRODUCT() The SUMPRODUCT() function multiplies corresponding elements in two or more arrays and returns the sum of those products. This is particularly useful for computing dot products or weighted sums. For example, =SUMPRODUCT(A1:A10, B1:B10) computes $\sum_{i=1}^{10} A_i B_i$.

COUNT() The COUNT() function counts the number of numeric entries in a range. For example, =COUNT(A1:A10) returns the number of numeric cells in the range A1 to A10.

COUNTIF() The COUNTIF() function counts the number of cells in a range that meet a specific condition. For example, =COUNTIF(A1:A10, ">5") counts the number of cells in A1:A10 with values greater than 5.

DUPLICATING FORMULAS ACROSS ROWS OR COLUMNS To apply a formula to multiple cells, you can drag the fill handle (a small square at the bottom-right corner of the selected cell) across the desired range. Excel automatically adjusts cell references based on the relative position (relative referencing). To keep a reference constant, use the dollar sign (\$) in the cell reference (absolute referencing). For example, \$A\$1 always refers to cell A1.

CONDITIONAL FORMATTING Conditional formatting allows you to highlight cells that meet specific criteria. For example, you can format cells with values above a threshold in bold or a particular color to identify important data visually.

DATA VALIDATION Data validation helps restrict the type of data entered in a cell. For example, you can set a rule that only allows numbers within a specific range or text from a predefined list.

NAMING RANGES Naming ranges makes formulas easier to read and maintain. Instead of =SUM(A1:A10), you can name the range "Sales" and write =SUM(Sales).

USING LOGICAL FUNCTIONS Logical functions like IF(), AND(), and OR() are helpful for decision-making within formulas. For example, =IF(A1>10, High, Low) returns High if A1 is greater than 10, and Low otherwise.

With these foundational tools, setting up optimization problems becomes much more manageable. In the next section, we will explore how to use these functions in combination with Excel Solver to define and solve optimization models.

Logical functions like IF(), AND(), and OR() are helpful for decision-making within formulas. For example, =IF(A1>10, High, Low) returns High if A1 is greater than 10, and Low otherwise.

With these foundational tools, setting up optimization problems becomes much more manageable. In the next section, we will explore how to use these functions in combination with Excel Solver to define and solve optimization models.

4.1.2. How to Use Excel Solver

Excel Solver is a powerful add-in that allows users to define and solve optimization problems. This section provides a step-by-step guide on how to effectively layout data and use Solver for optimization.

STEP 1: LAYOUT THE DATA Organizing your data clearly is crucial for using Solver effectively. Use color coding to distinguish between different types of cells:

- **Data:** Cells containing fixed input values, such as coefficients or constants, should be highlighted in **light blue**.
- **Variables:** Cells representing decision variables should be highlighted in **yellow**.
- **Formulas:** Cells with formulas, such as the objective function or summarizing constraints values should be highlighted in **light green**.

Other color combinations are also acceptable, as long as they consistently distinguish between these components.

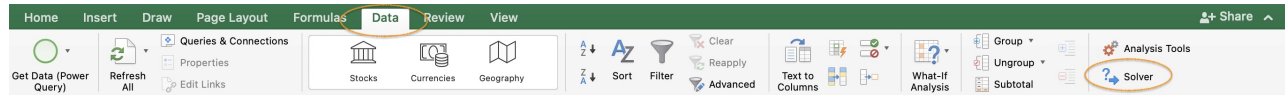
Objective	=SUMPRODUCT(B6:C6,B8:C8)					
Variables	x1	x2				
obj. coeff	7	6				
			Resources Used			
constraint 1	3	1	=SUMPRODUCT(\$B\$6:\$C\$6,B11:C11)	<=	120	
constraint 2	1	2	=SUMPRODUCT(\$B\$6:\$C\$6,B12:C12)	<=	160	
constraint 3	1	0	=SUMPRODUCT(\$B\$6:\$C\$6,B13:C13)	<=	35	

STEP 2: DEFINE THE OBJECTIVE FUNCTION Identify the cell that represents the objective function. This cell should contain a formula that calculates the value to be maximized or minimized, such as total profit or cost.

STEP 3: SET UP CONSTRAINTS Clearly specify the constraints of the problem. Constraints can be represented using formulas that involve the variables, coefficients, and right-hand side values. For example, a constraint ensuring that total production does not exceed capacity can be written as `=SUMPRODUCT(Production, Coefficients) <= Capacity`.

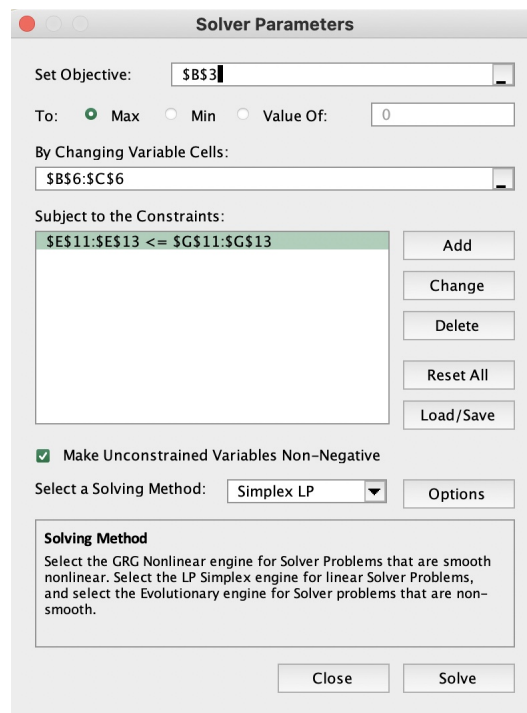
STEP 4: OPEN SOLVER Ensure Solver is enabled in Excel. If not, you can activate it by navigating to **File > Options > Add-Ins > Manage Excel Add-ins** and checking the box for Solver.

Once Solver is enabled, go to **Data > Solver** to open the Solver Parameters dialog box.



STEP 5: DEFINE SOLVER PARAMETERS In the Solver Parameters dialog box:

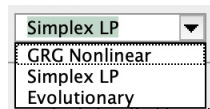
- Set the **Objective** to the cell containing the objective function.
- Choose **Max** or **Min** to indicate whether the objective is to be maximized or minimized.
- Specify the **Variable Cells** by selecting the range of cells representing decision variables.
- Add **Constraints** by specifying the relationship (e.g., `<=`, `>=`, `=`) and the cells involved.
- When the variables are non-negative, you can select **Make Unconstrained Variables Non-Negative**.



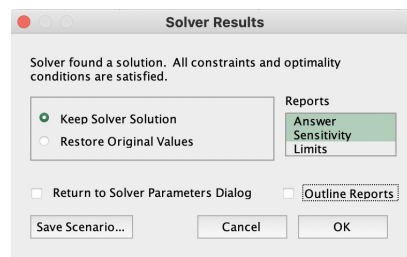
STEP 6: CHOOSE A SOLVING METHOD Solver provides three solving methods:

- **GRG Nonlinear:** For smooth nonlinear problems.
- **Simplex LP:** For linear programming problems.
- **Evolutionary:** For non-smooth problems or those with integer constraints.

Select the method that matches the problem type.



STEP 7: SOLVE THE PROBLEM Click **Solve** to run Solver. If Solver finds a solution, it will display the results and allow you to keep or discard them. Analyze the solution to ensure it meets the problem's requirements.



STEP 8: INTERPRET THE RESULTS Review the values of the decision variables and the objective function. Check that all constraints are satisfied and assess the solution's feasibility and quality.

- Answer report

	A	B	C	D	E	F	G															
1	Microsoft Excel 16.77 Answer Report																					
2	Worksheet: [toymaker (1).xlsx]Sheet1																					
3	Report Created: 1/17/25 4:27:45 AM																					
4	Result: Solver found a solution. All constraints and optimality conditions are satisfied.																					
5	Solver Engine																					
9	Solver Options																					
12																						
13																						
14	Objective Cell (Max)																					
15	<table><tr><th>Cell</th><th>Name</th><th>Original Value</th><th>Final Value</th></tr><tr><td>\$B\$3</td><td>Objective</td><td>0</td><td>544</td></tr></table>							Cell	Name	Original Value	Final Value	\$B\$3	Objective	0	544							
Cell	Name	Original Value	Final Value																			
\$B\$3	Objective	0	544																			
17																						
18																						
19	Variable Cells																					
20	<table><tr><th>Cell</th><th>Name</th><th>Original Value</th><th>Final Value</th><th>Integer</th></tr><tr><td>\$B\$6</td><td>x1</td><td>0</td><td>16</td><td>Contin</td></tr><tr><td>\$C\$6</td><td>x2</td><td>0</td><td>72</td><td>Contin</td></tr></table>							Cell	Name	Original Value	Final Value	Integer	\$B\$6	x1	0	16	Contin	\$C\$6	x2	0	72	Contin
Cell	Name	Original Value	Final Value	Integer																		
\$B\$6	x1	0	16	Contin																		
\$C\$6	x2	0	72	Contin																		
23																						
24																						
25	Constraints																					
26	<table><tr><th>Cell</th><th>Name</th><th>Cell Value</th><th>Formula</th><th>Status</th><th>Slack</th></tr><tr><td colspan="6">\$E\$11:\$E\$13 <= \$G\$11:\$G\$13</td></tr></table>							Cell	Name	Cell Value	Formula	Status	Slack	\$E\$11:\$E\$13 <= \$G\$11:\$G\$13								
Cell	Name	Cell Value	Formula	Status	Slack																	
\$E\$11:\$E\$13 <= \$G\$11:\$G\$13																						
27																						

- Sensitivity report

	A	B	C	D	E	F	G	H
1	Microsoft Excel 16.77 Sensitivity Report							
2	Worksheet: [toymaker (1).xlsx]Sheet1							
3	Report Created: 1/17/25 4:27:46 AM							
4								
5								
6	Variable Cells							
7				Final	Reduced	Objective	Allowable	Allowable
8	Cell	Name		Value	Cost	Coefficient	Increase	Decrease
9	\$B\$6	x1		16	0	7	11	4
10	\$C\$6	x2		72	0	6	8	3.666666667
11								
12	Constraints							
13				Final	Shadow	Constraint	Allowable	Allowable
14	Cell	Name		Value	Price	R.H. Side	Increase	Decrease
15	\$E\$11:\$E\$13 <= \$G\$11:\$G\$13							
16								
17								
18								
19								

- Solution on spreadsheet

	A	B	C	D	E	F	G
1							
2							
3	Objective	544					
4							
5	Variables	x1	x2				
6		16	72				
7							
8	obj. coeff	7	6				
9							
10					Resources Used		
11	constraint 1	3	1		120	<=	120
12	constraint 2	1	2		160	<=	160
13	constraint 3	1	0		16	<=	35
14							
15							

Resources

Examples and guides

- *Installing Excel Solver on Windows*
- <https://www.excel-easy.com/data-analysis/solver.html>
- *Excel Solver - Introduction on Youtube*
- *Some notes from MIT*

Some videos

- *Solving a linear program*
- *Optimal product mix*
- *Set Cover*
- *Introduction to Designing Optimization Models Using Excel Solver*
- *Traveling Salesman Problem*
- *Also Travelin Salesman Problem*
- *Multiple Traveling Salesman Problem*
- *Shortest Path*

Some other links

- *Loan Example*
- *Several Examples including TSP*

4.2 Exercises

Exercise 4.1 Consider the following table indicating the nutritional value of different food types:

	Price (\$) per serving	Calories per serving	Fat (g) per serving	Protein (g) per serving	Carbohydrate (g) per serving
<i>Raws carrots</i>	0.14	23	0.1	0.6	6
<i>Baked potatoes</i>	0.12	171	0.2	3.7	30
<i>Wheat bread</i>	0.2	65	0	2.2	13
<i>Cheddar cheese</i>	0.75	112	9.3	7	0
<i>Peanut butter</i>	0.15	188	16	7.7	2

You need to decide how many servings of each food to buy each day so that you minimize the total cost of buying your food while satisfying the following daily nutritional requirements:

- calories must be at least 2000,
- fat must be at least 50 g,
- protein must be at least 100 g,
- carbohydrates must be at least 250 g.

Write an LP that will help you decide how many servings of each of the aforementioned foods are needed to meet all the nutritional requirement, while minimizing the total cost of the food (you may buy fractional numbers of servings).

Solve this problem using Excel Solver. Provide a screenshot of your spreadsheet with the optimal solution.

5. Modeling with Compact Notation

Outcomes

- Review summation notation
- Work on Sets, Parameters, Variables, and Model
- Write compact formulations for linear programs

A Brief Review of Summation Notation and Its Relation to Vector Products

Summation notation is a concise way to express the addition of a series of terms. It is commonly written as

$$\sum_{i=m}^n \text{expression involving } i,$$

where i is the index of summation, m is the starting index, and n is the ending index.

1. Sum of x_i :

$$\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n.$$

In vector terminology, if we let $\mathbf{x}$ be an n -dimensional vector,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then $\sum_{i=1}^n x_i$ represents the sum of all components of $\mathbf{x}$.

2. Sum of $a_i x_i$:

$$\sum_{i=1}^n a_i x_i = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n.$$

This expression is the dot product (or inner product) of two n -dimensional vectors $\mathbf{a}$ and $\mathbf{x}$, where

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

In vector notation, the dot product is written as:

$$\mathbf{a}^\top \mathbf{x} = \mathbf{a} \cdot \mathbf{x} = \sum_{i=1}^n a_i x_i.$$

This product yields a scalar value that combines the corresponding components of the two vectors.

5.0.1. Notes to expand on

SETS. . Sets are important to help see what constraints and variables might iterating over.

We typically denote sets with capital letters and the index as the corresponding lower case letter.

For example:

Let I be the set of months in the year.

then we will index the months with the index i .

If

$$I = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

then we can write a summation over this index set in a few ways:

$$\sum_{i \in I} a_i x_i \quad \text{or} \quad \sum_{i=1}^{12} a_i x_i.$$

Suppose we have another set $J = \{'red', 'green', 'blue'\}$.

And suppose we have a variable y_{ij} indexed by both $i \in I$ and $j \in J$.

Consider for each color $j \in J$, we must sum over all the months and set that equal to 1. We can write this as

$$\sum_{i \in I} y_{ij} = 1 \quad \forall j \in J.$$

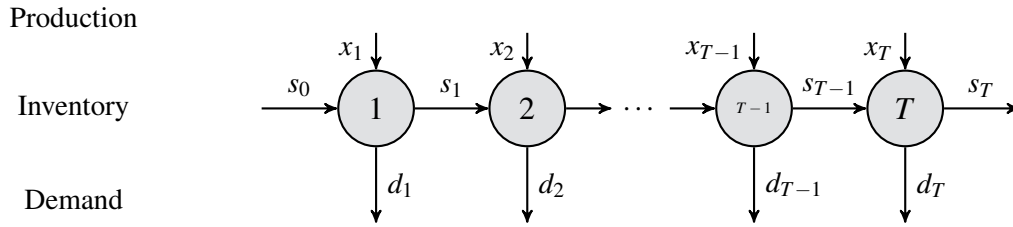
What does this encapsulate? It is shorthand for 3 constraints where in each constraint 12 variables are summed to equal 1.

IMPORTANT! . Every index used needs to appear in either a sum or a forall statement.

In the above example, we use the variable y_{ij} in the constraint. This means that in order to make the expression make sense mathematically, we need to define what i and j mean in the constraint. This is done by the summation $\sum_{i \in I}$ iterating over the i index, while the $\forall j \in J$ iterates over the j index.

5.1 Production Planning Models

Production Planning Model (for T Periods)



Sets:

- $t \in \{1, 2, \dots, T\}$: Periods in the planning horizon.

Parameters:

- c_t^{prod} : Production cost per unit in period t (for $t = 1, \dots, T$).
- c^{inv} : Inventory holding cost per unit carried from one period to the next.
- d_t : Demand in period t .
- s_0 : Initial inventory available at the start of period 1.

Decision Variables:

- x_t : Number of units produced in period t .
- s_t : Inventory level at the end of period t .

Objective and Constraints:

$$\min_{x,s} \quad z = \sum_{t=1}^T \left(c_t^{prod} x_t + c^{inv} s_t \right) \quad (\text{Total Cost})$$

$$\text{s.t.} \quad s_0 + x_1 = d_1 + s_1, \quad (\text{Period 1 Balance})$$

$$s_{t-1} + x_t = d_t + s_t, \quad \forall t = 2, \dots, T \quad (\text{Inventory Balance})$$

$$x_t \geq 0, \quad s_t \geq 0, \quad \forall t = 1, \dots, T \quad (\text{Non-negativity})$$

Example 5.1: Production Planning with 10 Periods ($T = 10$)

Excel PuLP Gurobipy

Initial Inventory:

$$s_0 = 5 \text{ units}$$

Data: (Download the data)

Parameter	1	2	3	4	5	6	7	8	9	10
Production Cost (\$ per unit)	10	12	14	16	18	20	22	24	26	28
Demand (units)	8	6	9	7	10	5	8	6	4	7

Inventory Holding Cost:

$$c^{inv} = 5 \text{ per unit}$$

Decision Variables:

- x_t : Number of units produced in period t .
- s_t : Inventory level at the end of period t .

Mathematical Formulation:

$$\begin{aligned}
 \min_{x,s} \quad & z = 10x_1 + 12x_2 + 14x_3 + 16x_4 + 18x_5 + 20x_6 + 22x_7 + 24x_8 + 26x_9 + 28x_{10} \\
 & + 5(s_1 + s_2 + s_3 + s_4 + s_5 + s_6 + s_7 + s_8 + s_9 + s_{10}) \\
 \text{s.t.} \quad & 5 + x_1 = 8 + s_1, \quad (\text{Period 1}) \\
 & s_1 + x_2 = 6 + s_2, \quad (\text{Period 2}) \\
 & s_2 + x_3 = 9 + s_3, \quad (\text{Period 3}) \\
 & s_3 + x_4 = 7 + s_4, \quad (\text{Period 4}) \\
 & s_4 + x_5 = 10 + s_5, \quad (\text{Period 5}) \\
 & s_5 + x_6 = 5 + s_6, \quad (\text{Period 6}) \\
 & s_6 + x_7 = 8 + s_7, \quad (\text{Period 7}) \\
 & s_7 + x_8 = 6 + s_8, \quad (\text{Period 8}) \\
 & s_8 + x_9 = 4 + s_9, \quad (\text{Period 9}) \\
 & s_9 + x_{10} = 7 + s_{10}, \quad (\text{Period 10}) \\
 & x_t \geq 0, \quad s_t \geq 0, \quad \forall t = 1, \dots, 10.
 \end{aligned}$$

Production Planning with Overtime

A manufacturing firm seeks to optimize its production schedule over a T -day planning horizon to meet daily demand while minimizing costs. The firm starts with an initial inventory of s_0 units. Production can be carried out during regular time and via overtime. The costs are as follows:

- **Regular production cost:** c_t^{prod} dollars per unit on day t .
- **Overtime production cost:** c_t^{over} dollars per unit on day t .
- **Inventory cost:** c^{inv} dollars per unit carried over from one day to the next.

The goal is to determine the number of units to produce during regular time and overtime on each day, and the inventory levels, in order to minimize the total cost while satisfying daily demand.

Sets:

- Days of production = $\{1, 2, \dots, T\}$.

Parameters:

- c_t^{prod} : Regular production cost per unit on day t (for $t = 1, \dots, T$).
- c_t^{over} : Overtime production cost per unit on day t (for $t = 1, \dots, T$).
- c^{inv} : Inventory holding cost per unit carried to the next day.
- d_t : Demand on day t .
- s_0 : Initial inventory at the start of day 1.

Decision Variables:

- x_t : Number of units produced in regular time on day t .
- y_t : Number of units produced in overtime on day t .
- s_t : Inventory at the end of day t .

Objective and Constraints:

$$\begin{aligned}
 \min \quad & z = \sum_{t=1}^T \left(c_t^{prod} x_t + c_t^{over} y_t + c^{inv} s_t \right) \quad (\text{Total Cost}) \\
 \text{s.t.} \quad & s_0 + x_1 + y_1 = d_1 + s_1 \quad (\text{Day 1 balance}) \\
 & s_{t-1} + x_t + y_t = d_t + s_t, \quad t = 2, \dots, T \quad (\text{Inventory balance}) \\
 & x_t, y_t, s_t \geq 0, \quad t = 1, \dots, T \quad (\text{Non-negativity}).
 \end{aligned}$$

Example 5.2:

Excel PuLP Gurobipy

Consider a 10-day planning horizon ($T = 10$) with the following data:

Data:

Parameter	1	2	3	4	5	6	7	8	9	10	
Regular Production Cost (\$)	10	12	14	16	18	20	22	24	26	28	data
Overtime Production Cost (\$)	16	18	20	22	24	26	28	30	32	34	
Demand (units)	8	6	9	7	10	5	8	6	4	7	

Assume the following additional parameters:

- Inventory cost: $c^{inv} = 5$ dollars per unit.
- Initial inventory: $s_0 = 0$ units.

Decision Variables:

- x_t : Regular production on day t .
- y_t : Overtime production on day t .
- s_t : Inventory at the end of day t .

Mathematical Formulation:

$$\begin{aligned}
\min \quad & z = 10x_1 + 16y_1 + 5s_1 \\
& + 12x_2 + 18y_2 + 5s_2 \\
& + 14x_3 + 20y_3 + 5s_3 \\
& + 16x_4 + 22y_4 + 5s_4 \\
& + 18x_5 + 24y_5 + 5s_5 \\
& + 20x_6 + 26y_6 + 5s_6 \\
& + 22x_7 + 28y_7 + 5s_7 \\
& + 24x_8 + 30y_8 + 5s_8 \\
& + 26x_9 + 32y_9 + 5s_9 \\
& + 28x_{10} + 34y_{10} + 5s_{10} \\
\text{s.t.} \quad & s_0 + x_1 + y_1 = 8 + s_1 \quad (\text{Day 1 balance}) \\
& s_1 + x_2 + y_2 = 6 + s_2 \quad (\text{Day 2 balance}) \\
& s_2 + x_3 + y_3 = 9 + s_3 \quad (\text{Day 3 balance}) \\
& s_3 + x_4 + y_4 = 7 + s_4 \quad (\text{Day 4 balance}) \\
& s_4 + x_5 + y_5 = 10 + s_5 \quad (\text{Day 5 balance}) \\
& s_5 + x_6 + y_6 = 5 + s_6 \quad (\text{Day 6 balance}) \\
& s_6 + x_7 + y_7 = 8 + s_7 \quad (\text{Day 7 balance}) \\
& s_7 + x_8 + y_8 = 6 + s_8 \quad (\text{Day 8 balance}) \\
& s_8 + x_9 + y_9 = 4 + s_9 \quad (\text{Day 9 balance}) \\
& s_9 + x_{10} + y_{10} = 7 + s_{10} \quad (\text{Day 10 balance}) \\
& x_t, y_t, s_t \geq 0, \quad t = 1, \dots, 10.
\end{aligned}$$

In this 10-day example, the decision is how many units to produce on regular time (x_t) and overtime (y_t) on each day, and what the end-of-day inventory levels (s_t) should be, so as to satisfy daily demand at minimum total cost.

5.2 Assignment Problem

The *assignment problem* (machine/person to job/task assignment) seeks to assign tasks to machines in a way that is most efficient. This problem can be thought of as having a set of machines that can complete various tasks (textile machines that can make t-shirts, pants, socks, etc) that require different amounts of time to complete each task, and given a demand, you need to decide how to allocate your machines to tasks.

Alternatively, you could be an employer with a set of jobs to complete and a list of employees to assign to these jobs. Each employee has various abilities, and hence, can complete jobs in differing amounts of time. And each employee's time might cost a different amount. How should you assign your employees to jobs in order to minimize your total costs?

Assignment Problem:

Given m machines and n jobs, find a least cost assignment of jobs to machines. The cost of assigning job j to machine i is c_{ij} .

Example 5.3: Machine Assignment

Gurobipy

Sets:

- Let $I = \{0, 1, 2, 3\}$ set of machines.
- Let $J = \{0, 1, 2, 3\}$ be the set of tasks.

Parameters:

- c_{ij} - the cost of assigning machine i to job j

Variables:

- Let

$$x_{ij} = \begin{cases} 1 & \text{if machine } i \text{ assigned to job } j \\ 0 & \text{otherwise.} \end{cases}$$

Model:

$$\begin{aligned} \min \quad & \sum_{i \in I, j \in J} c_{ij} x_{ij} && \text{(Minimize cost)} \\ \text{s.t.} \quad & \sum_{i \in I} x_{ij} = 1 && \text{for all } j \in J \quad \text{(All jobs are assigned one machine)} \\ & \sum_{j \in J} x_{ij} = 1 && \text{for all } i \in I \quad \text{(All machines are assigned to a job)} \\ & x_{ij} \in \{0, 1\} \forall i \in I, j \in J \end{aligned}$$

Example 5.4: School Bus Routing Problem

PuLP

A school district has a set of schools I and a fleet of school buses J . Each bus needs to be assigned to a school every morning to pick up students. The cost c_{ij} represents the fuel cost for bus j to reach school i and complete the route. The district aims to minimize the total fuel cost while ensuring that each school is served by exactly one bus and each bus is assigned to exactly one school.

The fuel costs can be found in the following table. They are also in csv file format [here](#).

Bus/School	School A	School B	School C	School D	School E
Bus 1	50	80	60	90	100
Bus 2	60	85	55	70	110
Bus 3	75	65	50	85	90
Bus 4	70	90	55	80	95
Bus 5	80	70	60	75	85

We can model this as follows:

Indices:

- i - Index for schools, $i \in I$
- j - Index for buses, $j \in J$

Parameters:

- c_{ij} - Fuel cost for bus j to serve school i .

Decision Variables:

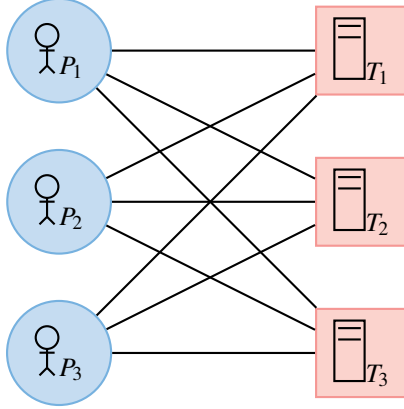
- x_{ij} - 1 if bus j is assigned to school i , 0 otherwise.

Model:

$$\begin{aligned}
 \min \quad & \sum_{i \in I, j \in J} c_{ij} x_{ij} && \text{(Minimize total fuel cost)} \\
 \text{s.t.} \quad & \sum_{i \in I} x_{ij} = 1 && \text{for all } j \in J \quad \text{(Each bus is assigned to one school)} \\
 & \sum_{j \in J} x_{ij} = 1 && \text{for all } i \in I \quad \text{(Each school is served by one bus)} \\
 & x_{ij} \in \{0, 1\} \forall i \in I, j \in J
 \end{aligned}$$

5.3 Assignment Problem

The **assignment problem** is a fundamental optimization problem in combinatorial optimization. It involves assigning a set of n agents (or workers) to n tasks (or jobs) such that each agent is assigned exactly one task, and each task is assigned to exactly one agent. The goal is to minimize the total assignment cost.



Sets:

- I – Set of agents (or workers), indexed by i .
- J – Set of tasks (or jobs), indexed by j .

Parameters:

- C_{ij} – Cost associated with assigning agent i to task j .

Decision Variables:

- x_{ij} – Binary decision variable, where:

$$x_{ij} = \begin{cases} 1, & \text{if agent } i \text{ is assigned to task } j, \\ 0, & \text{otherwise.} \end{cases}$$

Mathematical Model:

$$\min \sum_{i \in I} \sum_{j \in J} C_{ij} x_{ij} \quad (5.1)$$

$$\text{s.t. } \sum_{j \in J} x_{ij} = 1, \quad \forall i \in I \quad (\text{Each agent gets exactly one task}) \quad (5.2)$$

$$\sum_{i \in I} x_{ij} = 1, \quad \forall j \in J \quad (\text{Each task is assigned to exactly one agent}) \quad (5.3)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in I, j \in J. \quad (5.4)$$

Explanation:

- **Objective function** (5.1): Minimizes the total assignment cost based on given costs C_{ij} .
- **Constraints** (5.2): Each agent is assigned to exactly one task.
- **Constraints** (5.3): Each task is assigned to exactly one agent.
- **Binary constraint** (5.4): Ensures each assignment is either made ($x_{ij} = 1$) or not ($x_{ij} = 0$).

This problem can be efficiently solved using algorithms such as the **Hungarian method** for the classical case or integer programming solvers like **branch and bound** for larger instances.

Example 5.5: Hiring for tasks

Excel PuLP Gurobipy

Recall Example 3.12. We define the following sets, parameters, and variables to construct the mathematical model.

Sets:

- $I = \{1, 2, 3\}$, the set of people.
- $J = \{1, 2, 3\}$, the set of tasks.

Parameters:

- C_{ij} , the cost of assigning person $i \in I$ to task $j \in J$. The costs are given in the following table:

$$C = \begin{bmatrix} 40 & 47 & 80 \\ 72 & 36 & 58 \\ 24 & 61 & 71 \end{bmatrix}$$

Variables:

- $x_{ij} = \begin{cases} 1 & \text{if person } i \text{ is assigned to task } j \\ 0 & \text{otherwise} \end{cases}$ for all $i \in I, j \in J$.

Model:

The objective is to minimize the total cost of assignments:

$$\text{Minimize } z = \sum_{i \in I} \sum_{j \in J} C_{ij} x_{ij} \quad (5.5)$$

Subject to the constraints:

1. Each person is assigned to exactly one task:

$$\sum_{j \in J} x_{ij} = 1 \quad \forall i \in I \quad (5.6)$$

2. Each task is assigned to exactly one person:

$$\sum_{i \in I} x_{ij} = 1 \quad \forall j \in J \quad (5.7)$$

3. Binary constraints on the variables:

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, \forall j \in J \quad (5.8)$$

This model ensures that each person is assigned to exactly one task, each task is assigned to exactly one person, and the total cost of the assignments is minimized.

The assignment problem has an integrality property, such that if we remove the binary restriction on the x variables (now just non-negative, i.e., $x_{ij} \geq 0$) then we still get binary assignments, despite the fact that it is now an LP. This property is very interesting and useful.

Of course, the objective function might not quite what we want, we might be interested ensuring that the team with the worst assignment is as good as possible (a fairness criteria). One way of doing this is to modify the assignment problem using a max-min objective:

Min-max Assignment-like Formulation

$$\begin{aligned} \min z \\ \text{s.t. } z &\geq \sum_{i \in I} C_{ij} x_{ij}, & \forall j \in J \\ \sum_{i \in I} x_{ij} &= 1, & \forall j \in J \\ \sum_{j \in J} x_{ij} &= 1, & \forall i \in I \\ x_{ij} &\geq 0, & \forall i \in I, j \in J \end{aligned}$$

Does this formulation have the integrality property? (It is not an assignment problem.)

Consider a simple example where two teams are to be assigned to two projects, and the teams provide the following rankings (lower ranking is better):

	Project 1	Project 2
Team 1	2	1
Team 2	2	1

Both teams prefer Project 2.

If we remove the binary restriction on the x -variable, the values can range between 0 and 1. For the original assignment problem, the optimal solution has $z = 2$, and fractional x -values do not improve z .

For the min-max assignment problem, however, this is not the case. The optimal solution has $z = 1.5$, occurring when each team is assigned half of each project (i.e., for Team 1, we have $x_{11} = 0.5$ and $x_{21} = 0.5$). By fractionalizing assignments, the worst-case project cost is minimized.

Case Study:

Preference-Based Housing Assignment in Uruguayan Cooperatives

Housing cooperatives in Uruguay play a crucial role in providing affordable housing to their members. Traditionally, once a new cooperative building was completed, the assignment of individual housing units was conducted through a random draw. This method, while simple, ignored individual preferences such as sunlight orientation, floor level, and proximity to shared facilities. Recognizing the potential for improved member satisfaction, researchers developed an alternative assignment method based on **mixed-integer programming (MIP)**.

The Challenge

In Uruguayan housing cooperatives, every member is entitled to a housing unit. Since the assignment process happens only once per building, it is crucial to ensure fairness while maximizing overall satisfaction. The existing lottery-based system failed to account for members' preferences, sometimes leading to dissatisfaction and informal exchanges of units after the assignment. The challenge was to develop a new assignment process that:

- Ensured fairness among all members,
- Incorporated individual preferences,
- Was easy to explain and implement,
- Could be executed in real time during cooperative meetings.

The Solution: A Two-Stage Optimization Model

A team of researchers proposed a **two-stage mixed-integer programming (MIP) model** to address these concerns. The assignment problem was formulated with the following structure:

Stage 1 – Ensuring Fairness: The model maximizes the satisfaction of the least satisfied member. This ensures that no one is assigned an extremely undesirable unit.

Stage 2 – Optimizing Global Satisfaction: Given the fairness constraints from Stage 1, the model then seeks to optimize the total satisfaction across all cooperative members.

The assignment process was implemented using a software tool called **MTAV (Mejor Tecnología de Asignación de Viviendas – A Better Technology for Housing Assignment)**, designed to be transparent and user-friendly.

Implementation and Results

The first real-world test of the model occurred in **2016** with the **Virazón housing cooperative**, consisting of **50 members**. Before deployment, the cooperative members reviewed and voted to approve the new method. Their individual housing preferences were collected and input into the system.

On the day of the assignment:

- The housing allocation was carried out in **real-time** during a cooperative meeting.

- The results were projected on a large screen for transparency.
- A notary was present to certify the fairness of the process.
- The optimization model took only a few **seconds** to compute the assignments.

The outcomes were striking:

- **Four-bedroom apartments:** Both members received their first-choice units.
- **Three-bedroom apartments:** More than 37% of members received their first-choice unit, and the average satisfaction ranking was significantly better than a random assignment.
- **Two-bedroom apartments:** Over **half** of the members received one of their top four choices, with all assignments being significantly better than expected under a random draw.

The cooperative members overwhelmingly supported the new system, reporting higher satisfaction compared to previous assignment methods.

Broader Impact

Since the successful implementation at Virazón, **over 30 housing cooperatives** in Uruguay have adopted the model. Further developments have included:

- The introduction of a **graphical user interface** to make data input more intuitive.
- Support for **partial order preferences**, allowing members to indicate that they have no strong preference between certain units.
- Exploration of additional fairness criteria and trade-offs between individual and collective satisfaction.

Beyond its technical achievements, this case study highlights how **operations research techniques** can be applied to solve real-world social challenges. By integrating mathematical optimization with participatory decision-making, the project has significantly improved the way housing is assigned in cooperative communities, setting a precedent for future applications in housing policy and beyond.

Optimization models

The optimization model required the following input parameters and variables:

Sets:

- I : Set of cooperative members needing housing assignments.
- J : Set of available housing units.

Parameters:

- p_{ij} : Preference score of member i for housing unit j , where lower values indicate higher preference (e.g., 1 = most preferred).
- Some versions of the model supported **partial order preferences**, allowing members to express equal preference for multiple units.

Decision Variables:

- $x_{ij} \in \{0, 1\}$: Binary variable indicating whether member i is assigned to unit j :

$$x_{ij} = \begin{cases} 1, & \text{if member } i \text{ is assigned to unit } j \\ 0, & \text{otherwise} \end{cases}$$

Two-Stage Optimization Approach

Stage 1: Ensuring Fairness

- Define S as the satisfaction level of the least satisfied member:

$$S = \min_{i \in I} \sum_{j \in J} p_{ij} x_{ij}$$

- Ensure each member is assigned exactly one unit and each unit is assigned to exactly one member.

This is achieved by the model

$$S = \min z \quad (5.9)$$

$$\text{s.t. } z \geq \sum_{j \in J} p_{ij} x_{ij}, \quad \forall i \in I \quad (5.10)$$

$$\sum_{j \in J} x_{ij} = 1, \quad \forall i \in I \quad (5.11)$$

$$\sum_{i \in I} x_{ij} = 1, \quad \forall j \in J \quad (5.12)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in I, j \in J \quad (5.13)$$

$$z \in \mathbb{R} \quad (5.14)$$

Stage 2: Optimizing Global Satisfaction

- Given the fairness threshold S from Stage 1, optimize total satisfaction:

$$\min \sum_{i \in I} \sum_{j \in J} p_{ij} x_{ij}$$

- Subject to the fairness constraint:

$$\sum_{j \in J} p_{ij} x_{ij} \leq S, \quad \forall i \in I$$

- Ensures that no member receives a unit less desirable than the worst outcome allowed by Stage 1.

This is achieved by the model

$$\min \sum_{i \in I} \sum_{j \in J} p_{ij} x_{ij} \quad (5.15)$$

$$\text{s.t. } \sum_{j \in J} x_{ij} = 1, \quad \forall i \in I \quad (5.16)$$

$$\sum_{i \in I} x_{ij} = 1, \quad \forall j \in J \quad (5.17)$$

$$\sum_{j \in J} p_{ij} x_{ij} \leq S, \quad \forall i \in I \quad (5.18)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in I, j \in J \quad (5.19)$$

Implementation Notes

- The model was formulated as a **mixed-integer programming (MIP)** model.
- The assignment was computed in **real-time** during cooperative meetings.
- The **MTAV** software was used to collect user preferences and compute assignments efficiently.
- The system ensured transparency, allowing cooperative members to verify their preferences before the final optimization.

References

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- M. Prino, E. Sánchez, H. Cancela. “Optimal distribution of habitational units in a cooperative: A mathematical application to optimize satisfaction” (text in Spanish). Proceedings of CLEI 2016 (Conferencia Latinoamericana de Informática), 2016. <https://doi.org/10.1109/CLEI.2016.7833357>

5.4 Modeling Tricks

5.4.1. Maximizing a minimum

When the constraints could be general, we will write $x \in X$ to define general constraints. For instance, we could have $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ or $X = \{x \in \mathbb{R}^n : Ax \leq b, x \in \mathbb{Z}^n\}$ or many other possibilities.

Consider the problem

$$\begin{array}{ll} \max & \min\{x_1, \dots, x_n\} \\ \text{such that} & x \in X \end{array}$$

Having the minimum on the inside is inconvenient. To remove this, we just define a new variable y and enforce that $y \leq x_i$ and then we maximize y . Since we are maximizing y , it will take the value of the smallest x_i . Thus, we can recast the problem as

$$\begin{array}{ll} \max & y \\ \text{such that} & y \leq x_i \text{ for } i = 1, \dots, n \\ & x \in X \end{array}$$

Example 5.1: Minimizing an Absolute Value

Note that

$$|t| = \max(t, -t),$$

Thus, if we need to minimize $|t|$ we can instead write

$$\min \quad z \tag{5.1}$$

$$\text{such that } t \leq z \tag{5.2}$$

$$-t \leq z. \tag{5.3}$$

5.5 Network Flow Models and Applications

Network flow models are used to optimize the movement of commodities, resources, or information through a network while satisfying capacity and demand constraints. These models have a broad range of applications in various fields:

- **Transportation and Logistics:** Optimizing the movement of goods through a supply chain while minimizing transportation costs.
- **Telecommunications:** Managing data traffic flow in networks to minimize congestion and latency.
- **Water and Energy Distribution:** Balancing water flow in pipeline systems or optimizing power grid distribution to minimize energy loss.
- **Project Scheduling:** Allocating tasks to workers while minimizing delays and costs.
- **Assignment and Matching Problems:** Assigning resources to tasks, such as employees to projects or students to schools, in an optimal manner.

To begin a discussion on Network flow, we first need to discuss graphs.

5.5.1. Graphs

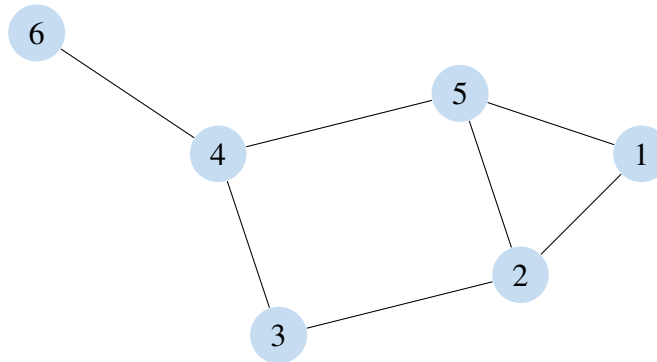
5.5.1.1. Undirected Graphs

Definition 5.2

A (undirected) graph $G = (V, E)$ is defined by a set of vertices V and a set of edges E that contains pairs of vertices.

For example, the following graph G can be described by:

- the vertex set $V = \{1, 2, 3, 4, 5, 6\}$ and
- the edge set $E = \{(4, 6), (4, 5), (5, 1), (1, 2), (2, 5), (2, 3), (3, 4)\}$.



In an undirected graph, we do not distinguish the direction of the edge. That is, for two vertices $i, j \in V$, we can equivalently write (i, j) or (j, i) to represent the edge.

5.5.1.2. Directed Graphs

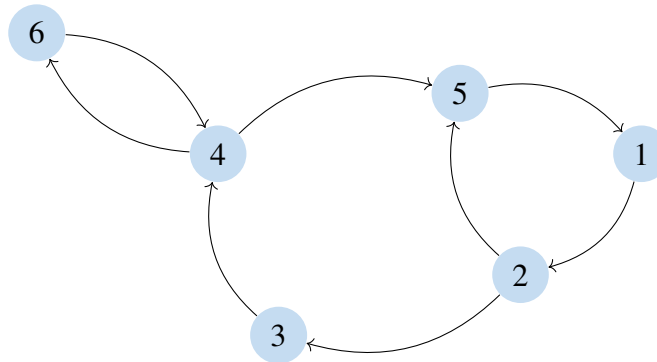
Alternatively, we will want to consider directed graphs.

Definition 5.3

A directed graph (or *di-graph* for short) is denoted as $G = (V, A)$ where V is a set of vertices and A is a set of ordered pairs of vertices called arcs. That is, an arc is like an edge, but the direction of it matters.

For example, the following directed graph G can be described by

- the vertex set $V = \{1, 2, 3, 4, 5, 6\}$ and
- the arc set $A = \{(4, 6), (6, 4), (4, 5), (5, 1), (1, 2), (2, 5), (2, 3), (3, 4)\}$.



5.5.2. Example: Minimum-Cost Flow in a Transportation Network

Example 5.6:

Excel PuLP Gurobipy

To illustrate the minimum-cost network flow problem, consider a company that wants to transport goods from two warehouses to two retail stores at the lowest possible cost. The company can send goods through a network of transportation routes, each with a given capacity and per-unit transportation cost.

The transportation network consists of:

- **Two warehouses** (supply nodes) that store goods:
 - **Warehouse 1** can supply up to 20 units.
 - **Warehouse 2** can supply up to 30 units.
- **Two retail stores** (demand nodes) that require goods:
 - **Store 1** requires 25 units.
 - **Store 2** requires 25 units.
- Transportation routes (**arcs**) connecting warehouses to stores, each with a transportation cost per unit and capacity limit.

Graph Representation

The transportation network can be represented as a directed graph $G = (V, A)$, where:

- $V = \{W_1, W_2, S_1, S_2\}$ (two warehouses and two stores).
- $A = \{(W_1, S_1), (W_1, S_2), (W_2, S_1), (W_2, S_2)\}$ (directed transportation routes).
- Each arc (i, j) has a given transportation cost C_{ij} and capacity u_{ij} .

The network data is summarized in the following table:

Route	Cost per unit (c_{ij})	Capacity (u_{ij})
Warehouse 1 to Store 1	4	15
Warehouse 1 to Store 2	6	10
Warehouse 2 to Store 1	5	20
Warehouse 2 to Store 2	2	20

Mathematical Formulation

Let x_{ij} represent the amount of goods transported from warehouse i to store j . The problem is formulated as:

$$\min \quad 4x_{W_1,S_1} + 6x_{W_1,S_2} + 5x_{W_2,S_1} + 2x_{W_2,S_2} \quad (5.1)$$

$$\text{s.t.} \quad x_{W_1,S_1} + x_{W_1,S_2} \leq 20, \quad (\text{Supply constraint for Warehouse 1}) \quad (5.2)$$

$$x_{W_2,S_1} + x_{W_2,S_2} \leq 30, \quad (\text{Supply constraint for Warehouse 2}) \quad (5.3)$$

$$x_{W_1,S_1} + x_{W_2,S_1} = 25, \quad (\text{Demand constraint for Store 1}) \quad (5.4)$$

$$x_{W_1,S_2} + x_{W_2,S_2} = 25, \quad (\text{Demand constraint for Store 2}) \quad (5.5)$$

$$0 \leq x_{ij} \leq u_{ij}, \quad \forall (i,j) \in A. \quad (5.6)$$

Solution

Solving the linear program, we obtain the following optimal flow values:

Route	Optimal Flow (x_{ij})
Warehouse 1 to Store 1	15
Warehouse 1 to Store 2	5
Warehouse 2 to Store 1	10
Warehouse 2 to Store 2	20

The total minimum cost is given by:

$$(4 \times 15) + (6 \times 5) + (5 \times 10) + (2 \times 20) = 60 + 30 + 50 + 40 = 180.$$

Interpretation of the Solution

- **Warehouse 1** sends **15 units** to Store 1 and **5 units** to Store 2.
- **Warehouse 2** sends **10 units** to Store 1 and **20 units** to Store 2.
- The **total transportation cost is minimized to 180**.

This solution meets all constraints while ensuring goods are transported at the lowest possible cost.

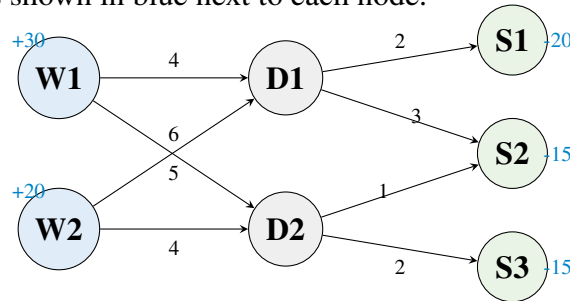
5.5.3. Minimum Cost Network Flow Problem

The *minimum cost network flow problem* generalizes basic flow models by incorporating arc capacities, arc costs, and supply/demand at the nodes. The goal is to determine how to route flow through a network at the lowest possible cost while meeting demand and respecting all constraints.

Real-World Example: Amazon's Distribution Network. Amazon must ship goods from large regional warehouses to local retail stores through intermediate distribution centers. Each warehouse has a certain supply capacity, each store has a demand, and each route incurs a transportation cost (fuel, time, and labor). Amazon wants to decide how to route products to minimize cost, while ensuring demand is met and no transportation link exceeds its capacity.

The figure below shows a small example. There are:

- Two warehouses (W1, W2) with a total of 50 units of supply,
- Two distribution centers (D1, D2),
- Three stores (S1, S2, S3) that need 20, 15, and 15 units, respectively,
- Arc costs labeled in black,
- Node demands/supplies shown in blue next to each node.



The problem is to determine how many units to send along each route to minimize the total shipping cost.

Sets:

- V : set of nodes in the network (warehouses, distribution centers, stores)
- A : set of directed arcs (i, j) representing shipment routes

Parameters:

- c_{ij} : cost per unit shipped on arc $(i, j) \in A$
- u_{ij} : maximum capacity of arc $(i, j) \in A$
- d_i : net demand at node $i \in V$ (positive = demand, negative = supply)

Decision Variables:

x_{ij} : amount of flow shipped along arc $(i, j) \in A$

Objective and Constraints:

$$\begin{aligned}
 \min \quad & \sum_{(i,j) \in A} c_{ij} \cdot x_{ij} \\
 \text{s.t.} \quad & \sum_{i:(i,k) \in A} x_{ik} - \sum_{j:(k,j) \in A} x_{kj} = d_k \quad \forall k \in V \quad (\text{flow balance}) \\
 & 0 \leq x_{ij} \leq u_{ij} \quad \forall (i,j) \in A \quad (\text{capacity constraints})
 \end{aligned}$$

This problem can be solved using linear programming, and efficient specialized algorithms exist due to its structure. Applications span:

- Freight transportation,
- Supply chain optimization,
- Scheduling and telecommunications.

5.5.4. (Unstructured) Minimum Cost Network Flow Problem

The minimum cost network flow problem is an extension of the network flow problem that considers not only the capacities of the arcs, but also the costs associated with sending flow along the arcs and the demands at the nodes. The objective is to find the flow of minimum cost that satisfies the demands at the nodes and the capacity constraints on the arcs.

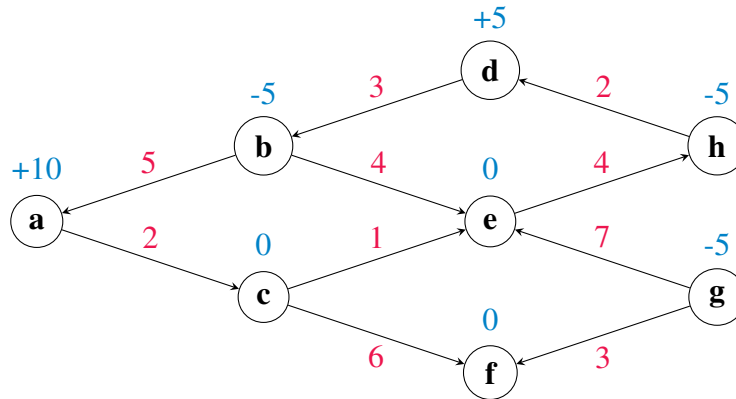
A network can be described as a directed graph $G = (V, A)$, where V is the set of nodes and A is the set of arcs. Each arc $(i, j) \in A$ has a capacity u_{ij} and a cost c_{ij} . Each node $i \in V$ has a demand d_i , which can be positive, negative, or zero. A positive demand means that the node requires that amount of flow, a negative demand means that the node supplies that amount of flow, and a zero demand means that the node neither requires nor supplies flow.

Consider, for example: The set of vertices is given by:

$$V = \{a, b, c, d, e, f, g, h\}$$

The set of directed arcs is:

$$A = \{(b, a), (a, c), (b, d), (b, e), (c, e), (c, f), (h, d), (g, e), (e, h), (g, f)\}$$



From	To	Cost
b	a	5
a	c	2
d	b	3
b	e	4
c	e	1
c	f	6
h	d	2
g	e	7
e	h	4
g	f	3

Table 5.1: Arc Costs in the Network Flow Problem

Node	Net Flow
a	+10
b	-5
c	0
d	+5
e	0
f	0
g	-5
h	-5

Table 5.2: Net Flow at Each Node

The minimum cost network flow problem can be defined as follows: Given a network $G = (V, A)$, capacities u_{ij} and costs c_{ij} on the arcs, and demands d_i at the nodes, find the flow x_{ij} on the arcs that minimizes the total cost of the flow, while satisfying the demands at the nodes and the capacity constraints on the arcs.

Sets:

- V : Set of nodes in the network.
- A : Set of arcs in the network.

Parameters:

- u_{ij} : Capacity of arc $(i, j) \in A$.
- c_{ij} : Cost per unit of flow on arc $(i, j) \in A$.
- d_i : Change at node $i \in V$.

Variables:

- x_{ij} : Flow on arc $(i, j) \in A$.

Model: The minimum cost network flow problem can be formulated as a linear programming problem as follows:

$$\begin{aligned}
 &\text{Minimize} && \sum_{(i,j) \in A} c_{ij}x_{ij} \\
 &\text{Subject to} && \sum_{i:(i,k) \in A} x_{ik} - \sum_{j:(k,j) \in A} x_{kj} = d_k, \quad \forall k \in V \\
 &&& 0 \leq x_{ij} \leq u_{ij}, \quad \forall (i, j) \in A
 \end{aligned}$$

Objective Function - The objective is to minimize the total cost of the flow.

Constraints - Flow Conservation: The first set of constraints ensures that the flow into each node minus the flow out of the node is equal to the demand at the node.

- **Capacity Constraints:** The second set of constraints ensures that the flow on each arc does not exceed its capacity.

Minimum-cost network flow problems are widely used in transportation planning, supply chain logistics, and production scheduling, where the goal is to efficiently allocate resources at the lowest possible cost.

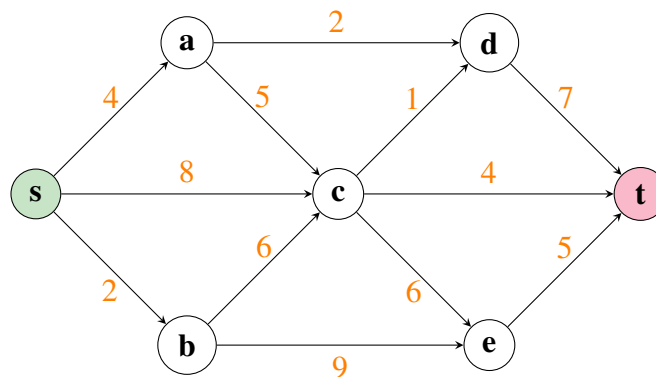
5.5.5. Maximum flow

The maximum flow problem is a fundamental problem in operations research and computer science, with applications in a wide variety of fields. It involves finding the maximum amount of flow that can be sent from a source node to a sink node in a network, without exceeding the capacities of the individual arcs, and ensuring that the flow on each arc is non-negative. This problem arises in many real-world situations, such as in transportation, where it can be used to find the maximum amount of goods that can be transported from a factory to a warehouse, or in telecommunications, where it can be used to find the maximum amount of data that can be sent from one server to another. Other applications include supply chain optimization, water distribution, and electrical power grid optimization. In this section, we will describe a linear programming formulation for the network flow problem.

A network can be described as a directed graph $G = (V, A)$, where V is the set of nodes and A is the set of arcs. Each arc $(i, j) \in A$ has a capacity u_{ij} , which is the maximum amount of flow that can traverse the arc from node i to node j .

The network flow problem can be defined as follows: Given a network $G = (V, A)$, a source node s , a sink node t , and capacities u_{ij} on the arcs, find the maximum flow from s to t that respects the capacity constraints.

For example, consider the directed graph here with capacities on the arcs.



Example 5.7: Airline Transfer Network: Maximizing Passenger Throughput

Excel PuLP

Gurobipy

An airline is trying to reroute as many stranded passengers as possible from an origin city (s) to a destination city (t) using a network of connecting flights. Each flight has a known number of available seats, and passengers may transfer at intermediate airports. The objective is to determine the maximum number of passengers that can be routed from origin to destination, using available seat capacity and respecting flight routes.

The network includes:

- Origin airport s ,
- Intermediate hubs: a, b, c ,
- Regional transfer airports: d, e ,
- Final destination airport t ,
- Directed flight legs (arcs) with known seat availability.

Sets and Parameters:

- $V = \{s, a, b, c, d, e, t\}$: set of airports.
- $A = \{(s, a), (s, b), (s, c), (a, c), (a, d), (b, c), (b, e), (c, d), (c, e), (c, t), (d, e), (e, t)\}$: set of flight legs.
- u_{ij} : number of available seats on flight from airport i to j for all $(i, j) \in A$.

Decision Variables:

x_{ij} : number of passengers assigned to fly from airport i to j for all $(i, j) \in A$

Objective and Constraints:

$$\begin{aligned}
 \max \quad & x_{sa} + x_{sb} + x_{sc} && \text{(max passengers from origin)} \\
 \text{s.t.} \quad & x_{ac} + x_{ad} - x_{sa} = 0 && \text{(flow balance at } a) \\
 & x_{bc} + x_{be} - x_{sb} = 0 && \text{(at } b) \\
 & x_{cd} + x_{ce} + x_{ct} - x_{ac} - x_{bc} - x_{sc} = 0 && \text{(at } c) \\
 & x_{de} - x_{ad} - x_{cd} = 0 && \text{(at } d) \\
 & x_{et} - x_{be} - x_{ce} - x_{de} = 0 && \text{(at } e) \\
 & 0 \leq x_{sa} \leq 4, \quad 0 \leq x_{sb} \leq 2, \quad 0 \leq x_{sc} \leq 8 && \text{(available seats on initial flights)} \\
 & 0 \leq x_{ac} \leq 5, \quad 0 \leq x_{ad} \leq 2, \quad 0 \leq x_{bc} \leq 6, \quad 0 \leq x_{be} \leq 9 \\
 & 0 \leq x_{cd} \leq 1, \quad 0 \leq x_{ce} \leq 3, \quad 0 \leq x_{ct} \leq 4, \quad 0 \leq x_{de} \\
 & \leq 7, \quad 0 \leq x_{et} \leq 5
 \end{aligned}$$

In compact summation notation form, we have

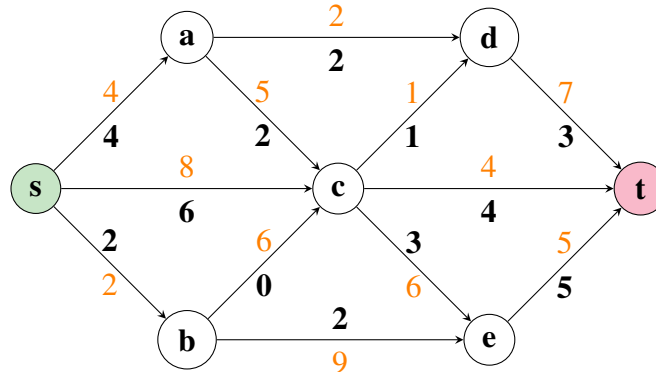
$$\begin{aligned}
& \max \sum_{j \in V: (s,j) \in A} x_{sj} \\
& \text{s.t.} \quad \sum_{j \in V: (k,j) \in A} x_{kj} - \sum_{i \in V: (i,k) \in A} x_{ik} = 0 \quad \forall k \in V \setminus \{s, t\} \\
& \quad 0 \leq x_{ij} \leq u_{ij} \quad \forall (i, j) \in A
\end{aligned}$$

This model helps the airline determine how to assign passengers to flight legs in a way that makes maximum use of the network's available capacity. It is especially useful during disruption scenarios like:

- severe weather cancellations,
- emergency rerouting due to maintenance issues,
- high-demand events requiring temporary overflow routing.

The same framework can be extended to include priorities, time windows, or rebooking costs.

The solution is shown here in bold values on the graph. There is a total flow of 12.



The maximum flow problem can be written mathematically in the following way.

Sets:

- V : Set of nodes in the network.
- A : Set of arcs in the network.

Parameters:

- u_{ij} : Capacity of arc $(i, j) \in A$, which is the maximum amount of flow that can traverse the arc from node i to node j .
- s : Source node.
- t : Sink node.

Variables:

- x_{ij} : Flow on arc $(i, j) \in A$.

Model: The network flow problem can be formulated as a linear programming problem as follows:

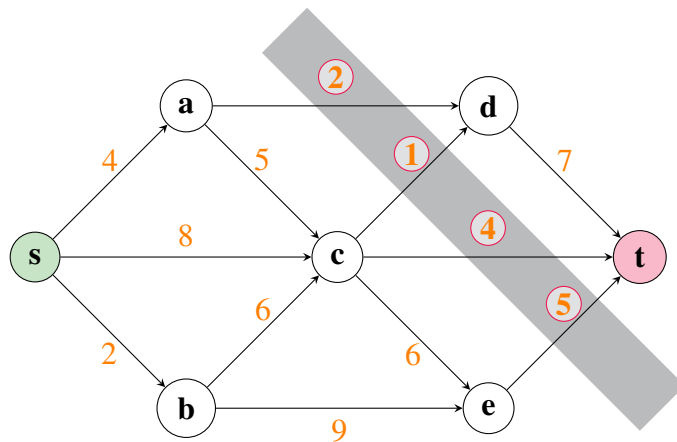
$$\begin{aligned}
 &\text{Maximize} && \sum_{j:(s,j) \in A} x_{sj} \\
 &\text{Subject to} && \sum_{i:(i,k) \in A} x_{ik} - \sum_{j:(k,j) \in A} x_{kj} = 0, \quad \forall k \in V \setminus \{s, t\} \\
 &&& 0 \leq x_{ij} \leq u_{ij}, \quad \forall (i, j) \in A
 \end{aligned}$$

Objective Function - The objective is to maximize the total flow from the source node s to the sink node t .

Constraints - Flow Balance: The first set of constraints ensures that the flow into each node is equal to the flow out of the node, except for the source and sink nodes.

- Capacity Constraints: The third set of constraints ensures that the flow on each arc does not exceed its capacity.

We will discuss duality later. But duality is a concept that helps create bounds on a problem. For this particular problem, we can think of a dual as a *cut* in the graph - something that separates s from t . For example:



The selected arcs are a cut that separates s from t . The sum of their weights is $2 + 1 + 4 + 5 = 12$, which implies that there is not a flow more than 12 units from s to t .

This bound is a form of *duality*. We will explore the concept more in later chapters. This duality is captured in the following theorem.

Theorem 5.4

The maximum flow amount is equal to the size of the minimum cut.

5.6 Exercises

Exercise 5.8 [Fair Housing Assignment for 12 Families - part 1]

A new apartment complex has been completed, and 12 families need to be assigned to 12 available apartments. Each family has ranked the apartments (1 = most preferred, 12 = least preferred).

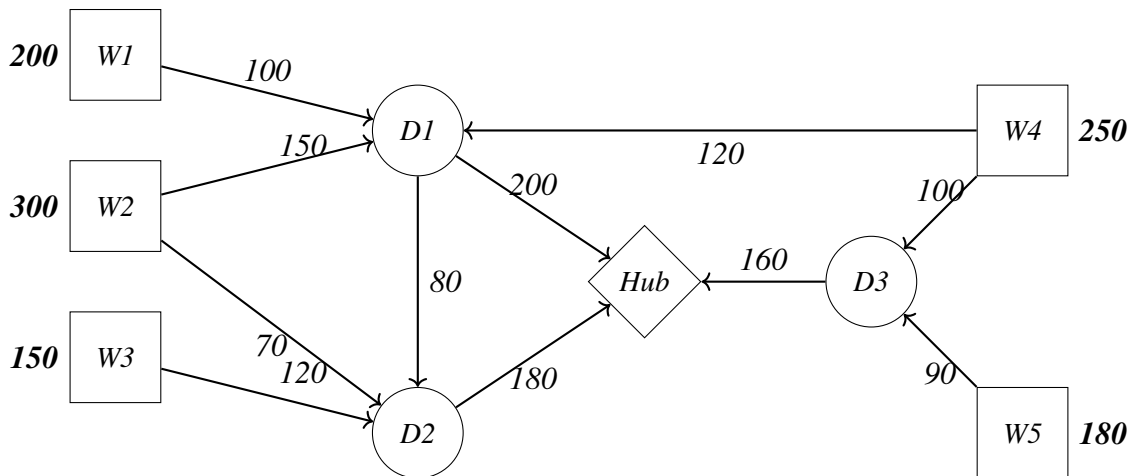
1. Use Excel Solver to find an assignment that minimizes the sum of preferences.
2. What units were assigned to which families, and what was their preference score?
3. What is the worst preference score assigned?

Download the data for this problem [here](#) data.

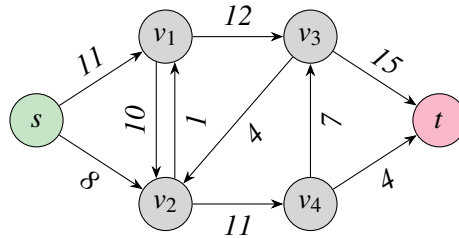
Exercise 5.9 Maximum Network Flow to a Single Node A disaster relief effort is underway to transport supplies to a central distribution hub in a flooded region. The network consists of multiple warehouses (nodes) that currently have varying inventory levels of emergency supplies. Each transportation route (arc) between warehouses is serviced by a single truck with a limited carrying capacity. The goal is to determine the optimal transportation plan that maximizes the amount of supplies delivered to the central hub while respecting inventory constraints and truck capacities on each route.

Formulate a max-flow optimization problem for this scenario. Define your decision variables, objective function, and constraints explicitly. Assume that you do not want to leave any inventory at the distribution centers.

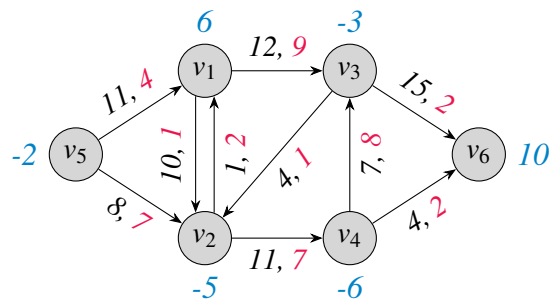
Node	Type	Inventory (if applicable)	Connected To	Capacity
W1	Warehouse	200	D1	100
W2	Warehouse	300	D1	150
W3	Warehouse	150	D2	120
W4	Warehouse	250	D3	100
W5	Warehouse	180	D3	90
D1	Distribution	N/A	Hub, D2	200, 80
D2	Distribution	N/A	Hub	180
Hub	Central Hub	N/A	D3	160
D3	Distribution	N/A	W4, W5	100, 90



Exercise 5.10 *Max flow* Write out the explicit constraints that define the max flow problem given in the picture. Solve this problem as a linear program using Excel.



Exercise 5.11 *Min Cost Network Flow* Use the picture below to formulate the corresponding capacitated min cost network flow. Solve this problem using Excel.



6. Graphically Solving Linear Programs

Outcomes

- A. Learn how to plot the feasible region and the objective function.
- B. Identify and compute extreme points of the feasible region.
- C. Find the optimal solution(s) to a linear program graphically.
- D. Classify the type of result of the problem as infeasible, unbounded, unique optimal solution, or infinitely many optimal solutions.

Resources

- Youtube! - Modeling with Linear Programming - Watch 4:20 - End
- Youtube! - Watch 1:43 - 3:43

Linear Programs (LP's) with two variables can be solved graphically by plotting the feasible region along with the level curves of the objective function.¹ We will show that we can find a point in the feasible region that maximizes the objective function using the level curves of the objective function.

We will begin with an easy example that is bounded and investigate the structure of the feasible region. We will then explore other examples.

6.1 Nonempty and Bounded Problem

Consider the problem

$$\begin{array}{ll}\max & 2x_1 + 5x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 16 \\ & 5x_1 + 3x_2 \leq 45 \\ & x_1, x_2 \geq 0\end{array}$$

We want to start by plotting the *feasible region*, that is, the set points (x_1, x_2) that satisfy all the constraints.

We can plot this by first plotting the four lines

¹Special thanks to Joshua Emmanuel and Christopher Griffin for sharing their content to help put this section together. Proper citations and referenes are forthcoming.

- $x_1 + 2x_2 = 16$
- $5x_1 + 3x_2 = 45$
- $x_1 = 0$
- $x_2 = 0$

and then shading in the side of the space cut out by the corresponding inequality.

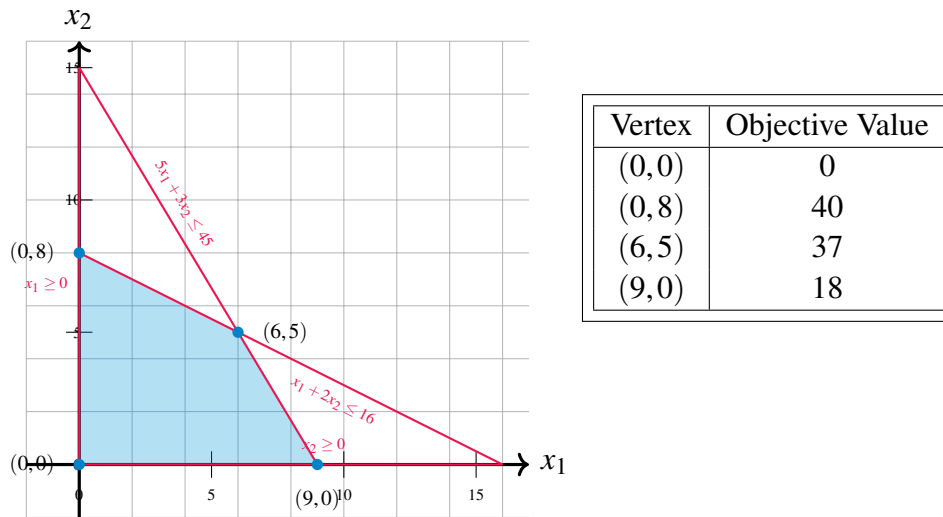


Figure 6.1: Desmos: Interactive plot!

The resulting feasible region can then be shaded in as the region that satisfies all the inequalities.

Notice that the feasible region is nonempty (it has points that satisfy all the inequalities) and also that it is bounded (the feasible points don't continue infinitely in any direction).

We want to identify the *extreme points* (i.e., the corners) of the feasible region. Understanding these points will be critical to understanding the optimal solutions of the model. Notice that all extreme points can be computed by finding the intersection of 2 of the lines. But! Not all intersections of any two lines are feasible.

We will later use the terminology *basic feasible solution* for an extreme point of the feasible region, and *basic solution* as a point that is the intersection of 2 lines, but is actually infeasible (does not satisfy all the constraints).

Theorem 6.1: Optimal Solution at a Vertex

If the feasible region of the linear program is nonempty and bounded, then there exists an optimal solution at vertex of the feasible region.

This theorem implies the following algorithm.

Algorithm 1 Enumerate Vertices Algorithm for Solving a Linear Programming Problem

1. **Identify the Feasible Region:** Determine the set of constraints defining the feasible region in the problem.
2. **Enumerate Vertices:** Find all the vertices (corner points) of the feasible region. This can be done by solving pairs of constraint equations to find their intersections, keeping only those that satisfy all constraints.
3. **Compute Objective Values:** Evaluate the objective function at each vertex of the feasible region.
4. **Compare Objective Values:** For a maximization problem, identify the vertex with the highest objective value. For a minimization problem, identify the vertex with the lowest objective value.
5. **Determine the Optimal Solution:** The vertex with the optimal objective value is the solution to the linear programming problem.

We will explore why Theorem 6.1 is true, and also what happens when the feasible region does not satisfy the assumptions of either nonempty or bounded.

6.2 Graphical Approach

We illustrate the idea first using the problem from Example 3.1. In a later section, we will formalize the mathematics to explain this result.

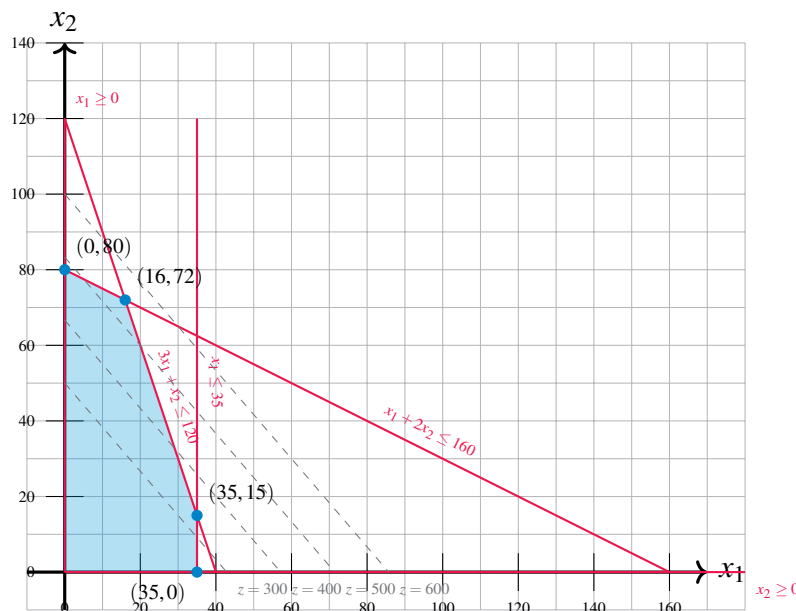


Figure 6.2: Feasible Region and Level Curves of the Objective Function: The shaded region in the plot is the feasible region and represents the intersection of the five inequalities constraining the values of x_1 and x_2 . On the right, we see the optimal solution is the “last” point in the feasible region that intersects a level set as we move in the direction of increasing profit.

Example 6.2: Continuation of ToyMaker Example

Let's continue the example of the Toy Maker begin in Example 3.1. Solve this problem graphically.

Solution. To solve the linear programming problem graphically, begin by drawing the feasible region. This is shown in the blue shaded region of Figure 6.2.

After plotting the feasible region, the next step is to plot the level curves of the objective function. In our problem, the level sets will have the form:

$$7x_1 + 6x_2 = z \implies x_2 = \frac{-7}{6}x_1 + \frac{z}{6}$$

This is a set of parallel lines with slope $-7/6$ and intercept $z/6$ where z can be varied as needed. The level curves for various values of z are parallel lines. In Figure 6.2 they are shown in colors ranging from red to yellow depending upon the value of z . Larger values of z are more yellow.

To solve the linear programming problem, follow the level sets along the gradient (shown as the black arrow) until the last level set (line) intersects the feasible region. If you are doing this by hand, you can draw a single line of the form $7x_1 + 6x_2 = z$ and then simply draw parallel lines in the direction of the gradient $(7, 6)$. At some point, these lines will fail to intersect the feasible region. The last line to intersect the feasible region will do so at a point that maximizes the profit. In this case, the point that maximizes $z(x_1, x_2) = 7x_1 + 6x_2$, subject to the constraints given, is $(x_1^*, x_2^*) = (16, 72)$.

Note the point of optimality $(x_1^*, x_2^*) = (16, 72)$ is at a corner of the feasible region. This corner is formed by the intersection of the two lines: $3x_1 + x_2 = 120$ and $x_1 + 2x_2 = 160$. In this case, the constraints

$$3x_1 + x_2 \leq 120$$

$$x_1 + 2x_2 \leq 160$$

are both *binding*, while the other constraints are non-binding. In general, we will see that when an optimal solution to a linear programming problem exists, it will always be at the intersection of several binding constraints; that is, it will occur at a corner of a higher-dimensional polyhedron. ♠

We can now define an algorithm for identifying the solution to a linear programming problem in two variables with a *bounded* feasible region.

(First Try) Algorithm for Solving a Linear Programming Problem Graphically

Bounded Feasible Region, Unique Solution

1. Plot the feasible region defined by the constraints.
 2. Plot the level sets of the objective function.
 3. For a maximization problem, identify the level set corresponding the greatest (least, for minimization) objective function value that intersects the feasible region. This point will be at a corner.
 4. The point on the corner intersecting the greatest (least) level set is a solution to the linear programming problem.
-

Example 6.3: Lemonade Vendor

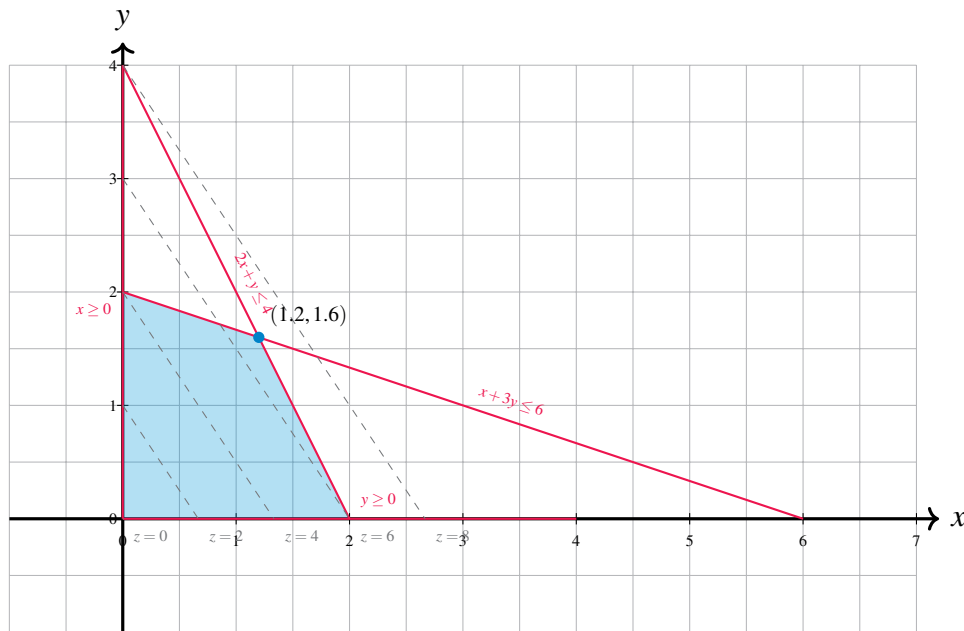
Say you are a vendor of lemonade and lemon juice. Each unit of lemonade requires 1 lemon and 2 litres of water. Each unit of lemon juice requires 3 lemons and 1 litre of water. Each unit of lemonade gives a profit of three dollars. Each unit of lemon juice gives a profit of two dollars. You have 6 lemons and 4 litres of water available. How many units of lemonade and lemon juice should you make to maximize profit?

If we let x denote the number of units of lemonade to be made and let y denote the number of units of lemon juice to be made, then the profit is given by $3x + 2y$ dollars. We call $3x + 2y$ the objective function. Note that there are a number of constraints that x and y must satisfy. First of all, x and y should be nonnegative. The number of lemons needed to make x units of lemonade and y units of lemon juice is $x + 3y$ and cannot exceed 6. The number of litres of water needed to make x units of lemonade and y units of lemon juice is $2x + y$ and cannot exceed 4. Hence, to determine the maximum profit, we need to maximize $3x + 2y$ subject to x and y satisfying the constraints $x + 3y \leq 6$, $2x + y \leq 4$, $x \geq 0$, and $y \geq 0$.

A more compact way to write the problem is as follows:

$$\begin{array}{llll} \text{maximize} & 3x & + & 2y \\ \text{subject to} & x & + & 3y \leq 6 \\ & 2x & + & y \leq 4 \\ & x & & \geq 0 \\ & & & y \geq 0. \end{array}$$

Solution. We can solve this maximization problem graphically as follows. We first sketch the set of (x, y) satisfying the constraints, called the feasible region, on the (x, y) -plane. We then take the objective function $3x + 2y$ and turn it into an equation of a line $3x + 2y = z$ where z is a parameter. Note that as the value of z increases, the line defined by the equation $3x + 2y = z$ moves in the direction of the normal vector $(3, 2)$. We call this direction the direction of improvement. Determining the maximum value of the objective function, called the optimal value, subject to the constraints amounts to finding the maximum value of z so that the line defined by the equation $3x + 2y = z$ still intersects the feasible region.



In the figure above, the lines with z at 0, 4 and 6.8 have been drawn. From the picture, we can see that if z is greater than 6.8, the line defined by $3x + 2y = z$ will not intersect the feasible region. Hence, the profit cannot exceed 6.8 dollars.

As the line $3x + 2y = 6.8$ does intersect the feasible region, 6.8 is the maximum value for the objective function. Note that there is only one point in the feasible region that intersects the line $3x + 2y = 6.8$, namely $(x, y) = (1.2, 1.6)$. In other words, to maximize profit, we want to make 1.2 units of lemonade and 1.6 units of lemon juice. ♠

The example linear programming problem presented in this has a single optimal solution. In general, the following outcomes can occur in solving a linear programming problem:

1. The linear programming problem has a unique solution. (We've already seen this.)
2. There are infinitely many alternative optimal solutions.
3. There is no solution and the problem's objective function can grow to positive infinity for maximization problems (or negative infinity for minimization problems).
4. There is no solution to the problem at all.

Case 3 above can only occur when the feasible region is unbounded; that is, it cannot be surrounded by a ball with finite radius. We will illustrate each of these possible outcomes in the next four sections. We will prove that this is true in a later chapter.

6.3 Infinitely Many Optimal Solutions

It can happen that there is more than one solution. In fact, in this case, there are infinitely many optimal solutions. We'll study a specific linear programming problem with an infinite number of solutions by modifying the objective function in Example 3.1.

Example 6.4: Toy Maker Alternative Solutions

Suppose the toy maker in Example 3.1 finds that it can sell planes for a profit of \$18 each instead of \$7 each. The new linear programming problem becomes:

$$\begin{aligned}
 \max \quad & z(x_1, x_2) = 18x_1 + 6x_2 \\
 \text{s.t.} \quad & 3x_1 + x_2 \leq 120 \\
 & x_1 + 2x_2 \leq 160 \\
 & x_1 \leq 35 \\
 & x_1 \geq 0 \\
 & x_2 \geq 0
 \end{aligned} \tag{6.1}$$

Solution. Applying our graphical method for finding optimal solutions to linear programming problems yields the plot shown in Figure 6.3. The level curves for the function $z(x_1, x_2) = 18x_1 + 6x_2$ are *parallel* to one face of the polygon boundary of the feasible region. Hence, as we move further up and to the right in the direction of the gradient (corresponding to larger and larger values of $z(x_1, x_2)$) we see that there is not *one* point on the boundary of the feasible region that intersects that level set with greatest value, but instead a side of the polygon boundary described by the line $3x_1 + x_2 = 120$ where $x_1 \in [16, 35]$.

Let:

$$P = \left\{ (x_1, x_2) \mid \begin{array}{l} 3x_1 + x_2 \leq 120, \\ x_1 + 2x_2 \leq 160, \\ x_1 \leq 35, \\ x_1, x_2 \geq 0 \end{array} \right\}$$

that is, P is the feasible region of the problem. Then for any value of $x_1^* \in [16, 35]$ and any value x_2^* so that $3x_1^* + x_2^* = 120$, we will have $z(x_1^*, x_2^*) \geq z(x_1, x_2)$ for all $(x_1, x_2) \in P$. Since there are infinitely many values that x_1 and x_2 may take on, we see this problem has an infinite number of alternative optimal solutions.

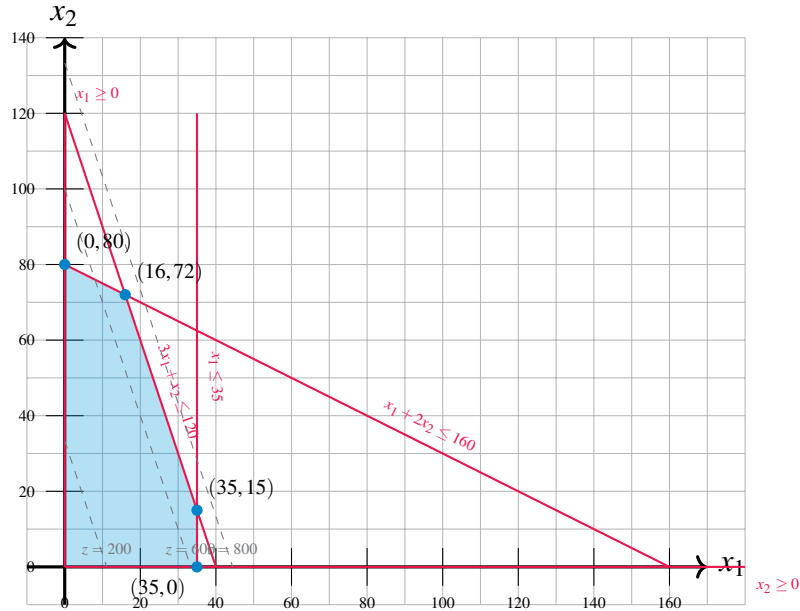


Figure 6.3: An example of infinitely many alternative optimal solutions in a linear programming problem. The level curves for $z(x_1, x_2) = 18x_1 + 6x_2$ are *parallel* to one face of the polygon boundary of the feasible region. Moreover, this side contains the points of greatest value for $z(x_1, x_2)$ inside the feasible region. Any combination of (x_1, x_2) on the line $3x_1 + x_2 = 120$ for $x_1 \in [16, 35]$ will provide the largest possible value $z(x_1, x_2)$ can take in the feasible region S .



Based on the example in this section, we can modify our algorithm for finding the solution to a linear programming problem graphically to deal with situations with an infinite set of alternative optimal solutions:

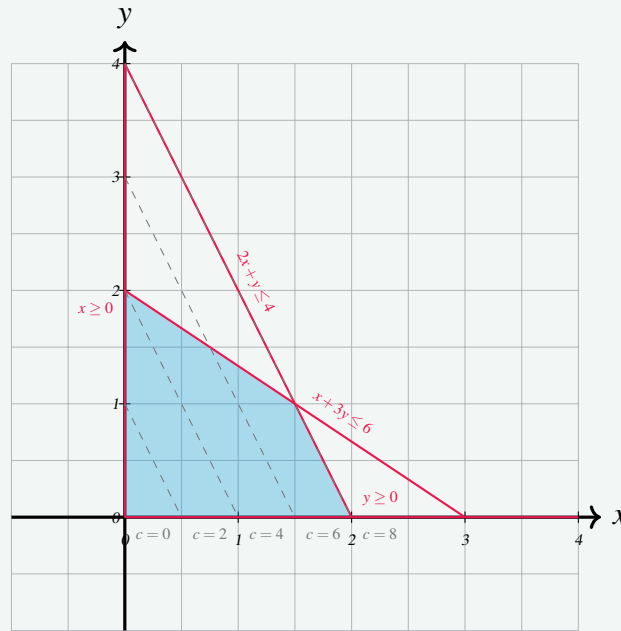
(Second try) Algorithm for Solving a Linear Programming Problem Graphically

Bounded Feasible Region

1. Plot the feasible region defined by the constraints.
 2. Plot the level sets of the objective function.
 3. For a maximization problem, identify the level set corresponding the greatest (least, for minimization) objective function value that intersects the feasible region. This point will be at a corner.
 4. The point on the corner intersecting the greatest (least) level set is a solution to the linear programming problem.
 5. **If the level set corresponding to the greatest (least) objective function value is parallel to a side of the polygon boundary next to the corner identified, then there are infinitely many alternative optimal solutions and any point on this side may be chosen as an optimal solution.**
-

Exercise 6.5

Consider the graphical representation below. How many optimal solutions are there if we are maximizing? How about if we are minimizing?



6.4 Problems with No Solution

Recall for *any* mathematical programming problem, the feasible set or region is simply a subset of $\mathbb{R}^n$. If this region is empty, then there is no solution to the mathematical programming problem and the problem is said to be *over constrained*. In this case, we say that the problem is *infeasible*. We illustrate this case for linear programming problems with the following example.

Example 6.6: Infeasible Problem

Consider the following linear programming problem:

$$\begin{aligned}
 \max \quad & z(x_1, x_2) = 3x_1 + 2x_2 \\
 \text{s.t.} \quad & \frac{1}{40}x_1 + \frac{1}{60}x_2 \leq 1 \\
 & \frac{1}{50}x_1 + \frac{1}{50}x_2 \leq 1 \\
 & x_1 \geq 30 \\
 & x_2 \geq 20
 \end{aligned} \tag{6.1}$$

Solution. The level sets of the objective and the constraints are shown in Figure 6.4.

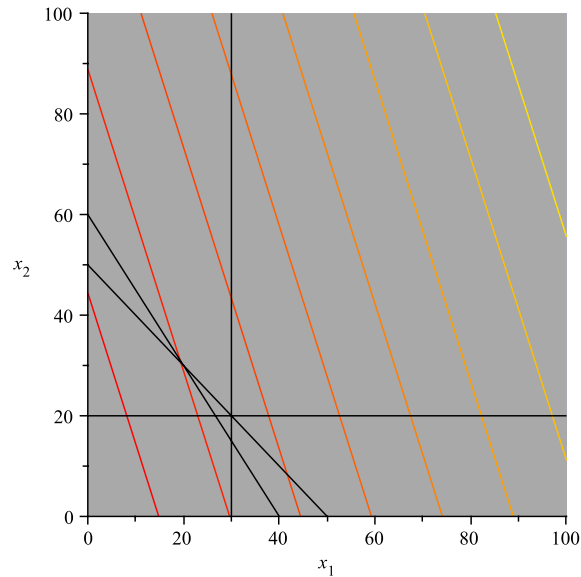


Figure 6.4: A Linear Programming Problem with no solution. The feasible region of the linear programming problem is empty; that is, there are no values for x_1 and x_2 that can simultaneously satisfy all the constraints. Thus, no solution exists.

The fact that the feasible region is empty is shown by the fact that in Figure 6.4 there is no blue region—i.e., all the regions are gray indicating that the constraints are not satisfiable. ♠

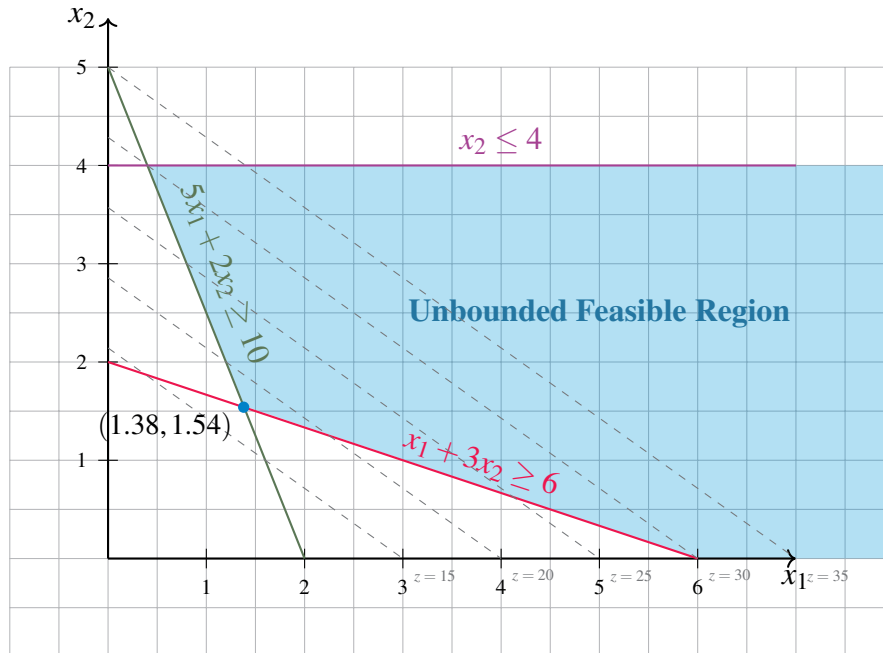
Based on this example, we will modify our previous algorithm for finding the solution to linear programming problems graphically to consider the case that it is empty.

Before stating complete algorithm, we also consider the unbounded case.

6.5 Problems with Unbounded Feasible Regions

Consider the problem

$$\begin{array}{ll}
 \min & z = 5x_1 + 7x_2 \\
 \text{s.t.} & x_1 + 3x_2 \geq 6 \\
 & 5x_1 + 2x_2 \geq 10 \\
 & x_2 \leq 4 \\
 & x_1, x_2 \geq 0
 \end{array}$$



As you can see, the feasible region is *unbounded*. In particular, from any point in the feasible region, one can always find another feasible point by increasing the x coordinate (i.e., move to the right in the picture). However, this does not necessarily mean that the optimization problem is unbounded.

Indeed, the optimal solution is at $\mathbf{x}^* = (1.38, 1.54)$, the extreme point in the lower left hand corner, with optimal objective value

$$z^* = 5 \cdot 1.38 + 7 \cdot 1.54 = 17.7.$$

Consider however, if we consider a different problem where we try to maximize the objective

$$\begin{array}{ll} \text{max} & z = 5x_1 + 7x_2 \\ \text{s.t.} & x_1 + 3x_2 \geq 6 \\ & 5x_1 + 2x_2 \geq 10 \\ & x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{array}$$

Solution. This optimization problem is unbounded! For example, notice that the point $(x_1, x_2) = (n, 0)$ is feasible for all $n = 1, 2, 3, \dots$. Then the objective function $z = 5n + 0$ follows the sequence $5, 10, 15, \dots$, which diverges to infinity. ♠

Let's see another example.

Example 6.7

Consider the linear programming problem below:

$$\begin{aligned}
 \max \quad & z(x_1, x_2) = 2x_1 - x_2 \\
 \text{s.t.} \quad & x_1 - x_2 \leq 1 \\
 & 2x_1 + x_2 \geq 6 \\
 & x_1, x_2 \geq 0
 \end{aligned} \tag{6.1}$$

Solution. The feasible region and level curves of the objective function are shown in Figure 6.5.

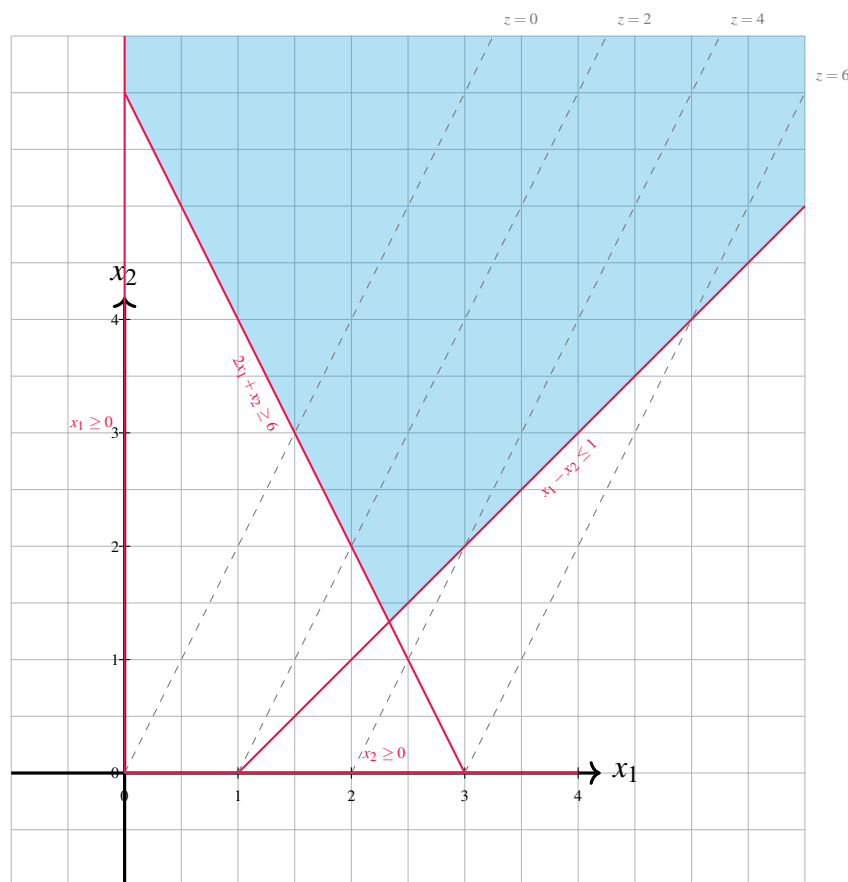


Figure 6.5: A Linear Programming Problem with Unbounded Feasible Region: Note that we can continue to make level curves of $z(x_1, x_2)$ corresponding to larger and larger values as we move down and to the right. These curves will continue to intersect the feasible region for any value of $v = z(x_1, x_2)$ we choose. Thus, we can make $z(x_1, x_2)$ as large as we want and still find a point in the feasible region that will provide this value. Hence, the optimal value of $z(x_1, x_2)$ subject to the constraints $+\infty$. That is, the problem is unbounded.

The feasible region in Figure 6.5 is clearly unbounded since it stretches upward along the x_2 axis infinitely far and also stretches rightward along the x_1 axis infinitely far, bounded below by the line $x_1 - x_2 = 1$. There is no way to enclose this region by a disk of finite radius, hence the feasible region is not bounded.

We can draw more level curves of $z(x_1, x_2)$ in the direction of increase (down and to the right) as long as we wish. There will always be an intersection point with the feasible region because it is infinite. That is, these curves will continue to intersect the feasible region for any value of $v = z(x_1, x_2)$ we choose. Thus, we can make $z(x_1, x_2)$ as large as we want and still find a point in the feasible region that will provide this value. Hence, the largest value $z(x_1, x_2)$ can take when (x_1, x_2) are in the feasible region is $+\infty$. That is, the problem is unbounded. ♠

Just because a linear programming problem has an unbounded feasible region does not imply that there is not a finite solution. We illustrate this case by modifying example 6.5.

Example 6.8: Continuation of Example

Consider the linear programming problem from Example 6.5 with the new objective function: $z(x_1, x_2) = (1/2)x_1 - x_2$. Then we have the new problem:

$$\begin{aligned} \max \quad & z(x_1, x_2) = \frac{1}{2}x_1 - x_2 \\ \text{s.t.} \quad & x_1 - x_2 \leq 1 \\ & 2x_1 + x_2 \geq 6 \\ & x_1, x_2 \geq 0 \end{aligned} \tag{6.2}$$

Solution. The feasible region, level sets of $z(x_1, x_2)$ and gradients are shown in Figure 6.6. In this case note, that the direction of increase of the objective function is *away* from the direction in which the feasible region is unbounded (i.e., downward). As a result, the point in the feasible region with the largest $z(x_1, x_2)$ value is $(7/3, 4/3)$. Again this is a vertex: the binding constraints are $x_1 - x_2 = 1$ and $2x_1 + x_2 = 6$ and the solution occurs at the point these two lines intersect.

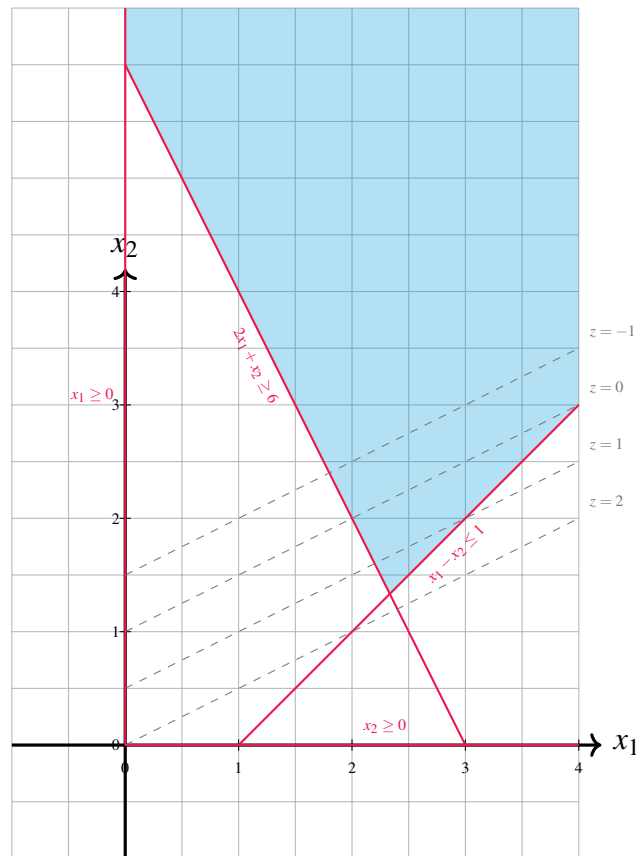


Figure 6.6: A Linear Programming Problem with Unbounded Feasible Region and Finite Solution: In this problem, the level curves of $z(x_1, x_2)$ increase in a more “southerly” direction than in Example 6.5—that is, *away* from the direction in which the feasible region increases without bound. The point in the feasible region with largest $z(x_1, x_2)$ value is $(7/3, 4/3)$. Note again, this is a vertex.



Based on these two examples, we can modify our algorithm for graphically solving a two variable linear programming problems to deal with the case when the feasible region is unbounded.

Algorithm 2 Algorithm for Solving a Two-Variable Linear Programming Problem Graphically

1. Plot the feasible region defined by the constraints.
2. If the feasible region is empty, there is no solution.
3. Plot the level sets of the objective function.
4. **Bounded Case:**
 - (a) Identify the corner point where the optimal level set (maximization: greatest; minimization: least) intersects the feasible region.
 - (b) If the level set is parallel to a side of the feasible region at this corner, any point on this side is an optimal solution (infinitely many solutions).
5. **Unbounded Case:**
 - (a) If the level sets intersect the feasible region indefinitely (maximization: larger; minimization: smaller), the problem is unbounded ($+\infty$ or $-\infty$).
 - (b) Otherwise, proceed as in the bounded case.

6.6 Exercises

Exercise 6.1 Consider the following system of constraints:

$$x + y \geq 5$$

$$x \geq 0, \quad y \geq 0$$

Which of the following statements is true about the feasible region?

1. The feasible region is a bounded polygon.
2. The feasible region is an unbounded region.
3. The feasible region consists of a single point.
4. The problem is infeasible.

Exercise 6.2 Sketch all $\begin{bmatrix} x \\ y \end{bmatrix}$ satisfying

$$x - 2y \leq 2$$

on the (x, y) -plane.

Exercise 6.3 Graphically Solving Graph the feasible region of the linear program:

$$\begin{aligned} \min \quad & x_1 - x_2 \\ & x_1 + x_2 \leq 6 \\ & x_1 - x_2 \geq 1 \\ & x_2 - x_1 \geq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Without considering the objective function, how many feasible solutions are there? Justify your answer.

Exercise 6.4 Graphical Method Use the graphical method to determine two optimal solutions to linear program:

$$\begin{aligned} \min \quad & 3x_1 + 5x_2 \\ & 3x_1 + 2x_2 \geq 36 \\ & 6x_1 + 10x_2 \geq 90 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Exercise 6.5 Graph the feasible region of the linear program:

$$\begin{aligned} \min \quad & x_1 - x_2 \\ & x_1 + x_2 \leq 6 \\ & x_1 - x_2 \geq 1 \\ & x_2 - x_1 \geq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Without considering the objective function, how many feasible solutions are there? Justify your answer.

Exercise 6.6 Determine the optimal value of

$$\begin{aligned} \text{Minimize} \quad & x + y \\ \text{Subject to} \quad & 2x + y \geq 4 \\ & x + 3y \geq 1. \end{aligned}$$

Exercise 6.7 Show that the problem

$$\begin{aligned} \text{Minimize} \quad & -x + y \\ \text{Subject to} \quad & 2x - y \geq 0 \\ & x + 3y \geq 3 \end{aligned}$$

is unbounded.

Exercise 6.8 Does the following problem have a bounded solution? Why?

$$\begin{aligned} \min \quad & z(x_1, x_2) = 2x_1 - x_2 \\ \text{s.t.} \quad & x_1 - x_2 \leq 1 \\ & 2x_1 + x_2 \geq 6 \\ & x_1, x_2 \geq 0 \end{aligned} \quad (6.1)$$

[Hint: Use Figure 6.6 and Algorithm 2.]

Exercise 6.9 Modify the objective function in Example 6.5 to produce a problem with an infinite number of solutions.

Exercise 6.10 Modify the objective function in Exercise 6.8 to produce a **minimization** problem that has a finite solution. Draw the feasible region and level curves of the objective to “prove” your example works.

[Hint: Think about what direction of increase is required for the level sets of $z(x_1, x_2)$.]

Exercise 6.11 Consider the following problem:

$$\begin{array}{llll} \text{minimize} & -2x_1 & + & x_2 \\ \text{subject to} & -x_1 & + & x_2 \leq 3 \\ & x_1 & - & 2x_2 \leq 2 \\ & x_1 & & \geq 0 \\ & & & x_2 \geq 0. \end{array}$$

1. Show that for any $t \geq 0$,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ t \end{bmatrix}$$

satisfies all the constraints. Show that the objective function is unbounded for this point as $t \rightarrow \infty$.

2. Show that

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2t+2 \\ t \end{bmatrix}$$

is feasible for any $t \geq 0$. What is the objective function value at this point? Show that the objective function is unbounded (yet again) by letting $t \rightarrow \infty$.

Exercise 6.12 Consider the following problem:

$$\begin{array}{llll} \text{maximize/minimize} & 2x & + & 2y \\ \text{subject to} & -x & + & 2y \leq 12 \\ & 2x & + & y \geq 4 \\ & x & + & 2y \geq 5 \\ & x & & \geq 0 \\ & & & y \geq 0. \end{array}$$

- Determine the minimum value of $2x + 2y$ by solving (graphically) the corresponding optimization problem. Show that the minimum is attained at a bounded feasible point.
- Show that the maximum value of $2x + 2y$ is unbounded by finding a feasible solution of the form

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

for any $t \geq 0$. Show that as $t \rightarrow \infty$, the objective function becomes unbounded.

3. Show that

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

is feasible for any $t \geq 0$. Compute the objective function value at this point and demonstrate that it is unbounded as $t \rightarrow \infty$.

Exercise 6.13 Suppose that you are shopping for dietary supplements to satisfy your required daily intake of 0.40mg of nutrient M and 0.30mg of nutrient N . There are three popular products on the market. The costs and the amounts of the two nutrients are given in the following table:

	Product 1	Product 2	Product 3
Cost	\$27	\$31	\$24
Daily amount of M	0.16 mg	0.21 mg	0.11 mg
Daily amount of N	0.19 mg	0.13 mg	0.15 mg

Table 6.1: Data for problem.

You want to determine how much of each product you should buy so that the daily intake requirements of the two nutrients are satisfied at minimum cost. Formulate your problem as a linear programming problem, assuming that you can buy a fractional number of each product.

Selected Solutions

Solution. (Exercise ??) In fact, constraints 2 and 3 point away from each other, and hence there are no feasible solutions. ♠

Solution. (Exercise 6.2)

The points (x, y) satisfying $x - 2y \leq 2$ are precisely those above the line passing through $(2, 0)$ and $(0, -1)$. ♠

Solution. (6.6)

We want to determine the minimum value z so that $x + y = z$ defines a line that has a nonempty intersection with the feasible region. However, we can avoid referring to a sketch by setting $x = z - y$ and substituting for x in the inequalities to obtain:

$$\begin{aligned} 2(z - y) + y &\geq 4 \\ (z - y) + 3y &\geq 1, \end{aligned}$$

or equivalently,

$$\begin{aligned} z &\geq 2 + \frac{1}{2}y \\ z &\geq 1 - 2y, \end{aligned}$$

Thus, the minimum value for z is $\min\{2 + \frac{1}{2}y, 1 - 2y\}$, which occurs at $y = -\frac{2}{5}$. Hence, the optimal value is $\frac{9}{5}$.

We can verify our work by doing the following. If our calculations above are correct, then an optimal solution is given by $x = \frac{11}{5}$, $y = -\frac{2}{5}$ since $x = z - y$. It is easy to check that this satisfies both inequalities and therefore is a feasible solution.

Now, taking $\frac{2}{5}$ times the first inequality and $\frac{1}{5}$ times the second inequality, we can infer the inequality $x + y \geq \frac{9}{5}$. The left-hand side of this inequality is precisely the objective function. Hence, no feasible solution can have objective function value less than $\frac{9}{5}$. But $x = \frac{11}{5}$, $y = -\frac{2}{5}$ is a feasible solution with objective function value equal to $\frac{9}{5}$. As a result, it must be an optimal solution.

Remark. We have not yet discussed how to obtain the multipliers $\frac{2}{5}$ and $\frac{1}{5}$ for inferring the inequality $x + y \geq \frac{9}{5}$. This is an issue that will be taken up later. In the meantime, think about how one could have obtained these multipliers for this particular exercise. ♠

Solution. (Exercise 6.7) We could glean some insight by first making a sketch on the (x, y) -plane.

The line defined by $-x + y = z$ has x -intercept $-z$. Note that for $z \leq -3$, $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -z \\ 0 \end{bmatrix}$ satisfies both inequalities and the value of the objective function at $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -z \\ 0 \end{bmatrix}$ is z . Hence, there is no lower bound on the value of the objective function. ♠

Solution. (Exercise 6.13) Let x_i denote the amount of Product i to buy for $i = 1, 2, 3$. Then, the problem can be formulated as

$$\begin{aligned} &\text{minimize} && 27x_1 &+& 31x_2 &+& 24x_3 \\ &\text{subject to} && 0.16x_1 &+& 0.21x_2 &+& 0.11x_3 &\geq 0.30 \\ &&& 0.19x_1 &+& 0.13x_2 &+& 0.15x_3 &\geq 0.40 \\ &&& x_1 &,& x_2 &,& x_3 &\geq 0. \end{aligned}$$

Remark. If one cannot buy fractional amounts of the products, the problem can be formulated as

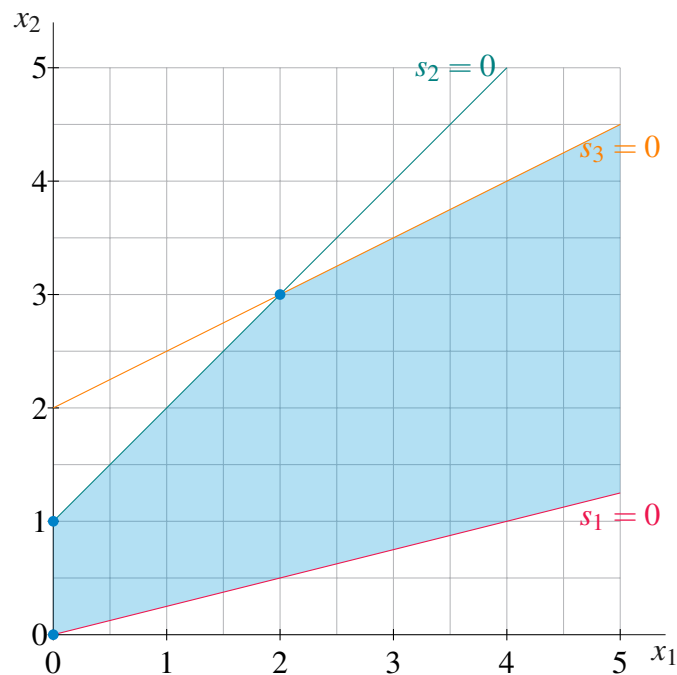
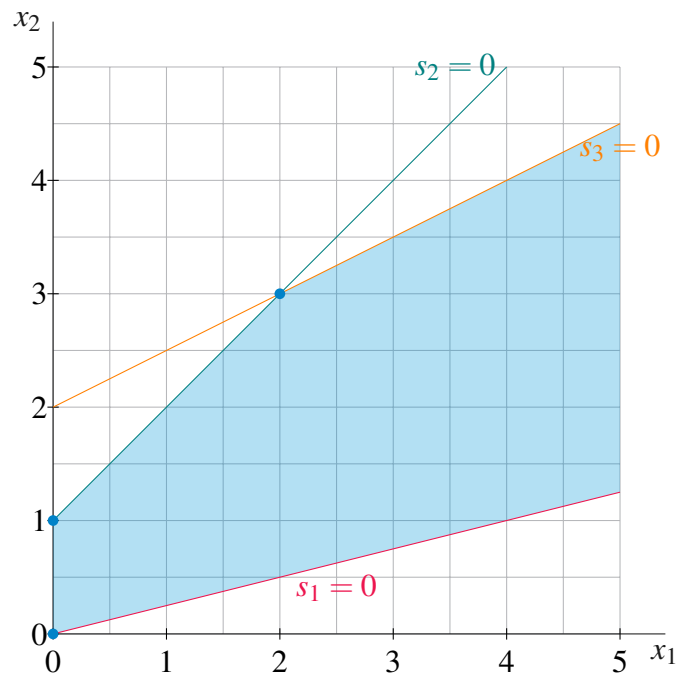
$$\begin{aligned} &\text{minimize} && 27x_1 &+& 31x_2 &+& 24x_3 \\ &\text{subject to} && 0.16x_1 &+& 0.21x_2 &+& 0.11x_3 &\geq 0.30 \\ &&& 0.19x_1 &+& 0.13x_2 &+& 0.15x_3 &\geq 0.40 \\ &&& x_1 &,& x_2 &,& x_3 &\geq 0. \\ &&& x_1 &,& x_2 &,& x_3 &\in \mathbb{Z}. \end{aligned}$$



6.7 Other Material: Extreme Directions

Let's define a procedure for finding the extreme directions, using the following LP's feasible region. Graphically, we can see that the extreme directions should follow the the $s_1 = 0$ (red) line and the $s_3 = 0$ (orange) line.

$$\begin{aligned} \max \quad & z = -5x_1 - x_2 \\ \text{s.t.} \quad & x_1 - 4x_2 + s_1 = 0 \\ & -x_1 + x_2 + s_2 = 1 \\ & -x_1 + 2x_2 + s_3 = 4 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$



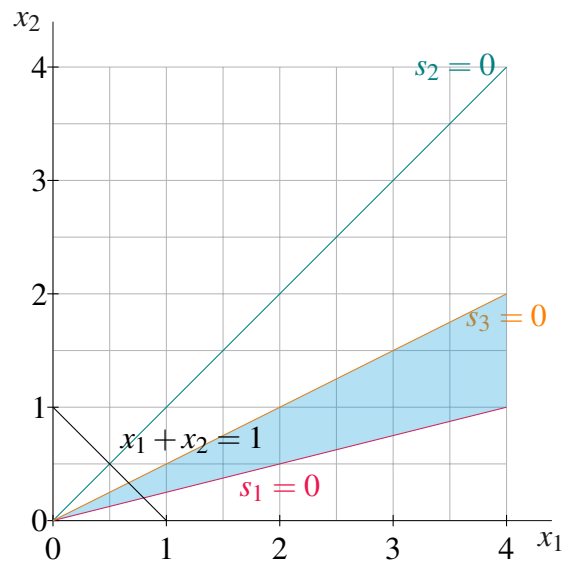
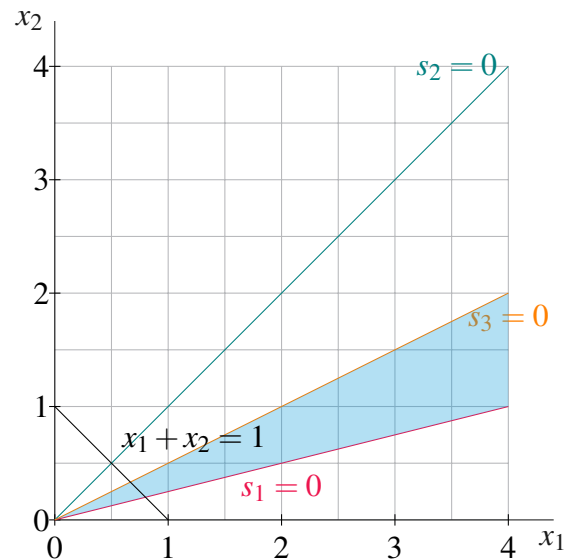
E.g., consider the $s_3 = 0$ (orange) line, to find the extreme direction start at extreme point (2,3) and find

another feasible point on the orange line, say (4,4) and subtract (2,3) from (4,4), which yields (2,1).

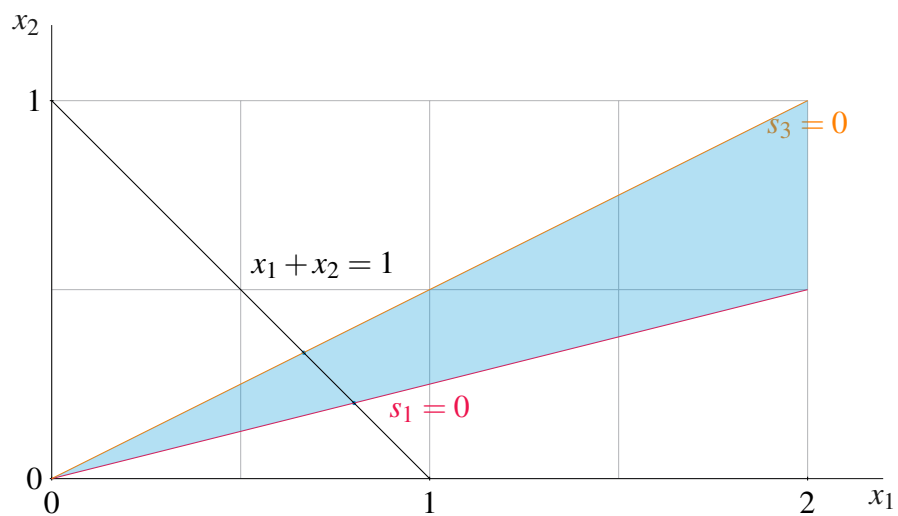
This is related to the slope in two-dimensions, as discussed in class, the rise is 1 and the run is 2. So this direction has a slope of 1/2, but this does not carry over easily to higher dimensions where directions cannot be defined by a single number.

To find the extreme directions we can change the right-hand-side to $\mathbf{b} = 0$, which forms a polyhedral cone (in yellow), and then add the constraint $x_1 + x_2 = 1$. The intersection of the cone and $x_1 + x_2 = 1$ form a line segment.

$$\begin{aligned} \max \quad & z = -5x_1 - x_2 \\ \text{s.t.} \quad & x_1 - 4x_2 + s_1 = 0 \\ & -x_1 + x_2 + s_2 = 0 \\ & -x_1 + 2x_2 + s_3 = 0 \\ & x_1 + x_2 = 1 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$



Magnifying for clarity, and removing the $s_2 = 0$ (teal) line, as it is redundant, and marking the extreme points of the new feasible region, $(4/5, 1/5)$ and $(2/3, 1/3)$, with red boxes, we have:



The extreme directions are thus $(4/5, 1/5)$ and $(2/3, 1/3)$.

7. Formal Mathematical Statements

Outcomes

- *Establish needed mathematical notation*
- *Understand structural results pertaining to linear programs*
- *Develop an intuition for why these results are true*

In this chapter, we formalize some of the mathematics about linear programs. We define some notation here and then state and prove several theorems.

CAUTION!

In this section, we provide rigorous mathematical proofs of several results. For this course, you do not need to be able to reproduce these proofs. However, try to understand the concepts and have a visual image in your head of why these results are true.

7.0.1. Vectors and Their Properties

Vectors: A vector is a mathematical object that represents a point in space or a direction. In n -dimensional real space ($\mathbb{R}^n$), a vector $\mathbf{v}$ is written as an ordered list of n elements:

$$\mathbf{v} = (v_1, v_2, \dots, v_n).$$

Vectors can be written in two forms: as a row vector $[v_1, v_2, \dots, v_n]$ or as a column vector:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

In linear programming, vectors are used to represent variables, constraints, and directions in optimization problems.

Vector Operations: Vector operations form the building blocks for linear programming, enabling the manipulation of decision variables and constraints.

Addition: Vectors of the same dimension can be added componentwise. For example, if $\mathbf{v} = (2, 3)$ and $\mathbf{u} = (3, 2)$, their sum is:

$$\mathbf{v} + \mathbf{u} = (2 + 3, 3 + 2) = (5, 5).$$

This operation is essential in modeling systems where contributions of multiple variables are combined.

Scalar Multiplication: A vector can be scaled by multiplying it with a scalar (constant). For example, if $k = 2$ and $\mathbf{v} = (2, 3)$, then:

$$k\mathbf{v} = (2 \cdot 2, 3 \cdot 2) = (4, 6).$$

Scalar multiplication adjusts the magnitude of a vector while retaining its direction, a key concept in scaling constraints or objectives in linear programming.

Inner (Dot) Product: The dot product of two vectors $\mathbf{v}$ and $\mathbf{u}$ of the same size is calculated as:

$$\mathbf{v} \cdot \mathbf{u} = \sum_{i=1}^n v_i u_i.$$

For example, if $\mathbf{v} = (2, 3)$ and $\mathbf{u} = (3, 2)$, then:

$$\mathbf{v} \cdot \mathbf{u} = 2 \cdot 3 + 3 \cdot 2 = 10.$$

The dot product is crucial in defining constraints and objective functions in linear programming, as it represents weighted sums of variables.

7.0.2. Linear Combinations and Related Concepts

Linear Combination: A vector $\mathbf{u}$ is a linear combination of vectors $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^k$ if:

$$\mathbf{u} = \sum_{i=1}^k \lambda_i \mathbf{v}^i,$$

where $\lambda_1, \lambda_2, \dots, \lambda_k$ are scalars. If all scalars are non-negative ($\lambda_i \geq 0$), the combination is called a *non-negative linear combination*. Linear combinations are used in linear programming to express feasible solutions as sums of variable contributions.

Convex Combination: A vector $\mathbf{u}$ is a convex combination of vectors $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^k$ if:

$$\mathbf{u} = \sum_{i=1}^k \lambda_i \mathbf{v}^i, \quad \text{where } \lambda_i \geq 0 \text{ and } \sum_{i=1}^k \lambda_i = 1.$$

Convex combinations are fundamental to linear programming because they describe feasible regions (e.g., convex polyhedra).

Definition 7.1: Linear Independence

A set of vectors $\{\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^k\} \subseteq \mathbb{R}^n$ is said to be linearly independent if the only solution to the equation

$$\lambda_1 \mathbf{v}^1 + \lambda_2 \mathbf{v}^2 + \dots + \lambda_k \mathbf{v}^k = \mathbf{0}$$

is $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$.

Equivalently, no vector in the set can be written as a linear combination of the others.

Examples:

- The vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in $\mathbb{R}^2$ are linearly independent.
- The vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ are linearly dependent, since the second is a scalar multiple of the first.

Definition 7.2: Linear Independence of Rows or Columns

Let $A \in \mathbb{R}^{m \times n}$ be a matrix.

- The rows of A are said to be linearly independent if no row can be written as a linear combination of the other rows.
- The columns of A are linearly independent if no column can be written as a linear combination of the other columns.
- In either case, the vectors are linearly independent if the only solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ (or $A^\top \mathbf{y} = \mathbf{0}$) is the trivial solution $\mathbf{x} = \mathbf{0}$ (or $\mathbf{y} = \mathbf{0}$).

Examples:

- The matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has both linearly independent rows and columns.
- The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has linearly dependent rows and columns; the second row is twice the first, and the second column is twice the first.

Definition 7.3: Rank of a Matrix

Let $A \in \mathbb{R}^{m \times n}$. The rank of A , denoted $\text{rank}(A)$, is the maximum number of linearly independent rows or columns of A .

Equivalently, $\text{rank}(A)$ is:

- the dimension of the row space of A ,
- the dimension of the column space of A ,
- the number of pivot columns in the reduced row echelon form of A ,
- the largest integer r such that A has an $r \times r$ submatrix with nonzero determinant.

Examples:

- The matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ has full rank 2.
- The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank 1, since the second row is a multiple of the first.

Definition 7.4: Linearly Independent Constraints

Let $A\mathbf{x} \leq \mathbf{b}$ be a system of linear inequalities, where each constraint corresponds to a row $A_i\mathbf{x} \leq b_i$. A subset of these constraints is said to be linearly independent at a point $\mathbf{x} \in \mathbb{R}^n$ if the set of row vectors $\{A_i : A_i\mathbf{x} = b_i\}$ (i.e., the tight constraints at $\mathbf{x}$) is linearly independent.

Equivalently, the tight constraints at $\mathbf{x}$ are linearly independent if the matrix A_I formed by stacking the tight rows satisfies

$$\text{rank}(A_I) = |I|,$$

where $I := \{i : A_i\mathbf{x} = b_i\}$.

Examples:

- Let $A\mathbf{x} \leq \mathbf{b}$ be the system:

$$x_1 \leq 1$$

$$x_2 \leq 1$$

$$x_1 + x_2 \leq 2$$

At $\mathbf{x} = (1, 1)$, all three constraints are tight. The first two have row vectors $[1, 0]$ and $[0, 1]$, which are linearly independent. The third is $[1, 1]$, which is a linear combination of the first two. So only two of the tight constraints are linearly independent.

- In contrast, for the system:

$$x_1 \leq 0$$

$$x_2 \leq 0$$

the point $\mathbf{x} = (0, 0)$ has both constraints tight, and their rows $[1, 0]$, $[0, 1]$ are linearly independent. Hence the tight constraints are linearly independent at that point.

7.1 Algebraic Proof - Exists Optimal Solution at a Vertex

Definition 7.5: Vertex

Let $P = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$ be a polyhedron. A point $\mathbf{x} \in P$ is called a vertex (or corner point) of P if it is the unique solution to a system of n linearly independent constraints that are tight at $\mathbf{x}$. In other words, $\mathbf{x}$ is a vertex if there exists a subset $I \subseteq \{1, \dots, m\}$ such that:

- $A_i\mathbf{x} = b_i$ for all $i \in I$,
- The rows $\{A_i : i \in I\}$ are linearly independent,
- $|I| = n$.

Theorem 7.6

Let $\max\{\mathbf{c}^\top \mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$ be a linear program with a nonempty, bounded feasible region P . If an optimal solution exists, then there exists an optimal solution that is a vertex of P .

Proof. Let $\mathbf{x}^* \in P$ be an optimal solution, and suppose for contradiction that $\mathbf{x}^*$ is *not* a vertex.

Define the index set of tight (active) constraints at $\mathbf{x}^*$ as

$$I := \{i \in \{1, \dots, m\} : A_i\mathbf{x}^* = b_i\},$$

where A_i is the i th row of A . Let A_I be the submatrix of A consisting of the rows indexed by I .

Since $\mathbf{x}^*$ is not a vertex, the tight constraints are linearly dependent: $\text{rank}(A_I) < n$. Therefore, the homogeneous system $A_I\mathbf{d} = 0$ has a nonzero solution $\mathbf{d} \in \mathbb{R}^n \setminus \{0\}$.

Consider the line $\mathbf{x}(t) := \mathbf{x}^* + t\mathbf{d}$ for small $t \in \mathbb{R}$. For all $i \in I$, we have:

$$A_i\mathbf{x}(t) = A_i\mathbf{x}^* + tA_i\mathbf{d} = b_i + 0 = b_i,$$

so the tight constraints remain tight along the line.

For any $i \notin I$, we have $A_i\mathbf{x}^* < b_i$. Since $A_i\mathbf{d}$ is bounded, we can choose a small enough $\varepsilon > 0$ such that for all $t \in [-\varepsilon, \varepsilon]$, the inequality $A_i\mathbf{x}(t) \leq b_i$ holds. Thus, for small t , $\mathbf{x}(t) \in P$.

Now define the objective value along this line:

$$z(t) := \mathbf{c}^\top \mathbf{x}(t) = \mathbf{c}^\top \mathbf{x}^* + t \cdot (\mathbf{c}^\top \mathbf{d}).$$

- If $\mathbf{c}^\top \mathbf{d} > 0$, then for small $t > 0$, $\mathbf{x}(t) \in P$ and $\mathbf{c}^\top \mathbf{x}(t) > \mathbf{c}^\top \mathbf{x}^*$, contradicting optimality of $\mathbf{x}^*$.
- If $\mathbf{c}^\top \mathbf{d} < 0$, then for small $t < 0$, same contradiction.
- If $\mathbf{c}^\top \mathbf{d} = 0$, then $\mathbf{x}(t)$ is also optimal for all small t , and we can move in both directions while maintaining feasibility and optimality.

In all cases, we can find another feasible optimal point along the line $\mathbf{x}(t)$, with the same or better objective value. Moreover, we can choose such a point with a strictly larger set of linearly independent tight constraints. Repeating this process, we eventually reach an optimal point at which the tight constraints form a full-rank system (rank n), which uniquely determines the point. Such a point is, by definition, a vertex.

This contradicts our assumption that no optimal vertex exists. ♠

7.2 Convexity

A key property that will enable efficient algorithms is *convexity*. This comes in the form of *convex sets* and *convex functions*. Then the constraints to an optimization problem form a convex set and the objective function is a convex function, then we say that it is a convex optimization problem.

We explain these definitions in the next two subsections.

7.2.1. Convex Sets

Definition 7.7: Convex Combination

Given two points $\mathbf{x}, \mathbf{y}$, a convex combination is any point $\mathbf{z}$ that lies on the line between $\mathbf{x}$ and $\mathbf{y}$. Algebraically, a convex combination is any point $\mathbf{z}$ that can be represented as $\mathbf{z} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ for some multiplier $\lambda \in [0, 1]$.



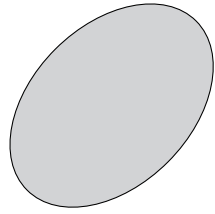
Figure 7.1: A convex combination of the points $\mathbf{x}$ and $\mathbf{y}$ is given by $\mathbf{z} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ with any $\lambda \in [0, 1]$. Here we demonstrate this using $\lambda = 2/3$.

Definition 7.8: Convex Set

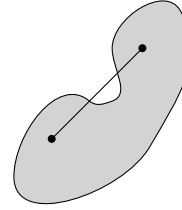
A set C is convex if it contains all convex combinations of points in C . That is, for any $\mathbf{x}, \mathbf{y} \in C$, it holds that $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C$ for all $\lambda \in [0, 1]$.

Some examples of convex sets are:

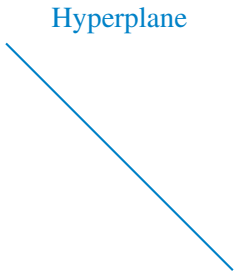
1. Hyperplane $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} = b\}$
2. Halfspace $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} \leq b\}$
3. Polyhedron $P = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A} \mathbf{x} \leq \mathbf{b}\}$



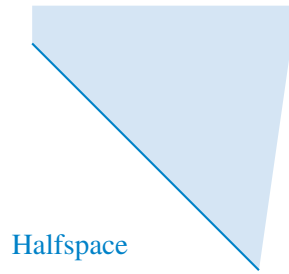
(a) Convex Set



(b) Non-convex Set



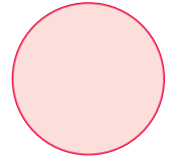
Hyperplane



Halfspace



Polyhedron



Ball

4. Ball $B = \{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 \leq 1\}$

Example 7.9: Proof that a Polyhedron is Convex

Let P be a polyhedron defined by:

$$P = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$$

where A is an $m \times n$ matrix and $\mathbf{b}$ is an $m \times 1$ vector.

To prove the convexity of P , we'll use the definition of convexity: For any $\mathbf{x}, \mathbf{y} \in P$ and any $\lambda \in [0, 1]$, the point $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$ must also belong to P .

Let $\mathbf{x}, \mathbf{y} \in P$. Then, by the definition of P , we have:

$$A\mathbf{x} \leq \mathbf{b} \quad \text{and} \quad A\mathbf{y} \leq \mathbf{b}$$

We want to demonstrate the inequality $A\mathbf{z} \leq \mathbf{b}$. We do this in just a few lines:

$$\begin{aligned} A\mathbf{z} &= A(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) \\ &= \lambda A\mathbf{x} + (1 - \lambda)A\mathbf{y} \\ &\leq \lambda\mathbf{b} + (1 - \lambda)\mathbf{b} \\ &= \mathbf{b} \end{aligned}$$

Substituting the expression for $\mathbf{z}$
Expanding the matrix-vector product
Explanation below

The inequality above uses the fact that $A\mathbf{x} \leq \mathbf{b}$ and $A\mathbf{y} \leq \mathbf{b}$, and since λ is in the interval $[0, 1]$, it follows that:

$$\lambda A\mathbf{x} \leq \lambda\mathbf{b} \quad \text{and} \quad (1 - \lambda)A\mathbf{y} \leq (1 - \lambda)\mathbf{b}$$

The computation shows that $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$ satisfies $A\mathbf{z} \leq \mathbf{b}$ and therefore $\mathbf{z} \in P$.

This shows that the polyhedron P is convex.

Lemma 7.10: Intersection of Convex Sets is Convex

Let C_1 and C_2 be convex sets. Then the intersection $C_1 \cap C_2$ is convex. In particular,

$$C_1 \cap C_2 := \{\mathbf{x} : \mathbf{x} \in C_1 \text{ and } \mathbf{x} \in C_2\}.$$

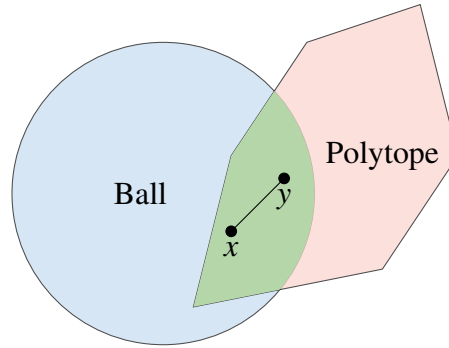


Figure 7.3: The green intersection of the convex sets that are the ball and the polytope is also convex. This can be seen by considering any points $x, y \in \text{Ball} \cap \text{Polytope}$. Since **Ball** is convex, the line segment between x and y is completely contained in **Ball**. And similarly, the line segment is completely contained in **Polytope**. Hence, the line segment is also contained in the intersection. This is how we can reason that the intersection is also convex.

Linear Independence: Vectors $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^k$ are linearly independent if the only solution to:

$$\sum_{i=1}^k \lambda_i \mathbf{v}^i = 0$$

is $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$. In $\mathbb{R}^2$, two vectors are linearly dependent if they lie on the same line. Linear independence is critical in linear programming to ensure constraints are non-redundant.

7.2.2. Spanning Sets and Basis

Spanning Set: A set of vectors $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^k$ spans $\mathbb{R}^m$ if any vector in $\mathbb{R}^m$ can be expressed as a linear combination of these vectors:

$$\sum_{i=1}^m \lambda_i \mathbf{v}^i.$$

Basis: A set of vectors $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^m$ is a basis for $\mathbb{R}^m$ if:

- The vectors span $\mathbb{R}^m$.
- The vectors are linearly independent.

In linear programming, the concept of a basis is used to identify corner points of the feasible region.

7.2.3. Convex and Polyhedral Sets

Convex Set: A set $C \subseteq \mathbb{R}^n$ is convex if, for any two points $\mathbf{v}^1, \mathbf{v}^2 \in C$, the line segment joining them lies entirely within C :

$$\lambda \mathbf{v}^1 + (1 - \lambda) \mathbf{v}^2 \in C, \quad \forall \lambda \in [0, 1].$$

Convex sets form the feasible regions in linear programming problems.

Polyhedral Set: A polyhedral set is the intersection of finitely many half-spaces, described as:

$$P = \{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq 0\}.$$

Polyhedral sets are convex and represent the feasible regions of linear programming models.

Extreme Points: An extreme point of a polyhedral set C is a point that cannot be written as a convex combination of other points in C . Linear programming solutions often occur at extreme points, as guaranteed by fundamental theorems of linear programming.

Lemma 7.11: Objective Values on a Line Segment

Given two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, a linear function evaluated at any point on the line segment between $\mathbf{x}$ and $\mathbf{y}$ is bounded above by the maximum of the evaluations at $\mathbf{x}$ and $\mathbf{y}$. Formally, for any $\mathbf{c} \in \mathbb{R}^n$,

$$\min\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\} \leq \mathbf{c}^\top \mathbf{z} \leq \max\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\}.$$

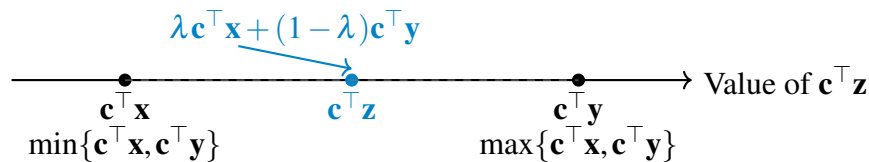
where $\mathbf{z} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ for some $\lambda \in (0, 1)$.

Either both of these inequalities are strict, or they are both held at equality.

Proof. Expanding $\mathbf{c}^\top \mathbf{z}$ based on the definition

$$\begin{aligned} \mathbf{c}^\top \mathbf{z} &= \mathbf{c}^\top (\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) && \text{Replace formula for } \mathbf{z} \\ &= \mathbf{c}^\top \lambda \mathbf{x} + \mathbf{c}^\top (1 - \lambda) \mathbf{y} && \text{use distributivity} \\ &= \lambda \mathbf{c}^\top \mathbf{x} + (1 - \lambda) \mathbf{c}^\top \mathbf{y} && \text{use linearity} \end{aligned}$$

This equation expresses $\mathbf{c}^\top \mathbf{z}$ as a convex combination of $\mathbf{c}^\top \mathbf{x}$ and $\mathbf{c}^\top \mathbf{y}$. Since $\lambda \in [0, 1]$, we know $\lambda \geq 0$ and $1 - \lambda \geq 0$, ensuring that the combination is convex.



By the properties of convex combinations, the value $\mathbf{c}^\top \mathbf{z}$ lies between $\mathbf{c}^\top \mathbf{x}$ and $\mathbf{c}^\top \mathbf{y}$. Formally:

$$\min\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\} \leq \mathbf{c}^\top \mathbf{z} \leq \max\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\}.$$

Therefore, the value of $\mathbf{c}^\top \mathbf{z}$ is bounded above by $\max\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\}$, as required. 

Theorem 7.12: Optimal Solution at an Extreme Point

Let $C \subseteq \mathbb{R}^n$ be a bounded, closed, and convex set. Let the objective be to maximize $\mathbf{c}^\top \mathbf{x}$ over C . Then, there exists an optimal solution $\mathbf{x}^ \in C$ such that $\mathbf{x}^*$ is an extreme point of C .*

Proof. Since C is bounded and closed, it is compact in $\mathbb{R}^n$. The linear function $\mathbf{c}^\top \mathbf{x}$ is continuous, and by the Extreme Value Theorem, it attains its maximum value on the compact set C . Let $\mathbf{x}^* \in C$ be such that:

$$\mathbf{c}^\top \mathbf{x}^* = \max\{\mathbf{c}^\top \mathbf{x} : \mathbf{x} \in C\}$$

Suppose $\mathbf{x}^*$ is not an extreme point of C . Then, $\mathbf{x}^*$ can be expressed as a convex combination of two distinct points $\mathbf{x}, \mathbf{y} \in C$:

$$\mathbf{x}^* = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}, \quad \text{where } \lambda \in (0, 1).$$

By Lemma 7.2.3, the value $\mathbf{c}^\top \mathbf{x}^*$ satisfies:

$$\min\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\} \leq \mathbf{c}^\top \mathbf{x}^* \leq \max\{\mathbf{c}^\top \mathbf{x}, \mathbf{c}^\top \mathbf{y}\}.$$

Since $\mathbf{x}^*$ maximizes $\mathbf{c}^\top \mathbf{x}$ over C , it must be that $\mathbf{c}^\top \mathbf{x} = \mathbf{c}^\top \mathbf{x}^*$ and $\mathbf{c}^\top \mathbf{y} = \mathbf{c}^\top \mathbf{x}^*$. Thus, both $\mathbf{x}$ and $\mathbf{y}$ also maximize $\mathbf{c}^\top \mathbf{x}$ over C .

If $\mathbf{x}^*$ is not an extreme point, we can repeat this argument for $\mathbf{x}$ and $\mathbf{y}$, expressing them as convex combinations of other points in C . However, since C is compact, this process must terminate after a finite number of steps, at which point we reach an extreme point that maximizes $\mathbf{c}^\top \mathbf{x}$.

Hence, there exists an optimal solution at an extreme point of C . 

Theorem 7.13: Extreme Points and Vertices of a Polyhedron

Let $P \subseteq \mathbb{R}^n$ be a polyhedron. Then, a point $\mathbf{v} \in P$ is an extreme point if and only if $\mathbf{v}$ is a vertex of P .

Proof. (If $\mathbf{v}$ is not a vertex, then it is not an extreme point):

Suppose $\mathbf{v} \in P$ is not a vertex. By definition, this means $\mathbf{v}$ is not the unique solution to any subset of n linearly independent constraints active at $\mathbf{v}$. Let $A'\mathbf{x} = \mathbf{b}'$ represent the active constraints at $\mathbf{v}$, where $A' \in \mathbb{R}^{m' \times n}$ ($m' \leq n$) is the matrix of active constraints, and assume the rows of A' are linearly independent. Since $\mathbf{v}$ is not a vertex, the null space of A' , denoted $\mathcal{N}(A')$, is nontrivial, i.e., there exists a nonzero direction $\mathbf{d} \in \mathbb{R}^n$ such that:

$$A'\mathbf{d} = \mathbf{0}.$$

Consider perturbing $\mathbf{v}$ in the directions $\mathbf{v} + \varepsilon \mathbf{d}$ and $\mathbf{v} - \varepsilon \mathbf{d}$, for some small $\varepsilon > 0$. Since $\mathbf{d} \in \mathcal{N}(A')$, the perturbed points satisfy the active constraints:

$$A'(\mathbf{v} + \varepsilon \mathbf{d}) = A'\mathbf{v} + \varepsilon A'\mathbf{d} = \mathbf{b}'.$$

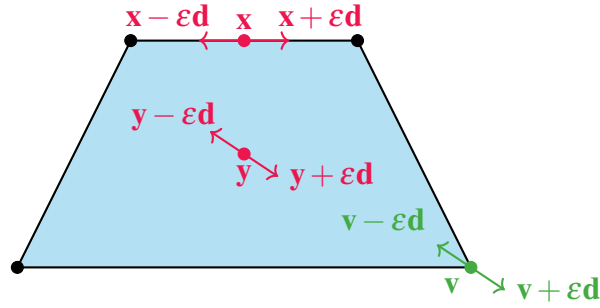
Similarly, $A'(\mathbf{v} - \varepsilon \mathbf{d}) = \mathbf{b}'$.

Furthermore, for sufficiently small $\varepsilon > 0$, the perturbed points also satisfy all other constraints defining P . This is because P is a polyhedron, and the constraints are continuous linear inequalities that remain satisfied under small perturbations of $\mathbf{v}$.

Now, we express $\mathbf{v}$ as a strict convex combination of $\mathbf{v} + \varepsilon \mathbf{d}$ and $\mathbf{v} - \varepsilon \mathbf{d}$:

$$\mathbf{v} = \frac{1}{2}(\mathbf{v} + \varepsilon \mathbf{d}) + \frac{1}{2}(\mathbf{v} - \varepsilon \mathbf{d}).$$

Since $\mathbf{v} + \varepsilon \mathbf{d} \neq \mathbf{v}$ and $\mathbf{v} - \varepsilon \mathbf{d} \neq \mathbf{v}$, and both points belong to P , this shows that $\mathbf{v}$ is not an extreme point of P .



(If $\mathbf{v}$ is a vertex, then it is an extreme point):

Assume $\mathbf{v} \in P$ is a vertex. By definition, there exists a subset of constraints, say $A'\mathbf{x} = \mathbf{b}'$, where $A' \in \mathbb{R}^{n \times n}$, such that $\mathbf{v}$ is the unique solution to this system.

Suppose, for the sake of contradiction, that $\mathbf{v}$ is not an extreme point. Then, $\mathbf{v}$ can be expressed as:

$$\mathbf{v} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2, \quad \mathbf{x}_1, \mathbf{x}_2 \in P, \lambda \in (0, 1),$$

where $\mathbf{x}_1 \neq \mathbf{v}$ and $\mathbf{x}_2 \neq \mathbf{v}$.

We claim that both $\mathbf{x}_1$ and $\mathbf{x}_2$ must satisfy $A'\mathbf{x} = \mathbf{b}'$. To see this, let $\mathbf{a}$ be a row of A' and b be the corresponding right hand side value in $\mathbf{b}'$. By Lemma ??, the value of $\mathbf{a}^\top \mathbf{z}$ satisfies:

$$\min\{\mathbf{a}^\top \mathbf{x}_1, \mathbf{a}^\top \mathbf{x}_2\} \leq \mathbf{a}^\top \mathbf{v} \leq \max\{\mathbf{a}^\top \mathbf{x}_1, \mathbf{a}^\top \mathbf{x}_2\}.$$

Since $\mathbf{v}$ satisfies $\mathbf{a}^\top \mathbf{v} = b$, this implies:

$$\mathbf{a}^\top \mathbf{v} = \lambda \mathbf{a}^\top \mathbf{x}_1 + (1 - \lambda) \mathbf{a}^\top \mathbf{x}_2 = b.$$

If either $\mathbf{a}^\top \mathbf{x}_1 \neq b$ or $\mathbf{a}^\top \mathbf{x}_2 \neq b$, the equation above cannot hold. Hence, both $\mathbf{x}_1$ and $\mathbf{x}_2$ must satisfy $\mathbf{a}^\top \mathbf{x} = b$.

Since $\mathbf{v}$ satisfies $A'\mathbf{x} = \mathbf{b}'$, both $\mathbf{x}_1$ and $\mathbf{x}_2$ must also satisfy $A'\mathbf{x} = \mathbf{b}'$, as $A'\mathbf{x} = \mathbf{b}'$ defines an affine subspace. However, this contradicts the assumption that $\mathbf{v}$ is the unique solution to $A'\mathbf{x} = \mathbf{b}'$.

Therefore, $\mathbf{v}$ cannot be expressed as a strict convex combination of distinct points in P , and $\mathbf{v}$ is an extreme point.

Conclusion: A point $\mathbf{v} \in P$ is an extreme point if and only if it is a vertex of P .



7.2.4. Unbounded Sets and Rays

Rays and Directions: A ray in $\mathbb{R}^n$ is a set of points:

$$\{\mathbf{x} : \bar{\mathbf{x}} + \lambda \mathbf{d}, \lambda \geq 0\},$$

where $\bar{\mathbf{x}}$ is the starting point and $\mathbf{d}$ is the direction. Rays describe unbounded feasible regions in linear programming models.

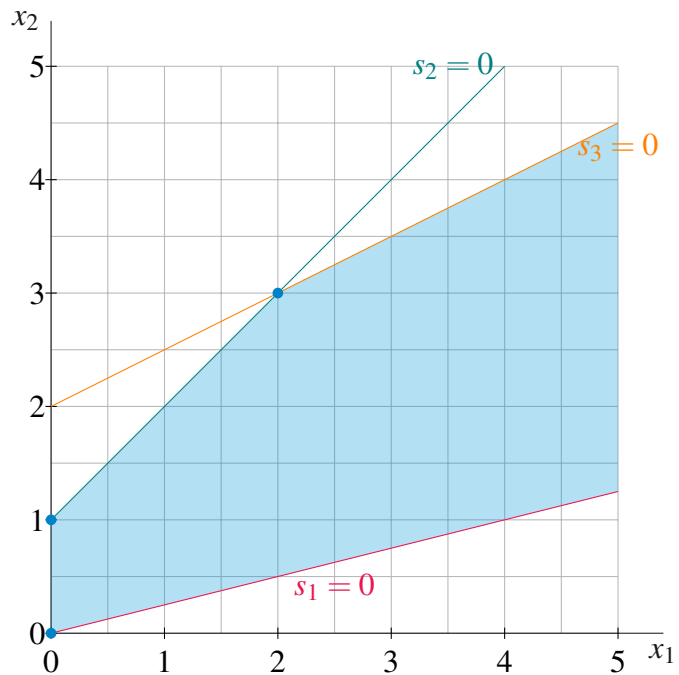
Extreme Directions: An extreme direction is one that cannot be written as a positive combination of other directions. In linear programming, extreme directions help characterize unbounded solutions.

This section provides the mathematical tools essential for understanding and formulating linear programming problems, emphasizing the geometric and algebraic structures that define feasible regions and optimal solutions.

Theorem 7.14: Representation Theorem

Let $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^k$ be the set of extreme points of P , and if P is unbounded, $\mathbf{d}^1, \mathbf{d}^2, \dots, \mathbf{d}^l$ be the set of extreme directions. Then any $\mathbf{x} \in P$ is equal to a convex combination of the extreme points and a non-negative linear combination of the extreme directions: $\mathbf{x} = \sum_{j=1}^k \lambda_j \mathbf{x}^j + \sum_{j=1}^l \mu_j \mathbf{d}^j$, where $\sum_{j=1}^k \lambda_j = 1$, $\lambda_j \geq 0$, $\forall j = 1, 2, \dots, k$, and $\mu_j \geq 0$, $\forall j = 1, 2, \dots, l$.

$$\begin{aligned}
 \max \quad & z = -5x_1 - x_2 \\
 \text{s.t.} \quad & x_1 - 4x_2 + s_1 = 0 \\
 & -x_1 + x_2 + s_2 = 1 \\
 & -x_1 + 2x_2 + s_3 = 4 \\
 & x_1, x_2, s_1, s_2, s_3 \geq 0.
 \end{aligned}$$



Represent point $(1/2, 1)$ as a convex combination of the extreme points of the above LP. Find λ s to solve the following system of equations:

$$\lambda_1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

7.3 Exercises

Exercise 7.1 *Vector Operations* Let $\mathbf{v} = [3, -2, 4]$ and $\mathbf{u} = [-1, 0, 2]$. Compute:

1. $\mathbf{v} + \mathbf{u}$
2. $2\mathbf{v}$
3. $\mathbf{v} \cdot \mathbf{u}$

Exercise 7.2 *Linear and Convex Combinations* Let $\mathbf{v}^1 = [1, 0]$, $\mathbf{v}^2 = [0, 1]$, and $\mathbf{v}^3 = [1, 1]$.

1. Which of the following are linear combinations of $\mathbf{v}^1$ and $\mathbf{v}^2$?
 - (a) $[2, 3]$
 - (b) $[1, 1]$
 - (c) $[1, -1]$

2. Is the vector $[1/2, 1/2]$ a convex combination of any two of these vectors?

Exercise 7.3 Linear Independence and Matrix Rank Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

1. Are the rows of A linearly independent?
2. Are the columns linearly independent?
3. What is $\text{rank}(A)$?

Exercise 7.4 Convex Combinations and Geometry Let P be the triangle with vertices at $(0,0)$, $(2,0)$, and $(0,2)$.

1. Plot the set P .
2. Determine whether the point $(1,1)$ is a convex combination of the vertices.
3. Determine whether the point $(1.5, 1.5)$ is in P .

Exercise 7.5 Halfspace Geometry and Convexity Consider the halfspace $H = \{\mathbf{x} \in \mathbb{R}^2 : x_1 + 2x_2 \leq 4\}$.

1. Plot the halfspace and the boundary.
2. Is the point $(2,1)$ in H ? What about $(1,2)$?
3. Verify that the set is convex by checking whether the midpoint between $(0,0)$ and $(2,1)$ lies in H .

8. Simplex Method

Outcomes

- Convert linear programs to standard form
- Reformulate linear programs from the perspective of different bases
- Define basic solutions and basic feasible solutions
- Describe optimality condition from the perspective of a reformulation
- Develop the Simplex Method that pivots through basic feasible solutions

In this section, we will develop the *Simplex Method*. This algorithm has been lauded as one of the most important algorithms of the 20th century. It will iteratively find better vertex solutions to a linear program until it can prove optimality.

To implement the simplex algorithm, we must first discuss a *Standard Form* with which to represent linear programs. After, we will discuss structure of linear programs in this form using matrix notation and demonstrate how a linear program can be transformed to new representations. These new representations will be the backbone of the simplex algorithm. We will *pivot* between vertices, and each vertex will have a new representation of the linear program that will help us understand how to choose a new vertex to move to.

The whole algorithm runs very fast in practice. Here, we will teach you a basic version of the algorithm, but note that state-of-the-art solvers will employ a number of tricks to make this process very fast. Linear programs can be solved with millions of variables!

8.1 Standard Form

To solve a linear program using methods such as the simplex algorithm, the problem must first be converted to a standard form. This ensures the problem adheres to a uniform structure, making it easier to analyze and solve algorithmically.

A linear program is in *standard form* if it satisfies all of the following:

- The objective is a maximization.
- All constraints are equalities (i.e., of the form $=$).
- All variables are bounded below by zero (i.e., $x_i \geq 0$).

Definition 8.1: Standard Form

A linear program is in standard form if is written as

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0. \end{aligned}$$

8.1.1. Converting to Standard Form

We now describe how to handle each of these aspects step-by-step, including common subcases and examples.

1. Objective: Convert Minimization to Maximization

If the linear program is given as a minimization, we convert it into a maximization by multiplying the objective function by -1 . This transformation does not affect the feasible region, and the maximizer of the new function will be the minimizer of the original.

Example 8.2: Convert Min to Max

Consider the following objective:

$$\min 3x_1 + 4x_2.$$

To convert this to a maximization, we multiply the entire expression by -1 :

$$\max (-3x_1 - 4x_2).$$

Now the problem can be treated using standard methods that assume a maximization objective.

2. Constraints: Convert Inequalities to Equalities

All inequality constraints must be transformed into equalities using slack or surplus variables, which represent the difference between the left- and right-hand sides of the constraint.

(a) **Convert $\leq$ to $=$ using a slack variable.**

A slack variable is added to "fill the gap" in a $\leq$ constraint and must be nonnegative.

Example 8.3: Slack Variable for $\leq$

Suppose we have the constraint:

$$x_1 + 2x_2 \leq 5.$$

We introduce a new variable $s_1 \geq 0$, called a slack variable, and rewrite the constraint as:

$$x_1 + 2x_2 + s_1 = 5.$$

Here, s_1 represents the unused portion of the right-hand side (i.e., how far the left-hand side is from 5).

(b) **Convert $\geq$ to $=$ using a surplus variable.**

A surplus variable is subtracted from the left-hand side of a $\geq$ constraint to convert it into an equality. This variable also must be nonnegative.

Example 8.4: Surplus Variable for $\geq$

Consider the constraint:

$$x_1 - x_2 \geq 3.$$

We introduce a new variable $s_2 \geq 0$, called a surplus variable, and rewrite the constraint as:

$$x_1 - x_2 - s_2 = 3.$$

In this case, s_2 accounts for the extra amount by which the left-hand side exceeds 3.

3. Variable Bounds: Ensure All Variables Are Nonnegative

The standard form requires all variables to satisfy $x_i \geq 0$. If any variable is unbounded below or is restricted in another way, we perform a variable substitution to enforce this condition.

(a) **Convert $x_i \leq 0$ to $x'_i \geq 0$ by renaming and negating.**

If a variable is constrained to be nonpositive, we can define a new nonnegative variable as its negation.

Example 8.5: Rename and Negate for ≤ 0

Suppose $x_3 \leq 0$. Define $x'_3 = -x_3$, which implies $x'_3 \geq 0$. Then substitute:

$$x_3 = -x'_3, \quad \text{where } x'_3 \geq 0.$$

This transformation allows us to enforce nonnegativity in the standard form.

(b) **Convert unrestricted variables into the difference of two nonnegative variables.**

If a variable is unrestricted in sign (i.e., it can be positive or negative), we express it as the difference of two new nonnegative variables.

Example 8.6: Unrestricted Variable to Two Variables

Suppose x_4 is unrestricted. We define:

$$x_4 = x_4^+ - x_4^-, \quad \text{with } x_4^+, x_4^- \geq 0.$$

Here, x_4^+ captures the positive part and x_4^- captures the negative part. At most one of them will be nonzero in a basic feasible solution.

After performing these transformations, the resulting linear program will be in standard form and ready for solution via the simplex method or other standard techniques.

Example 8.7: Complete Conversion to Standard Form

Consider the following linear program:

$$\begin{aligned} \min \quad & 2x_1 - 3x_2 + x_3 \\ \text{subject to} \quad & x_1 - x_2 \leq 4, \\ & 2x_1 + x_3 = 7, \\ & x_2 - x_3 \geq 2, \\ & x_1 \text{ is unrestricted, } x_2 \geq 0, \quad x_3 \text{ is unrestricted.} \end{aligned}$$

We will now convert this problem into standard form.

Step 1: Convert the objective to maximization.

Since the problem is a minimization, we multiply the objective by -1 :

$$\max (-2x_1 + 3x_2 - x_3).$$

Step 2: Replace unrestricted variables.

Both x_1 and x_3 are unrestricted. We express them as the difference of two nonnegative variables:

$$x_1 = x_1^+ - x_1^-, \quad x_3 = x_3^+ - x_3^-, \quad \text{with } x_1^+, x_1^-, x_3^+, x_3^- \geq 0.$$

Substitute these into the objective and constraints.

New objective:

$$\max [-2(x_1^+ - x_1^-) + 3x_2 - (x_3^+ - x_3^-)] = \max (-2x_1^+ + 2x_1^- + 3x_2 - x_3^+ + x_3^-).$$

New constraints:

$$(x_1^+ - x_1^-) - x_2 \leq 4 \quad (\text{original 1st constraint})$$

$$2(x_1^+ - x_1^-) + (x_3^+ - x_3^-) = 7 \quad (\text{original 2nd constraint})$$

$$x_2 - (x_3^+ - x_3^-) \geq 2 \quad (\text{original 3rd constraint})$$

Step 3: Convert inequalities to equalities.

- Add a slack variable $s_1 \geq 0$ to the first constraint:

$$x_1^+ - x_1^- - x_2 + s_1 = 4$$

- The second constraint is already an equality:

$$2x_1^+ - 2x_1^- + x_3^+ - x_3^- = 7$$

- Subtract a surplus variable $s_2 \geq 0$ from the third constraint:

$$x_2 - x_3^+ + x_3^- - s_2 = 2$$

Final standard form:

$$\begin{aligned} \max \quad & -2x_1^+ + 2x_1^- + 3x_2 - x_3^+ + x_3^- \\ \text{subject to} \quad & x_1^+ - x_1^- - x_2 + s_1 = 4 \\ & 2x_1^+ - 2x_1^- + x_3^+ - x_3^- = 7 \\ & x_2 - x_3^+ + x_3^- - s_2 = 2 \\ & x_1^+, x_1^-, x_2, x_3^+, x_3^-, s_1, s_2 \geq 0. \end{aligned}$$

This is now in standard form: a maximization problem with equality constraints and all variables nonnegative.

Remark 8.8: Variable Transformations Preserve Solution Meaning

To convert a linear program into standard form, we often introduce new variables:

- Slack variables to convert $\leq$ constraints to equalities,
- Surplus variables for $\geq$ constraints,
- Split unrestricted variables as $x_j = x_j^+ - x_j^-$,
- Negate variables with $x_j \leq 0$.

These transformations may change the number of variables and the way solutions are expressed, but:

1. Every feasible solution to the transformed problem corresponds to a feasible solution of the original problem.
2. Every optimal solution of the transformed problem yields an optimal solution to the original.

Thus, while the form of the solution may look different, the meaning and the value of the solution are preserved.

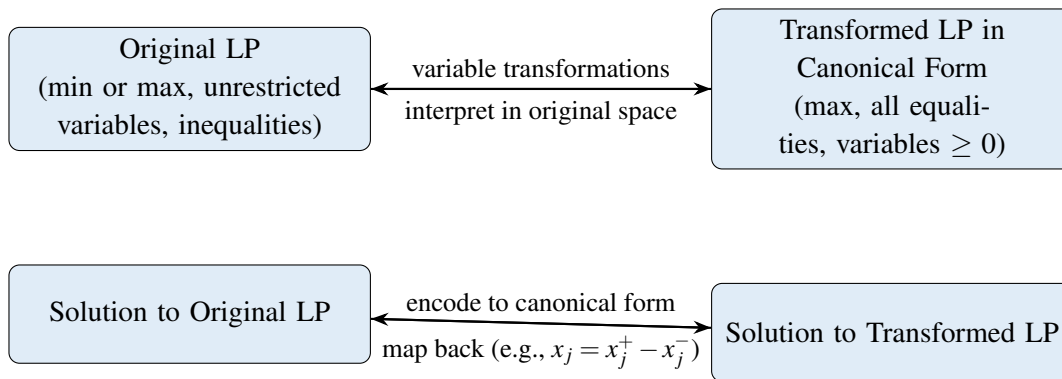


Figure 8.1: Correspondence between original and canonical form solutions.

Example 8.9: Transformation Preserves Solution Meaning

Consider the following linear program:

$$\begin{aligned}
 \min \quad & -x_1 + 2x_2 - 3x_3 \\
 \text{subject to} \quad & x_1 - x_2 + x_3 \geq 2, \\
 & x_2 \leq 0, \\
 & x_3 \text{ unrestricted.}
 \end{aligned}$$

Step 1: Convert to maximization.

Multiply the objective by -1 :

$$\max \quad x_1 - 2x_2 + 3x_3.$$

Step 2: Convert variables to be nonnegative.

- $x_2 \leq 0 \leadsto$ define $x_2 = -x'_2$, with $x'_2 \geq 0$.
- x_3 unrestricted $\leadsto$ write $x_3 = x_3^+ - x_3^-$, with $x_3^+, x_3^- \geq 0$.

Substitute into the objective and constraint:

$$\max x_1 - 2(-x'_2) + 3(x_3^+ - x_3^-) = x_1 + 2x'_2 + 3x_3^+ - 3x_3^-.$$

$$\text{Constraint: } x_1 + x'_2 + x_3^+ - x_3^- \geq 2.$$

Step 3: Convert constraint to equality using a surplus variable.

Introduce surplus variable $s_1 \geq 0$:

$$x_1 + x'_2 + x_3^+ - x_3^- - s_1 = 2.$$

Now the problem is in standard form:

$$\begin{aligned} \max \quad & x_1 + 2x'_2 + 3x_3^+ - 3x_3^- \\ \text{subject to} \quad & x_1 + x'_2 + x_3^+ - x_3^- - s_1 = 2, \\ & x_1, x'_2, x_3^+, x_3^-, s_1 \geq 0. \end{aligned}$$

Example 8.10: Example Continued

Step 4: Example solution in standard form.

Consider a solution of the canonical form

$$x_1 = 1, \quad x'_2 = 0, \quad x_3^+ = 1, \quad x_3^- = 0, \quad s_1 = 0.$$

Step 5: Map solution back to original variables.

$$x_2 = -x'_2 = 0, \quad x_3 = x_3^+ - x_3^- = 1.$$

So the solution to the original problem is:

$$x_1 = 1, \quad x_2 = 0, \quad x_3 = 1,$$

and the original objective value is:

$$-1(1) + 2(0) - 3(1) = -4.$$

Conclusion: The transformation allowed us to solve the LP using standard techniques, and we were able to directly interpret the solution in the space of the original variables. The solution meaning is preserved.

8.1.2. Basic Feasible Solutions for Standard Form

Consider a linear program in standard form:

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}, \\ & \mathbf{x} \geq 0, \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{c} \in \mathbb{R}^n$.

We assume throughout that:

- The matrix A has full row rank, i.e., $\text{rank}(A) = m$,
- The feasible region $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$ is nonempty.

Definition 8.11: Basic Feasible Solution

A vector $\mathbf{x} \in \mathbb{R}^n$ is called a basic feasible solution (BFS) if:

1. It satisfies all the constraints of the LP (i.e., it is feasible),
2. It has at most m positive components (i.e., $n - m$ components are set to zero),
3. These positive components correspond to a set of linearly independent columns of A .

Let $B \subseteq \{1, \dots, n\}$ be an index set of size m such that the submatrix A_B (formed from the columns of A indexed by B) is nonsingular. Then the corresponding basic solution is given by:

$$\mathbf{x}_B = A_B^{-1} \mathbf{b}, \quad \mathbf{x}_N = 0,$$

where $N = \{1, \dots, n\} \setminus B$. If $\mathbf{x}_B \geq 0$, the basic solution is feasible, and hence a basic feasible solution.

Basic feasible solutions play a central role in the simplex method, which moves from one BFS to another while improving the objective function value.

Example 8.12: Basic Feasible Solution

Consider the linear program:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4, \\ & 2x_1 + x_2 \leq 5, \\ & x_1, x_2 \geq 0. \end{aligned}$$

We first convert the inequalities to equalities by introducing slack variables s_1 and s_2 :

$$\begin{aligned} x_1 + x_2 + s_1 &= 4, \\ 2x_1 + x_2 + s_2 &= 5, \\ x_1, x_2, s_1, s_2 &\geq 0. \end{aligned}$$

This system has 4 variables and 2 equality constraints. A basic feasible solution corresponds to choosing any 2 variables (equal to the number of constraints) as basic and solving for them.

Let's choose s_1 and s_2 as the basic variables (so $x_1 = x_2 = 0$). Substituting into the equations:

$$\begin{aligned} s_1 &= 4, \\ s_2 &= 5. \end{aligned}$$

This gives the solution:

$$(x_1, x_2, s_1, s_2) = (0, 0, 4, 5),$$

which is a basic feasible solution. It is feasible because all variables are nonnegative, and basic because we set two variables (equal to the number of constraints) to nonzero values.

Geometrically, this solution corresponds to the intersection of the two axes (i.e., the origin) in the (x_1, x_2) plane — a corner point of the feasible region.

Proposition 8.13: Vertices and BFS

Let $P = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$ be the feasible region of a linear program in standard form, where $A \in \mathbb{R}^{m \times n}$ has full row rank $m \leq n$. Then every vertex of P is a basic feasible solution.

Proof. Let $\mathbf{x}^* \in P$ be a vertex of the feasible region P . By definition, this means that $\mathbf{x}^*$ cannot be expressed as a strict convex combination of two distinct points in P . We will show that $\mathbf{x}^*$ is a basic feasible solution.

Let $I := \{j \in \{1, \dots, n\} : x_j^* > 0\}$ be the support of $\mathbf{x}^*$, and let A_I denote the submatrix of A consisting of the columns indexed by I . Since $\mathbf{A}\mathbf{x}^* = \mathbf{b}$, and $x_j^* = 0$ for $j \notin I$, the vector $\mathbf{x}_I^* \in \mathbb{R}^{|I|}$ satisfies:

$$A_I \mathbf{x}_I^* = \mathbf{b}.$$

If the columns of A_I were linearly dependent, then there would exist a nonzero vector $\mathbf{d} \in \mathbb{R}^n$ with support in I such that $\mathbf{A}\mathbf{d} = 0$, and for small enough $\varepsilon > 0$, both $\mathbf{x}^* + \varepsilon\mathbf{d}$ and $\mathbf{x}^* - \varepsilon\mathbf{d}$ would remain in P , contradicting the fact that $\mathbf{x}^*$ is a vertex.

Therefore, the columns of A_I must be linearly independent, and since A has rank m , it follows that $|I| \leq m$. We can now construct a basic solution by selecting any set $B \subseteq \{1, \dots, n\}$ of indices such that:

- $B \supseteq I$,
- $|B| = m$,
- The columns A_B are linearly independent.

Let $\mathbf{x}_B = A_B^{-1}\mathbf{b}$, and set $\mathbf{x}_N = 0$, where $N = \{1, \dots, n\} \setminus B$. Then $\mathbf{x}$ is a basic solution. Since $\mathbf{x}_j^* = 0$ for $j \notin I \subseteq B$, and $\mathbf{x}_j^* > 0$ for $j \in I$, it follows that $\mathbf{x}^*$ coincides with this basic solution and is nonnegative — hence it is a basic feasible solution. 

8.2 Canonical Form

For our purposes, it's not quite enough to have a problem in standard form. We also want to have easy access to a basic feasible solution. Thus, we like to convert linear programs to *canonical form*.

Definition 8.14: Canonical Form of a Linear Program

A linear program is in canonical form if:

- it is in standard form:

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}, \\ & \mathbf{x} \geq 0, \end{aligned}$$

- the constraint matrix A contains the identity matrix I_m as a submatrix corresponding to some set of basic variables $B \subseteq \{1, \dots, n\}$,
- and the right-hand side $\mathbf{b} \geq 0$.

In this case, a basic feasible solution is immediately evident: set the basic variables $\mathbf{x}_B = \mathbf{b}$, and nonbasic variables to zero.

Example 8.15: Canonical Form from Slack Variables

Consider the linear program:

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4, \\ & x_1 + 2x_2 \leq 6, \\ & x_1, x_2 \geq 0. \end{aligned}$$

We convert the $\leq$ constraints to equalities by introducing slack variables s_1 and s_2 :

$$\begin{aligned}x_1 + x_2 + s_1 &= 4, \\x_1 + 2x_2 + s_2 &= 6, \\s_1, s_2 &\geq 0.\end{aligned}$$

Now the LP is in standard form with 4 variables x_1, x_2, s_1, s_2 . The constraint matrix is:

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

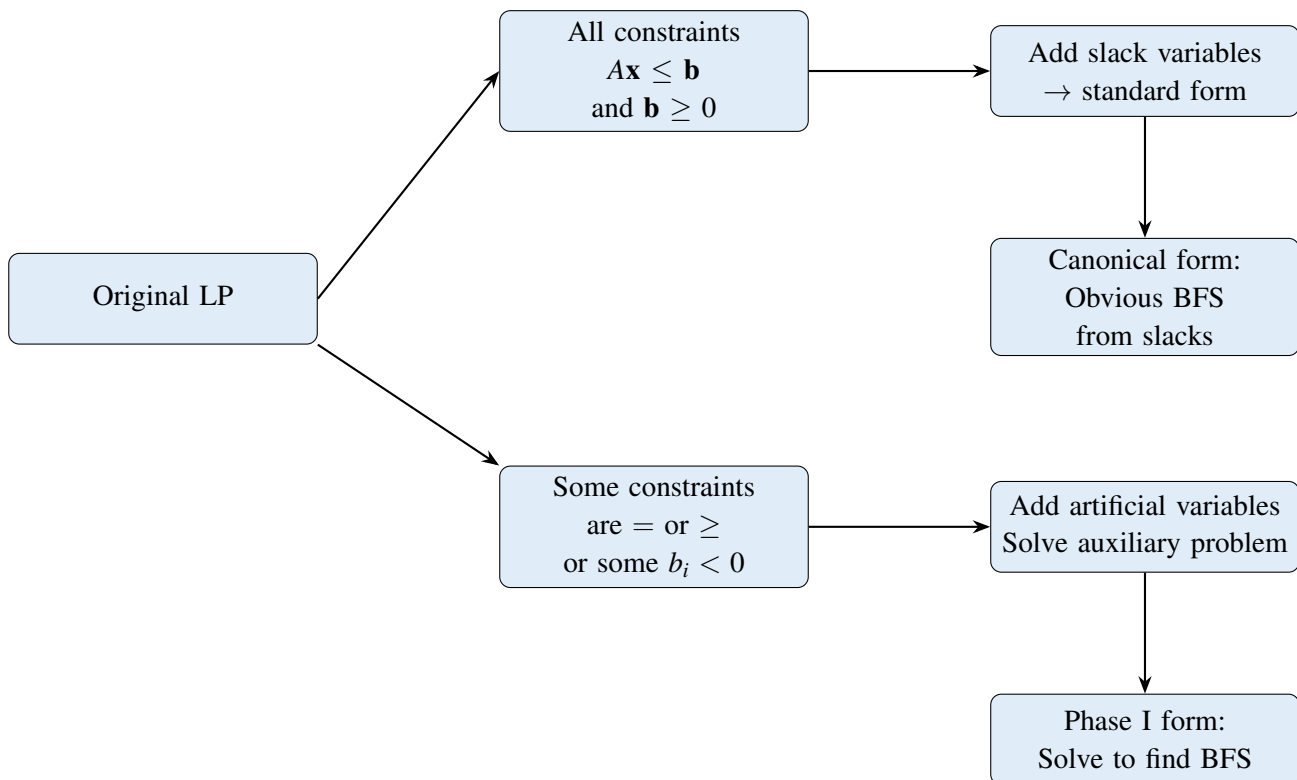
The columns corresponding to s_1 and s_2 form the identity matrix:

$$A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2, \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} \geq 0.$$

Hence, this LP is now in **canonical form**. The evident basic feasible solution is:

$$x_1 = 0, \quad x_2 = 0, \quad s_1 = 4, \quad s_2 = 6.$$

When Can You Use Canonical Form?



8.2.1. Augmenting system and Phase 1

We can summarize these tricks in the following table.

Type of Constraint	Form of Constraint	Constraint in Augmented Form	Constraint Boundary Equation
Nonnegativity	$x_j \geq 0$	$x_j \geq 0$	$x_j = 0$
Nonpositivity	$x_j \leq 0$	$x_j = -x'_j, x'_j \geq 0$	$x'_j = 0$
Unrestricted	$x_j \in \mathbb{R}$	$x_j = x_j^+ - x_j^-, x_j^+, x_j^- \geq 0$	$x_j^+ = x_j^-$
Functional ($\leq$)	$\sum_{j=1}^n a_{ij}x_j \leq b_i$	$\sum_{j=1}^n a_{ij}x_j + s_i = b_i$	$\sum a_{ij}x_j = b_i$
Functional ($=$)	$\sum_{j=1}^n a_{ij}x_j = b_i$	$\sum_{j=1}^n a_{ij}x_j + a_i = b_i$	$\sum a_{ij}x_j = b_i$
Functional ($\geq$)	$\sum_{j=1}^n a_{ij}x_j \geq b_i$	$\sum_{j=1}^n a_{ij}x_j - s_i + a_i = b_i$	$\sum a_{ij}x_j = b_i$

*Indicating variable = 0 $\implies$ constraint boundary is satisfied; indicating variable $\neq 0 \implies$ constraint boundary is not satisfied.

Table 8.1: Summary of constraint and domain transformations used to convert to standard/canonical form.

We end this subsection with a motivating result.

Proposition 8.16: Existence of Canonical Form for Feasible LPs

Let P be a linear program in standard form:

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}, \\ & \mathbf{x} \geq 0. \end{aligned}$$

If P is feasible (i.e., $\exists \mathbf{x} \geq 0$ such that $A\mathbf{x} = \mathbf{b}$), then there exists an equivalent linear program in canonical form — that is, a reformulation where:

- A contains the identity matrix I_m as a submatrix,
- The corresponding basic variables yield a basic feasible solution with $\mathbf{b} \geq 0$,
- The objective function is unchanged in value on feasible solutions.

Proof. [Idea of Proof] Start from any feasible solution $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{x} \geq 0$ and $A\mathbf{x} = \mathbf{b}$. Use the simplex method (or a pivoting method) to move to a basic feasible solution $\mathbf{x}^*$. Since A has full rank over the feasible region, the basis matrix A_B will be invertible. By reordering variables (i.e., choosing a new basis where $A_B = I_m$), we can express the LP in canonical form.

If no feasible solution exists, then Phase I must be used to find one (or certify infeasibility).



8.3 Pivoting and the Simplex Algorithm

Outcomes

- See an example of the simplex method in action
- Formally state the simplex algorithm
- See that changing the pivot rule can change the path to the optimal solution

Resources

Video

- Youtube! - Simplex Method - Watch 3:40 to 11:40

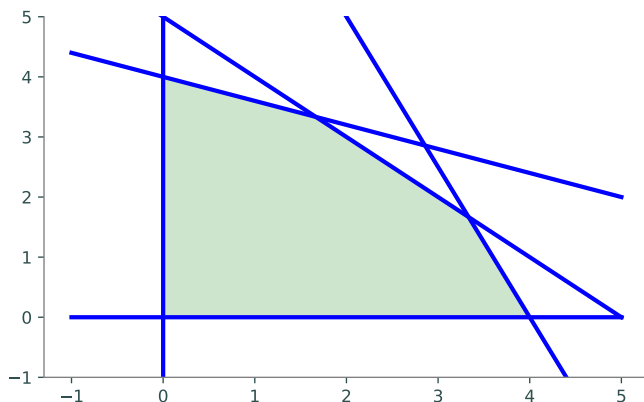
Interactive Simplex Method

- GILP: Interactive Simplex Pivoting with Dictionary - 2D example
- GILP: Interactive Simplex Pivoting with Dictionary - 3D example

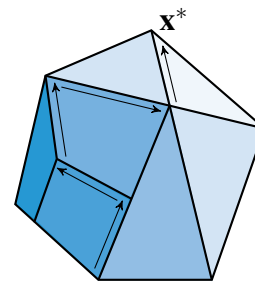
Simplex Dictionary Calculator Using Python + Jupyter

- Colab Version (online)
- Jupyter Notebook for Download

The Simplex Algorithm jumps among the vertices of the feasible region searching for an optimal point. It does this by moving along the edges of the feasible region in such a way that the objective function is always increased after each move.



(a) The feasible region for a linear program with 2-dimensional constraints.



(b) The feasible region for a linear program with 3-dimensional constraints.

Figure 8.2: If an optimal point exists, it is one of the vertices of the polyhedron. The simplex algorithm searches for optimal points by moving between adjacent vertices in a direction that increases the value of the objective function until it finds an optimal vertex.

To improve the solution, we:

1. Select an entering variable with a negative reduced cost.
2. Determine the leaving variable using the minimum ratio test.
3. Perform a pivot operation to update the basis.

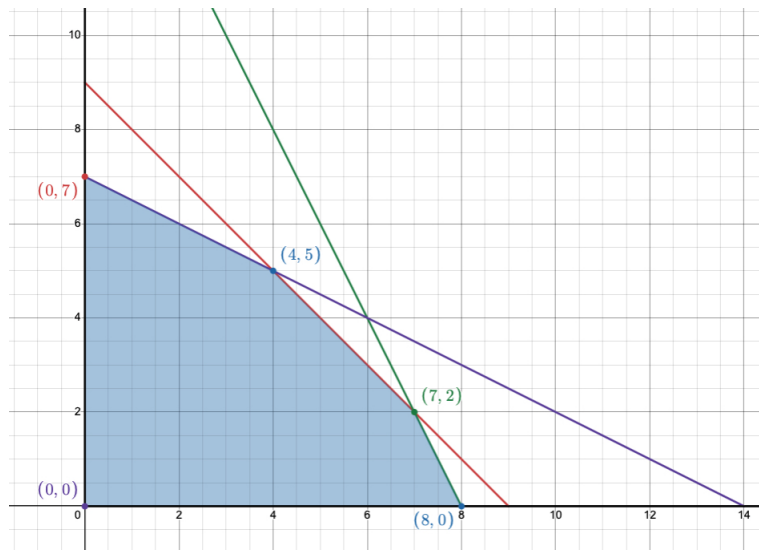
This process continues until no negative reduced costs remain, indicating an optimal solution.

8.3.1. Transforming the LP to Standard Form

We start with the following linear program:

Linear Program to Solve

$$\begin{array}{ll}
 \text{Maximize} & 2x + 3y \\
 \text{subject to} & x + y \leq 9, \\
 & 2x + y \leq 16, \\
 & x + 2y \leq 14, \\
 & x, y \geq 0.
 \end{array}$$



To convert these inequalities to equalities, we introduce slack variables s_1 , s_2 , and s_3 . Specifically, we rewrite each constraint as follows:

$$\begin{aligned}
 x + y + s_1 &= 9, \\
 2x + y + s_2 &= 16, \\
 x + 2y + s_3 &= 14,
 \end{aligned}$$

with the additional nonnegativity constraints $s_1, s_2, s_3 \geq 0$.

Hence, the linear program in standard form becomes:

Standard Form

$$\begin{array}{ll}
\text{Maximize} & 2x + 3y \\
\text{subject to} & x + y + s_1 = 9, \\
& 2x + y + s_2 = 16, \\
& x + 2y + s_3 = 14, \\
& x, y, s_1, s_2, s_3 \geq 0.
\end{array}$$

Rewriting the System with $\{s_1, s_2, s_3\}$ as the Basis

When we say $\{s_1, s_2, s_3\}$ is the basis, we solve each equation for the corresponding slack variable:

$$\begin{aligned}
s_1 &= 9 - x - y, \\
s_2 &= 16 - 2x - y, \\
s_3 &= 14 - x - 2y.
\end{aligned}$$

Our initial non-basic variables are x and y . We will call this representation of the linear program a *dictionary*.

Initial Dictionary

$$\begin{array}{ll}
\text{Maximize} & z = 0 + 2x + 3y \\
\text{subject to} & s_1 = 9 - x - y, \\
& s_2 = 16 - 2x - y, \\
& s_3 = 14 - x - 2y, \\
& x, y, s_1, s_2, s_3 \geq 0.
\end{array}$$

From here we can read off a *basic solution* that is

$$\text{Basic variables: } (s_1, s_2, s_3) = (9, 16, 14),$$

$$\text{Non-basic variables: } (x, y) = (0, 0)$$

with

$$\text{Objective value: } z = 0.$$

Since the basic variables are all non-negative (≥ 0), we call this a *basic feasible solution*.

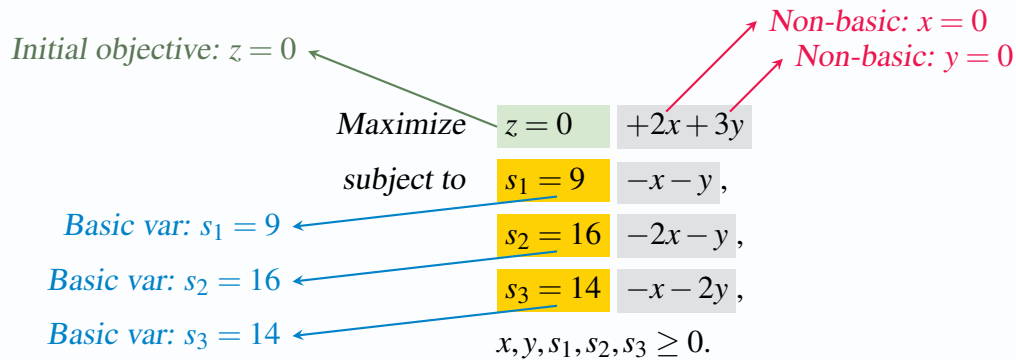
Remark 8.17: Reading the Dictionary

From the dictionary below, we can read off the corresponding **basic solution** and the **objective value**. The basic solution is obtained by:

- Setting all **nonbasic variables** (those on the right-hand side of the equations, i.e., x and y) equal to zero.
- Reading off the values of the **basic variables** (s_1, s_2, s_3) from the right-hand side constants.

The dictionary is said to be **feasible** if all basic variables have nonnegative values (which is true here). Thus, this corresponds to a basic feasible solution:

$$x = 0, \quad y = 0, \quad s_1 = 9, \quad s_2 = 16, \quad s_3 = 14, \quad z = 0.$$



At $(x, y, s_1, s_2, s_3) = (0, 0, 9, 16, 14)$, the objective value is $z = 0$.

Choosing the Entering Variable

Entering Variable

We examine the *objective row* to determine which nonbasic variable should enter the basis. At this point, the objective is:

$$z = 2x + 3y.$$

- The nonbasic variables are x and y .
- Both have **positive** coefficients, which means that increasing either variable (while staying feasible) will increase the objective value z .
- To decide between them, we apply a **pivot rule** — a strategy for selecting among eligible entering variables.

In this example, we use the *Steepest Ascent* rule: we choose the nonbasic variable with the largest positive coefficient in the objective row.

- Between x and y , the coefficient of y is larger ($3 > 2$).

Therefore, we choose y to **enter** the basis.

First Iteration. Ratio Test to Determine Who Leaves

First Iteration - Ratio Test

We hold $x = 0$ fixed and allow y to increase from 0. Then from the initial dictionary, the basic variables become (substituting $x = 0$):

$$\begin{aligned}s_1 &= 9 - y, \\s_2 &= 16 - y, \\s_3 &= 14 - 2y.\end{aligned}$$

To maintain feasibility, we require all basic variables to remain nonnegative:

$$\begin{aligned}s_1 \geq 0 &\implies 9 - y \geq 0 \implies y \leq 9, \\s_2 \geq 0 &\implies 16 - y \geq 0 \implies y \leq 16, \\s_3 \geq 0 &\implies 14 - 2y \geq 0 \implies y \leq 14/2 = 7.\end{aligned}$$

The *tightest constraint* is $y \leq 7$, which means s_3 reaches zero first as y increases. Therefore, s_3 will **leave the basis** in this pivot.

This is called the *Ratio Test* because we can compute these numbers as

$$9 - y = 0 \implies y = 9/1 = 9$$

$$16 - y = 0 \implies y = 16/1 = 16$$

$$14 - 2y = 0 \implies y = 14/2 = 7$$

If a coefficient on y was positive, we would ignore that constraint as increasing y would never violate the corresponding inequality.

The ratio test can also be thought of as changing the solution via a vector. In particular, we are at the basic feasible solution $(s_1, s_2, s_3, x, y) = (9, 16, 14, 0, 0)$. Now, as we increase y , we change the solution as

$$(s_1, s_2, s_3, x, y) = (9, 16, 14, 0, 0) + \Delta y \cdot (-1, -1, -2, 0, 1) = (9, 16, 14, 0, 0) + 7 \cdot (-1, -1, -2, 0, 1) = (2, 9, 0, 0, 7).$$

We will see these number appear as we pivot the dictionary to the new basis.

Pivoting on the s_3 -Equation Solved for y

Originally, $s_3 + x + 2y = 14$. Solve for y :

$$y = 7 - \frac{x}{2} - \frac{s_3}{2}.$$

Substitute into the other rows (and the objective row), isolating each *basic* variable on the left and all others on the right:

$$\begin{aligned} \text{Maximize } z &= 0 + 2x + 3y &&= 21 + \frac{1}{2}x - \frac{3}{2}s_3 \\ \text{subject to } s_1 &= 9 - x - y &&= 2 - \frac{1}{2}x + \frac{1}{2}s_3, \\ s_2 &= 16 - 2x - y &&= 9 - \frac{3}{2}x + \frac{1}{2}s_3, \\ s_3 &= 14 - x - 2y \Rightarrow y &&= 7 - \frac{1}{2}x - \frac{1}{2}s_3, \\ x, y, s_1, s_2, s_3 &\geq 0. \end{aligned}$$

Hence the new *basic variables* are $\{z, y, s_1, s_2\}$ if we were to place z in the dictionary as well. Typically, in a standard simplex dictionary, z is displayed in its own row, but it is not counted as a “basic variable” in the same sense as the constraints. The nonbasic variables are now $\{x, s_3\}$.

Updated Dictionary and Basic Feasible Solution

Second Dictionary

$$\begin{aligned} \text{Maximize } z &= 21 + \frac{1}{2}x - \frac{3}{2}s_3 \\ \text{subject to } s_1 &= 2 - \frac{1}{2}x + \frac{1}{2}s_3, \\ s_2 &= 9 - \frac{3}{2}x + \frac{1}{2}s_3, \\ y &= 7 - \frac{1}{2}x - \frac{1}{2}s_3, \\ x, y, s_1, s_2, s_3 &\geq 0. \end{aligned}$$

- Set $x = 0$ and $s_3 = 0$ (the new nonbasic variables).
- Then (y, s_1, s_2) become:

$$y = 7, \quad s_1 = 2, \quad s_2 = 9.$$

- Substitute in the objective row:

$$z = 21 + \frac{1}{2} \cdot 0 - \frac{3}{2} \cdot 0 = 21.$$

Thus the new BFS after the pivot is $(x, y, s_1, s_2, s_3) = (0, 7, 2, 9, 0)$ with $z = 21$.

Second Iteration

Since the objective function still has variables with positive coefficients in it, we will seek to pivot again to increase the objective.

Second Iteration. Identify the Entering Variable

Second Iteration - Entering Variable

We look at the coefficients of the nonbasic variables in the *objective* row,

$$z = 21 + \frac{1}{2}x - \frac{3}{2}s_3.$$

- x has a $\frac{1}{2}$ coefficient (which is **positive**),
- s_3 has a $-\frac{3}{2}$ coefficient (negative).

Since this is a *maximization* problem, increasing a variable with a **negative** objective coefficient would *decrease* z . Hence we choose x , the only nonbasic variable with a positive coefficient, to *enter* the basis.

Second Iteration. Ratio Test to Determine Who Leaves

Second Iteration - Ratio Test

We hold $s_3 = 0$ for the moment and let x increase from 0. Then from the above system, the three basic variables become (substituting $s_3 = 0$):

$$\begin{aligned} y &= 7 - \frac{1}{2}x, \\ s_1 &= 2 - \frac{1}{2}x, \\ s_2 &= 9 - \frac{3}{2}x. \end{aligned}$$

We require these to remain nonnegative:

$$\begin{aligned} y \geq 0 &\implies 7 - \frac{1}{2}x \geq 0 \implies x \leq 7/\frac{1}{2} = 14, \\ s_1 \geq 0 &\implies 2 - \frac{1}{2}x \geq 0 \implies x \leq 2/\frac{1}{2} = 4, \\ s_2 \geq 0 &\implies 9 - \frac{3}{2}x \geq 0 \implies x \leq 9/\frac{3}{2} = 6. \end{aligned}$$

The *smallest* upper bound is $x \leq 4$, which means s_1 will reach zero first when $x = 4$. Hence s_1 must *leave* the basis.

The Pivot: Solve the s_1 Equation for x

From

$$s_1 = 2 - \frac{1}{2}x + \frac{1}{2}s_3,$$

we isolate x :

$$s_1 - 2 - \frac{1}{2}s_3 = -\frac{1}{2}x \implies x = -2\left(s_1 - 2 - \frac{1}{2}s_3\right) = 4 + s_3 - 2s_1.$$

Rewriting,

$$x + 2s_1 - s_3 = 4.$$

This *becomes the new pivot row*, with x on the left side as a basic variable.

Eliminate x from the Other Rows

Next, we substitute $x = 4 + s_3 - 2s_1$ into the rows for $\{y, s_2, z\}$ to remove x . (In each row, replace x by $4 + s_3 - 2s_1$.)

UPDATING y : Originally

$$y = 7 - \frac{1}{2}x - \frac{1}{2}s_3.$$

Substitute x :

$$y = 7 - \frac{1}{2}(4 + s_3 - 2s_1) - \frac{1}{2}s_3 = 7 - 2 - \frac{1}{2}s_3 + s_1 - \frac{1}{2}s_3 = 5 + s_1 - s_3.$$

UPDATING s_2 : Originally

$$s_2 = 9 - \frac{3}{2}x + \frac{1}{2}s_3.$$

Substitute x :

$$s_2 = 9 - \frac{3}{2}(4 + s_3 - 2s_1) + \frac{1}{2}s_3 = 9 - 6 - \frac{3}{2}s_3 + 3s_1 + \frac{1}{2}s_3 = 3 + 3s_1 - s_3.$$

UPDATING THE OBJECTIVE ROW z : Originally

$$z = 21 + \frac{1}{2}x - \frac{3}{2}s_3.$$

Substitute x :

$$z = 21 + \frac{1}{2}(4 + s_3 - 2s_1) - \frac{3}{2}s_3 = 21 + 2 + \frac{1}{2}s_3 - s_1 - \frac{3}{2}s_3 = 23 - s_1 - s_3.$$

The Third Dictionary

Final Dictionary

After the pivot, the basic variables are now $\{x, y, s_2\}$, and the nonbasic variables are $\{s_1, s_3\}$. Our updated equations, in a clean dictionary form (basic on the left) are:

$$\begin{array}{ll}
 \text{Maximize} & z = 23 - s_1 - s_3 \\
 \text{subject to} & x = 4 - 2s_1 + s_3, \\
 & y = 5 + s_1 - s_3, \\
 & s_2 = 3 + 3s_1 - s_3, \\
 & x, y, s_1, s_2, s_3 \geq 0.
 \end{array}$$

Basic var: $x = 4$
 Basic var: $y = 5$
 Basic var: $s_2 = 3$

- Setting the new nonbasic variables $s_1 = 0$ and $s_3 = 0$ gives

$$x = 4, \quad y = 5, \quad s_2 = 3, \quad z = 23.$$

- Thus our new BFS is $(x, y, s_1, s_2, s_3) = (4, 5, 0, 3, 0)$ with objective value $z = 23$.

Remark 8.18: Reading the Final Dictionary

From the final dictionary below, we read the solution using the same process:

- Set all **nonbasic variables** equal to zero. Here, the nonbasic variables are s_1 and s_3 .
- Evaluate the values of the basic variables x, y, s_2 and the objective z .

This yields the final basic feasible solution:

$$s_1 = 0, \quad s_3 = 0, \quad x = 4, \quad y = 5, \quad s_2 = 3, \quad z = 23.$$

Final objective: $z = 23$

$$\begin{array}{ll}
 \text{Maximize} & z = 23 - s_1 - s_3 \\
 \text{subject to} & x = 4 - 2s_1 + s_3, \\
 & y = 5 + s_1 - s_3, \\
 & s_2 = 3 + 3s_1 - s_3, \\
 & x, y, s_1, s_2, s_3 \geq 0.
 \end{array}$$

Why is this optimal? This is a maximization problem, and the final dictionary expresses the objective as:

$$z = 23 - s_1 - s_3.$$

Both s_1 and s_3 are nonbasic and currently zero. Since increasing either would decrease the value of z , we cannot improve the objective by pivoting.

Hence, this solution is optimal.

8.3.2. Summarizing the steps taken

We now summarize the whole process:

1. Original LP

Linear Program to Solve

$$\begin{array}{ll}\text{Maximize} & 2x + 3y \\ \text{subject to} & x + y \leq 9, \\ & 2x + y \leq 16, \\ & x + 2y \leq 14, \\ & x, y \geq 0.\end{array}$$

2. Standard Form with Feasible Basis $\{s_1, s_2, s_3\}$

Standard Form

$$\begin{array}{ll}\text{Maximize} & 2x + 3y \\ \text{subject to} & x + y + s_1 = 9, \\ & 2x + y + s_2 = 16, \\ & x + 2y + s_3 = 14, \\ & x, y, s_1, s_2, s_3 \geq 0.\end{array}$$

3. Initial Dictionary with Basis (s_1, s_2, s_3)

Initial Dictionary

$$\begin{array}{ll}\text{Maximize} & z = 0 + 2x + 3y \\ \text{subject to} & s_1 = 9 - x - y, \\ & s_2 = 16 - 2x - y, \\ & s_3 = 14 - x - 2y. \\ & x, y, s_1, s_2, s_3 \geq 0\end{array}$$

Basic Feasible Solution: $(s_1, s_2, s_3) = (9, 16, 14)$, $(x, y) = (0, 0)$ Objective Value: $z = 0$.
Since the objective row has a positive coefficient, it is not optimal.

4. First Iteration:

(a) **Ratio Test:** y enters, s_3 leaves.

(b) **Dictionary after Pivoting y in, s_3 out****Second Dictionary**

$$\begin{aligned}
 \text{Maximize } z &= 21 + \frac{1}{2}x - \frac{3}{2}s_3 \\
 \text{subject to } s_1 &= 2 - \frac{1}{2}x + \frac{1}{2}s_3, \\
 s_2 &= 9 - \frac{3}{2}x + \frac{1}{2}s_3, \\
 y &= 7 - \frac{1}{2}x - \frac{1}{2}s_3.
 \end{aligned}$$

Basic Feasible Solution: $(s_1, s_2, y) = (2, 9, 7), (x, s_3) = (0, 0)$ Objective Value: $z = 21$.
 Since the objective row has a positive coefficient, it is not optimal.

5. Second Iteration

(a) **Ratio Test:** x enters, s_1 leaves.

(b) **Final Dictionary (Optimal) after pivoting x in, s_1 out.**

Final Dictionary

$$\begin{aligned}
 \text{Maximize } z &= 23 + s_1 + s_3 \\
 \text{subject to } x &= 4 - 2s_1 + s_3, \\
 y &= 5 + s_1 - s_3, \\
 s_2 &= 3 + 3s_1 - s_3.
 \end{aligned}$$

Basic Feasible Solution: $(x, y, s_2) = (4, 5, 3), (s_1, s_3) = (0, 0)$ Objective Value: $z = 23$.
 Since all coefficients in the objective row are non-positive, this is the optimal solution.

Resources

You can review these calculations using the code available here in Jupyter Notebook Format.

- *Colab Version (online)*
- *Jupyter Notebook for Download*

8.4 The Simplex Algorithm!

Here we formally describe the simplex algorithm.

Algorithm 3 Simplex Algorithm

- 1: **Input:** Linear program in standard form with a feasible starting basis (typically the slack variables).
- 2: **Initialization:** Reformulate the system in dictionary form:
 - Write the equations with the basic variables on the left and the nonbasic variables on the right.
 - Substitute the basic variables into the objective function.
- 3: **repeat**
- 4: **1. Choose the entering variable:** Select a nonbasic variable with a positive coefficient in the objective row.
- 5: **2. Determine the leaving variable:** For each basic variable, perform the ratio test by calculating

$$\text{ratio} = \frac{\text{current value of basic variable}}{\text{coefficient of entering variable (if positive)}}$$

The basic variable that reaches zero first (i.e., the smallest ratio) is chosen to leave the basis.

- 6: **3. Pivot:**
 - Rewrite the equation of the leaving variable so that it is expressed in terms of the entering variable.
 - Substitute this expression into all other equations and the objective function.
 - 7: **until** All coefficients in the objective row for nonbasic variables are non-positive (indicating that an optimal solution has been reached).
 - 8: **Output:** The optimal solution.
-

We dive a little deeper into what the ratio test is here:

- **The Ratio Test (Formal Definition)**

The Ratio Test is used to determine which basic variable should leave the basis when a nonbasic variable enters. It ensures that we maintain feasibility by preventing any basic variables from becoming negative.

Given a *pivot column* corresponding to the entering variable (say, x_e), the Ratio Test is defined as follows:

- Identify all constraints of the form:

$$s_i = b_i - a_{i,e}x_e$$

where b_i is the right-hand side value, and $a_{i,e}$ is the coefficient of the entering variable in row i .

- Compute the ratio for all constraints where $a_{i,e} > 0$:

$$\frac{b_i}{a_{i,e}}$$

- The *leaving variable* corresponds to the row that gives the *smallest positive ratio*. This ensures that the increase in x_e does not cause any basic variable to become negative.

Special Cases:

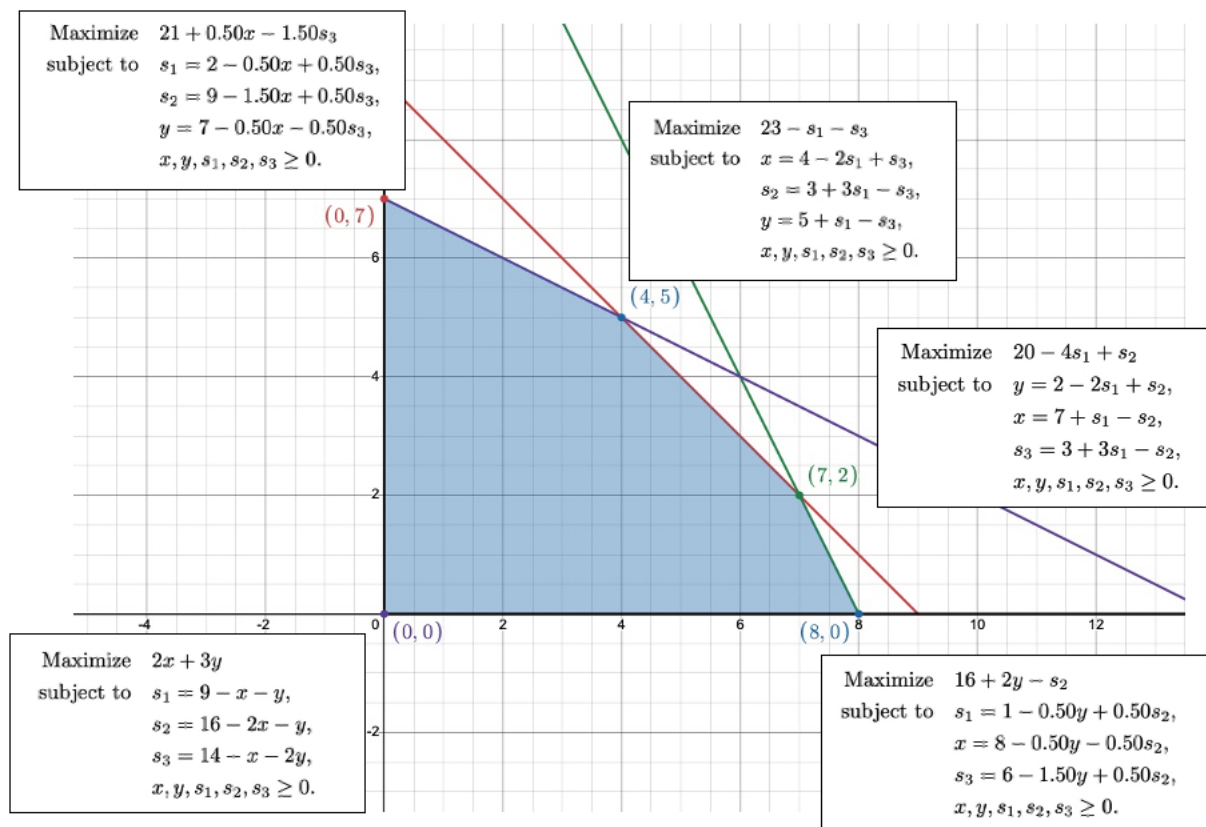
- If all $a_{i,e} \leq 0$, then the problem is *unbounded*, meaning the objective function can be increased indefinitely.
- If multiple rows attain the same minimum ratio, a *tie-breaking rule* is needed (e.g., Bland's Rule to avoid cycling).

The Ratio Test guarantees that we always select the proper constraint to maintain feasibility while pivoting towards an improved solution.

We will see another example of this, and then modify it to handle LPs that don't have a simple initial basic feasible solution.

In the example worked out in the prior section, we used the *steepest ascent* pivot rule, which means that we always used the variable in the objective with the largest positive coefficient to enter the basis. However, we could make other choices instead. The choice of how to pivot is left a little ambiguous in the simplex method and many different pivot rules have been proposed. See Exercise 8.6 to see how the simplex method plays out when pivoting in a different direction first.

In fact, in the figure below, we show the simplex dictionary at every single vertex. You can see that from any vertex, you can pivot along an edge of the feasible region to another vertex.



Remark 8.19: Reduced Costs

The coefficients of the nonbasic variables in the objective row are their reduced costs:

$$\text{Reduced cost of } s_1 = -1, \quad \text{Reduced cost of } s_3 = -1.$$

$$\begin{aligned} \text{Maximize } z &= 23 - s_1 - s_3 \\ \text{subject to } x &= 4 - 2s_1 + s_3, \\ y &= 5 + s_1 - s_3, \\ s_2 &= 3 + 3s_1 - s_3, \\ x, y, s_1, s_2, s_3 &\geq 0. \end{aligned}$$

- These values indicate how much the objective z would change per unit increase in the corresponding variable, assuming all others are held fixed and feasibility is maintained.
- Because this is a maximization problem, a negative reduced cost implies that increasing the variable would decrease the objective.

Optimality Condition: A basic feasible solution is optimal if and only if all reduced costs of nonbasic variables satisfy:

$$(\text{for maximization}): \quad \text{reduced cost} \leq 0.$$

In this dictionary:

$$\text{Reduced costs of } s_1, s_3 < 0 \quad \Rightarrow \quad \text{no nonbasic variable can enter to improve } z.$$

Therefore, the current solution is optimal.

Definition 8.20: Reduced Costs in an Arbitrary Dictionary

Let a linear program in standard form be given by

$$\text{Maximize } \mathbf{c}^\top \mathbf{x} \quad \text{subject to } A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0,$$

and let $B \subseteq \{1, \dots, n\}$ be a basis of linearly independent columns of A . The corresponding dictionary expresses the basic and objective variables in terms of the nonbasic variables $N = \{1, \dots, n\} \setminus B$:

$$z = z_0 + \sum_{j \in N} \bar{c}_j x_j$$

$$x_i = \bar{b}_i + \sum_{j \in N} \bar{a}_{ij} x_j \quad \text{for } i \in B$$

Here:

- x_i are the basic variables,
- x_j for $j \in N$ are the nonbasic variables,
- z_0 is the current value of the objective,
- $\bar{c}_j$ is the reduced cost of variable x_j , and
- $\bar{b}_i$ is the value of basic variable x_i when $x_j = 0$ for all $j \in N$.

8.5 No Feasible Initial Basis and the Big-M Method

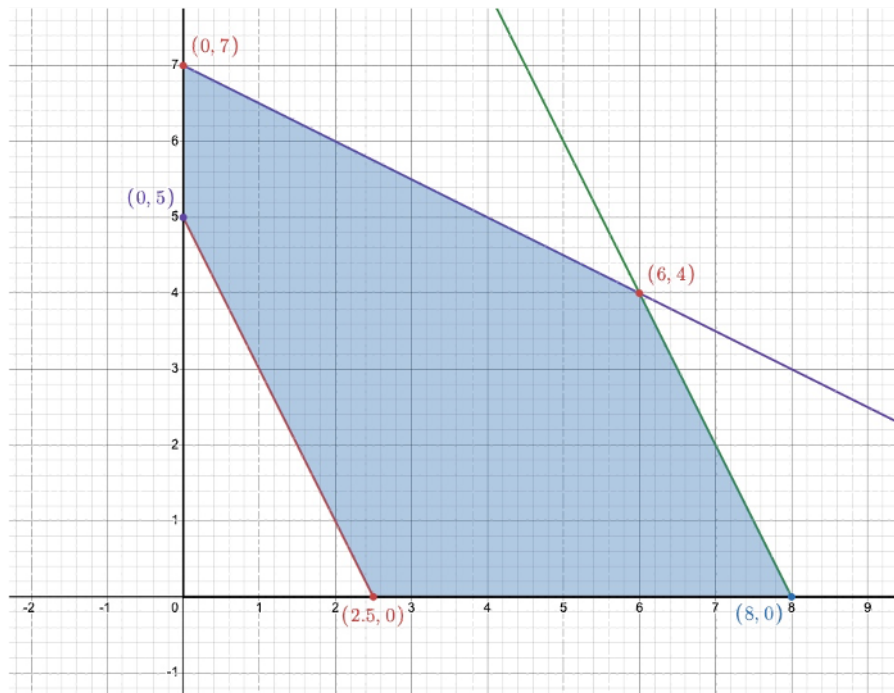
Outcomes

- Address case where initial basis is infeasible
- Modify the simplex algorithm using the Big-M method to find a feasible starting basis
 - Introduce artificial variables to rows that violate initial feasibility
 - Add large penalty (Big-M) to the objective on these artificial variables
 - Run Simplex and pivot until can remove artificial variables
 - Drop the auxiliary variables and proceed with the simplex algorithm

Consider the linear program

Original Linear Program

$$\begin{array}{ll} \text{Maximize} & 2x + 3y \\ \text{subject to} & 2x + y \geq 5, \\ & 2x + y \leq 16, \\ & x + 2y \leq 14, \\ & x, y \geq 0. \end{array}$$



Unfortunately, the origin is not feasible!!!

We rewrite the first inequality as

$$-2x - y \leq -5,$$

so that all constraints are of the form “ $\leq$ ”. Then, introducing slack variables, we have

Infeasible Dictionary!

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & s_1 = -5 + 2x + y, \\ & s_2 = 16 - 2x - y, \\ & s_3 = 14 - x - 2y, \\ & x, y, s_1, s_2, s_3 \geq 0. \end{aligned}$$

At the starting point $x = y = 0$, we have $s_1 = -5$, so the basic solution is infeasible!

Introducing an Artificial Variable

To obtain a feasible initial basis, we add an artificial variable $a_1 \geq 0$ to the first constraint:

$$s_1 = -5 + 2x + y + a_1.$$

This new variable can be thought of as altering the right hand side of the inequality as much as needed by "cheating" and making the initial basis feasible.

Simultaneously, we modify the objective function by adding a large penalty $-1000a_1$ (here, $M = 1000$) so that the optimal solution forces a_1 to zero:

Artificial Dictionary - Need to Pivot a_1 into basis

$$\begin{aligned}
&\text{Maximize} && 2x + 3y - 1000a_1 \\
&\text{subject to} && s_1 = -5 + 2x + y + a_1, \\
&&& s_2 = 16 - 2x - y, \\
&&& s_3 = 14 - x - 2y, \\
&&& x, y, s_1, s_2, s_3, a_1 \geq 0.
\end{aligned}$$

Changing the Basis

Next, we change the basis to one that is feasible. After some pivoting, we may arrive at a dictionary like:

Initial Artificial Dictionary

$$\begin{aligned}
&\text{Maximize} && -5000 + 2002x + 1003y - 1000s_1 \\
&\text{subject to} && a_1 = 5 - 2x - y + s_1, \\
&&& s_2 = 16 - 2x - y, \\
&&& s_3 = 14 - x - 2y, \\
&&& x, y, s_1, s_2, s_3, a_1 \geq 0.
\end{aligned}$$

With further pivoting, we obtain a dictionary where a_1 is nonbasic:

Second Artificial Dictionary

$$\begin{aligned}
&\text{Maximize} && 5 + 2y + s_1 - 1001a_1 \\
&\text{subject to} && x = 2.50 - 0.50y + 0.50s_1 - 0.50a_1, \\
&&& s_2 = 11 - s_1 + a_1, \\
&&& s_3 = 11.50 - 1.50y - 0.50s_1 + 0.50a_1, \\
&&& x, y, s_1, s_2, s_3, a_1 \geq 0.
\end{aligned}$$

Since a_1 is now nonbasic (and hence $a_1 = 0$ in the basic solution), the basis is feasible.

Remark 8.21: Artificial Variable Can Be Removed — Feasible Start for Phase II

*This is the second artificial dictionary from **Phase I** of the two-phase simplex method. Our goal was to eliminate the artificial variable a_1 while finding a feasible solution to the original problem.*

Let's analyze this dictionary:

- *The artificial variable a_1 appears in the system, but it is **not basic**.*

- Because a_1 is a nonbasic variable and its value is currently zero, it can be **removed** from the dictionary without violating feasibility.
- All basic variables (x, s_2, s_3) have nonnegative values when $a_1 = 0$, and all coefficients of a_1 can simply be deleted.

Conclusion: We now have a basic feasible solution to the original problem (with no artificial variables present), so we can begin **Phase II of the simplex method** using this dictionary as the starting point.

$$\begin{array}{ll}
 \text{Maximize} & 5 + 2y + s_1 - 1001a_1 \\
 \text{subject to} & x = 2.50 - 0.50y + 0.50s_1 - 0.50a_1, \\
 & s_2 = 11 - s_1 + a_1, \\
 & s_3 = 11.50 - 1.50y - 0.50s_1 + 0.50a_1, \\
 & x, y, s_1, s_2, s_3, a_1 \geq 0.
 \end{array}$$

Basic var: $s_2 = 11$
 Basic var: $s_3 = 11.50$

Non-basic: $a_1 = 0$

Removing the Artificial Variable

Once feasibility is achieved, we remove the artificial variable from the formulation. This yields:

First Feasible Dictionary!

$$\begin{array}{ll}
 \text{Maximize} & 5 + 2y + s_1 \\
 \text{subject to} & x = 2.50 - 0.50y + 0.50s_1, \\
 & s_2 = 11 - s_1, \\
 & s_3 = 11.50 - 1.50y - 0.50s_1, \\
 & x, y, s_1, s_2, s_3 \geq 0.
 \end{array}$$

Now, with a feasible starting basis in hand, the standard simplex method can be applied to find the optimal solution.

First Feasible Iteration - Choose entering variable

We choose y to enter the basis since it has the largest positive coefficient in the objective of 2.

Exercise 8.22: Complete the simplex method

Complete the simplex method for this problem by carrying out 3 more pivots using the steepest ascent pivot rule.

Big-M Method Algorithm

Algorithm 4 Big-M Method for Linear Programming

- 1: **Input:** Linear program in standard form (maximize), possibly with an infeasible initial basic solution.
 - 2: **Preprocessing:**
 - For each constraint that yields a negative right-hand side, multiply the constraint by -1 so that the right-hand side becomes positive.
 - For each constraint without an obvious feasible basic variable, add an artificial variable $a_i \geq 0$.
 - 3: **Modify the Objective Function:**
 - Add a penalty term $-Ma_i$ (with M a very large constant) for each artificial variable to the original objective.
 - 4: **Formulate the Initial Dictionary:**
 - Include slack (or surplus) variables and artificial variables.
 - Use the modified objective function.
 - 5: **Compute an Initial Basis:**
 - Choose basic variables so that the artificial variables are initially in the basis if necessary.
 - 6: **while** an artificial variable is in the basis or the current basic solution is infeasible **do**
 - 7: Perform a simplex pivot that moves at least one artificial variable out of the basis ▷ Pivot as in the standard simplex method but using the modified objective function.
 - 8: **Remove Artificial Variables:**
 - Once a feasible basis is obtained (i.e., all artificial variables are zero), remove the columns corresponding to the artificial variables.
 - 9: **Proceed with the Standard Simplex Method:**
 - Continue simplex iterations using the original objective function (without the artificial variables).
 - 10: **Output:** The optimal solution for the original problem, or a declaration that the problem is infeasible if any artificial variable remains positive.
-

Detecting Infeasibility via the Big-M Method

Consider the linear program

Original Linear Program

$$\begin{array}{ll}
 \text{Maximize} & 2x + 3y \\
 \text{subject to} & 2x + y \geq 22, \\
 & 2x + y \leq 16, \\
 & x + 2y \leq 14, \\
 & x, y \geq 0.
 \end{array}$$

Notice that the first two constraints on $2x + y$ are contradictory (the same expression is required to be both at least 22 and at most 16). This conflict suggests that the problem may be infeasible.

[Setup problem in standard form] We first rewrite the inequality $2x + y \geq 22$ as

$$-2x - y \leq -22.$$

Then, after introducing slack variables, the problem becomes

Infeasible Dictionary!

$$\begin{array}{ll}
 \text{Maximize} & 2x + 3y \\
 \text{subject to} & s_1 = -22 + 2x + y, \\
 & s_2 = 16 - 2x - y, \\
 & s_3 = 14 - x - 2y, \\
 & x, y, s_1, s_2, s_3 \geq 0.
 \end{array}$$

[Introduce artificial variable] Because the right-hand side here is negative for the natural starting solution $x = y = 0$ (giving $s_1 = -22$), we add an artificial variable a_1 to obtain

$$s_1 = -22 + 2x + y + a_1.$$

This produces the dictionary

Infeasible Artificial Dictionary

$$\begin{array}{ll}
 \text{Maximize} & +2x + 3y - 1000a_1 \\
 \text{subject to} & s_1 = -22 + 2x + y + a_1, \\
 & s_2 = 16 - 2x - y, \\
 & s_3 = 14 - x - 2y, \\
 & x, y, s_1, s_2, s_3, a_1 \geq 0.
 \end{array}$$

Pivot artificial variables into basis to create feasible basis] Next, we pivot to swap a_1 and s_1

First Feasible Artificial Dictionary

$$\begin{aligned}
&\text{Maximize} && -22000 + 2002x + 1003y - 1000s_1 \\
&\text{subject to} && a_1 = 22 - 2x - y + s_1, \\
& && s_2 = 16 - 2x - y, \\
& && s_3 = 14 - x - 2y, \\
& && x, y, s_1, s_2, s_3, a_1 \geq 0.
\end{aligned}$$

[Apply simplex pivots] Next, we hope to drive out the artificial variable. We pivot to let x enter the basis and s_2 leaves.

Second Artificial Dictionary

$$\begin{aligned}
&\text{Maximize} && -5984 + 2y - 1000s_1 - 1001s_2 \\
&\text{subject to} && a_1 = 6 + s_1 + s_2, \\
& && x = 8 - 0.50y - 0.50s_2, \\
& && s_3 = 6 - 1.50y + 0.50s_2, \\
& && x, y, s_1, s_2, s_3, a_1 \geq 0.
\end{aligned}$$

Finally, we pivot to let y enter the basis and s_3 leaves.

Final Artificial Dictionary

$$\begin{aligned}
&\text{Maximize} && -5976 - 1000s_1 - 1000.33s_2 - 1.33s_3 \\
&\text{subject to} && a_1 = 6 + s_1 + s_2, \\
& && x = 6 - 0.67s_2 + 0.33s_3, \\
& && y = 4 + 0.33s_2 - 0.67s_3, \\
& && x, y, s_1, s_2, s_3, a_1 \geq 0.
\end{aligned}$$

Interpreting the Final Dictionary

In the final dictionary, the artificial variable a_1 appears in the equation

$$a_1 = 6 + s_1 + s_2.$$

When the nonbasic variables s_1 and s_2 are set to zero (which is typical in a basic solution), we find

$$a_1 = 6.$$

Because $a_1 > 0$ in the optimal dictionary, this indicates that the original linear program has no feasible solution. The presence of a positive artificial variable in the final optimal dictionary is the standard signal

in the Big-M method that the original problem is infeasible.

In summary, the contradictory constraints (requiring $2x + y$ to be both at least 22 and at most 16) force the artificial variable a_1 to remain positive. Thus, the Big-M method confirms that the problem is infeasible.

Remark 8.23: Final Artificial Dictionary Indicates Infeasibility

This is the final dictionary at the end of **Phase I** of the two-phase simplex method, where the goal was to eliminate the artificial variable a_1 and reach a feasible solution to the original problem.

To interpret this:

- The artificial variable a_1 is still in the basis.
- Its value is expressed as $a_1 = 6 + s_1 + s_2$, and since all variables on the right-hand side are constrained to be nonnegative, we have $a_1 \geq 6$.
- Therefore, we cannot make $a_1 = 0$ — which is required to reach feasibility in Phase I.

Thus, there is no feasible solution to the original problem.

Conclusion: The original linear program is infeasible.

$$\begin{array}{ll}
 \text{Maximize} & -5976 - 1000s_1 - 1000.33s_2 - 1.33s_3 \\
 \text{subject to} & a_1 = 6 + s_1 + s_2, \\
 & x = 6 - 0.67s_2 + 0.33s_3, \\
 & y = 4 + 0.33s_2 - 0.67s_3, \\
 & x, y, s_1, s_2, s_3, a_1 \geq 0.
 \end{array}$$

a_1 still in basis $\leftarrow$ (points to a_1 in the constraint)

$\leftarrow$ All objective coefficients ≤ 0 (points to the objective function)

Theorem 8.24: Infeasibility Detection via the Big-M Method

Consider the linear programming problem

$$\begin{aligned} &\text{Maximize} && \mathbf{c}^\top \mathbf{x} \\ &\text{subject to} && \mathbf{Ax} \leq \mathbf{b}, \\ &&& \mathbf{x} \geq 0. \end{aligned}$$

After converting inequalities to equalities by introducing slack, surplus, and, if needed, artificial variables, the system takes the form

$$\mathbf{Ax} - I_a \mathbf{a} = \mathbf{b}, \quad \mathbf{x}, \mathbf{a} \geq 0,$$

where I_a is a submatrix of the identity matrix, with each column corresponding to an artificial variable introduced to enforce equality in a constraint that initially lacked a basic variable with a nonnegative right-hand side. The Big-M formulation is then given by

$$\text{Maximize} \quad \mathbf{c}^\top \mathbf{x} - M \sum_{i \in I} a_i.$$

If, in any optimal solution of the Big-M problem, there exists an artificial variable $a_i > 0$ for some $i \in I$, then the original linear programming problem is infeasible.

Proof. Assume that an optimal solution $(\mathbf{x}^*, \mathbf{a}^*)$ of the Big-M problem satisfies $a_i^* > 0$ for some $i \in I$.

Suppose, for contradiction, that the original linear programming problem is feasible. Then there exists some $\mathbf{x}' \geq 0$ such that $\mathbf{Ax}' \leq \mathbf{b}$. In the reformulation process, if a constraint lacked a basic variable with a nonnegative right-hand side, an artificial variable a_i was introduced, rewriting the constraint as

$$(\text{original constraint}) - a_i = b_i.$$

For any feasible $\mathbf{x}'$ of the original problem, we can set $\mathbf{a}' = 0$ for all artificial variables, and the equality constraints will still be satisfied.

Thus, the solution $(\mathbf{x}', 0)$ is feasible for the Big-M formulation, and its objective value is exactly $\mathbf{c}^\top \mathbf{x}'$. On the other hand, the optimal solution $(\mathbf{x}^*, \mathbf{a}^*)$ yields an objective value of

$$\mathbf{c}^\top \mathbf{x}^* - M \sum_{i \in I} a_i^*.$$

Since M is chosen to be a very large positive constant, any positive contribution from $\sum_{i \in I} a_i^*$ imposes a severe penalty. Consequently, the objective value of $(\mathbf{x}', 0)$ would be strictly greater than that of $(\mathbf{x}^*, \mathbf{a}^*)$ provided that M is sufficiently large.

This contradiction implies that no feasible solution $(\mathbf{x}', 0)$ exists for the original problem. In other words, the assumption that the original problem is feasible is false.

Hence, if in the optimal solution of the Big-M problem any artificial variable remains positive, the original linear programming problem must be infeasible. 

8.5.1. Let's see another example

Consider the linear program

$$\begin{array}{ll}\text{Maximize} & 3x + y \\ \text{subject to} & 2x + y \geq 5, \\ & 2x + y \leq 16, \\ & x + 2y \geq 3, \\ & x, y \geq 0.\end{array}$$

Again, the origin is not feasible!!!

We rewrite the inequalities to all be of the form “ $\leq$ ”:

$$\begin{array}{l} -2x - y \leq -5, \\ 2x + y \leq 16, \\ -x - 2y \leq -3. \end{array}$$

Now, we introduce slack and artificial variables to write the system as equalities:

$$\begin{array}{l} s_1 = -5 + 2x + y + a_1, \\ s_2 = 16 - 2x - y, \\ s_3 = -3 + x + 2y + a_3. \end{array}$$

The modified objective function with the Big-M penalty (using $M = 1000$) becomes:

$$\text{Maximize } 3x + y - 1000a_1 - 1000a_3.$$

So the initial system is:

$$\begin{array}{ll}\text{Maximize} & 3x + y - 1000a_1 - 1000a_3 \\ \text{subject to} & s_1 = -5 + 2x + y + a_1, \\ & s_2 = 16 - 2x - y, \\ & s_3 = -3 + x + 2y + a_3, \\ & x, y, s_1, s_2, s_3, a_1, a_3 \geq 0.\end{array}$$

We then swap the constraints with negative values for the basic variables (s_1, s_3) for our artificial variables.

$$\begin{array}{ll}\text{Maximize} & -8000 + 3003x + 1002y - 1000s_1 - 1000s_3 \\ \text{subject to} & a_1 = 5 - 2x - y + s_1, \\ & s_2 = 16 - 2x - y, \\ & a_3 = 3 - x + s_3, \\ & x, y, s_1, s_2, s_3, a_1, a_3 \geq 0.\end{array}$$

Now we can run the simplex method. Here is the sequence of pivots:

$$\begin{aligned} \text{Maximize} \quad & -492.50 - 499.50y + 501.50s_1 - 1000s_3 - 1501.50a_1 \\ \text{subject to} \quad & x = 2.50 - 0.50y + 0.50s_1 - 0.50a_1, \\ & s_2 = 8.50 + 0.50y - 1.50s_1 + 1.50a_1, \\ & a_3 = 0.50 + 0.50y - 0.50s_1 + s_3 + 0.50a_1, \\ & x, y, s_1, s_2, s_3, a_1, a_3 \geq 0. \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & 9 + 2y + 3s_3 - 1000a_1 - 1003a_3 \\ \text{subject to} \quad & x = 3 + s_3 - a_3, \\ & s_2 = 7 - y - 3s_3 + 3a_3, \\ & s_1 = 1 + y + 2s_3 + a_1 - 2a_3, \\ & x, y, s_1, s_2, s_3, a_1, a_3 \geq 0. \end{aligned}$$

Now that the artificial variables are out of the basis, we can remove them.

$$\begin{aligned} \text{Maximize} \quad & 9 + 2y + 3s_3 \\ \text{subject to} \quad & x = 3 + s_3, \\ & s_2 = 7 - y - 3s_3, \\ & s_1 = 1 + y + 2s_3, \\ & x, y, s_1, s_2, s_3, a_1, a_3 \geq 0. \end{aligned}$$

Now we can continue simplex until we reach an optimal solution.

$$\begin{aligned} \text{Maximize} \quad & 16 + y - s_2 \\ \text{subject to} \quad & x = 5.33 - 0.33y - 0.33s_2, \\ & s_3 = 2.33 - 0.33y - 0.33s_2, \\ & s_1 = 5.67 + 0.33y - 0.67s_2, \\ & x, y, s_1, s_2, s_3, a_1, a_2 \geq 0. \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & 23 - 2s_2 - 3s_3 \\ \text{subject to} \quad & x = 3 + s_3, \\ & y = 7 - s_2 - 3s_3, \\ & s_1 = 8 - s_2 - s_3, \\ & x, y, s_1, s_2, s_3, a_1, a_2 \geq 0. \end{aligned}$$

Since all coefficients in the objective are negative, we have reached the optimal solution!

8.6 Degeneracy, Cycling, and Bland's Rule

Definition 8.25: Degeneracy

A basic feasible solution (BFS) in the Simplex Method is said to be **degenerate** if one or more of its basic variables take a value of zero. Mathematically, a BFS corresponding to a basis B is degenerate if there exists at least one basic variable $\mathbf{x}_B$ such that $\mathbf{x}_B = 0$.

Remark 8.26: Degenerate Dictionary

The dictionary below represents a feasible solution, but one of the basic variables is equal to zero. This situation is called **degeneracy**.

What is degeneracy? A dictionary is called degenerate if **any basic variable is equal to zero**. This is important because:

- It can cause the simplex method to perform a pivot without changing the objective value.
- It can potentially lead to **cycling** — revisiting the same basic solution more than once.

In this dictionary:

- The basic variables are x_3, x_4, x_5, x_6 .
- When we set the nonbasic variables $x_1 = 0, x_2 = 0$, we find:

$$x_3 = 4, \quad x_4 = 2, \quad x_5 = 3, \quad x_6 = \boxed{0}, \quad z = 0.$$

- Since $x_6 = 0$ and it is in the basis, this is a **degenerate basic feasible solution**.

Maximize	$z = 0$	$+x_1 + 2x_2$	 Non-basic: $x_1 = x_2 = 0$
subject to	$x_3 = 4$	$-x_2$,	
Basic var: $x_3 = 4$ ←	$x_4 = 2$	$-x_1 + x_2$,	
Basic var: $x_4 = 2$ ←	$x_5 = 3$	$-x_1$,	
Basic var: $x_5 = 3$ ←	$x_6 = 0$	$+2x_1 - x_2$,	
Degenerate: $x_6 = 0$ ←	$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.$		

In this case, the basic feasible solution is (basic variables) $(x_3, x_4, x_5, x_6) = (4, 2, 3, 0)$ and (non-basic variables) $(x_1, x_2) = (0, 0)$.

8.6.1. Degeneracy and Unchanged BFS after Pivoting

Degeneracy can lead to cases where pivoting in the Simplex Method does not change the basic feasible solution. When a pivot is performed, the entering variable is increased until one of the current basic variables reaches zero, at which point that variable leaves the basis. However, if multiple constraints yield the same minimum ratio in the ratio test, then a tie-breaking rule must be applied. If the variable that leaves the basis was already zero, the BFS remains unchanged.

Notice in the prior example, we can try to let x_2 enter the basis. What happens? By the ratio test, we see that x_6 leaves the basis. This results in the following new dictionary:

Remark 8.27: Degenerate Pivot — No Change in BFS

After pivoting x_2 into the basis and x_6 out, we arrive at the dictionary below.

Let's examine what happened:

- Before the pivot, the basic variables were x_3, x_4, x_5, x_6 , and the nonbasic variables were $x_1 = 0, x_2 = 0$.
- After the pivot, x_2 enters and x_6 leaves. The new basic variables are x_3, x_4, x_5, x_2 .
- However, when we plug in the nonbasic values $x_1 = 0, x_6 = 0$, we get:

$$x_3 = 4, \quad x_4 = 2, \quad x_5 = 3, \quad x_2 = 0, \quad z = 0.$$

- This is **identical to the previous basic feasible solution**.

Conclusion: This is a case of a **degenerate pivot**: we performed a pivot step, but the resulting basic feasible solution did not change. The simplex algorithm moved to a new dictionary (i.e., a new basis), but the point in solution space stayed the same.

Objective: $z = 0$

Maximize $z = 0 + 5x_1 - 2x_6$

subject to

Basic var: $x_3 = 4$

Basic var: $x_4 = 2$

Basic var: $x_5 = 3$

New basic: $x_2 = 0$ (degenerate)

Non-basic: $x_1 = 0, x_6 = 0$

$$x_3 = 4 - 2x_1 + x_6,$$

$$x_4 = 2 + x_1 - x_6,$$

$$x_5 = 3 - x_1,$$

$$x_2 = 0 + 2x_1 - x_6,$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.$$

But! The basic feasible solution is the same!!!

Fortunately, the next pivot leads to a new solution.

However, this phenomenon can cause the algorithm to visit the same BFS multiple times, potentially leading to cycling.

See https://gilp.henryrobbins.com/en/latest/examples/3d/SQUARE_PYRAMID_3D_LP.html for a highly degenerate case where the first two pivots do not change the solution.

8.7 Cycling in the Simplex Method

In the Simplex Method, a pivot operation normally improves the objective function by moving from one basic feasible solution (BFS) to another. However, in rare cases, the method can revisit the *same BFS multiple times* without progress. This phenomenon is called **cycling**.

Cycling occurs due to:

- **Degeneracy**: when one or more basic variables are zero.
- **Unlucky pivot rules**: which cause the algorithm to rotate through different bases that correspond to the same BFS.

The following example demonstrates cycling. All pivots are chosen using the **standard minimum ratio rule** with **lexicographic tie-breaking**, which fails to prevent cycling here.

8.7.1. Cycling in the Simplex Method

Cycling occurs when the Simplex Method revisits the same set of basic variables repeatedly without making progress in improving the objective function. This happens when degeneracy is present and the choice of entering and leaving variables results in looping between a set of degenerate BFSs.

The following example demonstrates cycling in the Simplex Method.

Initial Dictionary

$$\begin{aligned}
 \text{Maximize } z &= 0 + 0.75x_1 - 20x_2 + 0.50x_3 - 6x_4 \\
 \text{subject to } s_1 &= 0 - 0.25x_1 + 8x_2 + x_3 - 9x_4, \\
 s_2 &= 0 - 0.50x_1 + 12x_2 + 0.50x_3 - 3x_4, \\
 s_3 &= 1 - x_3, \\
 x_1, x_2, x_3, x_4, s_1, s_2, s_3 &\geq 0.
 \end{aligned}$$

Pivot: x_1 enters, s_1 leaves

$$\begin{aligned}
 \text{Maximize } z &= 0 + 4x_2 + 3.50x_3 - 33x_4 - 3s_1 \\
 \text{subject to } x_1 &= 0 + 32x_2 + 4x_3 - 36x_4 - 4s_1, \\
 s_2 &= 0 - 4x_2 - 1.50x_3 + 15x_4 + 2s_1, \\
 s_3 &= 1 - x_3, \\
 x_1, x_2, x_3, x_4, s_1, s_2, s_3 &\geq 0.
 \end{aligned}$$

Pivot: x_2 enters, s_2 leaves

$$\begin{aligned}
\text{Maximize } & z = 0 + 2x_3 - 18x_4 - s_1 - s_2 \\
\text{subject to } & x_1 = 0 - 8x_3 + 84x_4 + 12s_1 - 8s_2, \\
& x_2 = 0 - 0.38x_3 + 3.75x_4 + 0.50s_1 - 0.25s_2, \\
& s_3 = 1 - x_3, \\
& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

Pivot: x_3 enters, x_1 leaves

$$\begin{aligned}
\text{Maximize } & z = 0 - 0.25x_1 + 3x_4 + 2s_1 - 3s_2 \\
\text{subject to } & x_3 = 0 - 0.12x_1 + 10.50x_4 + 1.50s_1 - s_2, \\
& x_2 = 0 + 0.05x_1 - 0.19x_4 - 0.06s_1 + 0.12s_2, \\
& s_3 = 1 + 0.12x_1 - 10.50x_4 - 1.50s_1 + s_2, \\
& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

Pivot: x_4 enters, x_2 leaves

$$\begin{aligned}
\text{Maximize } & z = +0.50x_1 - 16x_2 + s_1 - s_2 \\
\text{subject to } & x_3 = 0 + 2.50x_1 - 56x_2 - 2s_1 + 6s_2, \\
& x_4 = 0 + 0.25x_1 - 5.33x_2 - 0.33s_1 + 0.67s_2, \\
& s_3 = 1 - 2.50x_1 + 56x_2 + 2s_1 - 6s_2, \\
& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

Pivot: s_1 enters, s_3 leaves

$$\begin{aligned}
\text{Maximize } & z = +1.75x_1 - 44x_2 - 0.50x_3 + 2s_2 \\
\text{subject to } & s_1 = 0 + 1.25x_1 - 28x_2 - 0.50x_3 + 3s_2, \\
& x_4 = 0 - 0.17x_1 + 4x_2 + 0.17x_3 - 0.33s_2, \\
& s_3 = 1 - x_3, \\
& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

Pivot: s_2 enters, x_4 leaves

$$\begin{aligned}
\text{Maximize } & z = +0.75x_1 - 20x_2 + 0.50x_3 - 6x_4 \\
\text{subject to } & s_1 = 0 - 0.25x_1 + 8x_2 + x_3 - 9x_4, \\
& s_2 = 0 - 0.50x_1 + 12x_2 + 0.50x_3 - 3x_4, \\
& s_3 = 1 - x_3, \\
& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

8.7.2. Bland's Rule

Bland's Rule is a tie-breaking strategy used to prevent cycling in the Simplex Method. The rule ensures that:

1. The variable with the **smallest index** enters the basis when multiple choices exist.
2. The variable with the **smallest index** leaves the basis when multiple choices exist.

By enforcing this lexicographic ordering, the method guarantees progress and prevents the algorithm from revisiting the same basis indefinitely.

Theorem 8.28: Bland's Rule Prevents Cycling

If Bland's Rule is used to select the entering and leaving variables in the Simplex Method, then the algorithm will terminate in a finite number of steps, either by reaching an optimal solution or by detecting an unbounded problem.

Proof. The key idea behind the proof is that Bland's Rule ensures that the sequence of bases never repeats. By always choosing the smallest indexed variable for entry and exit, the algorithm guarantees a strictly lexicographic improvement in the sequence of dictionaries. Since there are a finite number of possible bases, the algorithm must eventually terminate.

If cycling were to occur, the method would need to revisit a previously seen basis. However, because Bland's Rule enforces a strict ordering, this would contradict the assumption that lexicographic improvement always occurs. Therefore, cycling is impossible under Bland's Rule. ♠

From the example above, following the same sequence, we arrive at this dictionary.

$$\begin{aligned}
 &\text{Maximize} && z = +0.50x_1 - 16x_2 + s_1 - s_2 \\
 &\text{subject to} && x_3 = 0 + 2.50x_1 - 56x_2 - 2s_1 + 6s_2, \\
 &&& x_4 = 0 + 0.25x_1 - 5.33x_2 - 0.33s_1 + 0.67s_2, \\
 &&& s_3 = 1 - 2.50x_1 + 56x_2 + 2s_1 - 6s_2, \\
 &&& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
 \end{aligned}$$

Now with Bland's rule, we pivot x_1 into the basis and s_3 leaves.

$$\begin{aligned}
 &\text{Maximize} && z = 0.20 - 4.80x_2 + 1.40s_1 - 2.20s_2 - 0.20s_3 \\
 &\text{subject to} && x_3 = 1 - s_3, \\
 &&& x_4 = 0.10 + 0.27x_2 - 0.13s_1 + 0.07s_2 - 0.10s_3, \\
 &&& x_1 = 0.40 + 22.40x_2 + 0.80s_1 - 2.40s_2 - 0.40s_3, \\
 &&& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
 \end{aligned}$$

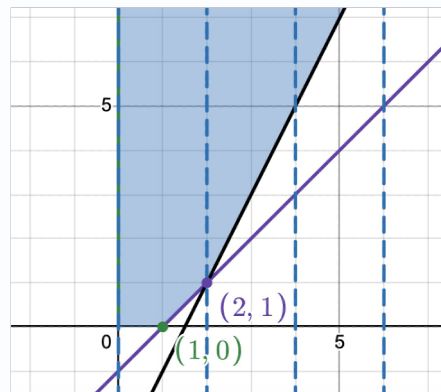
$$\begin{aligned}
&\text{Maximize} && 0.20 - 4.80x_2 + 1.40s_1 - 2.20s_2 - 0.20s_3 \\
&\text{subject to} && x_3 = 1 - s_3, \\
&&& x_4 = 0.10 + 0.27x_2 - 0.13s_1 + 0.07s_2 - 0.10s_3, \\
&&& x_1 = 0.40 + 22.40x_2 + 0.80s_1 - 2.40s_2 - 0.40s_3, \\
&&& x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0.
\end{aligned}$$

8.8 Simplex and Unbounded LPs

Example 8.29: Unbounded Linear Program

Consider the linear program:

$$\begin{aligned}
&\text{maximize} && x_1 \\
&\text{subject to} && x_1 - x_2 \leq 1 \\
&&& 2x_1 - x_2 \leq 3 \\
&&& x_1, x_2 \geq 0
\end{aligned}$$



We convert the inequalities into equalities using slack variables s_1 and s_2 :

$$\begin{aligned}
s_1 &= 1 - x_1 + x_2 \\
s_2 &= 3 - 2x_1 + x_2 \\
x_1, x_2, s_1, s_2 &\geq 0
\end{aligned}$$

Initial dictionary (dictionary form):

$$\text{Maximize } x_1 \text{ subject to } \begin{cases} s_1 = 1 - x_1 + x_2 \\ s_2 = 3 - 2x_1 + x_2 \end{cases}$$

Pivot Step 1: Enter x_1 , leave s_2 (smallest ratio rule: $\frac{3}{2} < \frac{1}{1}$).

Solving $s_2 = 3 - 2x_1 + x_2 \Rightarrow x_1 = \frac{3+x_2-s_2}{2}$, and substitute into the objective and other constraint.

$$\text{Maximize } 1 + x_2 - s_1 \text{ subject to } \begin{cases} x_1 = 1 + x_2 - s_1 \\ s_2 = 1 - x_2 + 2s_1 \end{cases}$$

Pivot Step 2: Enter s_1 , but observe: the variable s_1 appears in both equations with positive coefficients and no upper bound is imposed. That is, increasing s_1 increases the objective indefinitely:

$$\text{Maximize } 1 + x_2 - s_1 \Rightarrow \text{Maximize } 2 + s_1 - s_2$$

$$\text{subject to } \begin{cases} x_1 = 2 + s_1 - s_2 \\ x_2 = 1 + 2s_1 - s_2 \end{cases}$$

Since increasing s_1 increases the objective and does not violate any constraint (no upper bound in any equation), the linear program is **unbounded**.

Conclusion: This linear program is unbounded. Once s_1 enters the basis, there is no constraint limiting how large it can grow while maintaining feasibility. Therefore, the objective function can increase without bound.

Pioneer Spotlight:

George Dantzig

George Dantzig, known for spearheading linear programming with the simplex algorithm, was born in Portland, Oregon on November 8, 1914 [2, pp. 100–106]. George received his Bachelor of Arts in Mathematics and Physics from the University of Maryland, College Park. Gass [2] explained that after George married his wife, Anne Shmuner, he completed his Master of Arts in Mathematics at the University of Michigan in 1938. Before graduating with his Master’s degree, he began working at the United States Bureau of Labor Statistics as a junior statistician. Gass [2] described how George took interest in statistics through the work of Jerzy Neyman, which led him to get a Ph.D. in Statistics with Neyman’s guidance from the University of California, Berkeley in 1946. Before his dissertation, he began working with the Army Air Force Combat Analysis Branch of Statistical Control at the Pentagon, then transferred to the Department of the Air Force as a mathematical advisor. After his work at the Pentagon, he became a professor in operations research at the University of California, Berkeley, and Stanford. He went on to receive the National Medal of Science in 1975. He was also a member of the National Academy of Engineering and the National Academy of Sciences and the American Association of Arts and Sciences [3].

While in school at the University of California, Berkeley, George became widely known for solving two of the “unsolved” statistics problems, which he mistakenly thought were homework [3]. In 1947, Dantzig proposed the simplex algorithm, which started the field of linear programming [1]. Geometrically, the simplex method consists of traveling along the edges of a polyhedron toward its vertices. The optimal solution will be at a vertex if the feasible set is convex. The constraints should be in the following standard form:

$$\mathbf{Ax}_n = \mathbf{b}, \quad x_n = 1, \quad x_j \geq 0 \quad \text{for } j = 1, \dots, n.$$

The corresponding vertices of the feasible region are the basic feasible solutions. If a system is nondegenerate, the process of simplex pivoting can occur, where the previous solution is used to pivot towards a better feasible basis.

After Dantzig's development of the simplex algorithm, he presented a talk geared toward the application of his linear programming model in the digital world, including games and computers. Cottle, Johnson, and Wets [1] described how his algorithm was implemented in the animal feed industry and computing machines for the Air Force. It was also used to solve Stigler's "diet problem," a simple question with modern technology, but a long computation process in 1947. Dantzig's procedure has been utilized in countless industries, such as communications, railroads, petroleum, airlines, and the economy. His utilization of linear algebra theory and the creation of pivoting will continue to be used when solving complex linear problems.

According to Cottle, Johnson, and Wets [1], Dantzig also studied network flow problems, which led to the duality theorem in the 1950s. The dual theorem utilizes the costs as integers to create a pair that highlights integer basic solutions, which was used in the application of the "fleet assignment model." With a great understanding of real-world problems, Dantzig developed the recourse stochastic program, which accounted for uncertain constraints and utilized statistical tests to determine the likelihood of an optimal solution. Dantzig was responsible for the solution to countless military problems through his development of linear programming in the late-1940s.

This spotlight was contributed by Irma Adams

References

- [1] R. Cottle, E. Johnson, and R. Wets. *George B. Dantzig (1914-2005)*. Mar. 2007. URL: <https://www.ams.org/journals/notices/200703/fea-cottle.pdf?adat=March%202007&trk=200703fea-cottle&cat=feature&galt=feature> (visited on 04/12/2025) (cit. on pp. 167, 168).
- [2] S. I. Gass. *Memorial Tributes*. Vol. 12. National Academies Press, Oct. 2008, pp. 100–106. DOI: 10.17226/12473 (cit. on p. 167).
- [3] UC Berkeley Engineering. *George Dantzig: Operations research phenom*. Berkeley Engineering, Sept. 2018. URL: <https://engineering.berkeley.edu/george-dantzig-operations-research-phenom/> (visited on 03/13/2025) (cit. on p. 167).

8.9 Exercises

8.9.1. Exercises: Converting to Standard Form

Exercise 8.1 Convert the following linear program to standard form:

$$\begin{array}{ll}\max & 3x_1 - 5x_2 \\ \text{subject to} & 2x_1 + x_2 \leq 6, \\ & -x_1 + 3x_2 \geq 2, \\ & x_1, x_2 \geq 0.\end{array}$$

Exercise 8.2 Convert the following linear program to standard form:

$$\begin{array}{ll}\min & -x_1 + 2x_2 + x_3 \\ \text{subject to} & x_1 - x_2 + 2x_3 = 4, \\ & -2x_1 + x_2 \leq 1, \\ & x_3 \text{ is unrestricted, } x_1, x_2 \geq 0.\end{array}$$

Exercise 8.3 Convert the following linear program to standard form:

$$\begin{array}{ll}\max & 5x_1 + x_2 - 3x_3 \\ \text{subject to} & x_1 + 2x_2 - x_3 \geq 7, \\ & x_2 - 4x_3 = 3, \\ & x_1 \text{ is unrestricted, } x_2, x_3 \leq 0.\end{array}$$

Exercise 8.4 Convert the following linear program to standard form:

$$\begin{array}{ll}\min & 2x_1 + 4x_2 \\ \text{subject to} & x_1 + x_2 \geq 3, \\ & 2x_1 - x_2 \leq 5, \\ & x_1 \text{ is unrestricted, } x_2 \geq 0.\end{array}$$

Exercise 8.5 Convert the following linear program to standard form:

$$\begin{array}{ll}\min & 5y_1 - 3y_2 + y_3 \\ \text{subject to} & 2y_1 - y_2 + 3y_3 = 7, \\ & -y_1 + 4y_2 \leq 5, \\ & y_3 \text{ is unrestricted.}\end{array}$$

Selected Solutions**Solution.** (Exercise 8.4)

1. Convert the minimization problem to a maximization problem by negating the objective function:

$$\max \quad -2x_1 - 4x_2.$$

2. Convert the $\geq$ constraint into an equality by introducing a surplus variable $s_1 \geq 0$:

$$x_1 + x_2 - s_1 = 3.$$

3. Convert the $\leq$ constraint into an equality by introducing a slack variable $s_2 \geq 0$:

$$2x_1 - x_2 + s_2 = 5.$$

4. Since x_1 is unrestricted, replace it with $x_1 = x'_1 - x''_1$, where $x'_1, x''_1 \geq 0$.

Final Standard Form:

$$\begin{aligned} \max \quad & -2(x'_1 - x''_1) - 4x_2 \\ \text{subject to} \quad & (x'_1 - x''_1) + x_2 - s_1 = 3, \\ & 2(x'_1 - x''_1) - x_2 + s_2 = 5, \\ & x'_1, x''_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

**Solution.** (Exercise 8.5)

1. Convert the minimization problem to a maximization problem by negating the objective function:

$$\max \quad -5y_1 + 3y_2 - y_3.$$

2. Convert the $\leq$ constraint into an equality by introducing a slack variable $s_1 \geq 0$:

$$-y_1 + 4y_2 + s_1 = 5.$$

3. Since y_3 is unrestricted, replace it with $y_3 = y'_3 - y''_3$, where $y'_3, y''_3 \geq 0$.

Final Standard Form:

$$\begin{aligned} \max \quad & -5y_1 + 3y_2 - (y'_3 - y''_3) \\ \text{subject to} \quad & 2y_1 - y_2 + 3(y'_3 - y''_3) = 7, \\ & -y_1 + 4y_2 + s_1 = 5, \\ & y_1, y_2, y'_3, y''_3, s_1 \geq 0. \end{aligned}$$



8.9.2. Simplex Method

Exercise 8.6 [Pivoting in a different direction] We consider the example from Section 8.3. In that example, we pivoted first on the y variable. This choice of pivot called the steepest ascent rule since we chose the variable in the objective with the largest coefficient. However, we can consider pivoting on any variable with a positive coefficient in the objective. We will now pivot first on the x variable.

Starting Point

$$\begin{array}{ll}\text{Maximize} & 2x + 3y \\ \text{subject to} & s_1 = 9 - x - y, \\ & s_2 = 16 - 2x - y, \\ & s_3 = 14 - x - 2y, \\ & x, y, s_1, s_2, s_3 \geq 0.\end{array}$$

1. Let x increase first.
2. Determine which variable leaves the basis using the ratio test.
3. Pivot by rewriting the equation of the leaving variable in terms of x and substitute it into the other equations and the objective.
4. Write down the new dictionary.
5. Decide whether the resulting dictionary is optimal. If not, repeat the process.

Exercise 8.7 Solve the following linear programming problem using the simplex method:

$$\begin{array}{ll}\text{Maximize:} & z = 5x_1 + 3x_2 \\ \text{subject to:} & x_1 + x_2 \leq 12, \\ & 2x_1 + x_2 \leq 16, \\ & x_1, x_2 \geq 0.\end{array}$$

Exercise 8.8 Solve the following linear programming problem using the simplex method:

$$\begin{array}{ll}\text{Maximize:} & z = 5x_1 + 8x_2 \\ \text{subject to:} & x_1 + 2x_2 \leq 30, \\ & 3x_1 + x_2 \leq 30, \\ & x_1, x_2 \geq 0.\end{array}$$

Exercise 8.9 Solve the following linear programming problem using the simplex method:

$$\begin{array}{ll}\text{Maximize:} & z = 2x_1 + 3x_2 + x_3 \\ \text{subject to:} & 4x_1 + 2x_2 + 5x_3 \leq 32, \\ & 2x_1 + 4x_2 + 3x_3 \leq 28, \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Exercise 8.10 Solve the following linear programming problem using the simplex method:

$$\begin{aligned} \text{Maximize: } & z = 6x_1 + 8x_2 + 5x_3, \\ \text{subject to: } & 4x_1 + x_2 + x_3 \leq 1800, \\ & 2x_1 + 2x_2 + x_3 \leq 2000, \\ & 4x_1 + 2x_2 + x_3 \leq 3200, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Exercise 8.11 Solve the following linear programming problem using the simplex method:

$$\begin{aligned} \text{Minimize: } & z = 4x_1 + 6x_2 + 7x_3, \\ \text{subject to: } & x_1 + x_2 + 2x_3 \geq 20, \\ & x_1 + 2x_2 + x_3 \geq 30, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Exercise 8.12 Solve the following linear programming problem using the simplex method:

$$\begin{aligned} \text{Minimize: } & z = 40x_1 + 48x_2 + 30x_3, \\ \text{subject to: } & 2x_1 + 2x_2 + x_3 \geq 25, \\ & x_1 + 3x_2 + 2x_3 \geq 30, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Exercise 8.13 Conditions in the dictionary. The following is a dictionary of a maximization problem. State conditions (i.e., the range of values) on $a_1, a_2, b, c_1, c_5, c_6$ that are required to make the following true. Note: This does not require any simplex pivots.

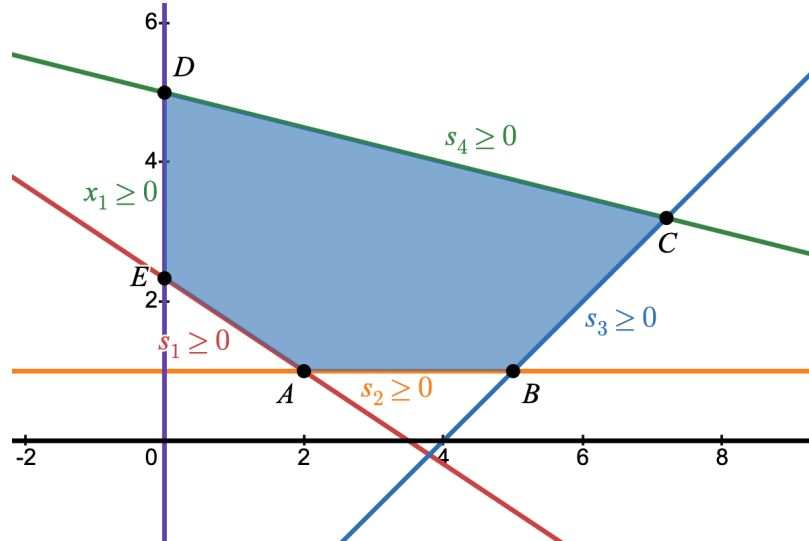
$$\begin{aligned} \max \quad & 0 + c_1x_1 + c_5x_5 + c_6x_6 \\ & x_2 = 1 + x_1 - a_1x_5 \\ & x_4 = 2 + x_1 + 2x_5 - x_6 \\ & x_3 = b - a_2x_1 - x_5 - x_6 \end{aligned}$$

1. The current solution is feasible and optimal.
2. The current basic solution is not feasible.
3. The current basic solution is degenerate.
4. The current basic solution is feasible, and the LP is unbounded.

Exercise 8.14

Consider the following linear programming feasible region in 2 variables that is depicted in the plot below.

Here is a graph of the feasible region:



The variables in the augmented linear program are $\{x_1, x_2, s_1, s_2, s_3, s_4\}$. In the context of the simplex method, answer the following questions:

(a) At the basic feasible solution D , which variables are non-basic and which variables are basic?

$$B = \{ \quad \quad \quad \}, \quad N = \{ \quad \quad \quad \}.$$

(b) If we pivot from the basic feasible solution at D to the basic feasible solution at C ,

- Which variable enters the basis?
- Which variable leaves the basis?

Solutions to selected exercises

Solution. [Solution for Exercise 8.6] We will have to do 3 pivots!

We begin with the original linear program:

$$\begin{aligned} &\text{Maximize} && 2x + 3y \\ &\text{subject to} && s_1 = 9 - x - y, \\ & && s_2 = 16 - 2x - y, \\ & && s_3 = 14 - x - 2y, \\ & && x, y, s_1, s_2, s_3 \geq 0. \end{aligned}$$

First Pivot: x Enters the Basis

Let x increase first. Applying the ratio test shows that a certain slack variable (here, for example, the calculations lead us to pivot so that the new dictionary is obtained) leaves the basis. After performing the necessary algebraic manipulations (i.e., rewriting the leaving variable's equation in terms of x and substituting into the other equations), we obtain:

$$\begin{array}{ll}\text{Maximize} & 16 + 2y - s_2 \\ \text{subject to} & s_1 = 1 - 0.50y + 0.50s_2, \\ & x = 8 - 0.50y - 0.50s_2, \\ & s_3 = 6 - 1.50y + 0.50s_2, \\ & x, y, s_1, s_2, s_3 \geq 0.\end{array}$$

Second Pivot: y Enters the Basis

Next, we pivot again. In this pivot, the calculations lead us to select y (or, equivalently, pivot so that the basic variable changes as indicated by the ratio test). After rewriting the appropriate equation in terms of the entering variable and substituting into the remaining equations and the objective function, the dictionary becomes:

$$\begin{array}{ll}\text{Maximize} & 20 - 4s_1 + s_2 \\ \text{subject to} & y = 2 - 2s_1 + s_2, \\ & x = 7 + s_1 - s_2, \\ & s_3 = 3 + 3s_1 - s_2, \\ & x, y, s_1, s_2, s_3 \geq 0.\end{array}$$

Third Pivot: s_2 Enters the Basis

The new dictionary is

$$\begin{array}{ll}\text{Maximize} & 23 - s_1 - s_3 \\ \text{subject to} & x = 4 - 2s_1 + s_3, \\ & s_2 = 3 + 3s_1 - s_3, \\ & y = 5 + s_1 - s_3, \\ & x, y, s_1, s_2, s_3 \geq 0.\end{array}$$

This is optimal!



9. Simplex - Matrix Calculations

9.1 Solutions to $A\mathbf{x} = \mathbf{b}$

Outcomes

- Explore how a system of equations can be rewritten via partitioning the variables
- See how we can also partition the matrix and rewrite the equation in a new format that solves for basic variables
- Familiarize with matrix notation
- Define terminology of a basic solution and a basic feasible solution

We will now explore ways to represent solutions to this linear system of equations. This is meant to give the theoretical backbone of the simplex algorithm.

We consider a linear system

$$A\mathbf{x} = \mathbf{b},$$

where $A \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{b} \in \mathbb{R}^m$. Explicitly,

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}.$$

Assumptions on A

We assume:

- $\text{rank}(A) = m$, i.e., the rows of A are linearly independent (full row rank);
- $m \leq n$, so the system is not overdetermined.

If A lacked full row rank, some equations would be redundant and removable without changing the solution set. When $m > n$, the system is typically overdetermined and inconsistent; if consistent, it can be reduced to $m \leq n$ via row operations.

Under these assumptions, $A\mathbf{x} = \mathbf{b}$ represents a system of m independent equations in n unknowns with a feasible (possibly infinite) solution space.

9.1.0.1. Example Problem - Small Bakery

A small bakery produces two products: *loaves of bread* (denoted by x) and *cakes* (denoted by y). Each loaf of bread contributes \$2 in profit, while each cake contributes \$3 in profit. The bakery must operate under the following daily constraints:

- **Oven Time:** The total number of bread loaves and cakes baked each day cannot exceed 9, due to limited oven time. This gives the constraint:

$$x + y \leq 9.$$

- **Flour Supply:** Bread uses more flour than cake. Each loaf uses 2 units of flour and each cake uses 1 unit. The bakery has a maximum of 16 units of flour available each day, leading to:

$$2x + y \leq 16.$$

- **Sugar Supply:** Cakes use more sugar than bread. Each loaf uses 1 unit of sugar and each cake uses 2 units. The daily sugar supply is limited to 14 units, so:

$$x + 2y \leq 14.$$

- **Non-negativity:** The bakery cannot produce a negative number of items, so:

$$x \geq 0, \quad y \geq 0.$$

Mathematical Formulation

The bakery wants to determine how many loaves of bread (x) and cakes (y) to produce each day in order to maximize total profit. The resulting linear program is:

Original Linear Program

$$\begin{array}{ll} \text{Maximize} & 2x + 3y \\ \text{subject to} & x + y \leq 9 \quad (\text{hours}) \\ & 2x + y \leq 16 \quad (\text{flour}) \\ & x + 2y \leq 14 \quad (\text{sugar}) \\ & x \geq 0, y \geq 0. \end{array}$$

We begin by adding *slack variables* s_1, s_2, s_3 to convert the inequalities to equalities:

Standard Form

$$\begin{array}{ll}
\text{Maximize} & 2x + 3y \\
\text{subject to} & x + y + s_1 = 9 \quad (\text{hours}) \\
& 2x + y + s_2 = 16 \quad (\text{flour}) \\
& x + 2y + s_3 = 14 \quad (\text{sugar}) \\
& x, y, s_1, s_2, s_3 \geq 0
\end{array}$$

It will be useful to also understand standard form from the matrix perspective.

Matrix-Vector Standard Form

In matrix-vector notation, define:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ s_1 \\ s_2 \\ s_3 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 9 \\ 16 \\ 14 \end{bmatrix}$$

The standard form linear program is then:

$$\begin{array}{ll}
\text{Maximize} & \mathbf{c}^\top \mathbf{x} \\
\text{subject to} & A\mathbf{x} = \mathbf{b}, \\
& \mathbf{x} \geq 0
\end{array}$$

Submatrix Notation

We extract submatrices of A by selecting columns corresponding to specific variable sets. For a set I , we let A_I denote the submatrix of A corresponding to the columns in I .

$$\begin{array}{ccccc}
& x & y & s_1 & s_2 & s_3 \\
\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix} & \Rightarrow & A_{\{x\}} = & \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}
\end{array}$$

$$\begin{array}{ccccc}
& x & y & s_1 & s_2 & s_3 \\
\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix} & \Rightarrow & A_{\{y, s_2\}} = & \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 2 & 0 \end{bmatrix}
\end{array}$$

$$\begin{array}{ccccc} x & y & s_1 & s_2 & s_3 \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix} & \Rightarrow & A_{\{s_1, s_2, s_3\}} = & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 1x + 1y + 1s_1 \\ 2x + 1y + 1s_2 \\ 1x + 2y + 1s_3 \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}}_{A_{\{x,y\}}(x,y)} + \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}}_{A_{\{s_1, s_2, s_3\}}(s_1, s_2, s_3)} \end{aligned}$$

$$A\mathbf{x} = \underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ s_2 \end{bmatrix}}_{A_{\{x,y,s_2\}}(x,y,s_2)} + \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s_1 \\ s_3 \end{bmatrix}}_{A_{\{s_1,s_3\}}(s_1,s_3)} = \begin{bmatrix} 9 \\ 16 \\ 14 \end{bmatrix}$$

This implies:

$$A_{\{x,y,s_2\}} \begin{bmatrix} x \\ y \\ s_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 16 \\ 14 \end{bmatrix} - A_{\{s_1,s_3\}} \begin{bmatrix} s_1 \\ s_3 \end{bmatrix}$$

So,

$$\begin{bmatrix} x \\ y \\ s_2 \end{bmatrix} = A_{\{x,y,s_2\}}^{-1} \left(\begin{bmatrix} 9 \\ 16 \\ 14 \end{bmatrix} - A_{\{s_1,s_3\}} \begin{bmatrix} s_1 \\ s_3 \end{bmatrix} \right)$$

9.1.1. Partitioning the Variables

In solving the system $A\mathbf{x} = \mathbf{b}$ in the context of linear programming, we introduce the concept of **partitioning the variables**. This approach is essential in the Simplex Method for identifying and analyzing basic solutions.

Definition 9.1: Basis

Given a matrix A of full row rank m , a basis B is an index set of m of the n columns such that the corresponding m of the matrix A are linearly independent. We then form the submatrix A_B that contains only those columns of A .

The remaining $n - m$ columns are referred to as the nonbasic columns, which form the matrix A_N .

Given a basis B and the nonbasic set N , this partitions the variable vector $\mathbf{x}$ into two sets:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{bmatrix},$$

where:

- $\mathbf{x}_B$ consists of the **basic variables**, associated with the basis matrix A_B ,
- $\mathbf{x}_N$ consists of the **nonbasic variables**, associated with the nonbasic matrix A_N .

Rewriting the system $A\mathbf{x} = \mathbf{b}$ in terms of these partitions gives:

$$A_B\mathbf{x}_B + A_N\mathbf{x}_N = \mathbf{b}. \quad (9.1)$$

If the basis matrix A_B is invertible, we can express $\mathbf{x}_B$ in terms of $\mathbf{x}_N$:

$$\mathbf{x}_B = A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N. \quad (9.2)$$

Definition 9.2: Basic solution and Basic Feasible Solution (BFS)

*A **basic solution** is obtained by setting $\mathbf{x}_N = 0$, yielding:*

$$\mathbf{x}_B = A_B^{-1}\mathbf{b}, \quad \mathbf{x}_N = 0. \quad (9.3)$$

*A **basic feasible solution** (BFS) is a basic solution that satisfies the nonnegativity constraint $\mathbf{x}_B \geq 0$, ensuring that all basic variables are nonnegative.*

Example 1: A 3-variable, 2-equation System

Consider the following system:

$$\begin{aligned}x_1 + 2x_2 + x_3 &= 4, \\3x_1 + x_2 + 2x_3 &= 5.\end{aligned}$$

Here, the matrix A and the variable vector x are:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Selecting $B = \{1, 2\}$ as the basis, the corresponding submatrix is:

$$A_B = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad A_N = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

If A_B is invertible, we compute:

$$A_B^{-1} = \begin{bmatrix} -1/5 & 2/5 \\ 3/5 & -1/5 \end{bmatrix}.$$

Setting $x_3 = 0$ (i.e., $\mathbf{x}_N = 0$), we solve for $\mathbf{x}_B$:

$$x_B = A_B^{-1} \mathbf{b} = \begin{bmatrix} -1/5 & 2/5 \\ 3/5 & -1/5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Since $\mathbf{x}_B \geq 0$, this is a **basic feasible solution**.

Example 2: A System with Slack Variables

Consider a linear program in inequality form:

$$\begin{aligned}x_1 + 3x_2 &\leq 6, \\2x_1 + x_2 &\leq 4.\end{aligned}$$

We introduce slack variables $s_1, s_2 \geq 0$ to convert inequalities into equalities:

$$\begin{aligned}x_1 + 3x_2 + s_1 &= 6, \\2x_1 + x_2 + s_2 &= 4.\end{aligned}$$

The augmented system has:

$$A = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

Choosing the slack variables as the basis ($B = \{s_1, s_2\}$), the basis matrix is:

$$A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_N = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}.$$

Since A_B is the identity matrix, its inverse is itself, and we obtain:

$$\mathbf{x}_B = A_B^{-1}\mathbf{b} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}, \quad \mathbf{x}_N = 0.$$

Since $\mathbf{x}_B \geq 0$, this is a **basic feasible solution**.

9.1.2. Another Example

To convert the given problem into standard form, we introduce **slack variables** s_1, s_2 to convert the inequalities into equalities.

$$\begin{aligned} \max \quad & 2x_1 + 5x_2 \\ \text{subject to} \quad & x_1 + 2x_2 + s_1 = 16, \\ & 5x_1 + 3x_2 + s_2 = 45, \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Now the system is in standard form:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 16 \\ 45 \end{bmatrix}, \quad x_1, x_2, s_1, s_2 \geq 0.$$

9.1.3. Selecting and Evaluating Different Bases

A basis consists of any two linearly independent columns from A . We select three different bases:

- Basis 1: $B = \{x_1, x_2\}$ (Feasible) The basis matrix is:

$$A_B = \begin{bmatrix} 1 & 2 \\ 5 & 3 \end{bmatrix}.$$

Computing A_B^{-1} :

$$A_B^{-1} = \frac{1}{(1)(3) - (2)(5)} \begin{bmatrix} 3 & -2 \\ -5 & 1 \end{bmatrix} = \frac{1}{3-10} \begin{bmatrix} 3 & -2 \\ -5 & 1 \end{bmatrix} = \frac{1}{-7} \begin{bmatrix} 3 & -2 \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} -3/7 & 2/7 \\ 5/7 & -1/7 \end{bmatrix}.$$

Computing the basic solution:

$$\mathbf{x}_B = A_B^{-1}\mathbf{b} = \begin{bmatrix} -3/7 & 2/7 \\ 5/7 & -1/7 \end{bmatrix} \begin{bmatrix} 16 \\ 45 \end{bmatrix} = \begin{bmatrix} (-3/7)(16) + (2/7)(45) \\ (5/7)(16) + (-1/7)(45) \end{bmatrix} = \begin{bmatrix} -48/7 + 90/7 \\ 80/7 - 45/7 \end{bmatrix} = \begin{bmatrix} 42/7 \\ 35/7 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}.$$

Since $\mathbf{x}_B = (6, 5)$ contains no negative values, this is actually a **feasible solution**, contradicting our initial assumption that it would be infeasible!

- Basis 2: $B = \{x_1, s_1\}$ (Infeasible) The basis matrix is:

$$A_B = \begin{bmatrix} 1 & 1 \\ 5 & 0 \end{bmatrix}.$$

Computing A_B^{-1} :

$$A_B^{-1} = \frac{1}{(1)(0) - (1)(5)} \begin{bmatrix} 0 & 1 \\ 5 & -1 \end{bmatrix} = \frac{1}{-5} \begin{bmatrix} 0 & 1 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -1/5 \\ -1 & 1/5 \end{bmatrix}.$$

Computing the basic solution:

$$\mathbf{x}_B = A_B^{-1} \mathbf{b} = \begin{bmatrix} 0 & -1/5 \\ -1 & 1/5 \end{bmatrix} \begin{bmatrix} 16 \\ 45 \end{bmatrix} = \begin{bmatrix} (0)(16) + (-1/5)(45) \\ (-1)(16) + (1/5)(45) \end{bmatrix} = \begin{bmatrix} -9 \\ -16 + 9 \end{bmatrix} = \begin{bmatrix} -9 \\ -7 \end{bmatrix}.$$

Since the solution contains negative values, this basis is **infeasible**.

- Basis 3: $B = \{s_1, s_2\}$ (Feasible) The basis matrix is:

$$A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since this is the identity matrix, its inverse is itself. Thus, the basic solution is:

$$\mathbf{x}_B = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = A_B^{-1} \mathbf{b} = \begin{bmatrix} 16 \\ 45 \end{bmatrix}.$$

Since all basic variables are nonnegative, this is a **feasible solution**.

9.2 Optimality Conditions

Outcomes

- Define reduced costs at a basic feasible solution
- Establish optimality conditions via the reduced costs

The objective function of the linear program is given by:

$$z = \mathbf{c}^\top \mathbf{x}.$$

Substituting the partitioned variables $\mathbf{x}_B$ and $\mathbf{x}_N$,

$$z = \mathbf{c}_B^\top \mathbf{x}_B + \mathbf{c}_N^\top \mathbf{x}_N.$$

Expressing $\mathbf{x}_B$ in terms of $\mathbf{x}_N$, using the basic solution $\mathbf{x}_B = A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N$, we rewrite the objective function as:

$$z = \mathbf{c}_B^\top (A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N) + \mathbf{c}_N^\top \mathbf{x}_N. \quad (9.1)$$

Expanding and regrouping terms:

$$z = \mathbf{c}_B^\top A_B^{-1}\mathbf{b} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N)\mathbf{x}_N. \quad (9.2)$$

Definition 9.3: Reduced Costs

The term $\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N$ represents the reduced costs.

This is explained as the objective rewritten from the viewpoint of a basic feasible solution. In particular, these are the coefficients on the non-basic variables.

The system is rewritten as the dictionary form at basis B as in the following definition.

Definition 9.4: Revised Simplex Dictionary at basis B

The revised simplex dictionary at a basis B is

$$\begin{aligned} \max \quad & \mathbf{c}_B^\top A_B^{-1}\mathbf{b} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N)\mathbf{x}_N \\ \text{subject to} \quad & \mathbf{x}_B = A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N. \\ & \mathbf{x} \geq 0 \end{aligned} \quad (9.3)$$

Let's see this in action.

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & s_1 = 23 - 3x - y, \\ & s_2 = 45 - 5x - 3y, \\ & x, y, s_1, s_2 \geq 0. \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & 17 - 0.25s_1 - 0.25s_2 \\ \text{subject to} \quad & x = 6 - 0.75s_1 + 0.25s_2, \\ & y = 5 + 1.25s_1 - 0.75s_2, \\ & x, y, s_1, s_2 \geq 0. \end{aligned}$$

Theorem 9.5: Optimality Condition in Simplex Method

Given a linear program in standard form:

$$\max \mathbf{c}^\top \mathbf{x} \quad \text{subject to} \quad A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0,$$

let $\mathbf{b}$ be a basis corresponding to a basic feasible solution $\mathbf{x}_B = A_B^{-1}\mathbf{b}$, with nonbasic variables $\mathbf{x}_N = 0$.

If all reduced costs satisfy:

$$\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N \leq 0,$$

then the current basic feasible solution is optimal.

Proof. Let $\bar{\mathbf{x}} = (\bar{\mathbf{x}}_B, \bar{\mathbf{x}}_N) = (A_B^{-1}\mathbf{b}, 0)$.

If all reduced costs are nonnegative, i.e.,

$$\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N \leq 0,$$

then for any feasible choice of $\mathbf{x}_N \geq 0$, we have

$$\mathbf{c}^\top \mathbf{x} = \mathbf{c}_B^\top A_B^{-1} \mathbf{b} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N) \mathbf{x}_N \quad (9.4)$$

$$= \mathbf{c}^\top \bar{\mathbf{x}} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N) \mathbf{x}_N \quad (9.5)$$

$$\leq \mathbf{c}^\top \bar{\mathbf{x}}, \quad (9.6)$$

where the last line follows since $(\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N) \leq 0$ and $\mathbf{x}_N \geq 0$.

Thus, $\bar{\mathbf{x}}$ is an optimal solution! 

Example 9.6: Verifying Optimality Using Reduced Costs

We aim to verify that the basic feasible solutions $(6, 5)$ and $(0, 8)$ are optimal for the following linear program:

$$\begin{aligned} \max \quad & 2x_1 + 5x_2 \\ \text{subject to} \quad & x_1 + 2x_2 + s_1 = 16, \\ & 5x_1 + 3x_2 + s_2 = 45, \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Solution. Checking Optimality of $(6, 5)$

- Step 1: Identify the Basis At the point $(x, y) = (6, 5)$, we see that $(s_1, s_2) = 0$. Thus, we can choose $B = \{x_1, x_2\}$ and $N = \{s_1, s_2\}$, so:

$$A_B = \begin{bmatrix} 1 & 2 \\ 5 & 3 \end{bmatrix}, \quad \mathbf{c}_B^\top = [2 \quad 5].$$

The nonbasic variables are s_1, s_2 , so:

$$A_N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{c}_N^\top = \begin{bmatrix} 0 & 0 \end{bmatrix}.$$

- Step 2: Compute A_B^{-1}

$$A_B^{-1} = \frac{1}{(1)(3) - (2)(5)} \begin{bmatrix} 3 & -2 \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} -3/7 & 2/7 \\ 5/7 & -1/7 \end{bmatrix}.$$

Thus, the system can be written as

$$\mathbf{x}_B = A_B^{-1} \mathbf{b} - A_B^{-1} A_N \mathbf{x}_N,$$

that is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix} - \begin{bmatrix} -3/7 & 2/7 \\ 5/7 & -1/7 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}, \quad (9.7)$$

or equivalently

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix} - s_1 \begin{bmatrix} -3/7 \\ 5/7 \end{bmatrix} - s_2 \begin{bmatrix} 2/7 \\ -1/7 \end{bmatrix} \quad (9.8)$$

- Step 3: Compute Reduced Costs

$$A_B^{-1} A_N = \begin{bmatrix} -3/7 & 2/7 \\ 5/7 & -1/7 \end{bmatrix}.$$

$$\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N = \begin{bmatrix} 0 & 0 \end{bmatrix} - \begin{bmatrix} 2 & 5 \end{bmatrix} \begin{bmatrix} -3/7 & 2/7 \\ 5/7 & -1/7 \end{bmatrix}.$$

Computing:

$$[(2)(-3/7) + (5)(5/7) \quad (2)(2/7) + (5)(-1/7)] = [-6/7 + 25/7 \quad 4/7 - 5/7] = [19/7 \quad -1/7].$$

Since one reduced cost is negative, $(6, 5)$ is *not necessarily optimal* unless we reconsider the basis.

Verifying Optimality of $(0, 8)$

- Step 1: Identify the Basis

We choose $B = \{x_2, s_1\}$, so:

$$A_B = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}, \quad \mathbf{c}_B^\top = \begin{bmatrix} 5 & 0 \end{bmatrix}.$$

The nonbasic variables are x_1, s_2 , so:

$$A_N = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}, \quad \mathbf{c}_N^\top = \begin{bmatrix} 2 & 0 \end{bmatrix}.$$

- Step 2: Compute A_B^{-1}

$$A_B^{-1} = \frac{1}{(2)(0) - (1)(3)} \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 0 & -1/3 \\ -1 & 2/3 \end{bmatrix}.$$

- Step 3: Compute Reduced Costs

$$A_B^{-1}A_N = \begin{bmatrix} 0 & -1/3 \\ -1 & 2/3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 0(1) + (-1/3)(5) & 0(0) + (-1/3)(1) \\ -1(1) + (2/3)(5) & -1(0) + (2/3)(1) \end{bmatrix} = \begin{bmatrix} -5/3 & -1/3 \\ 7/3 & 2/3 \end{bmatrix}.$$

$$\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N = [2 \ 0] - [5 \ 0] \begin{bmatrix} -5/3 & -1/3 \\ 7/3 & 2/3 \end{bmatrix}.$$

Computing:

$$[(5)(-5/3) + (0)(7/3) \quad (5)(-1/3) + (0)(2/3)] = [-25/3 \quad -5/3].$$

Since both reduced costs are negative, this confirms that (0,8) is an optimal solution.

Conclusion

- (6,5) is *not necessarily optimal* because it has a negative reduced cost.
- (0,8) is *optimal* because all reduced costs are nonnegative.



CAUTION!

Still working on the associated figure.

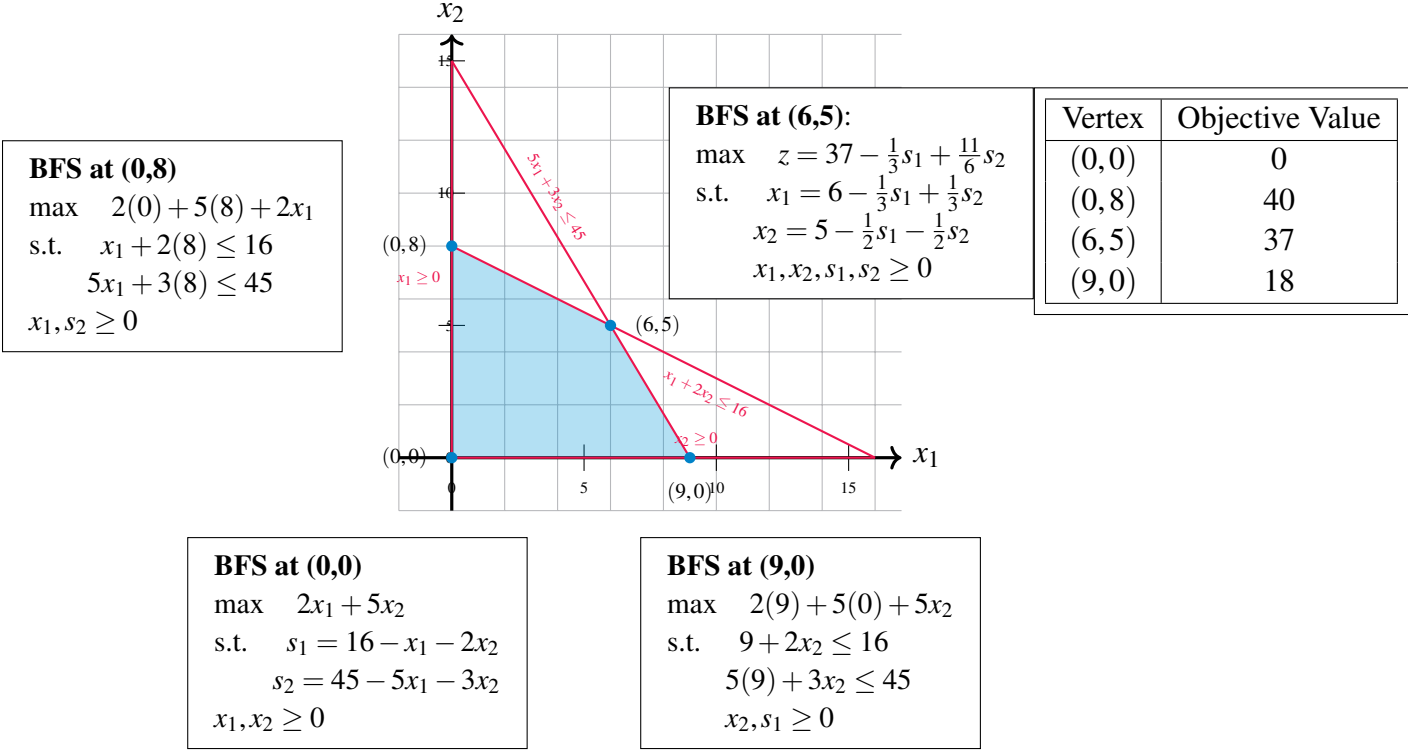


Figure 9.1: Desmos: Interactive plot! Showing the optimization problem from different BFS perspectives.

10. Sensitivity Analysis

Outcomes

- Understand what happens to the optimal solution as we slightly vary the parameters
- Calculate the range of parameters where the optimal solution (or optimal basis) remains optimal.
- Understand the change of objective function value as parameters are changed.

Sensitivity analysis examines how changes in the parameters (objective function coefficients, constraint right-hand sides) affect the optimal solution. This can be computed from the final simplex dictionary, often provided by optimization software such as Excel Solver.

We begin this section with an interactive desmos example. Go to [Desmos LP Sensitivity!](#).

10.0.1. Mathematical details of sensitivity analysis

We now analyze how changes in different parameters affect the optimal solution.

10.0.2. Effect of Changing an Objective Coefficient c_i

How it appears in the dictionary: The objective function in the revised dictionary is:

$$z = \mathbf{c}_B^\top A_B^{-1} \mathbf{b} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N) \mathbf{x}_N.$$

The coefficients of $\mathbf{x}_N$ are known as the **reduced costs**:

$$\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N.$$

Analysis:

- If i is a **basic variable**, c_i appears in $\mathbf{c}_B$, and since z^* contains $\mathbf{c}_B^\top A_B^{-1} \mathbf{b}$, a change in c_i directly affects the optimal objective function in a linear manner.
- If i is a **non-basic variable**, c_i appears in $\mathbf{c}_N$. The optimal basis remains unchanged, so the optimal objective function does **not** change. However, if the reduced cost $(\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N)$ changes sign, the variable may enter the basis.

Key Takeaways:

- Changing the coefficient of a **basic variable** shifts the objective function **linearly**.

- Changing the coefficient of a **non-basic variable** does not affect the objective value **unless it becomes negative**, triggering a basis change.

10.0.3. Effect of Changing the Right-Hand Side $\mathbf{b}$

How it appears in the dictionary: The basic solution is:

$$\mathbf{x}_B = A_B^{-1} \mathbf{b}.$$

Analysis:

- **Direct effect on feasibility:** Since $\mathbf{x}_B \geq 0$ is a feasibility condition, changes in $\mathbf{b}$ directly affect feasibility.
- **Basis stability:**
 - If $\mathbf{b}$ changes slightly, the basis may remain optimal, but $\mathbf{x}_B$ is rescaled accordingly.
 - If $\mathbf{b}$ changes significantly, one or more basic variables may become **negative**, requiring a pivot to restore feasibility.
- **Impact on objective function:**
 - Since $z^* = \mathbf{c}_B^\top A_B^{-1} \mathbf{b}$, the objective value scales with $\mathbf{b}$.
 - Small changes in $\mathbf{b}$ result in proportional changes in z^* , assuming the basis remains optimal.

Key Takeaways:

- Changing $\mathbf{b}$ **directly modifies** the basic solution.
- Large changes can force a **basis change** or make the problem **infeasible**.
- The **objective value scales** with $\mathbf{b}$.

10.0.4. Effect of Changing the Coefficients a_{ij} of the Constraint Matrix A

How it appears in the dictionary: The basic variables satisfy:

$$\mathbf{x}_B = A_B^{-1} \mathbf{b}.$$

If A changes, then A_B^{-1} changes, which affects:

1. **The basic solution $\mathbf{x}_B$.**
2. **The reduced costs $\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N$ in the objective function.**

Analysis:

- **Geometric Interpretation:** Changing a_{ij} **tilts** the corresponding constraint, altering the feasible region.

- **If i is a non-basic variable:**
 - The structure of A_B^{-1} remains unchanged.
 - The optimal basis remains unchanged.
 - The optimal solution **does not change**.
- **If i is a basic variable:**
 - Since A_B^{-1} depends on the inverse of the basis matrix, **even small changes in A_B can have large effects**.
 - The optimal solution and reduced costs change, potentially leading to a basis change.

Key Takeaways:

- If a_{ij} belongs to a **non-basic variable**, the optimal solution **remains unchanged**.
- If a_{ij} belongs to a **basic variable**, it **modifies A_B^{-1}** , causing significant changes to the optimal solution.
- The constraints are **tilted**, altering the feasible region.

10.0.5. Summary Table: Sensitivity Effects in the Revised Simplex Dictionary

Change Type	Effect on Basic Solution	Effect on Objective	Basis Change?
c_i (objective coefficient)	Only if i is in basis	Yes, if i is basic	If reduced cost changes sign
$\mathbf{b}$ (RHS vector)	Always changes	Always scales	If infeasibility occurs
a_{ij} (constraint matrix)	Only if i is in basis	Changes reduced costs	If basis becomes infeasible

10.0.6. Computing the Range of Parameters That Maintain the Optimal Basis

In this section, we analyze the sensitivity of the optimal basis to changes in different problem parameters. Specifically, we examine:

1. Changes in the right-hand side (RHS) value of the first constraint.
2. Changes in one of the objective function coefficients.

We determine the range of values these parameters can take while preserving the current optimal basis.

Note that we could also have written this as

$$\begin{aligned}
 &\text{Maximize} && \mathbf{c}^\top \mathbf{x} \\
 &\text{subject to} && A_x \mathbf{x} + I\mathbf{s} = \mathbf{b}, \\
 &&& \mathbf{x}, \mathbf{s} \geq 0
 \end{aligned}$$

where

$$A_x = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}, \text{ and } \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}$$

10.1 Sensitivity Analysis: Right-Hand Side Perturbations

After applying the simplex method, we obtain the following optimal dictionary:

Final Dictionary

$$\begin{aligned} \text{Maximize } z &= 23 - s_1 - s_3 \\ \text{subject to } x &= 4 - 2s_1 + s_3, \\ y &= 5 + s_1 - s_3, \\ s_2 &= 3 + 3s_1 - s_3, \\ x, y, s_1, s_2, s_3 &\geq 0. \end{aligned}$$

The current basis is $B = \{x, y, s_2\}$, with nonbasic variables $N = \{s_1, s_3\}$.

The optimal solution is:

$$(x, y, s_2, s_1, s_3) = (4, 5, 3, 0, 0) \quad \text{with} \quad z = 23.$$

Goal: Understanding Sensitivity to RHS Changes

In standard form, a linear program is written as $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$. Sensitivity analysis explores how the optimal solution and objective value change when we perturb entries of $\mathbf{b}$, while holding the objective coefficients and constraint matrix fixed.

In particular, we want to answer:

1. For which changes in the RHS does the current basis remain optimal?
2. How does the optimal solution vary with these changes?
3. How does the objective value change?

We consider changes to each component of the RHS independently, starting with the second constraint.

10.1.1. Perturbing b_2 : A Basic Slack Variable

The second constraint (corresponding to slack variable s_2) originally has RHS value $b_2 = 16$. We now perturb this to $16 + \Delta$, where $\Delta \in \mathbb{R}$, and investigate the effect.

Perturbed Standard Form (RHS of Constraint 2)

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & x + y + s_1 = 9 \quad (\text{hours}) \\ & 2x + y + s_2 = 16 + \Delta \quad (\text{flour}) \\ & x + 2y + s_3 = 14 \quad (\text{sugar}) \\ & x, y, s_1, s_2, s_3 \geq 0 \end{aligned}$$

To isolate the effect of Δ , define a new variable:

$$s_2 := s'_2 + \Delta \quad \Rightarrow \quad s'_2 = s_2 - \Delta.$$

Substituting into the system yields:

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & x + y + s_1 = 9 \\ & 2x + y + s_2 = 16 + \Delta \quad \Rightarrow \quad 2x + y + s'_2 = 16 \\ & x + 2y + s_3 = 14 \\ & x, y, s_1, s'_2, s_3 \geq 0 \end{aligned}$$

Adapting the Final Dictionary

Since the structure of the problem remains the same in terms of the new variable s'_2 , we can substitute into the final dictionary:

$$\begin{aligned} \text{Maximize} \quad & z = 23 - s_1 - s_3 \\ \text{subject to} \quad & x = 4 - 2s_1 + s_3 \\ & y = 5 + s_1 - s_3 \\ & s'_2 = 3 + 3s_1 - s_3 \quad \Rightarrow \quad s_2 - \Delta = 3 + 3s_1 - s_3 \\ & x, y, s_1, s'_2, s_3 \geq 0 \end{aligned}$$

To remain optimal, the dictionary must still satisfy feasibility: all basic variables must remain nonnegative when $s_1 = s_3 = 0$. So:

$$s'_2 = 3 + \Delta \geq 0 \quad \Longleftrightarrow \quad \boxed{\Delta \geq -3}.$$

Equivalently, since $b_2 = 16 + \Delta$, we require:

$$b_2 \geq 13.$$

Conclusion

The current basis $B = \{x, y, s_2\}$ remains optimal under the RHS perturbation $b_2 = 16 + \Delta$ as long as:

$$\Delta \geq -3 \Rightarrow b_2 \geq 13$$

Beyond this range, $s_2 < 0$, and the current dictionary becomes infeasible — triggering a new pivot in the simplex method.

10.1.2. Perturbing b_1 : A Nonbasic Slack Variable

The first constraint corresponds to slack variable s_1 , which is currently *nonbasic* and set to 0 in the optimal solution. Its original right-hand side is $b_1 = 9$. We now perturb it to $9 + \Delta$, where $\Delta \in \mathbb{R}$, and examine how this affects the dictionary.

Perturbed Standard Form (RHS of Constraint 1)

$$\begin{array}{ll} \text{Maximize} & 2x + 3y \\ \text{subject to} & x + y + s_1 = 9 + \Delta \quad (\text{hours}) \\ & 2x + y + s_2 = 16 \quad (\text{flour}) \\ & x + 2y + s_3 = 14 \quad (\text{sugar}) \\ & x, y, s_1, s_2, s_3 \geq 0 \end{array}$$

To isolate the effect of Δ , define a new variable:

$$s_1 := s'_1 - \Delta \quad \Rightarrow \quad s'_1 = s_1 + \Delta.$$

Substituting into the system yields:

$$\begin{array}{l} x + y + s_1 = 9 + \Delta \\ x + y + (s'_1 - \Delta) = 9 + \Delta \\ x + y + s'_1 = 9 \end{array}$$

So the perturbed system is equivalent to the original, but now expressed in terms of s'_1 .

Updated Dictionary

We substitute $s_1 = s'_1 - \Delta$ into the optimal dictionary:

$$\begin{aligned}\text{Maximize } z &= 23 - s_1 - s_3 = 23 - (s'_1 - \Delta) - s_3 = (23 + \Delta) - s'_1 - s_3 \\ x &= 4 - 2s_1 + s_3 = 4 - 2(s'_1 - \Delta) + s_3 = (4 + 2\Delta) - 2s'_1 + s_3 \\ y &= 5 + s_1 - s_3 = 5 + (s'_1 - \Delta) - s_3 = (5 - \Delta) + s'_1 - s_3 \\ s_2 &= 3 + 3s_1 - s_3 = 3 + 3(s'_1 - \Delta) - s_3 = (3 - 3\Delta) + 3s'_1 - s_3\end{aligned}$$

To ensure the basis remains feasible, set $s'_1 = s_3 = 0$ and check that the basic variables remain nonnegative:

$$x = 4 + 2\Delta, \quad y = 5 - \Delta, \quad s_2 = 3 - 3\Delta$$

Feasibility conditions:

$$\begin{aligned}x \geq 0 &\Rightarrow \Delta \geq -2 \\ y \geq 0 &\Rightarrow \Delta \leq 5 \\ s_2 \geq 0 &\Rightarrow \Delta \leq 1\end{aligned}$$

Thus, all conditions are satisfied when:

$$\boxed{-2 \leq \Delta \leq 1}$$

Conclusions

- The current basis $B = \{x, y, s_2\}$ remains optimal under the perturbation $b_1 = 9 + \Delta$ as long as:

$$\boxed{-2 \leq \Delta \leq 1} \quad \Rightarrow \quad \boxed{7 \leq b_1 \leq 10}$$

- Within this range, the objective value changes linearly as:

$$z = 23 + \Delta.$$

- Outside this range, one or more basic variables becomes negative, and the current solution becomes infeasible, prompting a new pivot in the simplex method.

10.1.3. Perturbing b_3 : Another Nonbasic Slack Variable

The third constraint corresponds to slack variable s_3 , which is currently *nonbasic* and equal to 0 in the optimal solution. Its original right-hand side is $b_3 = 14$. We now perturb it to $14 + \Delta$, where $\Delta \in \mathbb{R}$, and examine how this affects the dictionary.

Perturbed Standard Form (RHS of Constraint 3)

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & x + y + s_1 = 9 \quad (\text{hours}) \\ & 2x + y + s_2 = 16 \quad (\text{flour}) \\ & x + 2y + s_3 = 14 + \Delta \quad (\text{sugar}) \\ & x, y, s_1, s_2, s_3 \geq 0 \end{aligned}$$

To isolate the effect of Δ , define a new variable:

$$s_3 := s'_3 - \Delta \quad \Rightarrow \quad s'_3 = s_3 + \Delta.$$

Substituting into the system yields:

$$x + 2y + (s'_3 - \Delta) = 14 + \Delta \quad \Rightarrow \quad x + 2y + s'_3 = 14.$$

So the perturbed system returns to the original structure but in terms of s'_3 .

Updated Dictionary

Substitute $s_3 = s'_3 - \Delta$ into the optimal dictionary:

$$\begin{aligned} \text{Maximize} \quad & z = 23 - s_1 - s_3 = 23 - s_1 - (s'_3 - \Delta) = (23 + \Delta) - s_1 - s'_3 \\ & x = 4 - 2s_1 + s_3 = 4 - 2s_1 + (s'_3 - \Delta) = (4 - \Delta) - 2s_1 + s'_3 \\ & y = 5 + s_1 - s_3 = 5 + s_1 - (s'_3 - \Delta) = (5 + \Delta) + s_1 - s'_3 \\ & s_2 = 3 + 3s_1 - s_3 = 3 + 3s_1 - (s'_3 - \Delta) = (3 + \Delta) + 3s_1 - s'_3 \end{aligned}$$

Now test feasibility at the basic solution $s_1 = s'_3 = 0$:

$$x = 4 - \Delta, \quad y = 5 + \Delta, \quad s_2 = 3 + \Delta$$

Feasibility requires:

$$\begin{aligned} x \geq 0 &\Rightarrow \Delta \leq 4 \\ s_2 \geq 0 &\Rightarrow \Delta \geq -3 \end{aligned}$$

Thus, the allowable range for Δ is:

$$\boxed{-3 \leq \Delta \leq 4}$$

Conclusion

- The current basis $B = \{x, y, s_2\}$ remains optimal under the perturbation $b_3 = 14 + \Delta$ as long as:

$$\boxed{-3 \leq \Delta \leq 4} \quad \Rightarrow \quad \boxed{11 \leq b_3 \leq 18}$$

- Within this range, the objective value changes linearly as:

$$z = 23 + \Delta.$$

- Outside this range, $x < 0$ or $s_2 < 0$, and the dictionary becomes infeasible.

10.1.4. Perturbing c_1 and c_2 : Objective Coefficients of Basic Variables

We now examine how changes to the objective function coefficients affect the optimality of the current basis. In the original problem, the objective is:

$$\text{Maximize } 2x + 3y.$$

We now perturb these coefficients to:

$$\text{Maximize } (2 + \delta_1)x + (3 + \delta_2)y,$$

where $\delta_1, \delta_2 \in \mathbb{R}$.

From the final dictionary, we have:

$$z = 23 - s_1 - s_3$$

$$x = 4 - 2s_1 + s_3$$

$$y = 5 + s_1 - s_3$$

$$s_2 = 3 + 3s_1 - s_3,$$

with basis $B = \{x, y, s_2\}$ and nonbasic variables $N = \{s_1, s_3\}$.

Perturbing c_1 : Objective Coefficient of x

Let $c_1 = 2 + \delta_1$. Since x is basic, the perturbed objective becomes:

$$z = 23 + \delta_1 \cdot x = 23 + \delta_1(4 - 2s_1 + s_3) = (23 + 4\delta_1) + (-2\delta_1)s_1 + \delta_1 s_3.$$

Adding to the original dictionary $z = 23 - s_1 - s_3$, we obtain:

$$z = (23 + 4\delta_1) + (-1 - 2\delta_1)s_1 + (-1 + \delta_1)s_3.$$

To maintain optimality, the reduced costs of nonbasic variables s_1 and s_3 must remain ≤ 0 . Thus:

$$\begin{aligned} -1 - 2\delta_1 \leq 0 &\Rightarrow \delta_1 \geq -\frac{1}{2}, \\ -1 + \delta_1 \leq 0 &\Rightarrow \delta_1 \leq 1. \end{aligned}$$

So the allowable range for δ_1 is:

$$\boxed{-\frac{1}{2} \leq \delta_1 \leq 1} \Rightarrow \boxed{1.5 \leq c_1 \leq 3}.$$

Within this range, the objective value changes linearly as:

$$z = 23 + 4\delta_1.$$

Perturbing c_2 : Objective Coefficient of y

Let $c_2 = 3 + \delta_2$. Since y is also basic, the perturbed objective becomes:

$$z = 23 + \delta_2 \cdot y = 23 + \delta_2(5 + s_1 - s_3) = (23 + 5\delta_2) + \delta_2 s_1 - \delta_2 s_3.$$

Adding to the original dictionary $z = 23 - s_1 - s_3$, we obtain:

$$z = (23 + 5\delta_2) + (-1 + \delta_2)s_1 + (-1 - \delta_2)s_3.$$

To maintain optimality:

$$\begin{aligned} -1 + \delta_2 \leq 0 &\Rightarrow \delta_2 \leq 1, \\ -1 - \delta_2 \leq 0 &\Rightarrow \delta_2 \geq -1. \end{aligned}$$

So the allowable range for δ_2 is:

$$\boxed{-1 \leq \delta_2 \leq 1} \Rightarrow \boxed{2 \leq c_2 \leq 4}.$$

Within this range, the objective value changes linearly as:

$$z = 23 + 5\delta_2.$$

Summary of Sensitivity to Objective Coefficients

Perturbed Coefficient	Allowed Range	Effect on Objective
$c_1 = 2 + \delta_1$	$1.5 \leq c_1 \leq 3$	$z = 23 + 4\delta_1$
$c_2 = 3 + \delta_2$	$2 \leq c_2 \leq 4$	$z = 23 + 5\delta_2$

Outside these ranges, the reduced costs of nonbasic variables become positive, violating optimality. In such cases, the simplex method would require a new pivot.

10.2 Sensitivity with Matrix Notation

Final Dictionary in Matrix Notation

Let the basic variables be $\mathbf{x}_B = \begin{bmatrix} x \\ y \\ s_2 \end{bmatrix}$, and the non-basic variables be $\mathbf{x}_N = \begin{bmatrix} s_1 \\ s_3 \end{bmatrix}$. Then the dictionary can be written as:

$$\mathbf{x}_B = \mathbf{b}' - A'_N \mathbf{x}_N,$$

where:

$$\mathbf{b}' = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}, \quad A'_N = \begin{bmatrix} -2 & 1 \\ 1 & -1 \\ 3 & -1 \end{bmatrix}.$$

Thus, explicitly:

$$\begin{bmatrix} x \\ y \\ s_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} - \begin{bmatrix} -2 & 1 \\ 1 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} s_1 \\ s_3 \end{bmatrix}.$$

or equivalently

$$\begin{bmatrix} x \\ y \\ s_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} + s_1 \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix} + s_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

The basis matrix is

$$A_B = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \quad \text{and} \quad A_B^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & 1 & 1 \end{bmatrix}.$$

Remark 10.1: Recovering A_B^{-1}

$$A(x, s) = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$$

Partitioning into basic variables $x = (x, y)$ and slack variables $s = (s_1, s_2, s_3)$, we write:

$$A_x x + A_s s = b \quad \text{where} \quad A_x = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad A_s = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$$

Hence, since the slack columns came from the identity matrix in the original A , we conclude:

$$A_s = A_B^{-1}$$

10.2.0.1. 1. Changing the Right-Hand Side b_1 of the First Constraint

We modify the right-hand side of the first constraint and analyze its impact on feasibility:

$$b_1 \Rightarrow \begin{bmatrix} b_1 \\ 16 \\ 14 \end{bmatrix}.$$

Using the inverse basis matrix A_B^{-1} , the basic solution is updated as:

$$\mathbf{x}_B = A_B^{-1} \mathbf{b}.$$

Ensuring that all basic variables remain non-negative, we derive the feasibility conditions:

$$A_B^{-1} \mathbf{b} = A_B^{-1} \begin{bmatrix} b_1 \\ 16 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ 16 \\ 14 \end{bmatrix} = \begin{bmatrix} 2b_1 - 14 \\ -b_1 + 14 \\ -3b_1 + 30 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

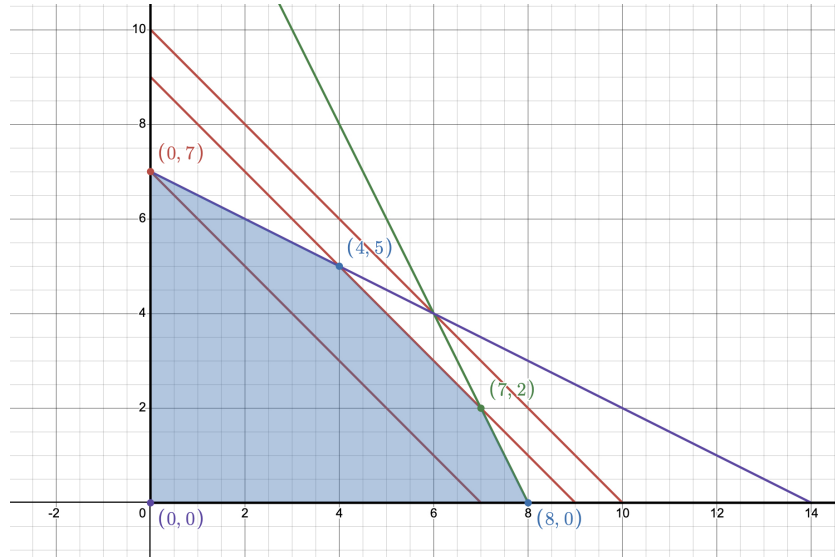
Simplifying, we have

$$\begin{bmatrix} b_1 \\ 14 \\ 10 \end{bmatrix} \geq \begin{bmatrix} 7 \\ b_1 \\ b_1 \end{bmatrix}$$

$$7 \leq b_1 \leq 10$$

is the feasible range for b_1 that preserves the optimal basis.

In the image here, we show changing b_1 to the values of 7, 9, 10.



10.2.0.2. Sensitivity Analysis on the Objective Coefficient c_x

To determine the range of values for c_x (the coefficient of x in the objective function) such that the current basis remains optimal, we analyze the simplex optimality conditions.

Step 1: Compute the Reduced Costs The objective function is initially:

$$z = 2x + 3y.$$

The cost vector for all variables is:

$$c = \begin{bmatrix} c_x \\ 3 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The cost vector corresponding to the basic variables x, y, s_2 (i.e., the basis) is:

$$c_B = [c_x \quad 3 \quad 0].$$

The reduced cost for any non-basic variable j is given by:

$$\bar{c}_j = c_j - c_B A_B^{-1} A_j.$$

The non-basic variables are s_1 and s_3 . We need to ensure that their reduced costs remain non-negative.

Step 2: Compute the Reduced Costs for s_1 and s_3 The constraint matrix is:

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix}.$$

Extracting the columns corresponding to s_1 and s_3 :

$$A_{s_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad A_{s_3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Using the inverse basis matrix:

$$A_B^{-1}A_{s_1} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix}.$$

$$A_B^{-1}A_{s_3} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

Thus, the reduced costs are:

$$\bar{c}_{s_1} = 0 - [c_x \quad 3 \quad 0] \cdot \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix} = 0 - (2c_x - 3).$$

$$\bar{c}_{s_3} = 0 - [c_x \quad 3 \quad 0] \cdot \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = 0 - (-c_x + 3).$$

To retain optimality, we want $\bar{c}_{s_1}$ and $\bar{c}_{s_3}$ to be ≤ 0 .

$$\begin{aligned} -2c_x + 3 &\leq 0 &\Rightarrow &c_x \geq \frac{3}{2} = 1.5. \\ c_x - 3 &\leq 0 &\Rightarrow &c_x \leq 3. \end{aligned}$$

Step 3: Feasible Range for c_x For the current basis to remain optimal, c_x must satisfy:

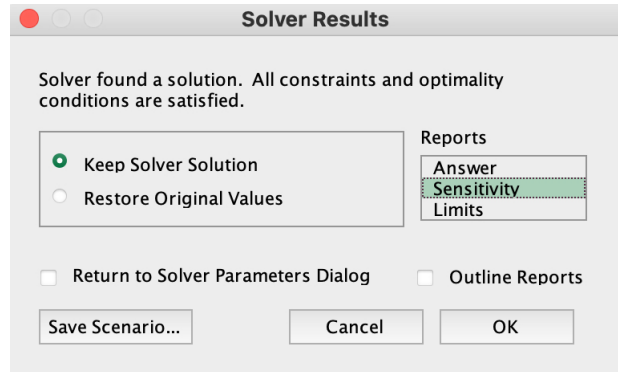
$$1.5 \leq c_x \leq 3.$$

Thus, we have confirmed that the coefficient c_x can range between 1.5 and 3 while maintaining the current optimal basis.



10.2.1. Excel Solver Sensitivity Analysis

When solving a problem in excel, you can ask for a sensitivity analysis.



This will generate a report that will show the following:

- **Allowable Increase/Decrease:** For each decision variable's coefficient in the objective function, the report provides the range (allowable increase and allowable decrease) within which the coefficient can change without altering the current optimal basis. This range indicates the stability of the optimal solution with respect to changes in that coefficient.
- **Right-Hand Side (RHS) Values:** For each constraint, the sensitivity report lists the *shadow price* (or dual value), which represents the rate of change in the objective function per unit increase in the RHS of that constraint. It also provides the allowable increase and allowable decrease for the RHS. These values indicate how robust the constraints are; small allowable changes might imply that the solution is sensitive to fluctuations in available resources or demand levels.
- **Constraint Status:** The report shows which constraints are binding (active) and which are non-binding. A binding constraint has a positive shadow price and restricts the optimal solution, while a non-binding constraint has a shadow price of zero.

	A	B	C	D	E	F	G	H
1	Microsoft Excel 16.78 Sensitivity Report							
2	Worksheet: [Book1]Sheet1							
3	Report Created: 3/17/25 12:29:15 AM							
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6	Variable Cells							
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11. Duality

Outcomes

You will learn

- How to derive a bound on the optimal objective value
- How to reformulate the problem into a dual with a different interpretation
- Correspondence between the primal (original) and dual problems

11.1 Duality Theory

Motivating Example

Imagine you run a small bakery that makes two products: cakes (x_1) and cookies (x_2). Your goal is to maximize your daily profit. Cakes yield a profit of \$3 each, and cookies yield \$2 each. You are limited by the ingredients you have on hand: you have at most 4 units of flour and 5 units of sugar available each day. Each cake requires 1 unit of flour and 2 units of sugar, while each cookie requires 1 unit of flour and 1 unit of sugar. We can write this scenario as a linear program:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 && \text{(maximize total profit)} \\ \text{subject to} \quad & x_1 + x_2 \leq 4 && \text{(flour constraint)} \\ & 2x_1 + x_2 \leq 5 && \text{(sugar constraint)} \\ & x_1, x_2 \geq 0. \end{aligned}$$

Here, x_1 is the number of cakes you bake, and x_2 is the number of cookies. The constraints limit how many of each product you can produce given your resources.

Now, suppose a friend is not sure that you are telling the truth about the maximum profit you can make. They ask, "Can you prove an upper bound on how much money you could possibly earn with these ingredient constraints, without actually solving the problem?"

Intuition Behind Duality

Consider the constraints in the problem. Each one puts a cap on how large the combination of x_1 and x_2 can be. If we think of them separately:

- The flour constraint: $x_1 + x_2 \leq 4$.
- The sugar constraint: $2x_1 + x_2 \leq 5$.

Now, what if we take some nonnegative multipliers y_1 and y_2 and combine these inequalities into one single inequality? For instance, multiply the flour constraint by $y_1 \geq 0$ and the sugar constraint by $y_2 \geq 0$ and add them up:

$$y_1(x_1 + x_2) + y_2(2x_1 + x_2) \leq y_1(4) + y_2(5).$$

Grouping the terms by x_1 and x_2 gives:

$$(x_1(y_1 + 2y_2)) + (x_2(y_1 + y_2)) \leq 4y_1 + 5y_2.$$

If we choose y_1 and y_2 cleverly, we can force the left-hand side to be an expression that is always *at least* as large as our profit function $3x_1 + 2x_2$. Specifically, we want

$$y_1 + 2y_2 \geq 3 \quad \text{and} \quad y_1 + y_2 \geq 2.$$

Why? Because if the coefficients on x_1 and x_2 in our combined inequality dominate those in our profit function, then the upper bound on the combined inequality's right-hand side ($4y_1 + 5y_2$) will also serve as an upper bound on the profit $3x_1 + 2x_2$. This ensures:

$$3x_1 + 2x_2 \leq (y_1 + 2y_2)x_1 + (y_1 + y_2)x_2 \leq 4y_1 + 5y_2.$$

Thus, by choosing y_1 and y_2 , we obtain a valid upper bound on the maximum profit. If we try to minimize $4y_1 + 5y_2$ over all $y_1, y_2 \geq 0$ that satisfy these inequalities, we find the best (smallest) upper bound on our original problem.

General Derivation of the Dual

We started with the primal problem in a general form:

$$\begin{aligned} & \max \mathbf{c}^\top \mathbf{x} \\ & \text{subject to } A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0. \end{aligned}$$

Each constraint $a^i \cdot \mathbf{x} \leq b_i$ (where a^i is the i -th row of A) is valid for all feasible $\mathbf{x}$. If we take a nonnegative combination of these constraints, say with multipliers $y_i \geq 0$, we get:

$$\sum_{i=1}^m y_i (a^i \cdot \mathbf{x}) \leq \sum_{i=1}^m y_i b_i.$$

This can be written as:

$$\left(\sum_{i=1}^m y_i \mathbf{a}^i \right) \cdot \mathbf{x} \leq \sum_{i=1}^m y_i b_i.$$

If we choose $\mathbf{y}$ such that $\sum_{i=1}^m y_i \mathbf{a}^i \geq \mathbf{c}$, then for all feasible $\mathbf{x}$ (since $\mathbf{x} \geq 0$):

$$\mathbf{c}^\top \mathbf{x} \leq \left(\sum_{i=1}^m y_i \mathbf{a}^i \right) \cdot \mathbf{x} \leq \sum_{i=1}^m y_i b_i.$$

So, $\mathbf{b}^\top \mathbf{y} = \sum_{i=1}^m y_i b_i$ is an upper bound on the maximum primal objective value. By minimizing over all such $\mathbf{y}$, we seek the tightest upper bound:

$$\begin{aligned} & \min \mathbf{b}^\top \mathbf{y} \\ & \text{subject to } A^\top \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq 0. \end{aligned}$$

This is the *dual* problem. In our bakery scenario, this dual formulation corresponds to adjusting the prices (the y values can be thought of as "shadow prices") you would assign to each ingredient such that if you charge yourself for using these ingredients at these shadow prices, you end up with the smallest possible "bill" that still always surpasses or equals your maximum possible profit. The beauty of the Strong Duality Theorem ensures that the optimal primal value equals the optimal dual value.

From the above reasoning, we have shown that if $\mathbf{x}$ is feasible for the primal and $\mathbf{y}$ is feasible for the dual, then $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$. In other words, the value of any feasible primal solution is always bounded above by the value of any feasible dual solution.

Theorem 11.1: Weak Duality

For any feasible $\mathbf{x}$ in the primal problem and any feasible $\mathbf{y}$ in the dual problem, we have:

$$\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}.$$

The weak duality theorem is not only specific to linear programming problems; it appears in many other classes of optimization problems. It establishes an important inequality between primal and dual objective values for any pair of feasible solutions.

However, in the special case of linear programs (and certain other problem classes), we can say even more. In these settings, not only does the dual objective always provide an upper bound on the primal objective, but there exist optimal solutions for both problems whose objective values match. This result is known as the *Strong Duality Theorem*.

11.2 Primal-Dual Pairs and Strong Duality

Outcomes

- Learn the two most important theorems in Linear Programming!!!

Consider the following pair of linear programs, known as a primal-dual pair. The primal problem (P) has n decision variables and m constraints, while the dual problem (D) has m decision variables and n constraints.

Primal (P)

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0 \\ & \text{(Here, } \mathbf{x} \in \mathbb{R}^n \text{ and } A \text{ is an } m \times n \text{ matrix.)} \end{aligned}$$

Dual (D)

$$\begin{aligned} \min \quad & \mathbf{b}^\top \mathbf{y} \\ \text{s.t.} \quad & A^\top \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq 0 \\ & \text{(Here, } \mathbf{y} \in \mathbb{R}^m, \text{ so the dual has } m \text{ variables and } n \text{ constraints.)} \end{aligned}$$

From weak duality, we know that for any feasible $\mathbf{x}$ in (P) and any feasible $\mathbf{y}$ in (D), it holds that:

$$\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}.$$

This indicates that the optimal value of the dual problem is always an upper bound on the optimal value of the primal problem.

Theorem 11.2: Weak Duality

For any feasible solution $\mathbf{x}$ of the primal problem (P) and any feasible solution $\mathbf{y}$ of the dual problem (D),

$$\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}.$$

Weak duality theorems appear in many classes of optimization problems. In the special case of linear programs (and some other structured problems), we also have a stronger result, known as the *Strong Duality Theorem*. This theorem completely characterizes the relationship between the primal and dual solutions:

Theorem 11.3: Strong Duality

For a primal-dual pair of linear programs:

$$\begin{array}{ll} \text{(P)} & \max \mathbf{c}^\top \mathbf{x} \\ & \text{subject to } A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq 0, \end{array} \quad \begin{array}{ll} \text{(D)} & \min \mathbf{b}^\top \mathbf{y} \\ & \text{subject to } A^\top \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq 0, \end{array}$$

the following cases hold:

1. If (P) is unbounded, then (D) must be infeasible.
2. If (D) is unbounded, then (P) must be infeasible.
3. If both (P) and (D) have feasible solutions and are bounded, then both have optimal solutions $\mathbf{x}^*$ and $\mathbf{y}^*$ respectively, and

$$\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*.$$

Example 11.4: Dual Infeasibility from Primal Unboundedness

Consider the following linear program:

$$\begin{array}{ll} \text{maximize} & x_1 \\ \text{subject to} & x_1 - x_2 \leq 1 \\ & 2x_1 - x_2 \leq 3 \\ & x_1, x_2 \geq 0 \end{array}$$

As shown in Example 8.8, this primal problem is unbounded. After applying the simplex method, we found that when s_1 enters the basis, there is no constraint preventing it from growing indefinitely, and the objective increases without bound. Therefore, the primal is unbounded.

Now, consider the dual of this problem:

$$\begin{array}{ll} \text{minimize} & y_1 + 3y_2 \\ \text{subject to} & y_1 + 2y_2 \geq 1 \\ & -y_1 - y_2 \geq 0 \\ & y_1, y_2 \geq 0 \end{array}$$

Conclusion via Strong Duality: Because the primal problem is unbounded and the primal-dual pair satisfies the usual assumptions of linear programming, **strong duality** implies that the dual must be **infeasible**.

This is confirmed by the Big-M simplex pivot steps, which fail to drive the artificial variable to zero. To solve the dual, we apply the simplex method using the Big-M technique to initialize feasibility. The auxiliary variable a_1 is introduced to handle the infeasibility of the initial basic solution.

First dictionary (with Big-M penalty):

$$\begin{aligned} &\text{Maximize } -y_1 - 3y_2 - 1000a_1 \\ &\text{subject to } \begin{cases} w_1 = -1 + y_1 + 2y_2 + a_1 \\ w_2 = 0 - y_1 - y_2 \end{cases} \end{aligned}$$

Pivot 1: Enter y_1 , leave a_1 . We rewrite in terms of the new basic variables.

Second dictionary:

$$\begin{aligned} &\text{Maximize } -1000 + 999y_1 + 1997y_2 - 1000w_1 \\ &\text{subject to } \begin{cases} a_1 = 1 - y_1 - 2y_2 + w_1 \\ w_2 = 0 - y_1 - y_2 \end{cases} \end{aligned}$$

Pivot 2: Enter y_2 , leave w_2 . Continue pivoting.

Final dictionary:

$$\begin{aligned} &\text{Maximize } -1000 - 998y_1 - 1000w_1 - 1997w_2 \\ &\text{subject to } \begin{cases} a_1 = 1 + y_1 + w_1 + 2w_2 \\ y_2 = 0 - y_1 - w_2 \end{cases} \end{aligned}$$

In this final dictionary, all variables in the objective function have non-positive coefficients, but the artificial variable a_1 remains in the basis with a positive value. Since $a_1 > 0$ at termination, we conclude that the original dual problem is infeasible.

11.2.1. Economic Interpretation of the Dual

Duality in linear programming offers a powerful economic interpretation: the dual variables represent the *value* of resources or constraints in the primal problem. These values are commonly known as **shadow prices**.

Definition 11.5: Shadow Price

The shadow price of a constraint in a linear program tells us how much the optimal objective value would change if we slightly increased the right-hand side of that constraint by one unit, while keeping everything else fixed.

In a maximization problem, a shadow price represents the **increase in value** per additional unit of a scarce resource. In a minimization problem, it represents the **decrease in cost**.

Interpretation: If a constraint represents the availability of a resource (e.g., time, material), then its shadow price reflects the marginal value of that resource — how much the objective would improve if you had one more unit of it.

Mathematically: If the constraint $a_i^\top \mathbf{x} \leq b_i$ has corresponding dual variable y_i^* , then:

$$\text{Shadow price} = \frac{dz^*}{db_i} = y_i^*$$

where z^* is the optimal value of the objective function.

UNITS AND INTERPRETATION: The units of a shadow price match the units of the objective function (e.g., dollars of profit) per unit of the resource (e.g., kilograms of flour, hours of labor). If $y_2 = 10$ in the bakery example, and the constraint corresponds to sugar (in kg), then each additional kg of sugar would allow the bakery to increase profit by \$10.

Example: The Bakery Revisited

Consider the primal problem:

$$\begin{array}{ll} \text{maximize} & 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 4 \quad (\text{Flour}) \\ & 2x_1 + x_2 \leq 5 \quad (\text{Sugar}) \\ & x_1, x_2 \geq 0 \end{array}$$

The dual problem is:

$$\begin{array}{ll} \text{minimize} & 4y_1 + 5y_2 \\ \text{subject to} & y_1 + 2y_2 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2 \geq 0 \end{array}$$

If the optimal dual solution is $y_1^* = 1, y_2^* = 1$, then:

- The shadow price of flour is 1 — each extra unit of flour increases profit by \$1.
- The shadow price of sugar is 1 — each extra unit of sugar increases profit by \$1.

Example: Bakery 2.0

The primal problem describes a bakery optimizing the production of cakes, cookies, and muffins to maximize profit. The constraints represent limited resources: flour, sugar, and baking time. The dual problem attaches a shadow price to each of these resources, reflecting their marginal value.

Primal Problem:

$$\begin{aligned}
 \max \quad & 60x_1 + 30x_2 + 20x_3 \\
 \text{s.t.} \quad & 8x_1 + 6x_2 + x_3 \leq 48 \quad (\text{Flour}) \\
 & 4x_1 + 2x_2 + 1.5x_3 \leq 20 \quad (\text{Sugar}) \\
 & 2x_1 + 1.5x_2 + 0.5x_3 \leq 8 \quad (\text{Baking Time}) \\
 & x_1, x_2, x_3 \geq 0
 \end{aligned}$$

Optimal Primal Solution:

$$x_1 = 2, \quad x_2 = 0, \quad x_3 = 8, \quad \text{Objective Value} = 280$$

Dual Problem:

$$\begin{aligned}
 \min \quad & 48y_1 + 20y_2 + 8y_3 \\
 \text{s.t.} \quad & 8y_1 + 4y_2 + 2y_3 \geq 60 \quad (\text{Cakes}) \\
 & 6y_1 + 2y_2 + 1.5y_3 \geq 30 \quad (\text{Cookies}) \\
 & y_1 + 1.5y_2 + 0.5y_3 \geq 20 \quad (\text{Muffins}) \\
 & y_1, y_2, y_3 \geq 0
 \end{aligned}$$

Optimal Dual Solution:

$$y_1 = 0, \quad y_2 = 10, \quad y_3 = 10, \quad \text{Objective Value} = 280$$

Interpretation of Shadow Prices

- $y_1 = 0$: The shadow price of flour is \$0. This implies the flour constraint is **non-binding** — the bakery is not using all 48 kg of flour. Increasing flour supply would not increase profit.
- $y_2 = 10$: The shadow price of sugar is \$10. Each additional kilogram of sugar allows the bakery to increase profit by \$10. The sugar constraint is **binding**.
- $y_3 = 10$: The shadow price of baking time is \$10 per hour. Baking time is fully used and limits production. Gaining one more hour would allow an increase in profit by \$10.

Complementary Slackness Verification

From the primal solution:

- The **flour constraint** is not tight $\Rightarrow y_1 = 0$
- The **sugar constraint** is tight $\Rightarrow y_2 > 0$
- The **baking time constraint** is tight $\Rightarrow y_3 > 0$

These relationships satisfy the *complementary slackness conditions*:

$$y_i > 0 \Rightarrow \text{Constraint } i \text{ is tight, } \text{Constraint } i \text{ not tight} \Rightarrow y_i = 0$$

Sensitivity Analysis: Marginal Value of Resources Assuming the current basis remains optimal, the increase in maximum profit from one additional unit of:

- Flour = \$0
- Sugar = \$10
- Baking Time = \$10

The dual solution not only validates the primal objective value (via strong duality), but also assigns economic meaning to each constraint. Shadow prices inform how much you'd be willing to pay for more of each resource — a foundational concept in operations, pricing, and planning.

WHEN ARE SHADOW PRICES MEANINGFUL? Shadow prices are meaningful only when:

- The constraint is *binding* (i.e., tight) at the optimal solution.
- The problem satisfies strong duality (which always holds for feasible and bounded linear programs).
- The right-hand side is perturbed slightly (local sensitivity).

If a constraint is not tight, then its shadow price is 0 — the resource is not limiting the optimal solution.

CONNECTION TO THE DUAL PROBLEM: In the dual problem:

- The *objective function* is the total value of resources, priced at shadow prices.
- The *constraints* ensure that no activity in the primal makes more profit than it costs, when inputs are valued at shadow prices.

This ensures equilibrium: no profitable arbitrage is possible, and every unit of constrained resource is priced at its true marginal value.

Summary of Shadow Price Properties:

- A shadow price represents how much you would *pay* for an extra unit of a constrained resource.
- It is equal to the dual variable corresponding to the constraint.
- It reflects the marginal value of relaxing that constraint.
- Shadow prices are only nonzero for **binding** constraints.
- If a resource is fully used and limits optimality, its shadow price is positive.
- If a resource is unused or abundant, its shadow price is zero.

11.3 Different Primal–Dual Forms

The form of a dual problem depends on the primal’s objective (maximize or minimize), inequality directions, and sign constraints on variables. The following layout uses paired boxes to clearly compare each primal and its dual.

1. Maximization with = Constraints (Standard Form)

Primal (P)	Dual (D)
$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$	$\begin{aligned} \min \quad & \mathbf{b}^\top \mathbf{y} \\ \text{s.t.} \quad & A^\top \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \text{ free} \end{aligned}$

2. Minimization with $\leq$ Constraints

Primal (P)

$$\begin{aligned} \min \quad & \mathbf{b}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{c} \\ & \mathbf{x} \geq 0 \end{aligned}$$

Dual (D)

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{y} \\ \text{s.t.} \quad & A^\top \mathbf{y} \leq \mathbf{b} \\ & \mathbf{y} \geq 0 \end{aligned}$$

3. Equality Constraints in the Primal

Primal (P)

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \text{ free} \end{aligned}$$

Dual (D)

$$\begin{aligned} \min \quad & \mathbf{b}^\top \mathbf{y} \\ \text{s.t.} \quad & A^\top \mathbf{y} = \mathbf{c} \\ & \mathbf{y} \geq 0 \end{aligned}$$

4. Sign Variations in Primal Variables and Constraints

- If a primal variable x_j is unrestricted (free), the corresponding dual constraint is an *equality*.
- If $x_j \leq 0$, the dual constraint becomes $\leq$ instead of $\geq$.
- If a primal constraint $a_i^\top \mathbf{x} \leq b_i$ is changed to $\geq b_i$, then the sign constraint on the corresponding dual variable y_i flips.

Example: Primal with Free Variables and Equalities

Primal (P)

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \text{ free} \end{aligned}$$

Dual (D)

$$\begin{aligned} \min \quad & \mathbf{b}^\top \mathbf{y} \\ \text{s.t.} \quad & A^\top \mathbf{y} = \mathbf{c} \\ & \mathbf{y} \geq 0 \end{aligned}$$

This structured approach helps you identify how each component in the primal affects the dual—particularly how inequality direction and sign restrictions control the dual variable domain and constraint form.

As you can see, changing the primal's constraints and variable domains changes the structure of the dual problem. Every such transformation has a corresponding primal-dual pair that maintains the duality principles. In each case, the weak and strong duality theorems still apply.

Deriving the Dual with Equality Constraints

Let us now consider a variation of our previous derivation of the dual problem. This time, our primal problem (P) will have equality constraints rather than inequalities, and all variables will remain nonnegative. For example, consider the following linear program with $m = 2$ equality constraints and $n = 4$ variables:

$$\begin{aligned} & \max 2x_1 + 3x_2 - x_3 + x_4 \\ & \text{subject to } x_1 + x_2 + 0 \cdot x_3 + x_4 = 10 \\ & \quad 2x_1 + x_2 - x_3 + 0 \cdot x_4 = 5 \\ & \quad x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

In matrix form, if we let

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 1 & -1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 5 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix},$$

our primal problem is:

$$\max \mathbf{c}^\top \mathbf{x} \quad \text{subject to } A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0.$$

Forming a Bound Using Combinations of Equalities

Previously, when we had inequalities, we formed upper bounds on the objective by taking nonnegative linear combinations of the constraints. Now, since we have *equalities*, we are free to choose any real multipliers for these constraints (positive, negative, or zero), and the resulting combination will still hold as an equality. This is because any scalar multiple of an equation remains a valid equation.

Let y_1 and y_2 be arbitrary real numbers, and form the linear combination:

$$y_1(x_1 + x_2 + x_4 = 10) \quad \text{and} \quad y_2(2x_1 + x_2 - x_3 = 5).$$

Adding these together gives:

$$y_1x_1 + y_1x_2 + y_1x_4 + y_2(2x_1 + x_2 - x_3) = 10y_1 + 5y_2.$$

Collecting terms by variable:

$$x_1(y_1 + 2y_2) + x_2(y_1 + y_2) + x_3(-y_2) + x_4(y_1) = 10y_1 + 5y_2.$$

Since $x \geq 0$, if we want the left-hand side to serve as an upper bound on our objective function $2x_1 + 3x_2 - x_3 + x_4$ for all feasible x , we need the coefficients of each x_j in the combined inequality to dominate those in the objective. Specifically, we want:

$$y_1 + 2y_2 \geq 2, \quad y_1 + y_2 \geq 3, \quad -y_2 \geq -1 \text{ (which implies } y_2 \leq 1), \quad y_1 \geq 1.$$

If these inequalities hold, then for every feasible x :

$$2x_1 + 3x_2 - x_3 + x_4 \leq (y_1 + 2y_2)x_1 + (y_1 + y_2)x_2 + (-y_2)x_3 + (y_1)x_4 = 10y_1 + 5y_2.$$

Thus, $10y_1 + 5y_2$ is an upper bound on the maximal value of $2x_1 + 3x_2 - x_3 + x_4$. Since y_1 and y_2 are unrestricted in sign, we can try to *minimize* $10y_1 + 5y_2$ subject to ensuring these dominance conditions.

The Dual Problem

In the general setting with equality constraints and nonnegative primal variables, the dual variables associated with these equality constraints are *free* (they can be positive, negative, or zero). The process we followed above leads us to the dual problem:

$$\min \mathbf{b}^\top \mathbf{y} = 10y_1 + 5y_2$$

subject to the condition that $A^\top \mathbf{y} \geq \mathbf{c}$, i.e.:

$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

This system (in our specific numeric example) determines a unique $\mathbf{y}$ that perfectly matches the $\mathbf{c}$ vector. More generally, the dual problem corresponding to

$$\max \mathbf{c}^\top \mathbf{x} \quad \text{subject to } A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0$$

is

$$\min \mathbf{b}^\top \mathbf{y} \quad \text{subject to } A^\top \mathbf{y} \geq \mathbf{c}, \mathbf{y} \text{ free.}$$

The absence of sign restrictions on $\mathbf{y}$ is a direct consequence of using equalities in the primal. Since no direction of inequality is fixed, you are free to "add" or "subtract" the given equations as you please, hence no sign constraints emerge on the dual variables.

Example with Inequalities and Free Variables

Now consider a primal problem with three inequality constraints and two free variables. Suppose:

Primal (P)	Dual (D)
$\begin{aligned} \max \quad & 3x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 10 \\ & 2x_1 - x_2 \leq 5 \\ & -x_1 + 3x_2 \leq 7 \\ & x_1 \text{ free}, x_2 \text{ free} \end{aligned}$	$\begin{aligned} \min \quad & 10y_1 + 5y_2 + 7y_3 \\ \text{s.t.} \quad & y_1 + 2y_2 - y_3 = 3 \quad (\text{coeff. of } x_1) \\ & y_1 - y_2 + 3y_3 = -2 \quad (\text{coeff. of } x_2) \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$

Here, x_1 and x_2 can be positive or negative.

we can write this in a compact matrix form. Let:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 5 \\ 7 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Then (P) is:

$$\max \mathbf{c}^\top \mathbf{x} \quad \text{subject to } A\mathbf{x} \leq \mathbf{b}, \text{ and } \mathbf{x} \text{ free.}$$

To create a bound on $3x_1 - 2x_2$, we take nonnegative combinations of the constraints. Let $y_1, y_2, y_3 \geq 0$ be the multipliers for the three constraints:

$$y_1(x_1 + x_2) + y_2(2x_1 - x_2) + y_3(-x_1 + 3x_2) \leq 10y_1 + 5y_2 + 7y_3.$$

Combining like terms in x_1 and x_2 :

$$x_1(y_1 + 2y_2 - y_3) + x_2(y_1 - y_2 + 3y_3) \leq 10y_1 + 5y_2 + 7y_3.$$

We want this inequality to hold for all x_1, x_2 *free*. If $y_1 + 2y_2 - y_3 > 3$, then by choosing x_1 large and positive, we could make the left side arbitrarily large, violating the bound. If $y_1 + 2y_2 - y_3 < 3$, by choosing x_1 large and negative, we break the bound. The only way to always hold the inequality is to have exact equality:

$$y_1 + 2y_2 - y_3 = 3.$$

Similarly, for x_2 :

If $y_1 - y_2 + 3y_3 > -2$, we could send x_2 to $-\infty$ to break the bound. If $y_1 - y_2 + 3y_3 < -2$, we could send x_2 to ∞ . Again, the only safe choice is equality:

$$y_1 - y_2 + 3y_3 = -2.$$

Thus, the dual constraints must be equalities. The dual problem is:

$$\begin{array}{ll} \text{Dual (D):} & \min 10y_1 + 5y_2 + 7y_3 \\ & \text{subject to } y_1 + 2y_2 - y_3 = 3 \\ & y_1 - y_2 + 3y_3 = -2 \\ & y_1, y_2, y_3 \geq 0. \end{array}$$

From our derivation, the dual problem (D) becomes:

$$\min \mathbf{b}^\top \mathbf{y} = 10y_1 + 5y_2 + 7y_3$$

subject to:

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix},$$

and

$$y_1, y_2, y_3 \geq 0.$$

Summarizing, we have

Primal (P)	Dual (D)
$\begin{array}{ll} \max & [3 \quad -2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \text{s.t.} & \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 10 \\ 5 \\ 7 \end{bmatrix} \\ & x_1, x_2 \text{ free} \end{array}$	$\begin{array}{ll} \min & [10 \quad 5 \quad 7] \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ \text{s.t.} & \begin{bmatrix} 1 & 2 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \\ & y_1, y_2, y_3 \geq 0 \end{array}$

Here we see the pattern: having $\mathbf{x}$ free in the primal forces the dual constraints to be equalities. In summary:

- Primal inequalities with $\mathbf{x}$ free $\Rightarrow$ Dual equality constraints.
- Primal equalities with $\mathbf{x} \geq 0 \Rightarrow$ Dual inequalities and $\mathbf{y}$ free.

These relationships ensure that by choosing appropriate dual multipliers, we can always create a valid bound on the primal objective.

The following table can help remember the above.

Primal (min)	Dual (max)
$\geq$ constraint	≥ 0 variable
$\leq$ constraint	≤ 0 variable
$=$ constraint	free variable
≥ 0 variable	$\leq$ constraint
≤ 0 variable	$\geq$ constraint
free variable	$=$ constraint

Complicated Dual

Consider the following primal linear program:

Primal (P)

$$\begin{array}{ll}
 \max & 2x_1 + x_2 \\
 \text{s.t.} & x_1 + x_2 = 2 \quad (1) \\
 & 2x_1 - x_2 \geq 3 \quad (2) \\
 & x_1 - x_2 \leq 1 \quad (3) \\
 & x_1 \geq 0, x_2 \text{ free}
 \end{array}$$

Dual (D)

Let $y_1 \in \mathbb{R}$, $y_2 \geq 0$, $y_3 \leq 0$ be dual variables for constraints (1), (2), and (3), respectively.

$$\begin{array}{ll}
 \min & 2y_1 + 3y_2 + 1y_3 \\
 \text{s.t.} & y_1 + 2y_2 + y_3 \geq 2 \quad (\text{from } x_1 \geq 0) \\
 & y_1 - y_2 - y_3 = 1 \quad (\text{from } x_2 \text{ free}) \\
 & y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0
 \end{array}$$

To derive the dual, we introduce a dual variable for each constraint:

- Let y_1 be the dual variable for the equality constraint $x_1 + x_2 = 2$
- Let y_2 be the dual variable for the $\geq$ constraint $2x_1 - x_2 \geq 3$
- Let y_3 be the dual variable for the $\leq$ constraint $x_1 - x_2 \leq 1$

Explanation of Dual Structure:

- The primal has one equality constraint. Therefore, the corresponding dual variable y_1 is unrestricted in sign.
- The second primal constraint is a $\geq$ inequality, so its corresponding dual variable y_2 must be ≥ 0 .
- The third primal constraint is a $\leq$ inequality, so its corresponding dual variable y_3 must be ≤ 0 .
- The dual constraints come from the coefficients of the primal variables. Since the primal is a maximization, the dual constraints are $\geq$.

- The dual objective coefficients come from the right-hand sides of the primal constraints: 2, 3, 1.

Summary: This example demonstrates how mixed constraint types in the primal ($=, \geq, \leq$) lead to a dual with mixed variable sign restrictions. The primal variable bounds also affect whether dual constraints are inequalities or equalities. Understanding these mappings is crucial for correctly constructing dual problems.

Complicated Dual Derivation: A Second Look via Aggregation

Consider the following primal linear program:

$$\begin{aligned}
 &\text{maximize} && 2x_1 + x_2 \\
 &\text{subject to} && x_1 + x_2 = 2 \quad (\text{constraint 1}) \\
 & && 2x_1 - x_2 \geq 3 \quad (\text{constraint 2}) \\
 & && x_1 - x_2 \leq 1 \quad (\text{constraint 3}) \\
 & && x_1 \geq 0
 \end{aligned}$$

We now derive the dual using the **aggregation method**.

Step 1: Multiply constraints by dual variables.

We introduce dual variables y_1, y_2, y_3 corresponding to constraints 1, 2, and 3, respectively.

We aggregate the constraints by forming a linear combination:

$$y_1(x_1 + x_2 = 2) + y_2(2x_1 - x_2 \geq 3) + y_3(x_1 - x_2 \leq 1)$$

Before combining, we must ensure all constraints are oriented as $\leq$ inequalities, because we're maximizing in the primal. So:

- Equality constraint: no change needed, but equality implies $y_1 \in \mathbb{R}$ - $\geq$: multiply by $y_2 \geq 0$ - $\leq$: multiply by $y_3 \leq 0$

Now aggregate:

$$\begin{aligned}
 &y_1(x_1 + x_2) + y_2(2x_1 - x_2) + y_3(x_1 - x_2) \\
 &= (y_1 + 2y_2 + y_3)x_1 + (y_1 - y_2 - y_3)x_2
 \end{aligned}$$

And the aggregated right-hand side is:

$$2y_1 + 3y_2 + 1y_3$$

So the aggregated constraint is:

$$(y_1 + 2y_2 + y_3)x_1 + (y_1 - y_2 - y_3)x_2 \leq 2y_1 + 3y_2 + y_3$$

Step 2: Ensure this constraint dominates the primal objective.

To upper-bound the primal objective $2x_1 + x_2$, we want:

$$(y_1 + 2y_2 + y_3)x_1 + (y_1 - y_2 - y_3)x_2 \geq 2x_1 + x_2 \quad \text{for all feasible } x_1, x_2 \geq 0$$

Since $x_1, x_2 \geq 0$, domination means:

$$y_1 + 2y_2 + y_3 \geq 2 \quad (\text{coefficient on } x_1)$$

$$y_1 - y_2 - y_3 \geq 1 \quad (\text{coefficient on } x_2)$$

Step 3: Formulate the dual.

The dual objective is the right-hand side of the aggregated constraint:

$$\text{minimize } 2y_1 + 3y_2 + y_3$$

With constraints:

$$y_1 + 2y_2 + y_3 \geq 2$$

$$y_1 - y_2 - y_3 \geq 1$$

$$y_2 \geq 0 \quad (\text{from } \geq \text{ primal constraint})$$

$$y_3 \leq 0 \quad (\text{from } \leq \text{ primal constraint})$$

$$y_1 \in \mathbb{R} \quad (\text{from } = \text{ constraint})$$

Conclusion:

This derivation shows how:

- *Constraint direction* in the primal determines the sign of the corresponding dual variable.
- *Sign of a primal variable* (e.g., $x_1 \geq 0$) determines the direction of the dual constraint: to ensure domination for non-negative variables, we require the dual combination to be $\geq$ the primal coefficients.
- The *dual objective* comes from aggregating the right-hand sides, weighted by the dual variables.

This example illustrates how careful sign and inequality management gives rise to a valid dual that bounds the primal objective from above.

Question: What happens if variable x_1 instead was constrained as $x_1 \leq 0$?

11.4 Complementary Slackness

So far, we have established that the primal and dual problems are closely linked, and their optimal objective values coincide under strong duality. However, **how can we identify optimal solutions for both problems simultaneously?** The answer lies in *complementary slackness*, a crucial condition for verifying optimality.

11.4.1. Definition of Complementary Slackness

Definition 11.6: Complementary Slackness

Complementary slackness conditions state that if $\mathbf{x}^*$ and $\mathbf{y}^*$ are optimal solutions for the primal and dual problems, then the following must hold:

1. For each primal constraint i , either the constraint is tight and/or the corresponding dual variable is zero:

$$\text{either } \mathbf{a}_i^\top \mathbf{x}^* = b_i \quad \text{or} \quad y_i^* = 0, \quad \forall i.$$

Equivalently,

$$y_i^* (\mathbf{a}_i^\top \mathbf{x}^* - b_i) = 0.$$

2. For each dual constraint j , either the constraint is tight and/or the corresponding primal variable is zero:

$$\text{either } (\mathbf{A}^\top \mathbf{y}^*)_j = c_j \quad \text{or} \quad x_j^* = 0, \quad \forall j.$$

Equivalently,

$$x_j^* (c_j - (\mathbf{A}^\top \mathbf{y}^*)_j) = 0.$$

Together, these conditions ensure that at optimality, only the active constraints have positive dual variables, and only the active dual constraints have positive primal variables.

Theorem 11.7: Primal Dual Optimality

A primal solution $\mathbf{x}^*$ and a dual solution $\mathbf{y}^*$ are **optimal** if and only if:

1. $\mathbf{x}^*$ is **feasible** in the primal problem.
2. $\mathbf{y}^*$ is **feasible** in the dual problem.
3. **Complementary slackness holds** between the primal and dual solutions.

This set of conditions allows us to efficiently check whether a given solution is optimal without needing to compare all feasible solutions explicitly.

11.4.2. Interpreting Complementary Slackness

Complementary slackness provides a way to determine **which constraints are binding** at the optimal solution. If we solve either the primal or the dual problem, we can use these conditions to deduce the optimal solution of the other problem.

Example 11.8: The Bakery Problem Revisited

Consider the bakery problem from the previous section. Recall the primal and dual formulations:

Primal Problem:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4 \quad (\text{flour constraint}) \\ & 2x_1 + x_2 \leq 5 \quad (\text{sugar constraint}) \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual Problem:

$$\begin{aligned} \min \quad & 4y_1 + 5y_2 \\ \text{subject to} \quad & y_1 + 2y_2 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2 \geq 0. \end{aligned}$$

Suppose we are given the optimal solutions:

$$x_1^* = 1, \quad x_2^* = 3, \quad y_1^* = 1, \quad y_2^* = 1.$$

To verify optimality, we check complementary slackness:

1. The first constraint in the primal ($x_1 + x_2 \leq 4$) is tight ($1 + 3 = 4$), so y_1^* can be positive.
2. The second constraint ($2x_1 + x_2 \leq 5$) is tight ($2(1) + 3 = 5$), so y_2^* can be positive.
3. Checking the dual constraints:

$$y_1 + 2y_2 = 1 + 2(1) = 3, \quad (\text{tight, so } x_1^* > 0 \text{ is allowed})$$

$$y_1 + y_2 = 1 + 1 = 2, \quad (\text{tight, so } x_2^* > 0 \text{ is allowed}).$$

Since complementary slackness holds, the given values are indeed optimal solutions.

11.4.3. Using Complementary Slackness to Solve Problems

Often, we do not have full information about an optimal solution. Instead, we might have partial information or only one of the optimal solutions (primal or dual). Complementary slackness allows us to deduce missing information.

Example 11.9: Finding Missing Dual Variables

Suppose we solve the primal problem and find the optimal solution:

$$x_1^* = 1, \quad x_2^* = 3.$$

Using complementary slackness, we determine y^* :

- $x_1^* > 0$, so we must have the first dual constraint **tight**, so $y_1 + 2y_2 = 3$.
- $x_2^* > 0$, so we must have the second dual constraint **tight**, so $y_1 + y_2 = 2$.

Solving the system:

$$\begin{aligned} y_1 + 2y_2 &= 3, \\ y_1 + y_2 &= 2. \end{aligned}$$

Solving for y_1 and y_2 gives $y_1^* = 1, y_2^* = 1$.

Thus, complementary slackness allowed us to determine the dual solution directly from the primal solution.

11.4.4. Summary of Key Takeaways

- Complementary slackness characterizes optimality in primal-dual pairs.
- If a primal constraint is not tight, the corresponding dual variable is zero.
- If a dual constraint is not tight, the corresponding primal variable is zero.
- These conditions allow us to find missing solutions and verify optimality efficiently.

Bakery 2.0 Example Revisited

Example 11.10: Verifying Optimality with Complementary Slackness

We are given the primal and dual optimal solutions:

- Primal: $x_1 = 2, x_2 = 0, x_3 = 8$
- Dual: $y_1 = 0, y_2 = 10, y_3 = 10$

We will verify that these satisfy complementary slackness.

Step 1: Check tightness of primal constraints

- **Flour:** $8x_1 + 6x_2 + x_3 = 8(2) + 6(0) + 8 = 24$. This is strictly less than 48, so the constraint is not tight $\Rightarrow$ corresponding dual variable $y_1 = 0$.
- **Sugar:** $4x_1 + 2x_2 + 1.5x_3 = 4(2) + 2(0) + 1.5(8) = 8 + 12 = 20$. Tight $\Rightarrow y_2$ can be positive.
- **Baking Time:** $2x_1 + 1.5x_2 + 0.5x_3 = 4 + 0 + 4 = 8$. Tight $\Rightarrow y_3$ can be positive.

Step 2: Check tightness of dual constraints where $x_j > 0$

- $x_1 = 2 > 0$: The corresponding dual constraint must be tight:

$$8y_1 + 4y_2 + 2y_3 = 0 + 40 + 20 = 60$$

- $x_2 = 0$: No condition required.
- $x_3 = 8 > 0$: The dual constraint for muffins must be tight:

$$y_1 + 1.5y_2 + 0.5y_3 = 0 + 15 + 5 = 20$$

Since all complementary slackness conditions are satisfied, both solutions are optimal and duality holds (objective values match at 280).

Example 11.11: Recovering Dual Variables via Complementary Slackness

Suppose we only know the optimal primal solution:

$$x_1 = 2, \quad x_2 = 0, \quad x_3 = 8.$$

We use complementary slackness to recover the optimal dual solution.

Step 1: Use tightness conditions from primal

- $x_1 > 0$: Dual constraint for cakes must be tight:

$$8y_1 + 4y_2 + 2y_3 = 60 \tag{1}$$

- $x_3 > 0$: Dual constraint for muffins must be tight:

$$y_1 + 1.5y_2 + 0.5y_3 = 20 \tag{2}$$

Step 2: Use slackness in primal constraints to find zero dual variables

- Flour constraint is not tight $\Rightarrow y_1 = 0$

Step 3: Substitute $y_1 = 0$ into equations (1) and (2):

$$\begin{aligned} 4y_2 + 2y_3 &= 60 \\ 1.5y_2 + 0.5y_3 &= 20 \end{aligned}$$

Step 4: Solve the system

Multiply (2') by 2:

$$3y_2 + y_3 = 40$$

Now solve with (1'):

$$\begin{aligned} 4y_2 + 2y_3 &= 60 \\ 3y_2 + y_3 &= 40 \end{aligned}$$

Multiply second equation by 2:

$$6y_2 + 2y_3 = 80$$

Subtract:

$$(6y_2 + 2y_3) - (4y_2 + 2y_3) = 80 - 60 \Rightarrow 2y_2 = 20 \Rightarrow y_2 = 10$$

Substitute back: $3(10) + y_3 = 40 \Rightarrow y_3 = 10$

Result: $y_1 = 0, y_2 = 10, y_3 = 10$

Thus, the dual solution is fully recovered using complementary slackness.

11.5 Exercises

Exercise 11.1 Bakery Production and Duality Interpretation A bakery produces cakes (x_1), cookies (x_2), and muffins (x_3) to maximize profit:

$$\text{maximize } 60x_1 + 30x_2 + 20x_3$$

Subject to the following resource constraints:

- **Flour (≤ 48 kg):** $8x_1 + 6x_2 + x_3 \leq 48$
- **Sugar (≤ 20 kg):** $4x_1 + 2x_2 + 1.5x_3 \leq 20$
- **Baking Time (≤ 8 hrs):** $2x_1 + 1.5x_2 + 0.5x_3 \leq 8$
- **Non-negativity:** $x_1, x_2, x_3 \geq 0$

The optimal primal solution is:

$$x_1 = 2, \quad x_2 = 0, \quad x_3 = 8$$

Dual Linear Program:

$$\begin{aligned} &\text{minimize } 48y_1 + 20y_2 + 8y_3 \\ &\text{subject to } \begin{aligned} &8y_1 + 4y_2 + 2y_3 \geq 60 \\ &6y_1 + 2y_2 + 1.5y_3 \geq 30 \\ &y_1 + 1.5y_2 + 0.5y_3 \geq 20 \\ &y_1, y_2, y_3 \geq 0 \end{aligned} \end{aligned}$$

The optimal dual solution is:

$$y_1 = 0, \quad y_2 = 10, \quad y_3 = 10$$

Answer the following questions:

1. **Feasibility Check:** Verify that the primal solution satisfies all constraints. Then check that the dual solution satisfies all dual constraints.
2. **Complementary Slackness:** Determine which constraints are tight in the primal and which are tight in the dual. Use complementary slackness to verify consistency between the primal and dual optimal solutions.
3. **Objective Value Check:** Compute the primal and dual objective values using the given solutions and verify that they are equal.
4. **Economic Interpretation (Shadow Prices):** Using the dual solution (y_1, y_2, y_3) , interpret the shadow price of each resource (flour, sugar, baking time). For example, what does $y_2 = 10$ tell you about the value of an additional kg of sugar?
5. **Sensitivity Analysis:** How would the optimal profit change if you had:
 - (a) One more kg of flour?
 - (b) One more kg of sugar?
 - (c) One more hour of baking time?

Use the shadow prices to justify your answers.

Exercise 11.2 Meal Kit Optimization and Duality You are the operations manager at a startup that assembles and ships healthy meal kits. Each kit appeals to different customer preferences: vegetarian, high-protein, quick-prep, etc. You want to decide how many of each kit to produce today to maximize profit.

There are six meal kit options:

Variable	Description	Profit per Kit
x_1	Classic Chicken	\$3
x_2	Vegan Delight	\$2
x_3	Protein Boost	\$4
x_4	Quick & Light	\$1
x_5	Gourmet Chef Special	\$5
x_6	Breakfast Box	\$2

Each kit requires time in two key areas:

- **Kitchen Prep Time (in hours):** chopping, marinating, pre-measuring, etc.
- **Packaging Labor (in hours):** boxing, labeling, sealing, etc.

Your available resources:

- 10 hours of Kitchen Time
- 12 hours of Packaging Labor

<i>Meal Kit</i>	<i>Kitchen Time</i>	<i>Packaging Time</i>
x_1 : <i>Classic Chicken</i>	1	2
x_2 : <i>Vegan Delight</i>	2	0
x_3 : <i>Protein Boost</i>	1	0
x_4 : <i>Quick & Light</i>	0	1
x_5 : <i>Gourmet Chef Special</i>	1	3
x_6 : <i>Breakfast Box</i>	0	1

The primal linear program is:

$$\begin{aligned}
 &\text{maximize} && 3x_1 + 2x_2 + 4x_3 + x_4 + 5x_5 + 2x_6 \\
 &\text{subject to} && x_1 + 2x_2 + x_3 + x_5 \leq 10 \\
 &&& 2x_1 + x_4 + 3x_5 + x_6 \leq 12 \\
 &&& x_1, x_2, x_3, x_4, x_5, x_6 \geq 0
 \end{aligned}$$

Answer the following:

1. **Formulate the dual linear program.** Identify what each dual variable represents and write down the dual objective and constraints explicitly.
2. **Graphically solve the dual problem.** Plot the dual constraints in y_1 - y_2 space and identify the feasible region. Use level curves to determine the optimal solution.
3. **Economic interpretation.** Describe the meaning of the dual solution in terms of the value of your resources.
4. **Use complementary slackness to determine the basic variables in the primal.** Based on the optimal dual solution and complementary slackness, identify which constraints in the primal are tight and hence which variables are basic.
5. **Recover an optimal primal solution.** Use the tight constraints and system of equations to solve for the primal basic variables and compute the optimal objective value.

Exercise 11.3 *Duality with Mixed Constraint Types: Eco-Friendly Packaging Startup* You are managing production at an eco-friendly packaging startup. Your team produces various biodegradable container sets for local farms and grocery stores. Each set is assembled using a mix of labor, recycled material, and specialized equipment.

You produce three types of packaging sets:

<i>Variable</i>	<i>Packaging Set Type</i>	<i>Retail Price</i>
x_1	<i>FreshBox Standard</i>	\$6
x_2	<i>FreshBox Premium</i>	\$9
x_3	<i>Market Crate</i>	\$8

You must manage the use of three critical resources:

- **Labor Hours:** at most 100 available

- **Recycled Material:** must exactly use 240 units
- **Retail Contract Requirement:** the total retail value must be at least \$80

The profit per unit for each set is:

Packaging Set	Profit per Unit
x_1 : FreshBox Standard	\$4
x_2 : FreshBox Premium	\$6
x_3 : Market Crate	\$5

The retail revenue constraint is:

$$6x_1 + 9x_2 + 8x_3 \geq 80$$

The resource requirements for each packaging set are:

Packaging Set	Labor	Material	Equipment Time
x_1 : FreshBox Standard	2	4	1
x_2 : FreshBox Premium	4	6	2
x_3 : Market Crate	3	4	3

The primal linear program is:

$$\begin{aligned}
 &\text{maximize} && 4x_1 + 6x_2 + 5x_3 \\
 &\text{subject to} && 2x_1 + 4x_2 + 3x_3 \leq 100 \quad (\text{Labor}) \\
 & && 4x_1 + 6x_2 + 4x_3 = 240 \quad (\text{Material}) \\
 & && 6x_1 + 9x_2 + 8x_3 \geq 80 \quad (\text{Retail Revenue Commitment}) \\
 & && x_1, x_2, x_3 \geq 0
 \end{aligned}$$

Tasks:

1. **Formulate the dual linear program.** Carefully handle the signs of each constraint and variable in the dual. What does each dual variable represent? What type of constraint does each dual constraint become?
2. **Solve the dual graphically or using software.** If the dual is two-dimensional, solve it by plotting its feasible region and finding the optimal value. Otherwise, use a solver like PuLP or graphical analysis of intersections.
3. **Use complementary slackness.** Based on the optimal dual solution, determine which primal constraints are tight. Use this to identify likely basic variables in the optimal primal solution.
4. **Recover the optimal primal solution.** Solve the resulting system using the tight constraints to find the primal solution, and compute the optimal total profit.

Exercise 11.4 Consider the optimization problem

$$\begin{array}{llllll}
\max & x_1 & + & 4x_2 & + & 9x_3 & + & 16x_4 \\
\text{s.t.} & x_1 & & & & & & \leq 1 & (\text{Constraint 1}) \\
& & & x_2 & & & & \leq 1 & (\text{Constraint 2}) \\
& & & & & x_3 & & \leq 1 & (\text{Constraint 3}) \\
& & & & & & x_4 & \leq 1 & (\text{Constraint 4}) \\
& x_1 & + & 2x_2 & + & 3x_3 & + & 4x_4 & \leq 8 & (\text{Constraint 5}) \\
& & & & & & & & & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0
\end{array}$$

(a.) What is the dual of this linear program?

(b.) Explain how the dual can be used to prove that a solution is optimal?

(c.) The optimal solution to this linear program is $(x_1, x_2, x_3, x_4) = (0, \frac{1}{2}, 1, 1)$ with objective value 27.

Using duality theory, prove that this is an optimal solution.

(Hint: Recall that complimentary slackness implies that dual variables corresponding to constraints 1 and 2 must be 0).

Exercise 11.5

Consider the following primal-dual pair:

Primal Problem:

$$\begin{array}{ll}
\text{Maximize} & 5x_1 + 4x_2 \\
\text{subject to} & 2x_1 + x_2 \leq 8 \\
& x_1 + 3x_2 \leq 9 \\
& x_1, x_2 \geq 0
\end{array}$$

Dual Problem:

$$\begin{array}{ll}
\text{Minimize} & 8y_1 + 9y_2 \\
\text{subject to} & 2y_1 + y_2 \geq 5 \\
& y_1 + 3y_2 \geq 4 \\
& y_1, y_2 \geq 0
\end{array}$$

(a) Suppose you are given a dual solution $\mathbf{y}^* = (1, 1)$. Use complementary slackness to determine the values of x_1^* and x_2^* .

(b) Is the resulting primal solution feasible?

(c) Does complementary slackness hold? Is this a valid optimal solution pair?

Exercise 11.6

Suppose a linear program has the following primal and dual formulations:

Primal:

$$\begin{array}{ll}\text{Maximize} & 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 5 \\ & x_1, x_2 \geq 0\end{array}$$

Dual:

$$\begin{array}{ll}\text{Minimize} & 4y_1 + 5y_2 \\ \text{subject to} & y_1 + 2y_2 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2 \geq 0\end{array}$$

Suppose the optimal primal solution is $x_1^* = 2, x_2^* = 0$.

- (a) Write down the complementary slackness conditions for this problem.
- (b) Use them to derive conditions on y_1 and y_2 , and solve for the dual optimal solution $\mathbf{y}^*$.
- (c) Verify whether the resulting $\mathbf{y}^*$ is dual feasible.